A PROOF OF THE REINHARDT CONJECTURE

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Abstract. In 1934 Reinhardt conjectured that the smoothed octagon minimizes the greatest packing density among centrally symmetric convex disks in the plane. This disk is a regular octagon whose corners are replaced by hyperbolic arcs, each asymptotic to the two neighbouring side lines. Its greatest packing density is $\delta_{\mathrm{oct}} = (8-4\sqrt{2}-\log 2)/(2\sqrt{2}-1) = 0.902414\ldots$. We prove the conjecture: every centrally symmetric convex disk has greatest packing density at least δ_{oct}, with equality exactly for the images of the smoothed octagon under invertible affine maps. The proof uses Hales's optimal control formulation, in which the cost of a trajectory is the area of the corresponding disk. Using Hales and Vajjha's results on minimizers, we reduce a minimizing trajectory to finitely many pieces of constant control. Five cases cover the possible switching patterns, and each case is excluded for a minimizer with six or more pieces. The remaining four- and five-piece motions can be compared exactly; the minimum is attained only by the octagon's motion. The main theorems are formalized in Lean 4 with Mathlib, and their proofs are checked by Lean's kernel.

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1. Introduction

A *convex disk* is a compact convex subset of the plane with nonempty interior. For a convex disk K let $\delta(K)$ be its greatest packing density: the supremum of the densities of the packings of the plane by congruent copies of K (Definition 2.1). The circular disk has greatest packing density $\pi/\sqrt{12} = 0.906899\ldots$. A disk is centrally symmetric if it is invariant under reflection through a point. In 1934 Reinhardt [28] asked which centrally symmetric convex disk minimizes $\delta(K)$. He proposed the *smoothed octagon* K_{oct}: the regular octagon with each corner replaced by an arc of the hyperbola tangent to the two incident sides and asymptotic to the two neighbouring side lines (Definition A.1, Figure 2). Its greatest packing density is

$$\delta_{\text{oct}} = \frac{8 - 4\sqrt{2} - \log 2}{2\sqrt{2} - 1} = 0.902414\ldots.$$

We prove that the smoothed octagon minimizes the greatest packing density, uniquely up to invertible affine maps. Such a map has the form $x \mapsto Lx + b$, where L is an invertible linear map and b is a translation vector.

Theorem 1.1. (a) *The smoothed octagon K_{oct} is a centrally symmetric convex disk with $\delta(K_{\text{oct}}) = \delta_{\text{oct}}$.*

(b) *Every centrally symmetric convex disk K satisfies $\delta(K) \geq \delta_{\text{oct}}$.*

(c) *A centrally symmetric convex disk K satisfies $\delta(K) = \delta_{\text{oct}}$ if and only if $K = A(K_{\text{oct}})$ for an invertible affine map A of the plane.*

Part (b) is Reinhardt's conjecture. Part (a), due to Reinhardt and Mahler [28, 22], is proved here in the form needed for the subsequent argument. Part (c) gives the uniqueness statement. Since δ is invariant under invertible affine maps, its *if* direction follows from (a).

1.1. History.

Minkowski [23, 24] studied the densest lattice packings of a convex body and described the relevant lattices for a convex disk through inscribed hexagons. The question of the least dense centrally symmetric disk appears in Blaschke's book [5]. According to Hales and Vajjha [13, §1.2], Blaschke credits it to Courant, who expected the circle to be the answer. Reinhardt [28] proved that a least dense disk exists and has no corners. He also showed that every boundary point is the midpoint of a side of a circumscribed hexagon of least area. His smoothed octagon packs worse than the circle, but he left open whether it is the least dense disk.

Mahler, unaware of Reinhardt's work, reached the same conjecture in three papers of 1947 [20, 21, 22]. He reduced the problem to one in the calculus of variations and proved that a solution exists and is not an ellipse. He expected the boundary of an extreme disk to consist of line segments and hyperbolic arcs [21, p. 117]. Hales and Vajjha call this Mahler's first conjecture. Mahler also showed that the regular octagon minimizes density among centrally symmetric octagons. He constructed the smoothed octagon and computed its density [22, §12]. Ledermann and Mahler [18] named the

smoothed octagon, and found that the smoothed decagon packs better, "a result which seems to support Reinhardt's conjecture".

These authors studied lattice packings. Fejes Tóth [10] and Rogers [29] proved that, for a centrally symmetric convex disk, packings by congruent copies and packings by translates have the same greatest density as lattice packings (Theorem 2.2). Reinhardt's lattice problem therefore extends to all packings; see the surveys of G. Fejes Tóth [9] and of Bezdek and Kuperberg [4]. Lower bounds for $\delta(K)$ followed: $\sqrt{3}/2 = 0.866\ldots$ by Mahler [19], 0.88 by Ennola [7], and 0.892656 by Tammela [30]. Nazarov [25] proved that the smoothed octagon is a local minimum of δ in the Hausdorff metric.

Hales recast the problem twice. In [14] he formulates it as a variational problem with an integral cost and prescribed boundary conditions. The corresponding Euler–Lagrange equations have only the circle as a solution, up to affine maps, but the circle is not a minimizer. He also proves that a piecewise analytic minimizer is a smoothed polygon. In [15] it is an optimal control problem: the motion of six boundary contact points is governed by differential equations, and the area of the disk is the quantity to be minimized. Hales identifies the motions associated with smoothed regular $(6k+2)$-gons, proves that the octagon is a strict local minimum, and gives conditions under which an optimal motion has only finitely many changes of control. Vajjha's thesis [32] and the book of Hales and Vajjha [13] continue this work. The book confines the state of a global minimizer to a compact domain (Theorem A.12) and proves Mahler's first conjecture: a worst disk is a smoothed polygon, with finitely many straight segments and hyperbolic arcs. It leaves the Reinhardt conjecture open, and notes that a classification of the possible optimal motions by the smoothed $(6k+2)$-gons would complete it [13, §1.3]; see also [3].

Related extremal packing problems have also been studied. In three dimensions, Kallus [16] proved that the ball is a local pessimum for packing. For convex disks that need not be centrally symmetric, Fáry [8] proved that triangles have the least lattice packing density, and Kallus [17] proved that the regular heptagon is a local minimum of the double-lattice packing density.

1.2. **The strategy of the lower bound.** The first reduction concerns *lattice packings*, in which the copies are translates of the disk and their centres are the integer linear combinations of two fixed independent vectors. The parallelogram spanned by those vectors is a *lattice cell*. For a centrally symmetric convex disk, the theorem of Fejes Tóth and Rogers says that a densest packing can be chosen in this form. Its density is the area of the disk divided by the cell area. An invertible linear map changes both areas by the same factor. We may therefore normalize the cell area to $\sqrt{12}$ and seek a lower bound for the area of the disk.

A *worst disk* is a centrally symmetric convex disk that minimizes the greatest packing density. Such a disk exists. Its densest lattice packings admit six boundary contact points that move continuously around the disk

while preserving the cell area. These points form an inscribed hexagon and together trace the entire boundary. Hales's control problem describes this motion by differential equations. The *control* specifies the allowed curvatures of the contact curves, and the *cost* of the motion is the area of the traced disk. An *admissible trajectory* is a motion satisfying these equations and the closing conditions: after one period, each contact point has reached the starting position of the next one. The precise formulation is in Definition A.4.

The smoothed octagon supplies an admissible trajectory of cost

$$A_{\mathrm{oct}} = \sqrt{12}\,\delta_{\mathrm{oct}}.$$

The geometric problem is equivalent to the following bound.

Theorem 1.2. *Every admissible trajectory of Hales's control problem has cost at least* A_{oct}.

The proof reduces the possible motions in three stages.

(1) *A minimizer has finitely many pieces.* A trajectory of least cost is a *global minimizer*. Its control switches finitely often among three distinguished choices, labelled 0, 1 and 2. Each interval of constant control is a *piece*, and a change between consecutive pieces is a *switch*. The equations have explicit solutions on each piece, so the motion is determined by a finite sequence of labels and durations.

(2) *Six or more pieces are impossible.* At each switch the label advances or retreats in the cyclic order $0, 1, 2$. We call these switches *forward* and *reverse*, respectively. A pattern with both directions is called *mixed*. The sequences fall into five classes: two with only one direction of switching and three mixed classes. We exclude each class for a minimizer with six or more pieces. Some patterns contradict local necessary conditions for optimality; the others force the cost of a complete period above the octagon's. These exclusions form the main part of the proof.

(3) *The remaining motions can be compared exactly.* With at most five pieces, the closing equations leave a four-piece family and a five-piece family. The four-piece family has least cost A_{oct}, attained only by the octagon's motion. Every member of the five-piece family costs strictly more.

For a worst disk K_0, the cost of its normalized motion is $\sqrt{12}\,\delta(K_0)$. The comparison therefore gives

$$\sqrt{12}\,\delta(K_0) \geq A_{\mathrm{oct}} = \sqrt{12}\,\delta_{\mathrm{oct}}. \tag{1.1}$$

Every other centrally symmetric convex disk has density at least $\delta(K_0)$, which proves the lower bound in Theorem 1.1(b).

The exclusions relate local switching conditions to the cost of a complete period. The estimates use monotonicity, convexity and integral bounds for the logarithm. Some mixed cases also require polynomial sign checks, detailed in the corresponding appendix.

1.3. **The strategy of the equality case.** For uniqueness, let K be a centrally symmetric convex disk with $\delta(K) = \delta_{\mathrm{oct}}$. The lower bound shows that, after translation of its centre to the origin, this disk is a worst disk. Its trajectory is therefore a global minimizer of cost A_{oct}.

The exclusions leave at most five pieces. The equality statement in the final comparison then identifies this trajectory with the octagon's motion, up to a choice of starting point and a cyclic permutation of the three control labels. Reversing these changes and the initial linear normalization shows that the given disk K is an invertible affine image of the smoothed octagon (Theorem 5.3). The argument applies to the trajectory of K itself and hence to every disk attaining equality.

Conversely, invertible affine maps preserve the greatest packing density of centrally symmetric convex disks (Proposition 5.4). Every affine image of the smoothed octagon consequently has density δ_{oct}. This proves the equality characterization in Theorem 1.1(c).

1.4. **Formal verification.** Theorems 1.1 and 1.2 are formalized in the proof assistant Lean 4 [6], using its mathematical library Mathlib [31]. The proofs are available in the repository [35], and Lean's kernel checks every step. The three parts of Theorem 1.1 are stated in Mathlib's vocabulary in a file that imports only Mathlib, and the proof uses only Lean's standard axioms. It builds on formalizations of the book of Hales and Vajjha [36] and of the Pontryagin maximum principle [34], and it is registered in the Palomar registry [26]. Appendix J describes the formal statements, their verification and the corresponding Lean declarations.

1.5. **Organization.** Section 2 explains the passage from packing density to the control problem. Section 3 reduces a minimizer to finitely many pieces and classifies its switches. Section 4 gives the proof idea for each of the five exclusions. Section 5 compares the remaining motions and completes the lower bound and the equality case.

The appendices supply the details in the same order. Appendix A gives the geometric formulation and the reduction to finite normal periods. Appendix B develops the formulas shared by the case arguments. Each of the five cases then has its own appendix: Appendices C to G. The two continuation arguments within the last mixed case are subsections of Appendix G. Appendices H and I prove the final comparisons and recover the equality disks. Appendix J describes the formal verification. A notation table is provided in Appendix A.1.7.

2. From packing density to the control problem

Fixing the area of a densest lattice cell reduces the packing problem to an area minimization. For a worst disk, the contact points of the corresponding packings form a moving hexagon of constant area. Its motion defines a

control problem whose cost is the area of the disk. The coordinate formulas and regularity conditions are given in Appendix A.1.

2.1. **Packing densities.** A *convex disk* is a compact convex set $K \subset \mathbb{R}^2$ with nonempty interior. It is *centrally symmetric* if it has a centre c, so that $2c - x \in K$ for every $x \in K$. A *congruent copy* of K is its image under a Euclidean isometry, which may reverse the orientation. A *packing* is a family of convex disks with pairwise disjoint interiors. Let B_r be the closed disk of radius r about the origin.

Definition 2.1 (Greatest packing density)**.** The *density* of a packing $\mathcal{P}$ exists if the limit

$$\delta(\mathcal{P}) = \lim_{r\to\infty} \frac{1}{\operatorname{area}(B_r)} \sum_{K'\in\mathcal{P}} \operatorname{area}(K' \cap B_r)$$

exists and is finite, that is, if the upper and lower limits are equal and finite. The *greatest packing density* $\delta(K)$ of a convex disk K is the supremum of $\delta(\mathcal{P})$ over the packings $\mathcal{P}$ by congruent copies of K whose density exists [13, Def. 2.1.1].

A *lattice* is a discrete subgroup $L = \mathbb{Z}b_0 + \mathbb{Z}b_1$ of $\mathbb{R}^2$ with b_0, b_1 linearly independent; its determinant is $\det L = |\det(b_0, b_1)|$. If the translates $K + v$, $v \in L$, form a packing, its density exists and equals $\operatorname{area}(K)/\det L$. The *greatest lattice packing density* $\delta_L(K)$ is the supremum of these values.

Theorem 2.2 (Fejes Tóth [10], Rogers [29])**.** *For every centrally symmetric convex disk K, $\delta(K) = \delta_L(K)$.*

It therefore suffices to consider lattice packings of centrally symmetric disks. A linear map carries lattice packings to lattice packings and multiplies areas and determinants by the same factor, so $\delta(K)$ is invariant under invertible affine maps (Proposition 5.4). The *minimax density* $\delta_{\min}$ is the infimum of $\delta(K)$ over all centrally symmetric convex disks, and a *worst disk* is a centrally symmetric convex disk, centred at the origin, with $\delta(K) = \delta_{\min}$. Worst disks exist, by Blaschke's selection theorem and the continuity of δ_L in the Hausdorff metric (Reinhardt [28], Mahler [21]; [13, §2.2]). In this language the conjecture says that $\delta_{\min} = \delta_{\text{oct}}$ and that the worst disks are the linear images of the smoothed octagon.

2.2. **A moving hexagon on the boundary.** A densest lattice packing can be described by six contact points on the boundary of the disk. These occur in opposite pairs and form an *affinely regular hexagon*, the image of a regular hexagon under an invertible affine map. For a worst disk, Reinhardt's theory supplies a continuous family of such hexagons whose vertices trace the entire boundary. This family describes the disk through the motion of six points.

Normalize one contact hexagon to the regular hexagon with vertices

$$s_j = (\cos(j\pi/3), \sin(j\pi/3)), \qquad j = 0, \dots, 5.$$

The associated packing lattice has cell area $\sqrt{12}$. Write $\sigma_j(t)$ for the six moving contact points, with t a parameter along the motion. Their hexagon has constant area. There is therefore a unique area-preserving linear map $g(t)$ carrying every s_j to $\sigma_j(t)$. We call $g(t)$ the *frame*. In matrix notation it belongs to $\mathrm{SL}_2(\mathbb{R})$, the group of real 2×2 matrices with determinant one.

A suitable choice of the parameter gives a number $T > 0$ such that $\sigma_j(T) = s_{j+1}$, with the index read modulo six. Thus, during $[0, T]$, each contact point traces one part of the boundary. The six parts together trace the whole boundary. This is the period used in the control problem; the individual points return to their initial positions after six such periods. The exact conditions on the contact curves are expressed by a *balanced pair*, defined in Definition A.2.

2.3. **The state, the control and the cost.** The velocity of the frame is recorded by the matrix

$$X(t) = g(t)^{-1} g'(t), \qquad \text{so that} \qquad g'(t) = g(t) X(t).$$

This is the *state*. Since g preserves area, X has trace zero. We choose the time parameter so that $\det X = 1$. The geometric restrictions on the contact curves require X to lie in a specified open set of such matrices, called the *star states* (Appendix A.1.3).

Convexity requires nonnegative curvature of the contact curves. Hales expresses the allowed curvatures by a *control* $u(t) = (u_0(t), u_1(t), u_2(t))$, where

$$u_i(t) \geq 0, \qquad u_0(t) + u_1(t) + u_2(t) = 1.$$

The possible controls form a triangle, denoted U_T. Its three vertices are the coordinate vectors $\mathbf{e}_0 = (1, 0, 0)$, $\mathbf{e}_1 = (0, 1, 0)$ and $\mathbf{e}_2 = (0, 0, 1)$. A control determines the evolution of X through the second state equation in Definition A.4. At a vertex control, three contact curves trace straight segments and the other three trace hyperbolic arcs. This explains why sequences of vertex controls produce smoothed polygons.

Let I be the identity matrix and R the rotation through $\pi/3$. The closing conditions are

$$g(0) = I, \qquad g(T) = R, \qquad X(T) = R^{-1} X(0) R.$$

These conditions make the frame and its velocity compatible with relabeling the contact points. A solution of the state equations satisfying these conditions is an *admissible trajectory*, written (T, u, g, X). The terminal time T is allowed to vary, and the control is only required to be measurable. These conditions are stated precisely in Definition A.4.

The *cost* $\mathcal{C}(T, X)$ is the area integral of the six contact curves (Equation (A.2)). For a balanced disk K it satisfies

$$\mathcal{C}(T, X) = \mathrm{area}(K) = \sqrt{12}\, \delta(K).$$

Cost minimization therefore gives the least packing density in this normalization.

2.4. **The octagon as a reference motion.** Reinhardt's smoothed octagon K_{oct} has eight straight boundary segments joined tangentially by eight hyperbolic arcs. Its exact construction is given in Definition A.1. Hales represents an affine image of this disk by four pieces of constant control with labels $2, 0, 1, 2$. The pieces have equal durations. The resulting trajectory is admissible and has cost $A_{\mathrm{oct}} = \sqrt{12}\,\delta_{\mathrm{oct}} < \pi$ (Proposition A.7).

We denote the disk in this normalization by K_{H}. The disks K_{H} and K_{oct} differ by an explicit invertible linear map (Proposition I.4), so they have the same packing density. The octagon's trajectory consequently provides an upper bound for the cost of every global minimizer.

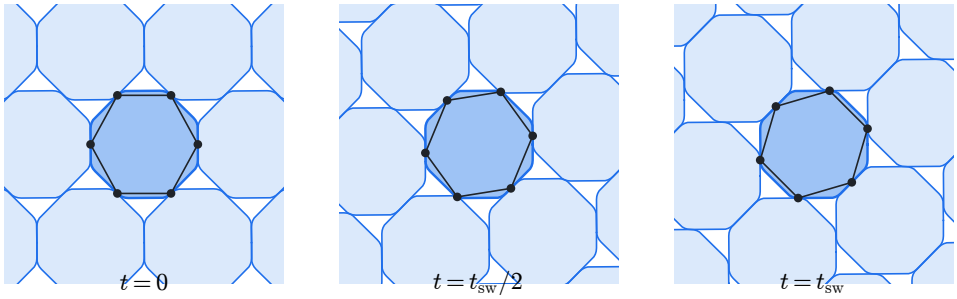
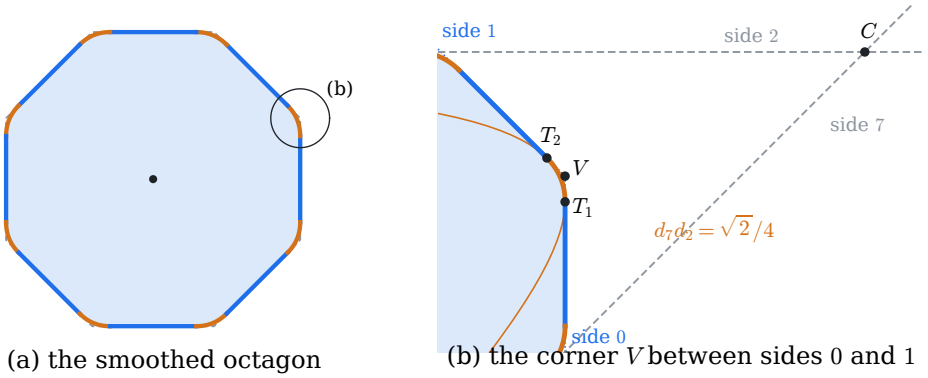


FIGURE 1. Three densest lattice packings of the smoothed octagon. The dark copy touches six neighbours in each packing. The contact points move along its boundary as the packing lattice varies, while the density remains δ_{oct}.



(a) the smoothed octagon

(b) the corner V between sides 0 and 1

FIGURE 2. (a) The smoothed octagon: straight boundary segments (blue) and hyperbolic arcs (orange) inside the regular octagon (dashed). (b) One corner is replaced by the arc from T_1 to T_2. The hyperbola is tangent to the two sides at these points and asymptotic to the neighbouring side lines, which meet at C. The coordinates are given in Appendix A.1.1.

2.5. **Equivalence of the geometric and control minima.** A *global minimizer* is an admissible trajectory whose cost is no greater than that of any other admissible trajectory, including trajectories with a different terminal time (Definition A.5). Two geometric comparisons show that the realization of a worst disk is a global minimizer.

First, a worst disk can be normalized and parametrized by an admissible trajectory whose cost is $\sqrt{12}$ times its density (Proposition A.8). Second, every admissible trajectory of cost below π traces a balanced disk with the same area (Proposition A.9). In this second comparison the cost bound ensures that the contact curves traverse the boundary once and bound a convex disk.

The octagon costs less than π. Any trajectory that could improve on the realization of a worst disk would therefore also have cost below π, and would produce a disk with a smaller packing density. This contradicts the choice of a worst disk. Its realization is consequently a global minimizer. The same comparisons prove the equivalence between the lower bound for all packing densities and the lower bound for all admissible costs (Theorem A.10).

Hales and Vajjha also show that the state of every global minimizer lies in a fixed compact set, denoted $\mathfrak{h}^{\star\star}$ (Theorem A.12). It therefore remains separated from the boundary of the allowed state domain. The last mixed case uses this restriction. We next reduce the control of a minimizer to a finite sequence of vertex choices.

3. From minimizers to finite switching patterns

An admissible control may take any value in the control triangle. A global minimizer, however, can be represented by finitely many pieces of constant vertex control. We establish this representation and classify its switching patterns. Excluding the longer patterns will leave at most five pieces.

3.1. **Finiteness and normality.** The *Pontryagin maximum principle* gives necessary conditions for an optimal control. It introduces auxiliary matrix-valued functions, called *costates*, that measure the response of the cost to variations of the trajectory. Together with the state and control, they form a *Pontryagin extremal* (Definition A.13). A scalar multiplier records the contribution of the cost. The extremal is *normal* when this multiplier is nonzero; it can then be normalized to -1.

These necessary conditions do not by themselves give a finite switching description. At certain states and costates, called the *singular locus*, they fail to distinguish the available controls. Such behaviour can lead to infinitely many switches in finite time. Our first estimate uses the bound $\mathcal{C} \leq A_{\mathrm{oct}} < \pi$ to keep a minimizing extremal away from this locus (Proposition A.20).

Away from the singular locus, one of the three vertex controls is locally excluded. The problem then reduces locally to an edge of the control triangle. An edge minimizer has at most four pieces (Proposition A.22). Covering the time interval by finitely many such neighbourhoods gives a finite number of

switches. After joining equal neighbouring pieces and choosing a switch as the starting time, the minimizer has a *vertex representation*: a finite sequence of constant vertex controls with unequal neighbouring labels, including across the closing seam. Finally, the closing conditions rule out a zero cost multiplier.

The resulting object is a *normal vertex period*: an admissible trajectory with a vertex representation and a normal Pontryagin extremal. It is *competitive* when its cost is at most A_{oct}. The finite normal representation theorem (Theorem A.30) gives such a period for every global minimizer. It has the same cost as the original trajectory and stays in $\mathfrak{h}^{\star\star}$; only its starting time has changed.

The equality case requires this representation of the original disk's motion, including its change of starting time. The precise reduction is proved in Appendix A.

3.2. **Data of a single piece.** Let N be the number of pieces. We write τ_k for their switching times and $i_k \in \mathbb{Z}/3\mathbb{Z}$ for their labels, where labels are read cyclically as $0, 1, 2$. The closing condition shifts every label by one after a period, so $i_{k+N} = i_k + 1$. For the octagon, the four labels are $2, 0, 1, 2$; the next label is 0, obtained by adding one to the first label.

The state has three positive coordinates $p_0(X), p_1(X), p_2(X)$, called *margins*, one for each vertex control (Lemma A.6). On a piece with label i, the coordinate $p_i(X)$ is constant. It is the *active margin* of that piece. Write a_k for this constant on piece k. It measures the scale of the state's motion during that piece. The other two margins, the frame and the costate all have explicit evolution formulas.

A piece can therefore be specified by its starting data and one positive duration. The state and frame must close after all the pieces have been composed. In these variables, closure is a finite system of equations, and the maximum principle becomes a finite system of inequalities. Appendix B derives the formulas and their consequences for the case arguments.

3.3. **The five switching patterns.** A switch is *forward* if $i_{k+1} = i_k + 1$ and *reverse* if $i_{k+1} = i_k - 1$. Denote its sign by $+$ or $-$, respectively. These are the only possibilities because consecutive labels are unequal. A *run* is a maximal consecutive block of switches with the same sign. The signs repeat after a period, so runs are understood cyclically. A period is *mixed* if it has both signs.

Definition 3.1 (The five classes)**.** A vertex period is in class

- R if every switch is reverse;
- F if every switch is forward;
- M3 if it is mixed and some reverse run has length other than two;
- M1 if it is mixed, every reverse run has length two and every forward run has length one, so its cyclic sign sequence is $(--+)^m$ for some positive integer m;

- M2 if it is mixed, every reverse run has length two and some forward run has length at least two.

Proposition 3.2 (Partition)**.** *Every vertex representation is in one of the classes* R*,* F*,* M3*,* M1*,* M2*.*

In a mixed period, the maximum principle excludes reverse runs whose length differs from two. The remaining patterns require comparisons with the cost of a complete period.

3.4. **A conserved quantity and the cost budget.** The costates supply a quantity that links these local and global conditions. For the first costate matrix $\Lambda_1(t)$, conjugation along the trajectory preserves its determinant. We use the scaled invariant

$$\gamma^2 = \frac{\det \Lambda_1(0)}{12},$$

called the *Casimir*. This notation does not assume positivity. Every mixed normal period has $\gamma^2 > 0$ (Proposition B.16).

The frame must turn through $\pi/3$ in a period. The cost of this turning gives an upper bound for the Casimir: every competitive mixed period satisfies

$$\gamma^2 < \frac{5}{27}.$$

The cost–Casimir inequality and this consequence are proved in Proposition B.14 and Corollary B.15. The entirely forward case uses a sharper bound of the same kind.

The invariant can also be computed from the data adjacent to a single piece. In two of the case arguments, these local data force a value of γ^2 exceeding the bound imposed by the cost. A third case compares the turning accumulated at successive pieces with that required by closure.

3.5. **The bound on the number of pieces.**

Theorem 3.3 (At most five pieces)**.** *A competitive normal vertex period whose state stays in the compact domain* $\mathfrak{h}^{\star\star}$ *has at most five pieces.*

Section 4 proves this theorem by excluding the five possible classes of longer periods. The remaining periods admit a direct comparison with the octagon.

4. Excluding six or more pieces

The five classes require distinct exclusions. Local necessary conditions for optimality exclude two classes for every normal vertex period. The other three require a comparison with the octagon's cost. We give the argument for each class; the corresponding appendix contains the calculations.

4.1. **Class R: every switch reverse.** In an entirely reverse period, choose a piece whose active margin is smallest. The pieces on either side also belong to the reverse pattern. The local switching conditions rule out a minimum in this position: among four pieces joined by three reverse switches, neither interior piece can have an active margin at most all the others. This is the *reverse-four principle* (Lemma C.1). Applying it at the chosen minimum gives the contradiction.

Theorem 4.1 (Class R)**.** *No normal vertex period has every switch reverse.*

The argument uses neither the cost nor the number of pieces. It excludes entirely reverse patterns from the final short-period comparison as well. Appendix C proves the local principle and its application to this case.

4.2. **Class M3: reverse runs of length other than two.** Every maximal reverse run in a mixed period is preceded and followed by a forward switch. At either end, two successive switches return to the same label. The active margins a, b, c of the three pieces then satisfy $ac < b^2$ (Lemma D.1). In particular the middle piece cannot have the smallest of these three margins.

This inequality forces the active margins to decrease as the run begins and to increase as it ends. A run of one reverse switch would require contradictory orderings at its two ends. A run of three or more would have an interior minimum, forbidden by the reverse-four principle. Only a run of two reverse switches remains.

Theorem 4.2 (Class M3)**.** *No normal vertex period is mixed with a maximal reverse run of length other than two.*

The piece between the two reverse switches is called the *centre*. Its active margin is strictly smaller than those of the surrounding pieces (Proposition D.2). This strict minimum supplies the local configuration for both remaining mixed cases. The argument and its consequences for the Casimir are proved in Appendix D.

4.3. **Class M1: isolated forward switches.** Here each reverse run has two switches and each forward run has one. The cyclic signs are therefore $(--+)^m$, where m is the number of reverse centres. This repeated pattern permits a comparison of the turning at individual centres with the turning required over a whole period.

Theorem 4.3. *No competitive normal vertex period belongs to class* M1*.*

The costate and frame determine an auxiliary vector at every switch, called a *switch vector*. Its direction is measured using a positive definite quadratic form supplied by the Casimir. Between centres these vectors only change sign, so their line directions do not change. All the turning is concentrated at the centres. The closing conditions fix the total turning as $(m+1)\pi/3$ (Proposition E.5). The local switching conditions and the Casimir cap require more than $2\pi/5$ at each centre (Proposition E.3).

For $m \geq 5$, the required total is too small to accommodate these local contributions. The closing condition on the labels leaves only $m = 2$ as another possibility. In that case the two centres must satisfy a rigid relation, and an exact comparison shows that their period costs more than A_{oct} (Proposition E.8). Thus this case is excluded without using the compact state domain or a lower bound on the number of pieces. The proof is in Appendix E.

4.4. **Class F: every switch forward.** The octagon itself has only forward switches. This case must therefore distinguish its four pieces from longer forward periods.

Theorem 4.4 (Class F)**.** *No competitive normal vertex period with at least six pieces has every switch forward.*

Closure requires the number of pieces to be 1 modulo 3. A forward period with six or more pieces consequently has at least seven. Its cost is a sum of terms determined by consecutive active margins. If all the margins were as large as the octagon's, these terms would exceed the available budget. A competitive period must therefore have a smaller minimum margin.

At a piece attaining that minimum, the switching conditions restrict the neighbouring data. The local formula for the Casimir then forces its value above the upper bound allowed by the cost of turning through $\pi/3$. These bounds contradict each other: the additional pieces force a small margin whose local switching conditions are incompatible with the available cost. Appendix F gives the cost estimates and the local comparison.

4.5. **Class M2: longer forward runs.** The last case is mixed, every reverse run has two switches, and at least one forward run has two or more. The forward runs need not have equal lengths, so there is no single repeated pattern to which the turning argument for M1 can be applied. Instead we study a few pieces around a smallest active margin.

Theorem 4.5 (Class M2)**.** *Let P be a competitive normal vertex period whose state stays in $\mathfrak{h}^{\star\star}$ and which has at least six pieces. Then the sign sequence of P is not of class* M2*.*

A smallest margin occurs at a reverse centre (Proposition G.3). Choose the neighbouring piece with the smaller active margin. After reflection and reversal of time, if necessary, this piece follows the centre. The forward run beginning there has either one switch or at least two. These possibilities require different comparisons.

If the run has at least two switches, a variation of one margin reduces the necessary switching inequalities to a simpler boundary configuration. A reverse continuation violates those inequalities directly. A forward continuation forces the Casimir above its cost bound. These comparisons are proved in Appendices G.1.6 to G.1.8.

If the run has one switch, it forms a short connection between two reverse runs. The proof first restricts the possible shape of this connection. It then shows that the closing and switching conditions force a definite deformation of the frame across the connection. This deformation is measured by a parameter called the *shear*. The cost of the connection together with the rest of the period exceeds A_{oct}. The estimates are organized as seven successive restrictions in Appendix G.3.

Both arguments use local data from at most seven pieces. The restrictions from the complete period are its cost bound, the compact domain and the turning identities. Appendix G contains the full proof of this case, with separate subsections for the two continuation arguments.

4.6. **Proof of Theorem 3.3.**

Proof of Theorem 3.3. Suppose the period has $N \geq 6$ pieces. By Proposition 3.2 it belongs to one of the five classes. Classes R and M3 are excluded by Theorems 4.1 and 4.2. Competitiveness excludes M1 by Theorem 4.3. The bound $N \geq 6$ also excludes F by Theorem 4.4. Finally, the compact-domain hypothesis allows Theorem 4.5 to exclude M2. This exhausts the classes. □

Only the forward case and the last mixed case use the assumption of six or more pieces. We now compare the periods left by this reduction.

5. The remaining periods and the main theorems

For periods with at most five pieces, the closing equations can be solved directly. Comparing their costs with the octagon's completes the density bound. To classify the disks attaining equality, the comparison must also identify their trajectories.

5.1. **The four- and five-piece comparisons.**

The closing equations leave four-piece periods with only forward switches and a two-parameter family of five-piece periods. Entirely reverse five-piece patterns are already excluded by Theorem 4.1; all other patterns with at most five pieces fail the closing equations (Proposition H.3).

The four-piece family contains the octagon. Its cost has a unique minimum at the symmetric configuration, where the four pieces have equal durations. The proof locates this minimum by monotonicity and an exact logarithmic inequality (Theorem H.6).

The five-piece family requires a strict comparison. It approaches the octagon when one piece shrinks to zero duration, so no uniform positive gap is possible near that boundary. A concave envelope gives the strict inequality throughout the family (Theorem H.10). The two comparisons yield the following theorem.

Theorem 5.1 (Terminal comparison)**.** *Every normal vertex period with at most five pieces costs at least* A_{oct}.

Equality holds only for the octagon's four-piece motion, up to a cyclic relabeling of the controls. Proposition H.11 states the precise identification of its frame and state. The comparisons and the equality statement are proved in Appendix H.

The cost comparison proves the lower bound. Identifying the minimizing trajectory then shows that its disk is an affine image of the smoothed octagon.

5.2. **The lower bound.**

Theorem 5.2 (Cost of a minimizer)**.** *Every global minimizer of the control problem costs at least* A_{oct}*.*

Proof. Let (T, u, g, X) be a global minimizer. The octagon's admissible trajectory has cost A_{oct} (Proposition A.7), so the minimizer is competitive. By Theorem A.30, choosing a switch as the starting time gives a normal vertex period with the same cost and with state in $\mathfrak{h}^{\star\star}$. It has at most five pieces by Theorem 3.3. Its cost is therefore at least A_{oct} by Theorem 5.1. □

Proof of Theorem 1.2. A global minimizer exists by Theorem A.10. Every admissible trajectory costs at least as much as this minimizer, which costs at least A_{oct} by Theorem 5.2. □

Proof of Theorem 1.1(b). Let K_0 be a worst disk. By Theorem A.10, its realization is a global minimizer of cost $\sqrt{12}\,\delta(K_0)$. Hence

$$\sqrt{12}\,\delta(K_0) \geq A_{\mathrm{oct}} = \sqrt{12}\,\delta_{\mathrm{oct}}.$$

Every centrally symmetric convex disk K, translated to the origin if necessary, has density at least $\delta(K_0)$. Translation preserves density, so $\delta(K) \geq \delta_{\mathrm{oct}}$. □

5.3. **Recovering the disk in the equality case.** Let a disk attain the least packing density. Apply the finite-period reduction to its realization as a global minimizer. The terminal equality statement identifies this trajectory with the octagon's after a cyclic relabeling.

To recover the given disk, we reverse the choices made in the reduction. The cyclic relabeling permutes the six contact curves, and the change of starting time changes their reference frame. Reversing the initial linear normalization then carries the octagon's boundary onto the boundary of the disk. Appendix I.1 gives the details and proves the following rigidity statement.

Theorem 5.3 (Rigidity)**.** *Every worst disk is the image of the octagon* K_{H} *under an invertible linear map of the plane.*

The converse follows from the lattice formulation of density. An invertible linear map multiplies the disk area and the lattice cell area by the same factor. Translation changes neither density.

Proposition 5.4 (Affine invariance)**.** *Let* K *be a convex disk symmetric about the origin,* A *an invertible linear map of the plane and* b *a vector. Then*

$\delta(AK) = \delta(K)$ and $\delta(K + b) = \delta(K)$. In particular $\delta(AK_{\mathrm{H}}) = \delta_{\mathrm{oct}}$ for every invertible linear map A.

Finally, Hales's octagon K_{H} is an invertible linear image of Reinhardt's K_{oct} (Proposition I.4). The two constructions therefore have exactly the same affine images. The explicit map and the proof of affine invariance are in Appendices I.2 and I.3.

5.4. **Completion of the main theorem.**

Proof of Theorem 1.1(a). The disk K_{H} is centrally symmetric and convex, with greatest packing density δ_{oct} (Proposition A.7). The invertible linear map of Proposition I.4 carries it onto K_{oct}. Central symmetry, convexity, compactness and nonempty interior pass to this image, and Proposition 5.4 preserves its density. □

Proof of Theorem 1.1(c). Suppose a centrally symmetric convex disk K has density δ_{oct}, and let c be its centre. By part (b), $K - c$ is a worst disk. Rigidity gives $K - c = LK_{\mathrm{H}}$ for an invertible linear map L. Thus K is an affine image of K_{H}, and hence of K_{oct} by Proposition I.4. Conversely, every invertible affine image of K_{oct} has density δ_{oct} by part (a) and Proposition 5.4. □

Appendix A. The control formulation and finite normal periods

This appendix gives the precise geometric and analytic setup behind Section 2 and proves the reduction used in Section 3. The first part records the coordinate formulas, regularity conditions and geometric comparisons. The remaining parts show that a global minimizer has a finite normal vertex representation.

A.1. **Geometric setup and coordinates.**

We use the packing densities of Section 2.1. The constructions below follow Reinhardt, Mahler and Hales, in the form developed in [13].

A.1.1. *Reinhardt's smoothed octagon.* For $k \in \mathbb{Z}/8$ let

$$d_k(x) = 1 - \big(x_0 \cos(k\pi/4) + x_1 \sin(k\pi/4)\big), \qquad x = (x_0, x_1) \in \mathbb{R}^2,$$

the signed distance from x to the line of the k-th side of the regular octagon of inradius 1, positive on the side of the origin.

Definition A.1 (The smoothed octagon)**.** Reinhardt's *smoothed octagon* is the set K_{oct} of the points $x \in \mathbb{R}^2$ such that, for every $k \in \mathbb{Z}/8$,

$$d_k(x) \geq 0 \qquad \text{and} \qquad d_k(x) + d_{k+1}(x) \leq \tfrac{3}{2} - \sqrt{2} \implies d_{k-1}(x)\, d_{k+2}(x) \geq \tfrac{\sqrt{2}}{4}.$$

The set K_{oct} is the regular octagon $\{d_k \geq 0\}$ with each corner cut off by an arc of a hyperbola (Figure 2). At the corner between sides k and $k+1$ the hyperbola is $d_{k-1}d_{k+2} = \sqrt{2}/4$. Its asymptotes are the lines of the next two sides, $k-1$ and $k+2$, and it touches sides k and $k+1$ at the two points where $d_k + d_{k+1} = \frac{3}{2} - \sqrt{2}$: along either side the product $d_{k-1}d_{k+2}$ is a concave quadratic whose largest value is $\sqrt{2}/4$, attained at the point of tangency. Beyond the chord $d_k + d_{k+1} = \frac{3}{2} - \sqrt{2}$ only the points between the hyperbola and the origin are kept. Each side of the octagon, of length $2(\sqrt{2}-1)$, keeps a straight middle part of length $2 - \sqrt{2}$. The boundary of K_{oct} is therefore made of eight segments and eight arcs of hyperbolas, and it is C^1, since the arcs leave the segments tangentially. Mahler [22] and Ledermann and Mahler [18] describe the same figure; Hales [14] and Hales and Vajjha [13, §7.5] obtain it, in a different linear position, from the control problem below (Propositions A.7 and I.4).

Coordinates for Figure 2. (a) Reinhardt's smoothed octagon K_{oct}: the regular octagon of inradius 1 (dashed) with its corners cut off; the straight parts of the sides are blue, the arcs of hyperbolas orange. (b) The corner $V = (1, \sqrt{2}-1)$ between side 0 ($x_0 = 1$) and side 1 ($x_0 + x_1 = \sqrt{2}$). The hyperbola $d_7 d_2 = \sqrt{2}/4$ (thin orange) has the lines of sides 7 and 2 as asymptotes, meeting at $C = (1+\sqrt{2}, 1)$, and touches the two sides at $T_1 = (1, 1 - 1/\sqrt{2})$ and $T_2 = (\sqrt{2} - \frac{1}{2}, \frac{1}{2})$, where $d_0 + d_1 = \frac{3}{2} - \sqrt{2}$. The arc from T_1 to T_2 replaces the corner. The straight part of each side has length $2 - \sqrt{2}$.

A.1.2. *Critical hexagons and balanced disks.* Let K be centrally symmetric about the origin. Two translates $K+v$ and $K+w$ have disjoint interiors exactly when $(v-w)/2$ is not an interior point of K, so $K+L$ is a packing exactly when the lattice $\frac{1}{2}L$ has no nonzero point in the interior of K. The densest lattice packings therefore come from the *critical* lattices of K, those of least determinant with no nonzero interior point. Reinhardt and Mahler described them through hexagons: a critical lattice has six points $\pm p, \pm q, \pm(q-p)$ on the boundary of K, the vertices of an affinely regular hexagon inscribed in K, and a least such hexagon gives a critical lattice [28, 22]; see [13, §2.2]. For a worst disk, Reinhardt showed that the least hexagons form a one-parameter family whose vertices sweep the whole boundary, and Hales normalized this family as follows.

The *sixth roots of unity* $s_j = (\cos(j\pi/3), \sin(j\pi/3))$, $j \in \mathbb{Z}/6$, satisfy $s_{j+3} = -s_j$, $s_0 + s_2 + s_4 = 0$ and $\det(s_j, s_{j+1}) = \sqrt{3}/2$. A *multi-point* is a map $j \mapsto \sigma_j \in \mathbb{R}^2$ with the same three properties, and six curves $\sigma_j(t)$ that form a multi-point at every time are a *multi-curve*. If the points $\sigma_j(t)$ lie on the boundary of K, the hexagon $\sigma_0(t), \dots, \sigma_5(t)$, of area $3\sqrt{3}/2$, is inscribed in K, and the lattice

$$L(t) = 2\mathbb{Z}\sigma_0(t) + 2\mathbb{Z}\sigma_1(t), \qquad \det L(t) = 4 \cdot \tfrac{\sqrt{3}}{2} = \sqrt{12},$$

contains the six vectors $2\sigma_j(t)$: the translate $K + 2\sigma_j(t)$ touches K at $\sigma_j(t)$.

Definition A.2 (Balanced pair [13, Def. 2.4.7])**.** A *balanced pair* (K, σ) consists of a convex disk K centred at the origin and a multi-curve σ such that each σ_j is C^1 and regular with Lipschitz derivative, the curvature $\det(\sigma_j', \sigma_j'')$ is nonnegative almost everywhere, each derivative $\sigma_j'(t)$ points into the open cone spanned at $\sigma_j(t)$ by $\sigma_{j+1}(t) - \sigma_j(t)$ and $\sigma_{j+1}(t)$ (the *star condition*), and the curves have a common period $6t_{\mathrm{f}}$ over which each of them traces the boundary of K once. It is in *circle representation* if $\sigma_j(0) = s_j$ for every j.

Proposition A.3 ([13, Thm. 2.2.2])**.** *If (K, σ) is a balanced pair, then $\delta(K) = \operatorname{area}(K)/\sqrt{12}$.*

For a balanced disk the lattices $L(t)$ thus have the determinant of a densest lattice packing. For the smoothed octagon the translates $K + L(t)$ are packings, and they form a one-parameter family of densest packings that slide continuously into one another (Figure 1). Baez [2] shows an animation of this family by Greg Egan.

A.1.3. *The control problem.* Let (K, σ) be a balanced pair in circle representation. A multi-point is determined by σ_0 and σ_1, and $\det(\sigma_0, \sigma_1) = \det(s_0, s_1)$, so there is a unique $g(t) \in \mathrm{SL}_2(\mathbb{R})$ with $\sigma_j(t) = g(t)s_j$ for every j. The matrix $X = g^{-1}g'$ is traceless, the star condition says that it lies in an open cone, and after a change of time $\det X = 1$. The six curves trace the boundary of K in turn, $\sigma_j(t + t_{\mathrm{f}}) = \sigma_{j+1}(t)$, which reads $g(t + t_{\mathrm{f}}) = g(t)R$ with R the rotation by $\pi/3$. The convexity of σ_0 becomes a constraint on X', which

Hales writes with a control in a triangle, and the area of K becomes an integral of X. Hales derived the problem that results in [14, 15]; [13, Ch. 3–4] gives the derivation in detail.

Write $\langle X, Y\rangle = \operatorname{tr}(XY)$ for the trace form on 2×2 matrices, and

$$J = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \qquad R = \exp(\pi J/3) = \begin{pmatrix} \frac{1}{2} & -\frac{\sqrt{3}}{2} \\ \frac{\sqrt{3}}{2} & \frac{1}{2} \end{pmatrix}.$$

The *control simplex* is $U_T = \{u = (u_0, u_1, u_2) : u_i \geq 0,\ u_0 + u_1 + u_2 = 1\}$, with vertices $\mathbf{e}_0, \mathbf{e}_1, \mathbf{e}_2$. A control $u \in U_T$ has the *control matrix*

$$Z_u = \begin{pmatrix} (u_1 - u_2)/\sqrt{3} & (u_0 - 2u_1 - 2u_2)/3 \\ u_0 & (u_2 - u_1)/\sqrt{3} \end{pmatrix}.$$

For $X = \left(\begin{smallmatrix} a & b \\ c & -a \end{smallmatrix}\right)$ the *star functions* are $\rho_0(X) = (c + \sqrt{3}a)/\sqrt{3}$, $\rho_1(X) = (c - \sqrt{3}a)/\sqrt{3}$ and $\rho_2(X) = -(3b + c)/(2\sqrt{3})$. A *star state* is a traceless matrix X with $\det X = 1$ and $\rho_0(X), \rho_1(X), \rho_2(X) > 0$.

Definition A.4 (The control problem; [13, Problem 3.6.2])**.** Let $T > 0$. A *state trajectory* on $[0, T]$ consists of a measurable control $u : [0, T] \to U_T$, a frame $g : [0, T] \to \mathrm{SL}_2(\mathbb{R})$ and a state X with values in the star states, such that

$$g' = gX, \qquad X' = [P_u, X], \qquad P_u = \frac{Z_u}{\langle Z_u, X\rangle}, \tag{A.1}$$

in the integral sense: the right-hand sides are integrable, and g and X are their indefinite integrals. It is *admissible* if

$$g(0) = I, \qquad g(T) = R, \qquad X(T) = R^{-1}X(0)R.$$

Its *cost* is

$$\mathcal{C}(T, X) = -\frac{3}{2}\int_0^T \langle J, X(t)\rangle \,\mathrm{d}t. \tag{A.2}$$

The terminal time T is free, and the control is assumed only to be measurable. A minimizer exists in this class [13, §6.1]; the comparison in Appendix A.1.5 therefore applies to the entire class.

Definition A.5 (Global minimizer)**.** A *global minimizer* is an admissible trajectory (T, u, g, X) such that $\mathcal{C}(T, X) \leq \mathcal{C}(T', X')$ for every admissible trajectory (T', u', g', X'), with any terminal time and any measurable control.

We use three coordinate descriptions of the star states. The first consists of three positive numbers, one for each vertex of the simplex.

Lemma A.6 (Margins)**.** *For a traceless matrix X let $p_i(X) = -\sqrt{3}\,\langle Z_{\mathbf{e}_i}, X\rangle$, $i \in \mathbb{Z}/3$, be its* margins*. Then $p_i = 2\rho_{2-i}$, and:*

(i) *a traceless X with $\det X = 1$ is a star state if and only if $p_0, p_1, p_2 > 0$;*

(ii) *$p_0p_1 + p_1p_2 + p_2p_0 = 4\det X$, which is 4 on star states;*

(iii) *$-\frac{3}{2}\langle J, X\rangle = \frac{\sqrt{3}}{2}(p_0 + p_1 + p_2)$, which is at least 3 on star states;*

(iv) *$p_i(R^{-1}XR) = p_{i-1}(X)$.*

Proof. For $X = \left(\begin{smallmatrix} a & b \\ c & -a \end{smallmatrix}\right)$ the margins are $p_0 = -(3b + c)/\sqrt{3}$, $p_1 = 2(c - \sqrt{3}a)/\sqrt{3}$ and $p_2 = 2(c + \sqrt{3}a)/\sqrt{3}$, which are $2\rho_2$, $2\rho_1$ and $2\rho_0$; this gives (i). Expanding, $p_0p_1 + p_1p_2 + p_2p_0 = 4(-a^2 - bc) = 4 \det X$. Since $Z_{\mathbf{e}_0} + Z_{\mathbf{e}_1} + Z_{\mathbf{e}_2} = J$, the sum of the margins is $-\sqrt{3}\langle J, X\rangle$, which is (iii); and $(p_0+p_1+p_2)^2 \geq 3(p_0p_1+p_1p_2+p_2p_0) = 12$ gives the lower bound $\frac{\sqrt{3}}{2} \cdot 2\sqrt{3} = 3$. Finally (iv) is a direct computation with the entries of $R^{-1}XR$. □

By (iii), every state trajectory on $[0, T]$ costs at least $3T$. The second coordinate is a point of the upper half-plane. For $z = x + iy$ with $y > 0$ put

$$\Phi(z) = \frac{1}{y} \begin{pmatrix} x & -(x^2 + y^2) \\ 1 & -x \end{pmatrix}.$$

Then $\Phi(z)$ is traceless with determinant one, and it is a star state exactly when z lies in the *star domain*

$$\mathfrak{h}^\star = \{x + iy : \ y > 0, \ |x| < 1/\sqrt{3}, \ x^2 + y^2 > 1/3\},$$

the ideal triangle with vertices $-1/\sqrt{3}$, $1/\sqrt{3}$ and ∞; every star state is $\Phi(z)$ for exactly one $z \in \mathfrak{h}^\star$ [13, Thm. 4.4.1]. In this coordinate the margins are

$$p_0 = \frac{\sqrt{3}\,(x^2 + y^2 - 1/3)}{y}, \qquad p_1 = \frac{2(1 - \sqrt{3}x)}{\sqrt{3}\,y}, \qquad p_2 = \frac{2(1 + \sqrt{3}x)}{\sqrt{3}\,y},$$

and each margin vanishes on one side of the ideal triangle. Conjugation by R acts on $\mathfrak{h}^\star$ by the Möbius transformation of R^{-1}, which permutes the three sides. The third coordinate, used from Appendix B on, is the triangular form of the state along a piece of constant control.

On an interval where the control is a vertex $\mathbf{e}_i$, the matrix $P_{\mathbf{e}_i}$ is a constant multiple of $Z_{\mathbf{e}_i}$, the state moves by conjugation by a one-parameter group, and the margin p_i stays constant (Lemma B.1). On such an interval three of the six contact curves trace segments of the boundary of the disk and the other three trace arcs of hyperbolas. A switch between two vertices is *forward* if it raises the label by one and *reverse* if it lowers it (Figure 3). The words of vertex controls are the subject of Appendices B to H.

A.1.4. *The smoothed octagon's trajectory.* Let

$$y_0 = \sqrt{\frac{2\sqrt{2} - 1}{3}}, \qquad t_{\mathrm{sw}} = \frac{\log 2}{2\sqrt{2\sqrt{2} - 1}}, \qquad t_{\mathrm{f}} = 4t_{\mathrm{sw}},$$

and let u_{oct} be the control that equals $\mathbf{e}_2, \mathbf{e}_0, \mathbf{e}_1, \mathbf{e}_2$ on the four intervals $[mt_{\mathrm{sw}}, (m + 1)t_{\mathrm{sw}})$, $m = 0, 1, 2, 3$. The *octagon's density* and *cost* are

$$\delta_{\mathrm{oct}} = \frac{8 - 4\sqrt{2} - \log 2}{2\sqrt{2} - 1} = 0.902414\ldots, \qquad A_{\mathrm{oct}} = \sqrt{12}\,\delta_{\mathrm{oct}} = 3.126054\ldots.$$

Proposition A.7 (The octagon's trajectory [13, §7.5])**.** *There is exactly one state trajectory* $(g_{\mathrm{oct}}, X_{\mathrm{oct}})$ *on* $[0, t_{\mathrm{f}}]$ *with control* u_{oct}*,* $g(0) = I$ *and*

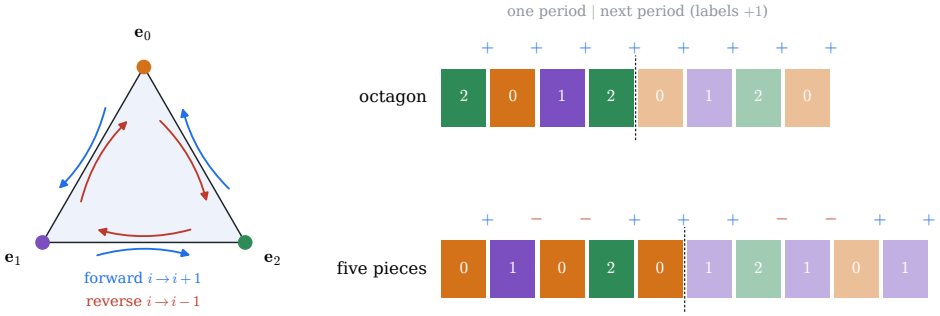


FIGURE 3. Left: the control simplex U_T with its vertices $\mathbf{e}_0, \mathbf{e}_1, \mathbf{e}_2$; a switch from $\mathbf{e}_i$ to $\mathbf{e}_{i+1}$ is forward (blue), one from $\mathbf{e}_i$ to $\mathbf{e}_{i-1}$ reverse (red). Right: two words of vertex labels over one period and the next, with the sign of each switch. The terminal condition $X(T) = R^{-1}X(0)R$ raises every label by one from one period to the next, so the switch across the seam (dashed) compares the last label of a period with the first label plus one. The octagon's word 2 0 1 2 has four forward switches; the word 0 1 0 2 0 has the signs $+\ -\ -\ +\ +$. In both cases the signs add up to 1 modulo 3 (Lemma B.6).

$X(0) = \Phi(iy_0)$. *It is admissible, its cost is* $A_{\mathrm{oct}} < \pi$, *and it is a Pontryagin extremal (Definition A.13). The convex hull*

$$K_{\mathrm{H}} = \operatorname{conv} \bigcup_{j \in \mathbb{Z}/6} \{g_{\mathrm{oct}}(t)s_j : \ 0 \le t \le t_{\mathrm{f}}\}$$

of its six arcs is a convex disk, symmetric about the origin, of area A_{oct} *and greatest packing density* $\delta(K_{\mathrm{H}}) = \delta_{\mathrm{oct}}$.

Sketch. On a piece of constant vertex control the state equations have explicit solutions (Lemma B.1). From $z_0 = iy_0$ the control $\mathbf{e}_2$ moves the state along $X(t) = \Phi((W-1)/\sqrt{3} + iy_0W)$, $W = \exp(\sqrt{3}y_0t)$, and the later pieces are conjugates of the first by powers of R. The choice of y_0 makes $\operatorname{tr}(R^{-1}g(t_{\mathrm{sw}})) = \sqrt{2}$, which gives $g(t_{\mathrm{f}}) = R$ and $X(t_{\mathrm{f}}) = R^{-1}X(0)R$; uniqueness holds because the right-hand sides are Lipschitz on each piece. The cost is four times an explicit integral over $[0, t_{\mathrm{sw}}]$, and it equals A_{oct} [13, §7.5]. The curves $\sigma_j(t) = g_{\mathrm{oct}}(t)s_j$, extended by $g(t + t_{\mathrm{f}}) = g(t)R$, form a balanced pair with the radial hull of σ_0, whose area is the cost; so its density is $A_{\mathrm{oct}}/\sqrt{12} = \delta_{\mathrm{oct}}$ by Proposition A.3. This radial hull is the convex hull of the six arcs, which together trace its boundary. Hales and Vajjha prove that the trajectory is a Pontryagin extremal [13, Thm. 7.5.6]. □

On each of the four pieces the factor $\exp(2t_{\mathrm{sw}}/a)$ of Lemma B.1, with a the active margin, equals $\sqrt{2}$ (Appendix H). Figure 4 shows the six arcs and the contact hexagons, and Figure 5 the state in the upper half-plane. The disk

K_{H} is a linear image of K_{oct}, namely $K_{\mathrm{H}} = A^{-1}K_{\mathrm{oct}}$ with $A = \mathrm{diag}(1, (2\sqrt{2}-1)/\sqrt{3})$ (Proposition I.4). Thus Hales's normalization compresses Reinhardt's octagon vertically by about five per cent.

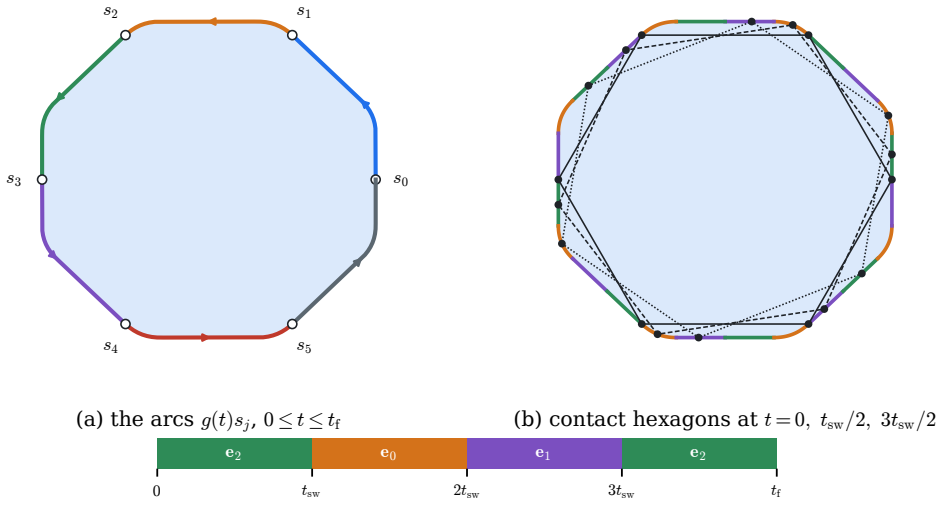


FIGURE 4. The octagon K_{H} of the trajectory. (a) As t runs over $[0, t_{\mathrm{f}}]$, the frame $g_{\mathrm{oct}}(t)$ carries each sixth root s_j (white) along an arc to s_{j+1}; the six arcs close up into the boundary of K_{H}, and K_{H} is their convex hull. (b) The boundary coloured by the vertex control that is active when σ_0 passes there (orange $\mathbf{e}_0$, purple $\mathbf{e}_1$, green $\mathbf{e}_2$), and the contact hexagons $\sigma_0(t), \dots, \sigma_5(t)$ at three times. Each hexagon has area $3\sqrt{3}/2$, opposite vertices are antipodal, and consecutive vertices span determinant $\sqrt{3}/2$. Below: the control word $\mathbf{e}_2\mathbf{e}_0\mathbf{e}_1\mathbf{e}_2$ on $[0, t_{\mathrm{f}}]$.

A.1.5. *From disks to trajectories and back.* The reformulation requires two comparisons. A worst disk is realized by an admissible trajectory of equal cost. Conversely, every admissible trajectory of cost below π gives a balanced disk of equal area, for any measurable control. This converse ensures that the realization of a worst disk minimizes cost even among trajectories with irregular controls.

Proposition A.8 (Realization of a worst disk)**.** *There are a worst disk K, a balanced pair (K, σ) in circle representation with period $6T$, and an admissible trajectory (T, u, g, X) with $\sigma_j(t) = g(t)s_j$ for $t \in [0, T]$, such that*

$$\mathcal{C}(T, X) = \mathrm{area}(K) = \sqrt{12}\,\delta(K) \le A_{\mathrm{oct}} < \pi, \qquad T < \pi/3.$$

Sketch. Take a worst disk and move it by a linear map, which does not change its density, into Reinhardt's normal position, in which a least-area

circumscribed hexagon has its side midpoints at the sixth roots of unity. These contact points form the inscribed hexagon used above. Every boundary point is then a contact point of a least hexagon [13, §2.2], and selecting the hexagon whose first contact point has a given polar angle gives six continuous simple closed curves that trace the boundary. The disk has a unique supporting line at each boundary point, so its radial parametrization is C^1 and regular; reparametrizing by arclength and then by the clock that makes $\det(g^{-1}g') = 1$ gives a balanced pair in circle representation. The contact map $\sigma_j \mapsto \sigma_{j+1}$ is conjugated by this clock to a translation by one sixth of the period, which gives $g(t+T) = g(t)R$. The curvatures determine the control; where the three curvatures do not form a nonzero nonnegative vector, which happens only on a null set, any point of the simplex may be chosen. The area formula for balanced disks gives $\mathcal{C} = \operatorname{area}(K)$, and Proposition A.3 gives $\delta(K) = \operatorname{area}(K)/\sqrt{12}$. Since K_{H} is a centrally symmetric convex disk with $\delta(K_{\mathrm{H}}) = \delta_{\mathrm{oct}}$, the worst disk has $\operatorname{area}(K) \le A_{\mathrm{oct}} < \pi$, and by Lemma A.6(iii), $3T \le \mathcal{C} < \pi$. □

Proposition A.9 (Reverse comparison)**.** *Let (T, u, g, X) be an admissible trajectory with $\mathcal{C}(T, X) < \pi$ and any measurable control. Then there is a balanced pair (K, σ) in circle representation with period $6T$ such that $\operatorname{area}(K) = \mathcal{C}(T, X)$ and $\delta(K) = \mathcal{C}(T, X)/\sqrt{12}$.*

Sketch. The state of an admissible trajectory is Lipschitz, since the control takes values in a compact set and the denominator $\langle Z_u, X\rangle$ stays away from zero on compact sets of star states; the frame is C^1. Extend the trajectory to all of $\mathbb{R}$ by $G(t+T) = G(t)R$ and $Y(t+T) = R^{-1}Y(t)R$; six copies close up because $R^6 = I$, and the integral equations glue across the seams. Let $\sigma_j(t) = G(t)s_j$ and let K be the radial hull of σ_0 over one full period $[0, 6T]$.

The terminal condition $g(T) = R$ alone allows the frame to wind around several times, and this is where the cost bound enters. The reflected first row $(g_{00}, -g_{01})$ of the frame turns with positive angular speed at most $-\langle J, X\rangle$, so a cost below π forces it to turn by exactly $\pi/3$ over $[0, T]$, with no extra multiple of 2π. The determinant-one identity keeps the first column $\sigma_0 = gs_0$ from ever pointing opposite to the reflected first row, so σ_0 also turns exactly once around the origin over $[0, 6T]$, and its positive angular speed makes it simple. The curvature condition of the control makes σ_0 locally convex, and a simple closed locally convex curve bounds a convex disk. The other conditions of a balanced pair follow from the state equations, and the area of the radial hull, $\frac{1}{2}\int_0^{6T} \det(\sigma_0, \sigma_0')$, is the cost. □

Theorem A.10 (Equivalence)**.** (i) *A worst disk exists, and the trajectory of Proposition A.8 is a global minimizer of the control problem, of cost $\sqrt{12}\,\delta_{\min}$.*

(ii) *Every global minimizer has cost $\sqrt{12}\,\delta_{\min}$.*

(iii) $\delta(K) \geq \delta_{\mathrm{oct}}$ *for every centrally symmetric convex disk* K *centred at the origin if and only if* $\mathcal{C}(T, X) \geq A_{\mathrm{oct}}$ *for every admissible trajectory* (T, u, g, X).

Proof. Let (T, u, g, X) and K be as in Proposition A.8, and let (T', u', g', X') be any admissible trajectory. If $\mathcal{C}(T', X') \geq \pi$, then $\mathcal{C}(T', X') > \mathcal{C}(T, X)$. Otherwise Proposition A.9 gives a balanced disk K' with $\mathcal{C}(T', X') = \operatorname{area}(K') = \sqrt{12}\,\delta(K') \geq \sqrt{12}\,\delta(K) = \mathcal{C}(T, X)$. This proves (i). If (T', u', g', X') is another global minimizer, each of the two costs is at most the other, which gives (ii). For (iii), suppose first that $\delta \geq \delta_{\mathrm{oct}}$ for every such disk. An admissible trajectory of cost at least π costs more than A_{oct}, and one of cost below π costs $\sqrt{12}\,\delta(K') \geq A_{\mathrm{oct}}$ by Proposition A.9. Conversely, if every admissible trajectory costs at least A_{oct}, then the worst disk of (i) has $\sqrt{12}\,\delta_{\min} = \mathcal{C}(T, X) \geq A_{\mathrm{oct}}$, so $\delta(K) \geq \delta_{\min} \geq \delta_{\mathrm{oct}}$ for every such disk. □

A.1.6. *The compact domain.* The star domain $\mathfrak{h}^\star$ is not compact, and near its three sides some margin is small. Hales and Vajjha showed that a balanced disk whose area does not exceed the octagon's has its state away from the sides and from the cusps.

Definition A.11 (The compact domain [13, Def. 5.0.1])**.** $\mathfrak{h}^{\star\star}$ is the set of the points $z = x + iy \in \mathfrak{h}^\star$ with $y \leq 9/2$, $(x \mp 1/\sqrt{3})^2 + y^2 \geq \frac{8}{27}\, y$ for both signs, and $\rho_k(\Phi(z)) \geq 1/15$ for $k = 0, 1, 2$. A star state X lies in $\mathfrak{h}^{\star\star}$ if $X = \Phi(z)$ with $z \in \mathfrak{h}^{\star\star}$.

The three conditions remove the horoball $y > 9/2$ at the cusp ∞, its two images under conjugation by $R^{\pm 1}$ (the open disks of radius $4/27$ tangent to the real line at $\pm 1/\sqrt{3}$), and three neighbourhoods of the sides of $\mathfrak{h}^\star$. By Lemma A.6, every margin of a state in $\mathfrak{h}^{\star\star}$ is at least $2/15$.

Theorem A.12 (The compact domain [13, Thm. 5.0.2])**.** *The state of every global minimizer lies in* $\mathfrak{h}^{\star\star}$ *at every time. In particular all its margins are at least* $2/15$*, at every time and for every vertex, active or not.*

Proof. Let (T, u, g, X) be a global minimizer. Its cost is at most that of the octagon's trajectory, which is $A_{\mathrm{oct}} < \pi$ (Proposition A.7), so Proposition A.9 applies: the cyclic extension of the trajectory is the boundary trajectory of a balanced disk K of area equal to the cost, with $X = \Phi \circ z$ for a curve z in $\mathfrak{h}^\star$. Hales and Vajjha prove that if $z(t_0)$ lies outside $\mathfrak{h}^{\star\star}$ at some time, then the area of such a disk is strictly larger than the area of the smoothed octagon [13, Thm. 5.0.2]: the area is bounded below by $\sqrt{12}$ times a density function of the single point $z(t_0)$ [13, Thm. 5.0.6], and this function exceeds δ_{oct} off $\mathfrak{h}^{\star\star}$. The cost would then exceed A_{oct}, the cost of an admissible trajectory, which contradicts minimality. □

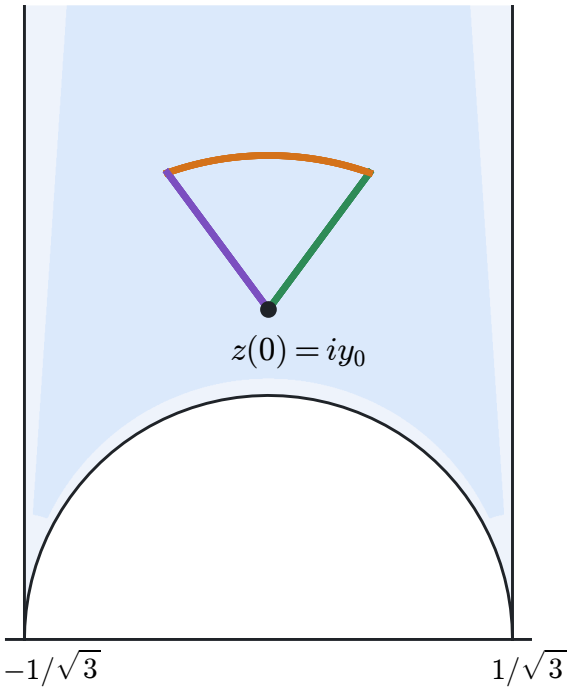


FIGURE 5. The star domain $\mathfrak{h}^\star$ (light), bounded by the lines $x = \pm 1/\sqrt{3}$ and the circle $x^2 + y^2 = 1/3$, and the compact domain $\mathfrak{h}^{\star\star}$ (darker; its cut at $y = 9/2$ lies above the frame). The curve is the state $z(t)$ of the octagon's trajectory, $X(t) = \Phi(z(t))$, over three periods $[0, 3t_{\mathrm{f}}]$, after which it returns to $z(0) = iy_0$; it is coloured by the active vertex control as in Figure 4. The curve stays in $\mathfrak{h}^{\star\star}$, so every margin along it exceeds $2/15$.

A.1.7. *Notation.* Table 1 lists the symbols that are used in more than one section. Symbols that belong to a single case are introduced where they are used.

Table 1: Notation.

Symbol	Meaning	Where
$\delta(K)$, $\delta_L(K)$	greatest packing density, greatest lattice packing density	§2.1
$\delta_{\min}$	least $\delta(K)$ over centrally symmetric convex disks	§2.1
K_{oct}, K_{H}	Reinhardt's smoothed octagon; the octagon of the trajectory	§A.1.1, §A.1.4

Symbol	Meaning	Where
$d_k(x)$	distance to the line of the k-th side of the regular octagon	§A.1.1
s_j, $\sigma_j(t)$	sixth roots of unity; contact curves $g(t)s_j$	§A.1.2
$\langle X, Y\rangle$	the trace form $\operatorname{tr}(XY)$	§A.1.3
J, R	rotation generator; rotation by $\pi/3$	§A.1.3
U_T, $\mathbf{e}_i$	control simplex and its vertices, $i \in \mathbb{Z}/3$	§A.1.3
Z_u, P_u	control matrix; $Z_u/\langle Z_u, X\rangle$	§A.1.3
g, X, T	frame in $\mathrm{SL}_2(\mathbb{R})$, star state, terminal time	§A.1.3
$\mathcal{C}(T, X)$	the cost $-\frac{3}{2}\int_0^T \langle J, X\rangle$	§A.1.3
ρ_i, p_i	star functions; margins $p_i = 2\rho_{2-i}$	§A.1.3
$\Phi(z)$, $\mathfrak{h}^\star$, $\mathfrak{h}^{\star\star}$	the state of $z = x+iy$; star domain; compact domain	§A.1.3, §A.1.6
δ_{oct}, A_{oct}	$(8-4\sqrt{2}-\log 2)/(2\sqrt{2}-1)$ and $\sqrt{12}\,\delta_{\mathrm{oct}}$	§A.1.4
y_0, t_{sw}, t_{f}	data of the octagon's trajectory	§A.1.4
Λ_1, Λ_R, λ	costates and cost multiplier	§A
H, H_i	Hamiltonian; vertex Hamiltonians	§A
N, τ_k, i_k	pieces per period, switching times, labels	§A
a_k, q_k	active margin and factor of piece k	§B
ε_k	sign of the switch after piece k: $+1$ forward, -1 reverse	§B
s_k, K_k	$a_k + a_{k+1}$ and $a_k^2 + 4$	§B
$T_i(q)$	tangent factor of a piece with label i	§B
R, F, M1, M2, M3	the five classes of sign sequences	§B
α_j, β_j, χ_j	costate coordinates; $\chi_j = 2\alpha_j + \beta_j = p_j H_j$	§B
U, μ_j, γ^2, κ	normal multiplier, its margins, the Casimir $\det U$, $\kappa = \sqrt{12\gamma^2}$	§B
y_k, d_k, e_k	switch vectors, switch norms, cross pairings	§B

A.2. **Outline of the finite reduction.** By Section 2, the control problem has a global minimizer whose state remains in the compact domain $\mathfrak{h}^{\star\star}$. After a change of time origin, its control takes only the three vertex values $\mathbf{e}_0, \mathbf{e}_1, \mathbf{e}_2$ and switches finitely often in a period. We also prove that it admits a Pontryagin lift with cost multiplier $\lambda = -1$. These conclusions give Theorem A.30, the starting point for the subsequent case analysis.

The maximum principle gives a lift $(\Lambda_1, \Lambda_R, \lambda)$ of the minimizer, possibly abnormal ($\lambda = 0$). At the singular locus, the Hamiltonian is independent of the control, so the maximum condition imposes no restriction on it. Near this locus Hales and Vajjha find Fuller's system [13, 12], whose optimal controls switch infinitely often in finite time. A budget inequality, which uses only that the cost is at most $A_{\mathrm{oct}} < \pi$, keeps every lift away from the singular locus in proportion to its size (Proposition A.20). Away from the singular

locus one vertex is excluded on each side of every time (Lemma A.23). On an interval where one vertex is excluded the control runs along an edge of the simplex, and an edge minimizer has at most four pieces (Proposition A.22). Compactness of $[0,T]$ then gives finitely many pieces (Proposition A.24). Merging equal neighbours and restarting the period at a genuine switch gives a vertex representation (Lemma A.27), and for a lift of a trajectory with a vertex representation the multiplier is nonzero (Theorem A.28).

The clearance and switch-count estimates are homogeneous in $(\lambda, \Lambda_1, \Lambda_R)$ and apply to both abnormal and normal lifts. The normality argument requires the costate values at finitely many switching times and uses the twisted closing condition. We therefore prove finiteness for an arbitrary lift before establishing normality.

A.3. **Pontryagin extremals.** The maximum principle is applied to the problem of Definition A.4 with free terminal time and with the closing conditions $g(0) = I$, $g(T) = R$, $X(T) = R^{-1}X(0)R$. Its conclusions, in the form of Hales and Vajjha's Problem 6.7.1 [13], define the following object.

Definition A.13 (Pontryagin extremal)**.** Let (T, u, g, X) be a state trajectory (Definition A.4) with $g(0) = I$, $g(T) = R$ and $X(T) = R^{-1}X(0)R$. A *Pontryagin lift* of it consists of two curves $\Lambda_1, \Lambda_R \colon [0,T] \to \mathfrak{sl}_2(\mathbb{R})$ and a constant $\lambda \le 0$, the *cost multiplier*, such that

(1) $\langle \Lambda_R(t), X(t) \rangle = 0$ and $(\lambda, \Lambda_1(t), \Lambda_R(t)) \neq 0$ for every $t \in [0,T]$;

(2) Λ_1 and Λ_R are absolutely continuous and satisfy, for almost every t,

$$\Lambda_1' = [\Lambda_1, X], \qquad \Lambda_R' = [P_u, \Lambda_R] - \langle \Lambda_R, P_u \rangle [P_u, X] + [-\Lambda_1 + \tfrac{3}{2}\lambda J, X], \tag{A.3}$$

with $P_u = Z_u / \langle Z_u, X \rangle$ as in Equation (A.1);

(3) for almost every $t \in (0,T)$ the control $u(t)$ maximizes over U_T the Hamiltonian

$$H(\Lambda_1, \Lambda_R, X; v) = \langle \Lambda_1 - \tfrac{3}{2}\lambda J, X \rangle - \frac{\langle \Lambda_R, Z_v \rangle}{\langle X, Z_v \rangle}, \tag{A.4}$$

evaluated at $(\Lambda_1(t), \Lambda_R(t), X(t))$, and the maximum is 0;

(4) $\Lambda_1(T) = R^{-1}\Lambda_1(0)R$ and $\Lambda_R(T) = R^{-1}\Lambda_R(0)R$.

The tuple $(T, u, g, X, \Lambda_1, \Lambda_R, \lambda)$ is a *Pontryagin extremal*. It is *normal* if $\lambda \neq 0$ and *abnormal* if $\lambda = 0$.

Here Λ_1 is the costate of the frame, read in the left trivialization of $\mathrm{SL}_2(\mathbb{R})$, and $\Lambda_R = [\Lambda_2, X]$ is the part of the costate of the state that the dynamics see. The conditions are positively homogeneous in $(\Lambda_1, \Lambda_R, \lambda)$, so a normal lift can be rescaled to $\lambda = -1$.

Theorem A.14 (Pontryagin lift)**.** *Every global minimizer (T, u, g, X) of the control problem (Definition A.5) has a Pontryagin lift $(\Lambda_1, \Lambda_R, \lambda)$.*

Hales and Vajjha use this statement in the proof of their Proposition 8.1.3. It follows from the maximum principle with free terminal time [27, 1], applied

on the open subset of $\mathbb{R}^{2\times 2} \times \mathfrak{sl}_2(\mathbb{R})$ cut out by $\det g > 0$ and the star inequalities, with the costates read through the left trivialization. The nontriviality of the reduced costate $(\lambda, \Lambda_1, \Lambda_R)$ needs a separate argument: if $\lambda = 0$ and Λ_1, Λ_R vanish at one time, then $\Lambda_1 \equiv 0$ by the conservation law below, and the equation for Λ_R becomes linear and homogeneous, so $\Lambda_R \equiv 0$.

Two consequences of the costate equations are used throughout. The first is the conservation law of Hales and Vajjha's Corollary 6.5.3,

$$\Lambda_1(t) = g(t)^{-1}\Lambda_1(0)\, g(t) \qquad (0 \le t \le T) \tag{A.5}$$

. In particular $\det \Lambda_1$ is constant. The second concerns the three *vertex Hamiltonians* $H_i(t) = H(\Lambda_1(t), \Lambda_R(t), X(t); \mathbf{e}_i)$. The denominator $-\langle X, Z_v\rangle$ is positive and affine in v, and $-\langle X, Z_v\rangle\big(H(\cdot\,; v) - c\big)$ is affine in v for every constant c. So the maximum of H over the simplex is attained at a vertex, and a vertex carrying positive weight in a maximizing control is itself maximizing.

Lemma A.15 (Vertex Hamiltonians)**.** *For a Pontryagin extremal the vertex Hamiltonians H_0, H_1, H_2 are continuous on $[0,T]$, and at every $t \in [0,T]$, not only almost every t, all three are ≤ 0 and one of them is 0.*

Proof. Continuity holds because X, Λ_1, Λ_R are continuous and $\langle X, Z_{\mathbf{e}_i}\rangle < 0$ on star states. By the maximum condition, $\max_i H_i(t) = 0$ for almost every t, hence for every t by continuity of $\max_i H_i$. □

Figure 6 shows the three vertex Hamiltonians along the smoothed octagon's period, for its normal lift.

The singular locus. If the three vertex Hamiltonians are equal at some time, then the maximum condition imposes no restriction on the control at that time. Hales and Vajjha's Lemma 8.2.1 identifies when this happens.

Lemma A.16 (Singular locus)**.** *Let X be a star state and Λ_R traceless with $\langle \Lambda_R, X\rangle = 0$. Then $H(\Lambda_1, \Lambda_R, X; \mathbf{e}_0) = H(\Lambda_1, \Lambda_R, X; \mathbf{e}_1) = H(\Lambda_1, \Lambda_R, X; \mathbf{e}_2)$ if and only if $\Lambda_R = 0$. If $\Lambda_R \neq 0$, there is a coordinate i such that $v_i = 0$ for every control v maximizing H over U_T.*

So all three vertices tie at a time where $\Lambda_R = 0$; if the Hamiltonian also vanishes there, then $\langle \Lambda_1 - \frac{3}{2}\lambda J, X\rangle = 0$. The *singular locus* is the set of costates with $\Lambda_1 = \frac{3}{2}\lambda J$ and $\Lambda_R = 0$; there $H(\cdot\,; v) = 0$ for every control v. Put

$$A = \Lambda_1 - \tfrac{3}{2}\lambda J, \qquad \|\mathcal{P}\|^2 = \lambda^2 + \|\Lambda_1\|^2 + \|\Lambda_R\|^2, \qquad Q^2 = \|A\|^2 + \|\Lambda_R\|^2,$$

with Frobenius norms. Here $\mathcal{P} = (\lambda, \Lambda_1, \Lambda_R)$ is the extended covector, and Q is its distance from the singular locus.

A.4. **Singular clearance.** The singular locus cannot be excluded by local considerations; the argument uses the whole period and its cost. Write $\omega(Y) = -\frac{1}{2}\langle J, Y\rangle$. For a star state $\omega(Y) \ge 1$, and the cost density is $-\frac{3}{2}\langle J, X\rangle = 3\omega(X)$, so $\mathcal{C}(T, X) = 3\int_0^T \omega(X)$. For a frame h with $\det h = 1$

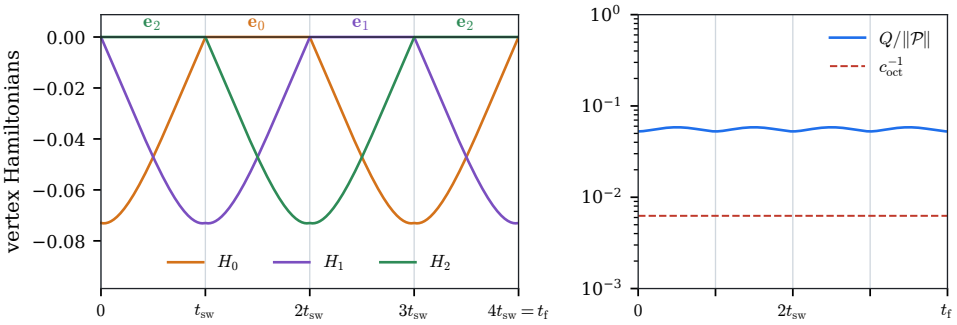


FIGURE 6. The normal Pontryagin lift ($\lambda = -1$) of the smoothed octagon's trajectory over one period $[0, t_f]$, with control word $\mathbf{e}_2\mathbf{e}_0\mathbf{e}_1\mathbf{e}_2$ and switching time $t_{sw} = t_f/4$; the lift is that of Hales and Vajjha's Theorem 7.5.1 [13], integrated numerically. Left: the vertex Hamiltonians H_0, H_1, H_2. All three are ≤ 0, the vertex of the current piece has $H_i = 0$, and at each switch the incoming vertex reaches 0. Right: the ratio $Q/\|\mathcal{P}\|$ of Proposition A.20, on a logarithmic scale. It stays above the uniform positive lower bound supplied by that proposition. The computation confirms the closing conditions $\Lambda_1(t_f) = R^{-1}\Lambda_1(0)R$ and $\Lambda_R(t_f) = R^{-1}\Lambda_R(0)R$ and every property stated here.

put $\omega_R(h, Y) = \omega(hYh^{-1})$, the same quantity for the state seen in the fixed frame.

Lemma A.17 (Winding)**.** *For every admissible trajectory (T, u, h, Y), with $h(0) = I$ and $h(T) = R$,*

$$\int_0^T \big(\omega(Y) + \omega_R(h, Y)\big)\,\mathrm{d}t \;\ge\; \frac{2\pi}{3}.$$

Proof. Write $\zeta(h) = \frac{1}{2}(h_{00} + h_{11}) + \frac{i}{2}(h_{10} - h_{01}) \in \mathbb{C}$, the coefficient of h on the rotations. Since $\det h = 1$, $|\zeta(h)| \ge 1$, and a direct computation gives $\omega(Y) + \omega_R(h, Y) = 2\,\mathrm{Im}(\overline{\zeta}\,\zeta')$ along $h' = hY$. Let $L(t) = \int_0^t \zeta'/\zeta$, so that $\zeta = e^L$. Then $\mathrm{Im}(\zeta'/\zeta) = (\omega + \omega_R)/(2|\zeta|^2) > 0$, and $\zeta(T) = \zeta(R) = e^{i\pi/3}$ forces $\mathrm{Im}\,L(T) \in \frac{\pi}{3} + 2\pi\mathbb{Z}$; being positive, it is at least $\pi/3$. Since $|\zeta| \ge 1$, $\int_0^T (\omega + \omega_R) = 2\int_0^T |\zeta|^2\,\mathrm{Im}(\zeta'/\zeta) \ge 2\,\mathrm{Im}\,L(T) \ge 2\pi/3$. □

The second ingredient is pointwise: at a star state the control u^* with $P_{u^*} = -X/2$ lies in the simplex and has $H(u^*) = \langle A, X\rangle$, so the maximum condition gives $\langle A, X\rangle \le 0$. The weights of u^* also bound Λ_R.

Lemma A.18 (Equilibrium inequality)**.** *Let X be a star state, Λ_R traceless with $\langle \Lambda_R, X\rangle = 0$, and suppose that every vertex Hamiltonian at $(\Lambda_1, \Lambda_R, X)$ is ≤ 0. Then $\langle A, X\rangle \le 0$. On each compact set of star states there is a constant C such that $\|\Lambda_R\| \le C(-\langle A, X\rangle)$.*

Proof. With $\varsigma_i = \sqrt{3}\langle \Lambda_R, Z_{\mathbf{e}_i}\rangle$ and the margins p_i of Lemma A.6, the vertex Hamiltonians are $H_i = \langle A, X\rangle + \varsigma_i/p_i$. The weights $\vartheta_i = p_i(p_j + p_k)/8$, $\{i,j,k\} = \{0,1,2\}$, sum to 1 by $p_0p_1 + p_1p_2 + p_2p_0 = 4$, and $\sum_i \vartheta_i\varsigma_i/p_i = \frac{1}{8}\sum_i (p_j + p_k)\varsigma_i = \frac{1}{2}\langle \Lambda_R, X\rangle = 0$. Hence $\langle A, X\rangle = \sum_i \vartheta_i H_i \le 0$. Put $m = -\langle A, X\rangle \ge 0$. From $H_i \le 0$, $(p_j + p_k)\varsigma_i \le 8m\vartheta_i$. Since the three products $(p_j + p_k)\varsigma_i$ sum to zero, each is at least the negative of the upper bounds for the other two, hence at least $-8m(1 - \vartheta_i) \ge -8m$; its upper bound is also at most $8m$. On a compact set of star states every $p_j + p_k$ has a positive lower bound. Thus every $|\varsigma_i|$ is bounded by a fixed multiple of m. The matrices $Z_{\mathbf{e}_i}$ span $\mathfrak{sl}_2(\mathbb{R})$, and the trace form on $\mathfrak{sl}_2(\mathbb{R})$ is nondegenerate, so the linear map $\Lambda_R \mapsto (\varsigma_0, \varsigma_1, \varsigma_2)$ is invertible. Its inverse has bounded norm, proving the assertion. □

Only the strict gap between the octagon's cost and the circular cost enters the next estimate. For $0 \le c < \pi$ write

$$b(c) = \frac{2\pi - c}{3(\pi - c)}, \qquad b_{\mathrm{oct}} = b(A_{\mathrm{oct}}).$$

The function b is increasing, and it is finite precisely while the cost stays below π.

Proposition A.19 (Budget inequality)**.** *Let (T, u, h, Y) be admissible with $\mathcal{C}(T, Y) \le A_{\mathrm{oct}}$, let $M \in \mathbb{R}^{2\times 2}$ and $\mu \in \mathbb{R}$, and suppose $\langle h(t)^{-1}Mh(t) + \frac{3}{2}\mu J, Y(t)\rangle \le 0$ for $t \in [0, T]$. Then $\mu \le b_{\mathrm{oct}}\,\|M + \frac{3}{2}\mu J\|$. Consequently every Pontryagin extremal with cost at most A_{oct} satisfies*

$$|\lambda| \le b_{\mathrm{oct}}\,\big\|\Lambda_1(t) - \tfrac{3}{2}\lambda J\big\| \qquad (0 \le t \le T).$$

Proof. Put $A_0 = M + \frac{3}{2}\mu J$, $a_0 = \|A_0\|$ and $V = hYh^{-1}$. The hypothesis reads $\langle A_0, V\rangle + 3\mu\,\omega_R \le 3\mu\,\omega$. By the Cauchy–Schwarz inequality and $\|V\|^2 = 4\omega_R^2 - 2$, $\langle A_0, V\rangle \ge -2a_0\omega_R$, so $(3\mu - 2a_0)\omega_R \le 3\mu\,\omega$ pointwise. Let $c = \mathcal{C}(T, Y)$, $I = \int \omega = c/3$ and $I_R = \int \omega_R$. If $3\mu \le 2a_0$, then $\mu \le 2a_0/3 \le b(c)a_0$. Otherwise $3\mu - 2a_0 > 0$, so integrating and using $I_R \ge 2\pi/3 - I$ from Lemma A.17 gives

$$(3\mu - 2a_0)(2\pi - c) \le 3\mu c, \qquad \text{hence} \qquad 3\mu(\pi - c) \le a_0(2\pi - c).$$

Since $c \le A_{\mathrm{oct}} < \pi$, division gives $\mu \le b(c)a_0 \le b_{\mathrm{oct}}a_0$. For the consequence, rebase the twisted extension of the extremal at t (Appendix A.7). The rebased trajectory is admissible with the same cost, its costate at time 0 is $\Lambda_1(t)$ by Equation (A.5), and Lemma A.18 gives the hypothesis with $M = \Lambda_1(t)$ and $\mu = -\lambda$. □

The positive gap $\pi - A_{\mathrm{oct}} > 0$ allows the budget inequality to bound the multiplier by the distance $\|A\|$ from Λ_1 to $\frac{3}{2}\lambda J$. Adding $\|\Lambda_R\|$ gives the clearance.

Put

$$c_{\mathrm{oct}} = \sqrt{b_{\mathrm{oct}}^2 + \big(1 + 3b_{\mathrm{oct}}/\sqrt{2}\big)^2}.$$

Proposition A.20 (Singular clearance)**.** *Let $(T, u, g, X, \Lambda_1, \Lambda_R, \lambda)$ be a Pontryagin extremal, normal or abnormal, with $\mathcal{C}(T, X) \le A_{\mathrm{oct}}$ and $X(t) \in \mathfrak{h}^{\star\star}$ for $t \in [0, T]$. Then*

$$\|\mathcal{P}(t)\| \le c_{\mathrm{oct}}\, Q(t) \qquad (0 \le t \le T).$$

In particular $Q(t) > 0$ for every t: the costate never meets the singular locus.

Proof. Write $\Lambda_1 = A + \frac{3}{2}\lambda J$. Since $\|J\| = \sqrt{2}$, the triangle inequality and Proposition A.19 give

$$\|\Lambda_1\| \le \|A\| + \frac{3}{\sqrt{2}}|\lambda| \le (1 + 3b_{\mathrm{oct}}/\sqrt{2})\|A\|.$$

Consequently $\|\mathcal{P}\|^2 \le c_{\mathrm{oct}}^2\|A\|^2 + \|\Lambda_R\|^2 \le c_{\mathrm{oct}}^2 Q^2$, since $c_{\mathrm{oct}} > 1$. The positivity of Q follows from the nontriviality $\|\mathcal{P}\| > 0$. □

Remark A.21. These estimates are uniform on each compact set K of star states. By Lemma A.18 and the boundedness of $\|X\|$ on K, there is $D_K \ge 0$ such that $\|\Lambda_R\|^2 \le D_K\|A\|^2$. Put $F = 2e^{2A_{\mathrm{oct}}/3}$. Along $g' = gX$,

$$\frac{\mathrm{d}}{\mathrm{d}t}\|g\|^2 = 2\,\mathrm{tr}(Xg^{\mathsf{T}}g) \le 2\omega(X)\|g\|^2,$$

so Gronwall's inequality gives $\|g(t)\|^2 \le F$. Since $\|g^{-1}\| = \|g\|$ for $\det g = 1$, spatial conservation (Equation (A.5)) bounds $\|\Lambda_1(t)\|^2$ by $F^2\|\Lambda_1(0)\|^2$ and conversely. Together with $\|A\|^2 \le 2\|\Lambda_1\|^2 + 9\lambda^2$, this yields

$$G_K^{-1}\|\mathcal{P}(0)\|^2 \le \|\mathcal{P}(t)\|^2 \le G_K\|\mathcal{P}(0)\|^2, \qquad G_K = (1+9D_K)+(1+2D_K)F^2.$$

The constant depends only on K and A_{oct}.

A.5. **The edge bound.** Suppose that on a time interval one coordinate of the control vanishes almost everywhere, say $u_0 = 0$. The trajectory then runs along the edge of the simplex opposite $\mathbf{e}_0$, and the control is determined by one number $v = u_1 \in [0, 1]$. A priori v could still alternate between the two remaining vertices infinitely often. The following bound excludes this for a global minimizer.

Proposition A.22 (Edge bound)**.** *Let (T, u, g, X) be a global minimizer, $0 \le t_0 < t_1 \le T$, and $j \in \{0, 1, 2\}$ with $u_j = 0$ almost everywhere on (t_0, t_1). Then on $[t_0, t_1]$ the control is almost everywhere a concatenation of at most four constant vertex pieces of positive length.*

Proof. Relabeling by a power of R (Appendix A.7) moves the missing vertex to $j = 0$ and keeps minimality. On the edge, a star state $X = \left(\begin{smallmatrix} a & b \\ c & -a \end{smallmatrix}\right)$ has $b = -(1 + a^2)/c$, and the star inequalities read $c > 0$, $3a^2 < c^2 < 3(1 + a^2)$. With $u = (0, v, 1 - v)$ and $\alpha = (2v - 1)/\sqrt{3}$, the state equation becomes the control system

$$a' = \frac{c}{c - 3\alpha a}, \qquad c' = \frac{3\alpha c}{c - 3\alpha a}, \qquad s' = c,$$

in the state (a, c, s), $s = \int c$, with cost density $\frac{3}{2}(c + (1 + a^2)/c)$. The frame is recovered from $x = a/c$ and s. Replacing the segment $[t_0, t_1]$ of the minimizer by any edge trajectory with the same end states and end frames, and translating the rest of the trajectory in time, gives an admissible trajectory whose cost changes by the difference of the segment costs. Since competitors with any terminal time are allowed, the segment minimizes the edge cost among edge trajectories with its end points and free duration.

The maximum principle with free time applies to this system on the open set given by the star inequalities. It gives a multiplier $\nu \leq 0$ and a covector $p = (p_a, p_c, p_s)$ with $(\nu, p(t)) \neq 0$ and $H = 0$, and p_s is constant since the system does not depend on s. The derivative of the Hamiltonian in α is $3c\varsigma/(c - 3\alpha a)^2$ with the *switching function* $\varsigma = ap_a + cp_c$, so $v = 1$ where $\varsigma > 0$ and $v = 0$ where $\varsigma < 0$. The adjoint equation, the homogeneity of the field in (a, c) and $H = 0$ give

$$\varsigma' = -c\,(2p_s + 3\nu(1 + x^2)), \qquad x = a/c, \quad x' = 1/c > 0.$$

Since x is strictly increasing, the quadratic $2p_s + 3\nu(1 + x^2)$ vanishes at most twice in time unless $\nu = p_s = 0$; then ς has at most three zeros by Rolle's theorem, and the control has at most four constant pieces. If $\nu = p_s = 0$, then ς is constant, and it is nonzero, because $\varsigma = 0$ and $H = 0$ would force $p = 0$. □

A.6. **Finitely many switches.**

Lemma A.23 (Local exclusion of a vertex)**.** *Let $(T, u, g, X, \Lambda_1, \Lambda_R, \lambda)$ be a Pontryagin extremal with $X(t) \in \mathfrak{h}^{\star\star}$ on $[0, T]$, and let $t_0 \in [0, T]$ with $Q(t_0) > 0$. There are $\delta > 0$ and vertices j, k such that $H_j < 0$ on $(t_0, t_0 + \delta) \cap [0, T]$ and $H_k < 0$ on $(t_0 - \delta, t_0) \cap [0, T]$.*

Proof. If $H_i(t_0) < 0$ for some i, continuity gives the claim with $j = k = i$. Otherwise all three vertex Hamiltonians vanish at t_0 (Lemma A.15), so $\Lambda_R(t_0) = 0$ by Lemma A.16 and $\langle A, X\rangle = 0$ there. Then $Q(t_0) > 0$ gives $A(t_0) \neq 0$, and $b = [X(t_0), A(t_0)] \neq 0$, since a traceless matrix commuting with X is a multiple of X and $\langle X, X\rangle = -2$. The control-dependent terms of Equation (A.3) are $O(\|\Lambda_R\|)$, so $\Lambda_R'(t_0) = b$, and $h_i = \langle \Lambda_R, Z_{\mathbf{e}_i}\rangle / \langle X, Z_{\mathbf{e}_i}\rangle$ has derivative $c_i = \langle b, Z_{\mathbf{e}_i}\rangle / \langle X, Z_{\mathbf{e}_i}\rangle$ at t_0. As b is traceless and orthogonal to X, Lemma A.16 applied to b shows that the c_i are not all equal. If $c_k < c_j$, then $H_j - H_k = h_k - h_j$ vanishes at t_0 with negative derivative, so $H_j < H_k \leq 0$ just after t_0 and $H_k < H_j \leq 0$ just before. □

Where $H_j < 0$ on an interval, the coordinate u_j of the control vanishes almost everywhere there, because a vertex with positive weight in a maximizing control is maximizing. So the edge bound applies on both sides of t_0.

Proposition A.24 (Finitely many switches)**.** *Let $(T, u, g, X, \Lambda_1, \Lambda_R, \lambda)$ be a Pontryagin extremal with $\mathcal{C}(T, X) \leq A_{\mathrm{oct}}$ and $X(t) \in \mathfrak{h}^{\star\star}$ on $[0, T]$, satisfying the clearance of Proposition A.20, and whose control has the property of*

Proposition A.22. Then there is n such that on $[0,T]$ the control is almost everywhere a concatenation of n constant vertex pieces of positive length.

Proof. By Proposition A.20, $Q(t_0) > 0$ at every t_0. Lemma A.23 and the edge bound give, around each t_0, a $\delta > 0$ such that every subinterval of $[0,T]$ inside $(t_0-\delta, t_0+\delta)$ has a finite vertex description: on the part after t_0 the vertex j is missing and Proposition A.22 gives at most four pieces, on the part before t_0 the vertex k is missing, and the two descriptions concatenate at t_0. A Lebesgue number of this cover of the compact interval $[0,T]$ and induction along a uniform partition give a finite description of $[0,T]$. □

A.7. **Vertex representations and rebasing.**

Definition A.25 (Vertex representation)**.** A *vertex representation* of a control u on $[0,T]$ consists of an integer $N \geq 1$, a strictly increasing sequence $(\tau_k)_{k\in\mathbb{Z}}$ of *switching times* with $\tau_0 = 0$ and $\tau_{k+N} = \tau_k + T$, and *labels* $i_k \in \mathbb{Z}/3$ with $i_{k+N} = i_k + 1$ and $i_{k+1} \neq i_k$ for every $k \in \mathbb{Z}$, such that for $0 \leq k < N$ the control equals $\mathbf{e}_{i_k}$ almost everywhere on (τ_k, τ_{k+1}). The interval (τ_k, τ_{k+1}) is the k-th *piece*, and N is the number of pieces of the period.

Definition A.26 (Normal period)**.** A *normal period* is a Pontryagin extremal $(T, u, g, X, \Lambda_1, \Lambda_R, \lambda)$ with $\lambda = -1$, together with a vertex representation of its control u. It is *competitive* if $\mathcal{C}(T, X) \leq A_{\mathrm{oct}}$, and it lies in the compact domain if $X(t) \in \mathfrak{h}^{\star\star}$ for $t \in [0,T]$. The *active margin* of the k-th piece is $a_k = p_{i_k}(X(\tau_k))$, with k read modulo N.

The pieces of Proposition A.24 may have equal neighbouring labels, and $t = 0$ need not be a switching time. Extending the trajectory periodically allows us to merge equal neighbours and choose a genuine switch as the time origin. The *twisted extension* of an admissible trajectory (T, u, g, X) is defined for $t \geq 0$ by

$$G(t) = g(t - nT)R^n, \qquad Y(t) = R^{-n}X(t - nT)R^n, \qquad n = \lfloor t/T \rfloor,$$

and with the control $v(t)$ equal to $u(t - nT)$ with its coordinates shifted n times, so that the vertex $\mathbf{e}_i$ becomes $\mathbf{e}_{i+n}$ and $Z_{v(t+T)} = R^{-1}Z_{v(t)}R$. The extension satisfies the state equations on every interval $[0, nT]$. The trajectory *rebased* at $s \in [0,T)$ is, for $0 \leq t \leq T$,

$$\hat{g}(t) = G(s)^{-1}G(s+t), \qquad \hat{X}(t) = Y(s+t), \qquad \hat{u}(t) = v(s+t).$$

It is again admissible, and it has the same cost because J commutes with R; so a rebased global minimizer is a global minimizer. Similarly the *relabeling* $(g, X) \mapsto (R^{-r}gR^r, R^{-r}XR^r)$, with the control coordinates shifted r times, preserves admissibility and cost.

In a vertex representation (Definition A.25), the condition $i_{k+1} \neq i_k$ for every k, including $k = N-1$, says that equal neighbours have been merged also across the seam $t = T$, where the label after the last piece is $i_N = i_0 + 1$.

Lemma A.27 (Merging and rebasing)**.** *If on $[0,T]$ the control u is almost everywhere a concatenation of n vertex pieces, there is $s \in [0,T)$ such that the rebased control $\hat{u}$ has a vertex representation with $N \le n$ pieces.*

Proof. Extend the pieces by the twist to a bi-infinite sequence with $\tau_{k+n} = \tau_k + T$ and labels $l_{k+n} = l_k + 1$. Call τ_k a genuine switch if $l_k \neq l_{k-1}$. The increments $l_k - l_{k-1}$ over one period sum to $1 \neq 0$ in $\mathbb{Z}/3$, so some increment is nonzero, and a genuine switch $s = \tau_{k_0}$ lies in $[0,T)$. Merge the pieces of $[s, s+T]$ with equal neighbouring labels, shift by $-s$, and extend the reduced word to $\mathbb{Z}$ by the twist; the seam condition is the genuine switch at s. □

A.8. **Normality.** Recall the normal periods of Definition A.26.

Theorem A.28 (Normality)**.** *Let (T, u, g, X) be a global minimizer with a Pontryagin lift $(\Lambda_1, \Lambda_R, \lambda)$, and suppose that u has a vertex representation. Then $\lambda < 0$, and $(\Lambda_1, \Lambda_R, \lambda)/(-\lambda)$ is a Pontryagin lift with multiplier -1.*

The proof reads an abnormal costate at the switching times, in coordinates in which every piece acts by an explicit linear map, and derives a contradiction with the twisted closing condition.

Coordinates. For $M \in \mathfrak{sl}_2(\mathbb{R})$ put $\mu_j(M) = \sqrt{3}\langle M, Z_{\mathbf{e}_j}\rangle$, so that the margins are $p_j = -\mu_j(X)$. With $\lambda = 0$ put

$$a = -X\Lambda_1 X, \qquad b = \Lambda_R + 2\Lambda_1 + \langle \Lambda_1, X\rangle X.$$

Using $X^2 = -I$ and $X\Lambda X = \langle \Lambda, X\rangle X + \Lambda$ one finds $p_j H_j = 2\mu_j(a) + \mu_j(b)$; this is the abnormal analogue of the switching functions χ_j of Definition B.7. Moreover $\langle b, X\rangle = \langle \Lambda_R, X\rangle = 0$, and $(a,b) = 0$ only if $(\Lambda_1, \Lambda_R) = 0$. For a label i and $q > 0$ let $L_i(q)$ be the linear map of $\mathbb{R}^3$ that fixes the coordinate i and sends

$$v_{i+1} \mapsto (v_i + v_{i+1})q - v_i, \qquad v_{i-1} \mapsto (v_i + v_{i-1})/q - v_i,$$

the one-arc margin map of Lemma B.1.

Lemma A.29 (Abnormal transport)**.** *Let the extremal have $\lambda = 0$, and let $u = \mathbf{e}_i$ almost everywhere on a piece (s_0, s_1), with $q = \exp(2(s_1 - s_0)/p_i(X(s_0)))$. Then $H_i(s_0) = 0$; the margins and the coordinates $\mu(b)$ at s_1 are those at s_0 transformed by $L_i(q)$; and the coordinates $\mu(a)$ are transformed by $L_i(q^2)$.*

Proof. Write $S_j = \sqrt{3}Z_{\mathbf{e}_j}$. On the piece $P_u = -S_i/p_i$ is constant, since the active margin p_i is constant. The state, a and b satisfy $f' = \kappa[P_u, f]$ with $\kappa = 1, 2, 1$ respectively; for b this uses that the active vertex Hamiltonian vanishes on the piece. The relation $[S_i, S_{i+1}] = 2(S_i + S_{i+1})$ and the invariance of the trace form make $\mu_i(f)$ constant and multiply $\mu_i(f) + \mu_{i\pm1}(f)$ by $q^{\pm\kappa}$. □

Proof of Theorem A.28. Suppose $\lambda = 0$. Continue the lift through the seams by the twist, using the closing conditions $X(T) = R^{-1}X(0)R$, $\Lambda_1(T) = R^{-1}\Lambda_1(0)R$, $\Lambda_R(T) = R^{-1}\Lambda_R(0)R$, which relabel the coordinates. At the

switching times τ_k, $k \geq 0$, this gives labels i_k with $i_{k+1} \neq i_k$ and $i_{k+N} = i_k + 1$, factors $q_k > 1$, positive margins, and coordinates $\alpha_k = \mu(a(\tau_k))$, $\beta_k = \mu(b(\tau_k))$, related by Lemma A.29, with $\sum_{j \neq l} \beta_{k,j} p_l = 0$, the active quantity $\chi_{i_k}(k) = 2\alpha_{k,i_k} + \beta_{k,i_k} = 0$, $\chi_j(k) = 2\alpha_{k,j} + \beta_{k,j} \leq 0$ for all j by Lemma A.15, and $(\alpha_k, \beta_k) \neq 0$. Four facts about such a sequence follow.

(1) No piece has a triple tie, $\chi_0 = \chi_1 = \chi_2 = 0$, at both of its ends. Indeed, for $j \neq i = i_k$ the pair sums $x = \alpha_i + \alpha_j$ and $y = \beta_i + \beta_j$ are multiplied by ϱ^2 and ϱ along the piece, $\varrho \in \{q_k, q_k^{-1}\}$. The ties $2x + y = 0$ at both ends give $x(\varrho^2 - \varrho) = 0$, so $x = y = 0$. Hence $\beta_i = -2\alpha_i$ and $\beta_j = 2\alpha_i$ for $j \neq i$, and the orthogonality relation becomes $4\alpha_i p_i = 0$. Then $\alpha = \beta = 0$, against nontriviality.

(2) If three consecutive pieces carry labels i, j, i, then χ_j vanishes at the start of the first and at the end of the third.

(3) No three consecutive switches have the same sign: the inequalities $\chi \leq 0$ for the missing vertex at the two inner switches, combined with the transports, force triple ties at both ends of the middle piece, against (1).

(4) From the second switch on, neighbouring switches have opposite signs, that is, $i_{k+2} = i_k$ for $k \geq 1$. This follows from (1)–(3).

By (4), $i_{1+2N} = i_1$. By the twist, $i_{1+2N} = i_{1+N} + 1 = i_1 + 2$. Since $2 \neq 0$ in $\mathbb{Z}/3$, this is a contradiction. □

The argument does not use the minimality of (T, u, g, X): an abnormal lift of any trajectory with a vertex representation contradicts the twisted closing condition.

A.9. **The finite normal representation.**

Theorem A.30 (Finite normal representation)**.** *Let (T, u, g, X) be a global minimizer of the control problem. There are a normal period $(T, \hat{u}, \hat{g}, \hat{X}, \Lambda_1, \Lambda_R, -1)$, with a vertex representation of $\hat{u}$, and a time $s \in [0, T)$ such that*

(1) $\hat{g}(t) = G(s)^{-1} G(s + t)$ and $\hat{X}(t) = Y(s + t)$ for $t \in [0, T]$, where (G, Y) is the twisted extension of (g, X);

(2) $\mathcal{C}(T, \hat{X}) = \mathcal{C}(T, X)$, and $(T, \hat{u}, \hat{g}, \hat{X})$ is a global minimizer;

(3) $\hat{X}(t) \in \mathfrak{h}^{\star\star}$ for every $t \in [0, T]$.

In particular the period is competitive, $\mathcal{C}(T, \hat{X}) \leq A_{\mathrm{oct}}$.

Proof of Theorem A.30. The state of the minimizer stays in $\mathfrak{h}^{\star\star}$ (Theorem A.12), and its cost is at most A_{oct} since the octagon is a competitor. Take a Pontryagin lift (Theorem A.14); it may be abnormal. It satisfies the clearance (Proposition A.20), and the minimizer has the edge property (Proposition A.22), so its control is a finite concatenation of vertex pieces (Proposition A.24). Merge and rebase at a genuine switch s (Lemma A.27). The rebased trajectory is a global minimizer with the same cost (Appendix A.7), so Theorem A.14 applies to it directly; Theorem A.28 makes the

new lift normal and rescales it to $\lambda = -1$, and Theorem A.12 applies to the rebased minimizer as well. □

Minimality allows a new Pontryagin lift of the rebased trajectory, so no costate transport is required in rebasing. The frame and state remain those of the original minimizer, restarted at s. The equality argument in Appendix I uses this identification. From Appendix B on, the hatted period is written (T, u, g, X) again, with switching times τ_k, labels i_k and active margins a_k.

Appendix B. Vertex periods

Theorem A.30 turns a global minimizer into a normal vertex period: a normal Pontryagin extremal whose control is a vertex of the simplex on each of finitely many pieces (Definitions A.25 and A.26). On a piece of constant vertex control the state, the frame and the costate solve linear differential equations with constant coefficients. Their solutions are explicit functions of the initial data. A period is therefore described by finitely many parameters satisfying algebraic and logarithmic identities, together with sign conditions imposed by the maximum principle. This section collects these formulas and conditions, which the case analysis of Appendices C to G uses to exclude periods with six or more pieces.

Throughout, P is a normal vertex period with N pieces, switching times τ_k, labels $i_k \in \mathbb{Z}/3$ and terminal time T, indexed by $k \in \mathbb{Z}$ through the twist $\tau_{k+N} = \tau_k + T$, $i_{k+N} = i_k + 1$ (Definition A.25). Its state X, frame g, costates Λ_1, Λ_R and cost multiplier $\lambda = -1$ are those of Definition A.26. We write $m_k(j) = p_j(X(\tau_k))$ for the margins at the switch τ_k $(0 \le k \le N)$. Since $p_j(R^{-1}XR) = p_{j-1}(X)$ and $X(T) = R^{-1}X(0)R$, the rule

$$m_{k+N}(j) = m_k(j-1) \tag{B.1}$$

extends them consistently to all $k \in \mathbb{Z}$. The *active margin* of piece k is $a_k = m_k(i_k)$, the margin of its own vertex at its start, and its *factor* is

$$q_k = \exp(2(\tau_{k+1} - \tau_k)/a_k) > 1,$$

so that the piece lasts $(a_k/2)\log q_k$. Both sequences are N-periodic. We also write

$$s_k = a_k + a_{k+1}, \qquad K_k = a_k^2 + 4.$$

B.1. **One piece of constant control.** For a label i and $w > 0$ let

$$T_i(w) = \frac{w + w^{-1}}{2}\, I + \frac{\sqrt{3}\,(w - w^{-1})}{2}\, Z_{\mathbf{e}_i}$$

be the *tangent factor* of the vertex $\mathbf{e}_i$. Since $Z_{\mathbf{e}_i}$ is traceless with determinant $-1/3$, $Z_{\mathbf{e}_i}^2 = I/3$, and $T_i(w)$ is the exponential of a multiple of $Z_{\mathbf{e}_i}$: $T_i(w)T_i(v) = T_i(wv)$, $\det T_i(w) = 1$, $T_i(w)^{-1} = T_i(w^{-1})$ and $R^{-1}T_i(w)R = T_{i+1}(w)$.

Lemma B.1 (One piece)**.** *Let (g, X) be a state trajectory on $[0, T]$ whose control is the vertex $\mathbf{e}_i$ almost everywhere on $(s_0, s_1) \subset [0, T]$, and let $(a, b, c) = (p_i, p_{i+1}, p_{i-1})(X(s_0))$. For $t \in [s_0, s_1]$ put $w = \exp((t - s_0)/a)$ and $q = w^2$. Then*

(a) *$X(t) = T_i(w)^{-1}X(s_0)T_i(w)$, and the tangent frame $V = gX$ satisfies $V(t) = V(s_0)\,T_i(q)$;*

(b) *the margins at time t are*

$$p_i = a, \qquad p_{i+1} = (a + b)q - a, \qquad p_{i-1} = \frac{a + c}{q} - a; \tag{B.2}$$

(c) *with* $q = \exp(2(s_1 - s_0)/a)$*, the cost of the piece is*

$$-\frac{3}{2}\int_{s_0}^{s_1} \langle J, X(t)\rangle \,\mathrm{d}t = \frac{\sqrt{3}}{4}\, a((a+b)(q-1)+(a+c)(1-q^{-1})-a\log q). \quad \text{(B.3)}$$

For a normal vertex period this gives, for every $k \in \mathbb{Z}$*,*

$$m_{k+1}(i_k) = a_k, \qquad m_{k+1}(i_k+1) = \big(a_k + m_k(i_k+1)\big)q_k - a_k,$$
$$m_{k+1}(i_k-1) = \frac{a_k + m_k(i_k-1)}{q_k} - a_k,$$

and the cost $\mathcal{C}(T, X)$ *of the period is the sum of* (B.3) *over the pieces* $0 \le k < N$*.*

Proof. The margin p_i is, up to a factor, the trace pairing with $Z_{\mathbf{e}_i}$, and $\langle Z_{\mathbf{e}_i}, [Z_{\mathbf{e}_i}, X]\rangle = 0$. Along the piece $P_{\mathbf{e}_i}(X) = Z_{\mathbf{e}_i}/\langle Z_{\mathbf{e}_i}, X\rangle$ is a multiple of $Z_{\mathbf{e}_i}$, so p_i is constant and $P_{\mathbf{e}_i}(X) = -(\sqrt{3}/a)Z_{\mathbf{e}_i}$ is a constant matrix. The state equation $X' = [P, X]$ then has the solution $X(t) = e^{(t-s_0)P}X(s_0)e^{-(t-s_0)P}$, and $e^{(t-s_0)P} = T_i(w)^{-1}$. The margins of this conjugate are a direct computation. Writing $g(t) = h(t)e^{-(t-s_0)P}$, the frame equation becomes $h' = h\,(X(s_0)+P)$ with a constant traceless matrix, whose exponential is explicit. Uniqueness of Carathéodory solutions identifies the actual trajectory with this flow, also when the control differs from $\mathbf{e}_i$ on a null set. Integrating $-\frac{3}{2}\langle J, X\rangle = \frac{\sqrt{3}}{2}(p_0+p_1+p_2)$ with (B.2) gives (B.3). □

Each margin stays positive along a piece, so a piece with factor q is admissible exactly when $1 < q < 1 + c/a$: the margin of the label $i-1$ decreases and must not reach 0. We call $i+1$ the *growing* label and $i-1$ the *decreasing* label of the piece.

The margins at a switch lie on the quadric $m_k(0)m_k(1) + m_k(1)m_k(2) + m_k(2)m_k(0) = 4$ (Lemma A.6), and two of them are active margins: at τ_k the label i_{k-1} has margin a_{k-1} (it was constant along piece $k-1$) and the label i_k has margin a_k. The third margin is therefore $(4 - a_{k-1}a_k)/s_{k-1}$, and the update rule of Lemma B.1 expresses each factor through the active margins alone.

Lemma B.2 (Factors from active margins)**.** *For every* k*, according to the signs of the switches before and after piece* k *(Definition B.5 below),*

forward, forward:	$q_kK_k = s_{k-1}s_k$,	*reverse, forward:*	$q_ks_{k-1} = s_k$,
reverse, reverse:	$q_ks_{k-1}s_k = K_k$,	*forward, reverse:*	$q_ks_k = s_{k-1}$

Proof. Consider two forward switches. The label $i_k - 1 = i_{k-1}$ of the previous piece has margin a_{k-1} at τ_k, so the growing label has margin $b = (4 - a_{k-1}a_k)/s_{k-1}$ and $a_k + b = K_k/s_{k-1}$. The next piece has the growing label, so $a_{k+1} = (a_k + b)q_k - a_k$ and $s_k = K_kq_k/s_{k-1}$. The other three cases are the same computation with the roles of the growing and the decreasing label exchanged. □

Up to the labels, a period is therefore determined by its signs and its active margins, subject to $q_k > 1$ and $a_k a_{k+1} < 4$. At a turn, $q_k > 1$ reads $a_{k+1} > a_{k-1}$ after a reverse switch and $a_{k+1} < a_{k-1}$ after a forward one. The cost takes a symmetric form in these variables. Let

$$G(a,b) = 3ab - 4 - (a^2+b^2)\log(a+b) + \tfrac{1}{2}a^2\log(a^2+4) + \tfrac{1}{2}b^2\log(b^2+4),$$

and $G_a(a,b) = \partial G/\partial a = 3b - 2a\log(a+b) - \frac{a^2+b^2}{a+b} + a\log(a^2+4) + \frac{a^3}{a^2+4}$.

Lemma B.3 (Signed action)**.** *With the switch signs $\varepsilon_k \in \{\pm 1\}$ of Definition B.5,*

$$\frac{8\,\mathcal{C}(T,X)}{\sqrt{12}} = \sum_{k=0}^{N-1} \varepsilon_k\, G(a_k, a_{k+1}).$$

Proof. Write the cost (B.3) of piece k, times $4/\sqrt{3}$, with the margins of Lemma B.2. It splits as $\varepsilon_k F_1(a_k, a_{k+1}) + \varepsilon_{k-1} F_1(a_k, a_{k-1})$, a term for the switch after the piece and one for the switch before it, with

$$F_1(a,n) = a\Big(n - \frac{4-an}{a+n}\Big) - a^2\log(a+n) + \frac{a^2}{2}\log(a^2+4).$$

The two terms that belong to the switch k add up to $\varepsilon_k(F_1(a_k,a_{k+1}) + F_1(a_{k+1},a_k)) = \varepsilon_k G(a_k,a_{k+1})$. □

The derivative of the sum in Lemma B.3 with respect to one active margin a_k, the others fixed, is $\varepsilon_{k-1}G_a(a_k,a_{k-1}) + \varepsilon_k G_a(a_k,a_{k+1})$. This quantity reappears in the stationarity identity (Lemma B.11).

B.2. **The tangent closure.** The tangent frame $V = gX$ is multiplied on the right by $T_{i_k}(q_k)$ along piece k (Lemma B.1(a)). It starts at $V(0) = X(0)$ and ends at $V(T) = R\,R^{-1}X(0)R = X(0)R$, and $X(0)$ is invertible. This gives one matrix equation that every period satisfies.

Lemma B.4 (Tangent closure)**.** *The factors and labels of a normal vertex period satisfy*

$$T_{i_0}(q_0)\,T_{i_1}(q_1)\cdots T_{i_{N-1}}(q_{N-1}) = R.$$

The closure is simplest in the *triangular coordinates* $Y \mapsto CYC^{-1}$, $C = \left(\begin{smallmatrix}\sqrt{3} & 1\\ \sqrt{3} & -1\end{smallmatrix}\right)$, in which

$$T_0(q) = \begin{pmatrix} q & 0 \\ 0 & q^{-1}\end{pmatrix}, \qquad T_1(q) = \begin{pmatrix} q^{-1} & q-q^{-1} \\ 0 & q\end{pmatrix},$$

$$T_2(q) = \begin{pmatrix} q^{-1} & 0 \\ q^{-1}-q & q\end{pmatrix}, \qquad R = \begin{pmatrix} 0 & 1 \\ -1 & 1\end{pmatrix}.$$

The change of coordinates has $\det C = -2\sqrt{3} < 0$: it reverses orientation, which matters for the sign of ellipticity in Appendix B.6. The labels, the factors and the margins at the switches, related by the update rule of Lemma B.1 and by the closure read from every switch, are all that Appendix H needs to solve the closure for $N \le 5$.

B.3. **Switch signs and the five classes.**

Definition B.5 (Signs, runs, turns, centres)**.** The switch k (from piece k to piece $k+1$) is *forward* if $i_{k+1} = i_k + 1$ and *reverse* if $i_{k+1} = i_k - 1$; its sign is $\varepsilon_k = +1$ or $\varepsilon_k = -1$. A *maximal reverse run* of length $L \geq 1$ is a block of reverse switches $k, \dots, k+L-1$ preceded and followed by forward switches; maximal forward runs are defined in the same way. A forward run of length one is *short*. Piece k is a *turn* if $\varepsilon_{k-1} \neq \varepsilon_k$, a *centre* if $\varepsilon_{k-1} = \varepsilon_k = -1$, and a *forward same-sign piece* if $\varepsilon_{k-1} = \varepsilon_k = +1$.

Consecutive labels are distinct elements of $\mathbb{Z}/3$, so each switch changes the label by one in either direction.

At a forward switch, the next control is the vertex whose margin increased along the preceding piece. At a reverse switch, it is the vertex whose margin decreased. Both kinds occur in extremals, but they behave differently: runs of reverse switches carry the first-order obstructions of Appendix C, while the forward switches of Appendix F are constrained only through the whole period.

Lemma B.6 (Sign sum)**.** $\sum_{k=0}^{N-1} \varepsilon_k \equiv 1 \pmod 3$. *In particular an entirely forward period has* $N \equiv 1 \pmod 3$ *and an entirely reverse period has* $N \equiv 2 \pmod 3$.

Proof. The label advances by ε_k at the switch k, and by one over a period: $i_N = i_0 + 1$ (Definition A.25). Hence $\sum_{k<N} \varepsilon_k \equiv 1$, and if all the signs are equal the sum is $\pm N$. □

The smoothed octagon's period has the word $\mathbf{e}_2\mathbf{e}_0\mathbf{e}_1\mathbf{e}_2$: three forward switches inside the period and the seam switch from $\mathbf{e}_2$ to $\mathbf{e}_{2+1} = \mathbf{e}_0$, which is forward as well (Figure 3). So $N = 4$ and the sign sum is $4 \equiv 1$.

Proof of Proposition 3.2. Every switch is forward or reverse. If both signs occur, the period is mixed, and the remaining alternatives are exhaustive by their definitions. □

B.4. **Costate coordinates.** The costate enters the case analysis only through the vertex Hamiltonians, whose signs the maximum principle controls. We use coordinates in which these Hamiltonians are linear and in which the costate equations along a piece decouple into scalar equations.

Definition B.7 (Costate coordinates)**.** For a star state X and costates Λ_1, Λ_R with $\lambda = -1$, let $L = \Lambda_1 + \frac{3}{2}J$ and

$$\hat{A} = XLX, \qquad \hat{B} = -\big(\Lambda_R + 2L + \langle L, X\rangle X\big),$$

and $\alpha_j = p_j(\hat{A})$, $\beta_j = p_j(\hat{B})$, where the margin $p_j(M) = -\sqrt{3}\langle Z_{\mathbf{e}_j}, M\rangle$ is used as a linear function of every matrix M. The *switching function* of the vertex $\mathbf{e}_j$ is $\chi_j = 2\alpha_j + \beta_j$.

The costate is recovered from (X, α, β) by $L = -X\hat{A}X$ and $\Lambda_R = -\hat{B} - 2L - \langle L, X\rangle X$, so these are coordinates.

Lemma B.8 (The costate along a piece)**.** *Let P be a normal vertex period, and write $\alpha_j(k), \beta_j(k), \chi_j(k)$ for the coordinates at the switch τ_k, extended to $k \in \mathbb{Z}$ by the twist like* (B.1)*.*

(a) *At every time, $p_j(X)\, H_j = \chi_j$ for each vertex Hamiltonian H_j, and $\langle \Lambda_R, X\rangle = 0$ reads $\sum_{j\neq l} \beta_j\, p_l = 0$.*

(b) *Along piece k, with $(a,b,c) = (a_k, m_k(i_k+1), m_k(i_k-1))$, $i = i_k$ and factor $q = q_k$, the coordinate α_i increases by $\mathrm{Act}(a,b,c,q)$ and β_i decreases by twice that amount, so that χ_i is constant, and*

$$\begin{aligned}
(\alpha_i + \alpha_{i+1})(k+1) &= q^2(\alpha_i + \alpha_{i+1})(k) + \mathrm{AG}(a,b,q),\\
(\alpha_i + \alpha_{i-1})(k+1) &= q^{-2}(\alpha_i + \alpha_{i-1})(k) + \mathrm{AD}(a,c,q),\\
(\beta_i + \beta_{i+1})(k+1) &= q\,(\beta_i + \beta_{i+1})(k) + \mathrm{BG}(a,b,q),\\
(\beta_i + \beta_{i-1})(k+1) &= q^{-1}(\beta_i + \beta_{i-1})(k) + \mathrm{BD}(a,c,q),
\end{aligned}$$

with the explicit functions $\mathrm{Act}, \mathrm{AG}, \mathrm{AD}, \mathrm{BG}, \mathrm{BD}$ *of Appendix C.3.1. The same formulas with an intermediate factor $s \in [1,q]$ in place of q give the coordinates inside the piece.*

(c) *$\chi_{i_k}(k) = \chi_{i_k}(k+1) = 0$ for every k; $\chi_j(k) \le 0$ for every j and k; and along piece k the switching functions of the growing and of the decreasing label are ≤ 0 at every intermediate factor $s \in [1, q_k]$.*

Proof. (a) is algebra with the identity $X\Lambda X = \langle \Lambda, X\rangle X + \Lambda$ for traceless Λ. For (b), on a piece the matrix $P = -(\sqrt{3}/a) Z_{\mathbf{e}_i}$ is constant and the costate equations of Definition A.13 become

$$\hat{A}' = 2[P, \hat{A}] - \tfrac{3}{2}[J, X], \qquad \hat{B}' = [P, \hat{B}] + 3[J, X] - H_i[P, X],$$

with $H_i = 0$ on the piece. In margin coordinates the bracket with $Z_{\mathbf{e}_i}$ annihilates p_i and sends $p_{i\pm1}$ to $\mp(2/\sqrt{3})(p_i + p_{i\pm1})$, while $\frac{3}{2}[J,F]$ has margins $\sqrt{3}\,(p_{j+1}(F) - p_{j-1}(F))$. Each of the six combinations in (b) therefore solves a scalar linear equation $y' = \kappa y + f$ with f explicit through (B.2), and an integrating factor solves it. (c) is (a) together with the maximum principle: every vertex Hamiltonian is ≤ 0 at every time and the active one vanishes on its piece (Lemma A.15), and the margins are positive. □

The proofs in Appendices C to G use the costate only through the data listed in Lemma B.8: the coordinates at the switches, their transport, the vanishing of the active switching functions, the sign conditions at the switches and along the pieces, and $\langle \Lambda_R, X\rangle = 0$. In particular no needle variation of the maximum principle is constructed: each first-order argument becomes an exact linear identity between switching functions at the switches (Appendix C).

B.5. **The normal multiplier.** The primary costate is conserved up to conjugation, $\Lambda_1(t) = g(t)^{-1}\Lambda_1(0)g(t)$, so its determinant is constant. Its margin coordinates in the moving frame of the state admit explicit transport formulas.

Definition B.9 (Normal multiplier)**.** Let $\mathcal{U}(t) = \frac{1}{\sqrt{12}} X(t)^{-1} \Lambda_1(t) X(t)$. At the switch k put

$$\mu_j(k) = -\frac{\alpha_j(k)}{2\sqrt{3}} - \frac{\big(m_k(0) + m_k(1) + m_k(2)\big)\, m_k(j)}{4} + \frac{1}{2};$$

at $k = 0$ these are the margins of $\mathcal{U}(0)$. The *Casimir* of P is

$$\gamma^2 = \tfrac{1}{4}(\mu_0\mu_1 + \mu_1\mu_2 + \mu_2\mu_0)(0) = \det \mathcal{U}(0) = \frac{\det \Lambda_1(0)}{12},$$

and when it is positive, $\kappa = \sqrt{12\gamma^2} = \sqrt{\det \Lambda_1}$. In triangular coordinates the traceless matrix with margins (p_0, p_1, p_2) is

$$Y(p) = \frac{1}{2}\begin{pmatrix} -p_0 & p_0 + p_2 \\ -p_0 - p_1 & p_0 \end{pmatrix}, \qquad \det Y(p) = \tfrac{1}{4}(p_0p_1 + p_1p_2 + p_2p_0).$$

The *normal multiplier* read in the frame of piece k is

$$M_k = Y\big(\mu_{i_k}(k), \mu_{i_k+1}(k), \mu_{i_k-1}(k)\big),$$

and $U = M_0$.

Along piece k (label i, factor q) the coordinate μ_i is constant, $\mu_i + \mu_{i+1}$ is multiplied by q^2 and $\mu_i + \mu_{i-1}$ by q^{-2}: the inhomogeneous terms of Lemma B.8(b) cancel against those of the margins. Hence the quadratic form $\frac{1}{4}(\mu_0\mu_1 + \mu_1\mu_2 + \mu_2\mu_0)$ takes the value γ^2 at every switch. In the triangular frame the transport is a conjugation, $M_{k+1} = A_k^{-1} M_k A_k$ with

$$A_k = T_0(q_k)\, R^{-\varepsilon_k}.$$

Put $P_n = A_0 A_1 \cdots A_{n-1}$, so that $M_n = P_n^{-1} U P_n$. Since $R^{-1} T_i R = T_{i+1}$ and $R^3 = -I$, the tangent closure (Lemma B.4) gives $P_N = \pm I$.

The multiplier enters the estimates through the symmetric matrix $S = -J_0 U$, with $J_0 = \left(\begin{smallmatrix} 0 & -1 \\ 1 & 0 \end{smallmatrix}\right)$, whose determinant is γ^2, and through three sequences of numbers. Let $w_k = -\mu_{i_k}(k)$ and

$$d_k = \frac{w_k + w_{k+1}}{2s_k^2}, \qquad e_k = \frac{w_k}{2K_k}, \qquad j_k = \frac{a_k^2 + a_k a_{k+1} + a_{k+1}^2}{4s_k^2}.$$

The number d_k is the *switch norm* of the switch k and e_k the *cross pairing* at piece k; the number $j_k \geq 3/16$ depends only on the active margins. The names come from the *switch vectors*. In the frame of piece k let the outgoing vector be $o_k = (q_k/s_k, 0)$ if the switch k is forward and $o_k = (0, 1/(q_k s_k))$ if it is reverse, and let the incoming vector be $\iota_k = (0, -1/s_{k-1})$ or $(-1/s_{k-1}, 0)$ according as the switch $k-1$ is forward or reverse. The switch vector of the switch $n \geq 0$ is $y_n = P_n\, o_n$; it is also $P_{n+1}\iota_{n+1}$.

Lemma B.10 (Switch vectors)**.** (a) $y_n^{\mathsf{T}} S y_n = d_n$ *for every* $n \geq 0$.

(b) *At a same-sign piece* $k \geq 1$, $\iota_k^{\mathsf{T}}(-J_0 M_k)\, o_k = e_k$ *and* $\det(\iota_k, o_k) = \varepsilon_k/K_k$*; hence* $y_{k-1}^{\mathsf{T}} S y_k = e_k$ *and* $\det(y_{k-1}, y_k) = \varepsilon_k/K_k$.

(c) *At a turn* k, $o_k = -\iota_k$*, so* $y_k = -y_{k-1}$.

(d) $y_{N+n} = \pm y_n$*, with the sign of* $P_N = \pm I$.

The entries of $-J_0 M_k$ are $-(\mu_i+\mu_{i+1})/2$, $\mu_i/2$, $\mu_i/2$ and $-(\mu_i+\mu_{i-1})/2$, with $i=i_k$, and (a)–(c) evaluate this quadratic form on the vectors above, using the factors of Lemma B.2 and the transport of μ.

The maximum principle imposes two kinds of conditions at a switch: the switching function of the new control vanishes there, and the switching function of the old control stays ≤ 0 just after it. The first gives an identity, the second an inequality.

Lemma B.11 (Stationarity)**.** *For every k,*

$$2\big(\varepsilon_{k-1}s_{k-1}d_{k-1}+\varepsilon_k s_k d_k-(\varepsilon_{k-1}+\varepsilon_k)\,a_k e_k\big) \\ =\frac{1}{8}\big(\varepsilon_{k-1}G_a(a_k,a_{k-1})+\varepsilon_k G_a(a_k,a_{k+1})\big).$$

At a centre k, with $a=a_k$, $p=a_{k-1}$, $n=a_{k+1}$, $r=a/(a+p)$, $t=a/(a+n)$ and $\rho=a^2/(a^2+4)$, this reads

$$16e_k=1+r+t-\rho+\log\frac{\rho}{rt}-\frac{1-8d_{k-1}}{r}-\frac{1-8d_k}{t}.$$

The right side is $1/8$ of the derivative of the signed action of Lemma B.3 with respect to a_k; the left side is the corresponding derivative of the closure condition, paired with the multiplier. The proof is local: by Lemma B.8, the difference of the two sides is an explicit nonzero multiple of the switching function of the next control at the end of piece k, which vanishes.

Lemma B.12 (Switch norms)**.** (a) *At a forward switch $d_k\ge j_k/2$; at a reverse switch $d_k\le j_k/2$.*

(b) *At a turn k, $d_{k-1}=d_k$ and $16(a_{k+1}-a_{k-1})\,d_k=G_a(a_k,a_{k+1})-G_a(a_k,a_{k-1})$. With $f(x)=x-x^{-1}-\log x$ this is the secant*

$$d_k=\frac{1}{8}\cdot\frac{f(1+a_{k+1}/a_k)-f(1+a_{k-1}/a_k)}{a_{k+1}/a_k-a_{k-1}/a_k},$$

and $d_k\ge 3/32$.

Proof. (a) At a reverse switch k the old label $i_k=i_{k+1}+1$ is the growing label of piece $k+1$. Its switching function vanishes at the start of that piece and is ≤ 0 along it. By Lemma B.8(b) it equals $2s(Y(s-1)+\frac{\sqrt{3}a}{2}F(s))$ at the intermediate factor s, where $a=a_{k+1}$, $Y=(\alpha_i+\alpha_{i+1})(k+1)$ with $i=i_{k+1}$, and F is the function of Appendix C.3.1, with $F(1)=F'(1)=0$. Its derivative at $s=1$ is $2Y$, so $Y\le 0$, and an exact identity gives $d_k-j_k/2=Y/(4\sqrt{3}s_k^2)$. A forward switch is the same with the decreasing label. (b) At a turn the transport of μ along the piece and the factor relation of Lemma B.2 give $d_{k-1}=d_k$, and then $\varepsilon_{k-1}+\varepsilon_k=0$ reduces Lemma B.11 to the secant. Since $f'(x)-\frac{3}{4}=(x-2)^2/(4x^2)\ge 0$, every secant slope of f is at least $3/4$. □

The inequalities in (a) prevent the preceding control from immediately having a larger Hamiltonian than the new control.

Lemma B.13 (Gram identity)**.** *At a centre or a forward same-sign piece k, $d_{k-1}d_k - e_k^2 = \gamma^2/K_k^2$.*

This is the Gram determinant of y_{k-1}, y_k for the form S: $\det S = \gamma^2$ and $\det(y_{k-1}, y_k)^2 = 1/K_k^2$ by Lemma B.10.

B.6. **Ellipticity and the Casimir cap.**

Proposition B.14 (Cost–Casimir inequality)**.** *Let P be a normal vertex period such that $\det \Lambda_1(0) > 0$ and the lower left entry of $\Lambda_1(0)$ is negative, and let $\kappa = \sqrt{\det \Lambda_1(0)}$. Then*

$$\mathcal{C}(T, X) \geq \frac{4\pi\kappa}{2\kappa + 3}.$$

Corollary B.15 (Casimir cap)**.** *Let P be a competitive normal vertex period ($\mathcal{C}(T, X) \leq A_{\mathrm{oct}}$) with $\gamma^2 > 0$ and $d_0 > 0$. Then $\gamma^2 < 5/27$. In particular this holds for every competitive mixed normal vertex period.*

Proposition B.16 (Ellipticity)**.** *Every mixed normal vertex period has $\gamma^2 > 0$.*

The circular trajectory, with $X \equiv J$ and $T = \pi/3$, has cost π. The lower bound $4\pi\kappa/(2\kappa + 3)$ in Proposition B.14 equals π at $\kappa = 3/2$. Consequently, a period of cost below π has $\kappa < 3/2$, with a quantitative bound determined by its cost. This conclusion requires Λ_1 to be negative elliptic.

The proof of Proposition B.16 is in Appendix D.3. It uses the run structure of mixed periods (Proposition D.2), which is proved from Lemma B.8 alone. For entirely forward periods the same argument gives $\gamma^2 > 0$.

When $\gamma^2 > 0$ and $d_0 > 0$, the symmetric form S has positive determinant and is positive on y_0, so it is positive definite. Then $\Lambda_1(0)$ is negative elliptic: $\det \Lambda_1(0) > 0$ and its lower left entry is negative. The positive definite S corresponds to a negative elliptic Λ_1 because the triangular coordinates reverse orientation: if Y is traceless with $\det Y > 0$ and A is invertible, the lower left entries of Y and of $AYA^{-1} \det A$ have the same sign, and $\det C < 0$.

Proof of Proposition B.14. A negative elliptic traceless matrix is $\Lambda_1(0) = -\kappa\, hJh^{-1}$ with $\det h = 1$. Put $B(t) = h^{-1}g(t)$, so that $B' = BX$, $\det B = 1$ and $B(T) = B(0)R$. Write $\rho(Y) = -\frac{1}{2}\langle J, Y\rangle$; the cost is $3\int_0^T \rho(X)\,\mathrm{d}t$. By conservation, $\langle \Lambda_1(t), X(t)\rangle = 2\kappa\,\rho(BXB^{-1})$. The maximum principle gives $\langle \Lambda_1 + \frac{3}{2}J, X\rangle \leq 0$ at every time: a positive combination of the three vertex Hamiltonians cancels their Λ_R terms. Hence

$$\rho(X) + \rho(BXB^{-1}) \leq \Big(1 + \frac{3}{2\kappa}\Big)\rho(X).$$

The complex number $z(B) = \frac{1}{2}(B_{11} + B_{22}) + \frac{i}{2}(B_{21} - B_{12})$ has $|z| \geq 1$, its argument increases at the rate $(\rho(X) + \rho(BXB^{-1}))/(2|z|^2)$, and $z(B(0)R) =$

$e^{i\pi/3}z(B(0))$. The argument therefore increases by at least $\pi/3$ over $[0,T]$, and

$$\int_0^T \left(\rho(X)+\rho(BXB^{-1})\right)\mathrm{d}t \geq \frac{2\pi}{3}.$$

Combining the two displays, $(1+\frac{3}{2\kappa})\,\mathcal{C}/3 \geq 2\pi/3$. □

The polar-angle argument also gives the winding bound behind the singular clearance (Proposition A.20).

Proof of Corollary B.15. By the discussion before Proposition B.14, $\Lambda_1(0)$ is negative elliptic with $\kappa^2 = 12\gamma^2$. Competitiveness, Proposition B.14 and Lemma C.5 give

$$\frac{4\kappa}{2\kappa+3} \leq \frac{\mathcal{C}}{\pi} \leq \frac{A_{\mathrm{oct}}}{\pi} < \frac{8\sqrt{5}}{9+4\sqrt{5}}.$$

Clearing the positive denominators gives $\kappa < 2\sqrt{5}/3$, and hence $\gamma^2 < 5/27$. For a mixed period, $\gamma^2 > 0$ by Proposition B.16, and $d_0 \geq 3/32$ by Corollary D.3. □

The exact threshold supplied by the cost inequality is $\kappa_* = 3A_{\mathrm{oct}}/(4\pi - 2A_{\mathrm{oct}})$ (Figure 7(b)). The cap $5/27$ matches the geometric comparison in class M1. For entirely forward periods Appendix F uses the stronger spectral bound $\mathcal{C} \geq \frac{\pi}{3}(\kappa+\frac{3}{2})$; the difference of the two bounds is $\pi(2\kappa-3)^2/(6(2\kappa+3)) \geq 0$.

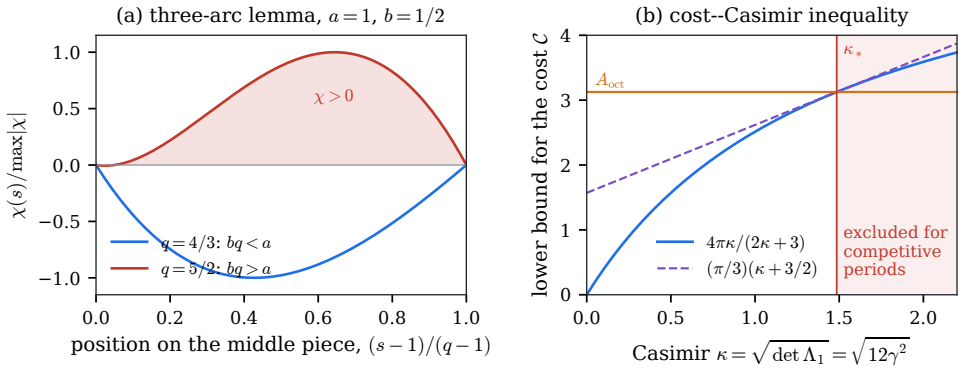


FIGURE 7. (a) The obstruction of the three-arc lemma (Lemma D.1), growing case, with active margin $a = 1$ of the middle piece and outer margin $b = a_k = 1/2$. The switching function of the outer label vanishes at both ends of the middle piece, and its value inside is fixed by this. For $q = 4/3$ ($bq < a$) it is negative, as the maximum principle requires; for $q = 5/2$ ($bq > a$, that is, $a_k a_{k+2} > a_{k+1}^2$) it becomes positive. The curves are normalized by their maximum modulus. (b) The cost–Casimir inequality of Proposition B.14 (solid) and the spectral bound of forward periods (dashed), against the octagon's cost A_{oct}. A competitive period has $\kappa < \kappa_* = 3A_{\mathrm{oct}}/(4\pi - 2A_{\mathrm{oct}})$; the proof of Corollary B.15 compares it with $2\sqrt{5}/3$. Both bounds equal π at $\kappa = 3/2$.

Appendix C. Class R: every switch reverse

This appendix proves Theorem 4.1. In an entirely reverse period, a piece of smallest active margin lies inside a sequence of reverse switches. The reverse-four principle rules out such a minimum. We first prove that local principle, then apply it to the period. The final part records the costate formulas and scalar estimates also used in Appendix D.

C.1. **The reverse-four principle.**

Lemma C.1 (Reverse-four principle)**.** *Let four consecutive pieces $k, \dots, k+3$ of a normal vertex period have three reverse switches between them. Then*

(a) *it is not the case that $a_{k+2} \le a_k$, $a_{k+2} \le a_{k+1}$ and $a_{k+2} \le a_{k+3}$;*
(b) *it is not the case that $a_{k+1} \le a_k$, $a_{k+1} \le a_{k+2}$ and $a_{k+1} \le a_{k+3}$.*

Consider four consecutive pieces $k, k+1, k+2, k+3$ with three reverse switches between them, so that their labels are i, $i-1$, $i+1$, i. Name the margins at the switch τ_{k+1} in the label order $(i, i+1, i-1)$ as (P, Q, Z), and the factors of the two interior pieces as $b = q_{k+1}$ and $c = q_{k+2}$. Then $P = a_k$ and $Z = a_{k+1}$. Along piece $k+1$ (label $i-1$) the label i grows and $i+1$ decreases, so at τ_{k+2}

$$A := m_{k+2}(i+1) = \frac{Z+Q}{b} - Z = a_{k+2}, \qquad B := m_{k+2}(i) = (Z+P)\,b - Z,$$

and along piece $k+2$ (label $i+1$) the label i decreases, so $a_{k+3} = (A+B)/c - A$.

Suppose the third piece is a weak minimum: $a_{k+2} \le a_k, a_{k+1}, a_{k+3}$. In these variables this reads

$$A \le Z, \qquad bA + (b-1)Z \le B, \qquad (2c-1)A \le B. \tag{C.1}$$

The obstruction is a variation of the period: insert short pieces of the two inactive controls at the two interior switches (the control $i+1$ at τ_{k+1} and the control i at τ_{k+2}), and change the logarithms of the factors of the four pieces at rates chosen so that the end state and the end frame do not move. To first order the normalized cost $\mathcal{C}/\sqrt{12}$ then changes by $\frac{Z}{4} N(A, B, Z, b, c)$ times the size of the variation, where

$$\begin{aligned} N(A, B, Z, b, c) = {} & b(b-1)A\big(Ac\,\eta(c) - B\,\zeta(c)\big) \\ & + (c-1)\big(bA(B+bZ)\,\eta(b) - B(bA+Z)\,\zeta(b)\big), \end{aligned} \tag{C.2}$$

with $\eta(t) = t - 1 - \log t$ and $\zeta(t) = t \log t - t + 1$, both positive for $t > 1$. Under the minimum hypothesis (C.1) this derivative is negative (Lemma C.2). The maximum principle, however, requires the inserted controls to have Hamiltonians no larger than those of the active controls, excluding a first-order cost decrease. We express this obstruction directly as an exact linear identity between switching functions at the switches (Lemma C.3). The proof uses the signs of these functions and does not require construction of the variation.

Lemma C.2 (Sign of the cost derivative)**.** *If $b, c > 1$, $A, Z > 0$ and (C.1) holds, then $N(A, B, Z, b, c) < 0$.*

The proof (Appendix C.3.3) rests on the inequality $(2t+1)\,\eta(t) < (t+2)\,\zeta(t)$ for $t > 1$, which is the Padé bound $\log t > 3(t^2-1)/(t^2+4t+1)$.

Lemma C.3 (The reverse-four identities)**.** *In the situation above, with $\chi_j(k)$ the switching functions at the switch τ_k (Definition B.7),*

$$B(c-1)\,\chi_{i+1}(k+1) + bD\,\chi_i(k+2) = -\sqrt{3}\,Z\,N(A, B, Z, b, c), \tag{C.3}$$

$$(b-1)B\,\chi_{i-1}(k+3) + cD\,\chi_i(k+2) = -\sqrt{3}\,A\,N(Z, B, A, c, b), \tag{C.4}$$

where $D = A(c-1) + Z(b-1)$.

Proof. Express the costate coordinates at τ_{k+2} and τ_{k+3} through those at τ_{k+1} by the transport of Lemma B.8(b) over the two interior pieces. The left sides are then linear in the six coordinates $\alpha_j(k+1), \beta_j(k+1)$, with coefficients rational in P, Q, Z, b, c and linear in $\log b$, $\log c$. Five linear relations hold among these coordinates: the switching functions of the active labels vanish at τ_{k+1} (labels i and $i-1$), at τ_{k+2} (label $i+1$) and at τ_{k+3} (label i), and $\langle \Lambda_R, X \rangle = 0$ at τ_{k+1}. Each identity is a combination of these five relations with polynomial coefficients, listed in Appendix C.3.2; after clearing denominators both sides are the same polynomial in the variables and in $\log b$, $\log c$. □

The coefficients $B(c-1)$ and bD in (C.3) are the weights of the two inserted pieces in the variation described above, and N is its cost derivative.

Proof of Lemma C.1. (a) The labels $i+1$ at τ_{k+1} and i at τ_{k+2} are inactive on both adjacent pieces, so their switching functions are ≤ 0 (Lemma B.8(c)). The weights $B(c-1)$ and bD are positive, so the left side of (C.3) is ≤ 0 and $N(A, B, Z, b, c) \ge 0$. The weak minimum at the third piece is (C.1), and Lemma C.2 gives $N < 0$.

(b) A weak minimum at the second piece reads $Z \le A$, $cZ + (c-1)A \le B$ and $(2b-1)Z \le B$: these are the conditions (C.1) with (A, Z, b, c) replaced by (Z, A, c, b), so $N(Z, B, A, c, b) < 0$. The switching functions $\chi_{i-1}(k+3)$ and $\chi_i(k+2)$ are ≤ 0, and (C.4) gives the contradiction as before. □

C.2. **Class R.**

Proof of Theorem 4.1. The active margins are N-periodic, so one of them, a_m, is a global minimum. Every switch is reverse, so the four pieces $m-2, \dots, m+1$ have three reverse switches between them, and $a_m \le a_{m-2}, a_{m-1}, a_{m+1}$. This contradicts Lemma C.1(a) for $k = m-2$. □

The window may run through the seam, and it may contain the same piece of the period twice when $N < 4$; the twisted indexing of Appendix B makes no distinction. The argument uses neither the cost nor the number of pieces, and Appendix H uses it also for the entirely reverse words with five pieces.

C.3. **Costate identities for reverse switches.** This subsection gives the explicit formulas behind Appendices B to D: the one-piece costate map, the two reverse-four identities, the sign of the cost derivative N, and the elementary inequalities of the three-arc lemma, the switch norms and the Casimir cap. Each identity below is an identity of rational functions in the displayed variables and the logarithms that occur; it is verified by clearing denominators and expanding.

C.3.1. *The one-piece costate map.* On a piece with active margin a, growing margin b and decreasing margin c at its start (Lemma B.1), the inhomogeneous terms of Lemma B.8(b) at the factor q are

$$\begin{aligned}
\mathrm{Act}(a,b,c,q) &= \tfrac{\sqrt{3}a}{2}\big((a+c)(1-q^{-1})-(a+b)(q-1)\big),\\
\mathrm{AG}(a,b,q) &= \tfrac{\sqrt{3}a}{2}\big(a(q^2-1)-(a+b)(q^2-q)\big),\\
\mathrm{AD}(a,c,q) &= \tfrac{\sqrt{3}a}{2}\big((a+c)(q^{-1}-q^{-2})-a(1-q^{-2})\big),\\
\mathrm{BG}(a,b,q) &= \sqrt{3}a\big((a+b)\,q\log q-2a(q-1)\big),\\
\mathrm{BD}(a,c,q) &= \sqrt{3}a\,q^{-1}\big(2a(q-1)-(a+c)\log q\big).
\end{aligned}$$

The logarithms come from the cost term $-\frac{3}{2}[J,X]$ of the costate equation, integrated against the exponentially moving margins.

These formulas give the switching functions of the two inactive labels inside the piece. Let i be the active label, and at the start of the piece let $Y=\alpha_i+\alpha_{i+1}$, $Y'=\alpha_i+\alpha_{i-1}$, $B=\beta_i+\beta_{i+1}$, $B'=\beta_i+\beta_{i-1}$. Since $\chi_i=0$ on the piece, at the intermediate factor $s\in[1,q]$

$$\chi_{i+1}(s)=2s^2Y+sB+\sqrt{3}\,a\,s\,F(s),\qquad \chi_{i-1}(s)=\frac{2Y'}{s^2}+\frac{B'}{s}-\frac{\sqrt{3}\,a}{s}\,h(s),$$

where

$$\begin{aligned}
F(s)&=(a+b)\log s-(a+b)(s-1)+a\,\frac{(s-1)^2}{s},\\
h(s)&=(a+c)\log s-a(s-1)-c\,(1-s^{-1}),
\end{aligned}$$

because $2\mathrm{AG}+\mathrm{BG}=\sqrt{3}asF$ and $2\mathrm{AD}+\mathrm{BD}=-\sqrt{3}a\,h/s$. Both F and h vanish to second order at $s=1$: $F(1)=F'(1)=h(1)=h'(1)=0$. When the growing label was active on the previous piece, $\chi_{i+1}(1)=2Y+B=0$ and

$$\chi_{i+1}(s)=2s\Big(Y(s-1)+\frac{\sqrt{3}a}{2}F(s)\Big);$$

when the decreasing label was, $B'=-2Y'$ and

$$\chi_{i-1}(s)=-\frac{2}{s}\Big(Y'(1-s^{-1})+\frac{\sqrt{3}a}{2}h(s)\Big).$$

These are the forms used in Lemma B.12(a) and in Lemma D.1. In each case the derivative at $s=1$ is $\pm2Y$ or $\pm2Y'$, which is how the sign condition at the start of a piece fixes the sign of Y or Y'.

C.3.2. *The reverse-four identities.* In the notation of Appendix C.1, let x_j, y_j be the coordinates α, β at the switch τ_{k+1} in the label order $(i, i+1, i-1)$, so that $j = 0, 1, 2$ stands for the labels $i, i+1, i-1$, and let u_j, v_j and r_j, s_j be those at τ_{k+2} and τ_{k+3} in the same order. The transport over piece $k+1$ (active margin Z, growing margin P, decreasing margin Q, factor b) and over piece $k+2$ (active margin A, growing margin Z, decreasing margin B, factor c) gives

$$\begin{aligned}
u_2 &= x_2 + \mathrm{Act}(Z, P, Q, b),\\
u_0 &= b^2(x_2 + x_0) + \mathrm{AG}(Z, P, b) - u_2,\\
u_1 &= b^{-2}(x_2 + x_1) + \mathrm{AD}(Z, Q, b) - u_2,\\
r_1 &= u_1 + \mathrm{Act}(A, Z, B, c),\\
r_0 &= c^{-2}(u_1 + u_0) + \mathrm{AD}(A, B, c) - r_1,\\
r_2 &= c^2(u_1 + u_2) + \mathrm{AG}(A, Z, c) - r_1,
\end{aligned}$$

and the same with $(\alpha, \mathrm{Act}, \mathrm{AG}, \mathrm{AD})$ replaced by $(\beta, -2\mathrm{Act}, \mathrm{BG}, \mathrm{BD})$ and the powers $b^{\pm 2}, c^{\pm 2}$ by $b^{\pm 1}, c^{\pm 1}$. The five relations are

$$h_1 = 2x_0 + y_0, \quad h_2 = 2x_2 + y_2, \quad h_3 = 2u_1 + v_1, \quad h_4 = 2r_0 + s_0,$$
$$h_5 = y_0(Q + Z) + y_1(P + Z) + y_2(P + Q),$$

all equal to 0: the active switching functions at the three switches, and $\langle \Lambda_R, X \rangle = 0$ at τ_{k+1}. Then, with $D = A(c-1) + Z(b-1)$ and as identities in all the variables,

$$\begin{aligned}
&B(c-1)(2x_1 + y_1) + bD(2u_0 + v_0) + \sqrt{3}ZN(A, B, Z, b, c)\\
&\quad = b^2(c-1)(Q - Zb + Z)\, h_1 + (b^2 - 1)(c-1)(Pb + Q)\, h_2\\
&\qquad + b(c-1)(Pb^2 + Zb^2 + Zbc - Zc - Z)\, h_3 + Zbc^2(b-1)\, h_4\\
&\qquad - b(b-1)(c-1)\, h_5,
\end{aligned}$$

and

$$\begin{aligned}
&(b-1)B(2r_2 + s_2) + cD(2u_0 + v_0) + \sqrt{3}AN(Z, B, A, c, b)\\
&\quad = bc(c-1)(Q - Zb + Z)\, h_1\\
&\qquad + \tfrac{c(b-1)}{b}(Pb^2c + Qbc - Qb + Qc - Q + Zb^2 - Zbc + Zc - Z)\, h_2\\
&\qquad + (b-1)(c^2 - 1)(Pb + Zb + Zc - Z)\, h_3 + Zc^3(b-1)\, h_4\\
&\qquad - c(b-1)(c-1)\, h_5,
\end{aligned}$$

with $A = (Z + Q)/b - Z$ and $B = (Z + P)b - Z$. Setting the five relations to zero gives Lemma C.3.

C.3.3. *The sign of the cost derivative.* Let $\eta(t) = t - 1 - \log t$ and $\zeta(t) = t \log t - t + 1$. For $t > 1$ both are positive.

Lemma C.4 (Padé bound)**.** *For $t > 1$, $3(t^2 - 1) < (t^2 + 4t + 1)\log t$; equivalently $(2t + 1)\,\eta(t) < (t + 2)\,\zeta(t)$. In particular $\eta(t) < \zeta(t)$.*

Proof. The function $\psi(t) = \log t - 3(t^2-1)/(t^2+4t+1)$ vanishes at 1 and $\psi'(t) = (t-1)^4/(t(t^2+4t+1)^2) > 0$ for $t > 1$. The second form follows from $(t+2)\zeta(t) - (2t+1)\eta(t) = (t^2+4t+1)\log t - 3(t^2-1)$, and the third because $(2t+1)/(t+2) > 1$ and $\eta(t) > 0$. □

Proof of Lemma C.2. Recall (C.2) and the hypotheses $b, c > 1$, $A, Z > 0$, $A \le Z$, $bA + (b-1)Z \le B$ and $(2c-1)A \le B$. In particular $B > cA > 0$.

The first bracket of N is negative: $Ac\,\eta(c) < Ac\,\zeta(c) < B\,\zeta(c)$.

The second bracket, $T(B) = bA(B+bZ)\,\eta(b) - B(bA+Z)\,\zeta(b)$, is affine in B with slope $bA\,\eta(b) - (bA+Z)\,\zeta(b) < 0$, so $T(B) \le T(B_0)$ with $B_0 = bA + (b-1)Z$. At B_0 we have $B_0 + bZ = bA + (2b-1)Z$ and $B_0(bA+Z) = b^2A^2 + b^2AZ + (b-1)Z^2 > 0$, and Lemma C.4 at $t = b$ gives

$$\begin{aligned}(b+2)\,T(B_0) &< \eta(b)\big[(b+2)\,bA\big(bA + (2b-1)Z\big)\\ &\qquad - (2b+1)\big(b^2A^2 + b^2AZ + (b-1)Z^2\big)\big]\\ &= -(b-1)\,\eta(b)\big(b^2A^2 + 2bZ(Z-A) + Z^2\big),\end{aligned}$$

which is ≤ 0 because $Z \ge A$. So $T(B) < 0$, and $N = b(b-1)A\cdot(\text{first bracket}) + (c-1)\,T(B) < 0$. □

C.3.4. *Elementary inequalities. Logarithms.* For $q > 1$, $2(q-1) < (q+1)\log q$ and $q - 1 < q\log q$: both differences vanish at $q = 1$, with derivatives $\log q + q^{-1} - 1 > 0$ and $\log q > 0$. They give the signs in the proof of Lemma D.1.

The turn secant. With $f(x) = x - x^{-1} - \log x$,

$$f'(x) - \tfrac{3}{4} = \frac{(x-2)^2}{4x^2}, \qquad 1 - f'(x) = \frac{x-1}{x^2}.$$

So every secant slope of f on $(1,\infty)$ lies in $[3/4, 1)$, and the switch norm at a turn, $d = \frac{1}{8}\times(\text{secant slope})$, satisfies $3/32 \le d < 1/8$. At a forward switch the lower bound $d_k \ge 3/32$ comes instead from $j_k/2 - 3/32 = (a_k - a_{k+1})^2/(32s_k^2) \ge 0$.

The cross pairing bound. With $a, p, n > 0$ and $j(a,b) = (a^2+ab+b^2)/(4(a+b)^2)$,

$$\begin{aligned}&\frac{1}{2a}\Big((a+p)\,\frac{j(p,a)}{2} + (a+n)\,\frac{j(a,n)}{2} - \frac{G_a(a,p) + G_a(a,n)}{16}\Big)\\ &\qquad = \frac{1}{16}\Big(-1 + \frac{2a}{a+p} + \frac{2a}{a+n} - \frac{a^2}{a^2+4} + \log\frac{(a+p)(a+n)}{a^2+4}\Big).\end{aligned}$$

This is the quantity X_k of Lemma D.5: by Lemma B.11, at a same-sign piece e_k is an increasing affine function of d_{k-1} and d_k, and X_k is its value at $d = j/2$.

Lemma C.5 (Octagon cost comparison)**.** *The octagon cost satisfies*

$$\frac{A_{\mathrm{oct}}}{\pi} < \frac{203}{204} < \frac{8\sqrt{5}}{9+4\sqrt{5}}, \qquad A_{\mathrm{oct}} < \frac{161\pi}{162} + \frac{9\sqrt{3}}{4000}.$$

Proof. We use positive integral remainders in the exact formula for A_{oct}. Set $s = \sin(\pi/12) = (\sqrt{6} - \sqrt{2})/4$ and put

$$P = 12s\left(1 + \frac{s^2}{6} + \frac{3s^4}{40} + \frac{5s^6}{112}\right), \qquad L = 2\sum_{j=0}^{4} \frac{1}{(2j+1)3^{2j+1}}.$$

For $p(t) = 1 + t^2/2 + 3t^4/8 + 5t^6/16$, the identity

$$1 - (1-t^2)p(t)^2 = \frac{t^8(140 + 56t^2 + 35t^4 + 25t^6)}{256} > 0 \qquad (0 < t < 1)$$

and positivity of $p(t)$ and $\sqrt{1-t^2}$ give $p(t) < 1/\sqrt{1-t^2}$. Integrating yields

$$\pi = 12\int_0^s \frac{\mathrm{d}t}{\sqrt{1-t^2}} > 12\int_0^s \left(1 + \frac{t^2}{2} + \frac{3t^4}{8} + \frac{5t^6}{16}\right)\mathrm{d}t = P.$$

Similarly, the geometric remainder gives

$$\log 2 = 2\int_0^{1/3} \frac{\mathrm{d}t}{1-t^2} > 2\int_0^{1/3} \sum_{j=0}^{4} t^{2j}\,\mathrm{d}t = L.$$

Thus

$$A_{\mathrm{oct}} < U := \frac{\sqrt{12}(8 - 4\sqrt{2} - L)}{2\sqrt{2} - 1}.$$

We compare U directly with $qP + b\sqrt{3}$, for the two pairs

$$(q, b) \in \left\{\left(\frac{203}{204}, 0\right), \left(\frac{161}{162}, \frac{9}{4000}\right)\right\}.$$

Write

$$m = 1 + \frac{1}{12} + \frac{21}{640} + \frac{65}{3584}, \qquad n = \frac{1}{24} + \frac{3}{160} + \frac{75}{7168},$$
$$p = 3(m+n), \qquad r = 3(m+3n).$$

The bracket in P expands to $m - n\sqrt{3}$, so $P = p\sqrt{6} - r\sqrt{2}$. With

$$A = 4qp - 16 + 2L - b, \qquad B = 8 - qp + 2b, \qquad C = qr,$$

we have

$$(qP + b\sqrt{3} - U)(2\sqrt{2} - 1) = (A + B\sqrt{2})\sqrt{3} - C(4 - \sqrt{2}).$$

For each pair, direct rational substitution gives $B > 0$, $A + B > 0$, $C > 0$ and

$$R := 18C^2 - 3A^2 - 6B^2 > 0, \qquad T := 6AB + 8C^2 > 0,$$
$$2T^2 - R^2 > 0.$$

The quantities $(A + B\sqrt{2})\sqrt{3}$ and $C(4 - \sqrt{2})$ are positive. Their squared difference is $T\sqrt{2} - R > 0$, proving $U < qP + b\sqrt{3}$. Since $q > 0$ and $P < \pi$, both cost bounds follow. Finally, $\lambda = \sqrt{5} - 2 = 1/(\sqrt{5} + 2) < 1/4$ gives

$$\frac{8\sqrt{5}}{9 + 4\sqrt{5}} = 1 - \lambda^4 > 1 - 4^{-4} > \frac{203}{204}.$$

□

Appendix D. Class M3: reverse runs of length other than two

This appendix proves Theorem 4.2. The reverse-four principle of Appendix C prevents an interior minimum in a long reverse run. A second local inequality controls the ends of a run. Together they force every reverse run in a mixed normal period to have exactly two switches. The resulting strict minimum at each centre also gives the positivity of the Casimir, proved at the end of the appendix.

D.1. **Three pieces with alternating labels.**

Lemma D.1 (Three-arc log-concavity and turn ordering)**.** *Let $k, k+1, k+2$ be consecutive pieces of a normal vertex period.*

(a) *If $i_{k+2} = i_k$, then $a_k a_{k+2} < a_{k+1}^2$; in particular a_{k+1} is not at most both a_k and a_{k+2}.*

(b) *If the switch k is reverse and the switch $k+1$ forward, then $a_k < a_{k+2}$; if the switch k is forward and the switch $k+1$ reverse, then $a_{k+2} < a_k$.*

When the switch after piece k is followed by a switch back to the label of piece k, the middle piece is a turn, and its two outer pieces have the same control. The switching function of that control vanishes at both ends of the middle piece and is ≤ 0 in between. This forces a relation between the three active margins.

Proof of Lemma D.1. (b) After a reverse switch the label i_k is the growing label of piece $k+1$, so $a_{k+2} = (a_{k+1} + a_k)\, q_{k+1} - a_{k+1} > a_k$ by Lemma B.1; after a forward switch it is the decreasing label and $a_{k+2} = (a_{k+1} + a_k)/q_{k+1} - a_{k+1} < a_k$.

(a) Write $a = a_{k+1}$ and $q = q_{k+1}$. Suppose first that the switch k is reverse, so that the outer label $i = i_k$ is the growing label of the middle piece, with margin $b = a_k$ at its start, and $a_{k+2} = (a+b)q - a$. Since

$$a^2 - b\big((a+b)q - a\big) = (a+b)(a - bq),$$

the claim is $bq < a$. By Lemma B.8 and Appendix C.3.1, the switching function of the label i at the intermediate factor $s \in [1, q]$ is

$$\chi_i(s) = 2s\Big(Y(s-1) + \frac{\sqrt{3}a}{2} F(s)\Big),$$

$$F(s) = (a+b)\log s - (a+b)(s-1) + a\frac{(s-1)^2}{s},$$

for a number Y. It vanishes at $s = 1$ and $s = q$ and is ≤ 0 in between. The vanishing at q fixes Y, and $\chi_i \le 0$ just before q makes the derivative at q nonnegative. Together these say $(q-1)F'(q) \ge F(q)$, that is,

$$(q-1)(2aq - a + bq) \ge (a+b)\, q^2 \log q.$$

If $bq \ge a$, this fails: the right side minus the left side equals $a\big(q^2 \log q - (q-1)(2q-1)\big) + b\big(q^2 \log q - q(q-1)\big)$; the second bracket is positive because

$q\log q > q-1$; if the first bracket is negative, replace a by the larger bq, which leaves $bq^2\big((q+1)\log q - 2(q-1)\big) > 0$. So $bq < a$.

If the switch k is forward, the outer label is the decreasing label of the middle piece, with margin $c = a_k$, and $a_{k+2} = (a+c)/q - a$. Now $a^2 - c\big((a+c)/q - a\big) = (a+c)(a - c/q)$, and the claim is $c < aq$. Here $\chi_i(s) = -(2/s)\big(Y(1-s^{-1}) + \frac{\sqrt{3}a}{2}h(s)\big)$ with $h(s) = (a+c)\log s - a(s-1) - c(1-s^{-1})$, and $h(1) = h'(1) = 0$. The sign at the start of the piece gives $Y \geq 0$. If $aq \leq c$, then

$$h(q) = (c - aq)\big(\log q - (1 - q^{-1})\big) + a\big((q+1)\log q - 2(q-1)\big) > 0,$$

so $\chi_i(q) < 0$, contradicting $\chi_i(q) = 0$. □

The proof uses one end of the middle piece in each case: the right end when the outer label grows, the left end when it decreases. Figure 7(a) shows the switching function of the growing case for an admissible factor and for one with $bq > a$.

D.2. **Reverse runs.**

Proposition D.2 (Run structure)**.** *Let P be a mixed normal vertex period, and let the switches $k, \dots, k+L-1$ form a maximal reverse run. Then $L = 2$, and the active margin of the centre piece $k+1$ is strictly smaller than those of the four pieces around it:*

$$a_{k+1} < a_{k-1}, \qquad a_{k+1} < a_k, \qquad a_{k+1} < a_{k+2}, \qquad a_{k+1} < a_{k+3}.$$

Proof of Proposition D.2. The run is preceded by the forward switch $k-1$ and followed by the forward switch $k+L$.

At the entry, the switches $k-1, k$ are forward then reverse, so $a_{k+1} < a_{k-1}$ by Lemma D.1(b), and $i_{k+1} = i_{k-1}$. If $a_k \leq a_{k+1}$, then a_k would be at most both a_{k-1} and a_{k+1}, against Lemma D.1(a). So $a_{k+1} < a_k$.

At the exit, the switches $k+L-1, k+L$ are reverse then forward, so $a_{k+L-1} < a_{k+L+1}$, and Lemma D.1(a) on the pieces $k+L-1, k+L, k+L+1$ gives $a_{k+L-1} < a_{k+L}$.

If $L = 1$, the entry gives $a_{k+1} < a_k$ and the exit $a_k < a_{k+1}$. If $L \geq 3$, let m be a piece of $k, \dots, k+L$ with the smallest active margin. It is not k, since $a_{k+1} < a_k$, nor $k+L$, since $a_{k+L-1} < a_{k+L}$. If $m = k+1$, the pieces $k, \dots, k+3$ have three reverse switches and the second one is a weak minimum, against Lemma C.1(b). If $k+2 \leq m \leq k+L-1$, the pieces $m-2, \dots, m+1$ have three reverse switches and the third one is a weak minimum, against Lemma C.1(a). So $L = 2$. The entry and the exit give the first three inequalities, and Lemma D.1(b) at the switches $k+1, k+2$ gives $a_{k+1} < a_{k+3}$. □

The proof does not use mixedness beyond the existence of the run: a maximal reverse run is bounded by forward switches by definition, so a period that has one is mixed.

Proof of Theorem 4.2. This is the first conclusion of Proposition D.2. □

In a mixed period every reverse switch therefore belongs to a pattern $+\,-\,-+$ of switch signs, whose middle piece is a centre, and every centre is a strict local minimum of the active margins. The other classes use these facts through the following corollary.

Corollary D.3 (Mixed periods)**.** *Let P be a mixed normal vertex period.*

(a) *Every reverse switch k is the first or the second switch of a pattern of signs $+\,-\,-+$: either the switches $k-1,k,k+1,k+2$ or the switches $k-2,k-1,k,k+1$ have these signs.*

(b) *At a centre k, $a_k < a_{k-1}$ and $a_k < a_{k+1}$.*

(c) *Every switch norm satisfies $d_k \geq 3/32$.*

Proof. (a) and (b) restate Proposition D.2. For (c), at a forward switch $d_k \geq j_k/2 \geq 3/32$ by Lemma B.12(a). By (a), a reverse switch is the outgoing switch of a turn (the piece k after the forward switch $k-1$) or the incoming switch of a turn (the piece $k+1$ before the forward switch $k+1$), and Lemma B.12(b) gives $d_k \geq 3/32$ in both cases. □

D.3. **Proof of ellipticity.** We prove Proposition B.16: a mixed normal vertex period has $\gamma^2 > 0$. Suppose $\gamma^2 \leq 0$. We use a projective coordinate on the switch vectors y_n of Lemma B.10 that changes monotonically from one switch to the next. Because $y_{N+n} = \pm y_n$, the coordinate returns to its value after one period; but at every centre it strictly increases, and nowhere does it decrease.

Lemma D.4 (Three Gram vectors)**.** *Let S be a symmetric 2×2 matrix with $\det S \leq 0$, and let y, y', y'' be vectors with $y^{\mathsf T}Sy$, $y'^{\mathsf T}Sy'$, $y''^{\mathsf T}Sy''$ all positive. Then $(y^{\mathsf T}Sy')(y^{\mathsf T}Sy'')(y'^{\mathsf T}Sy'') > 0$.*

Proof. For two vectors, $(y^{\mathsf T}Sy)(y'^{\mathsf T}Sy') - (y^{\mathsf T}Sy')^2 = \det S \cdot \det(y,y')^2 \leq 0$. The Gram determinant of three vectors of the plane vanishes: $d\,d'\,d'' - d\,c^2 - d'\,b^2 - d''a^2 + 2abc = 0$ with the quadratic values on the diagonal and the bilinear pairings off the diagonal. With the three two-vector inequalities, $2abc \geq d\,d'\,d'' > 0$. □

Lemma D.5 (Signs of the cross pairings)**.** *Suppose $\gamma^2 \leq 0$. At a forward same-sign piece k, $e_k > 0$. At a centre k with $a_k < a_{k-1}$, $a_k < a_{k+1}$ and $d_{k-1}, d_k \geq 3/32$, $e_k < 0$.*

Proof. At a same-sign piece, Lemma B.11 expresses e_k through d_{k-1}, d_k and the active margins. At a forward same-sign piece both switches are forward, and the lower bounds $d \geq j/2$ of Lemma B.12(a) give $e_k \geq X_k$ with

$$X_k = \frac{1}{16}\Big(-1 + 2a\Big(\frac{1}{a+p} + \frac{1}{a+n}\Big) - \frac{a^2}{a^2+4} + \log\frac{(a+p)(a+n)}{a^2+4}\Big),$$

$a = a_k$, $p = a_{k-1}$, $n = a_{k+1}$. Here the logarithm is $\log q_k > 0$ by Lemma B.2, and $a^2/(a^2+4) < 2a/(a+p)$ because $ap < 4$; so $X_k > -1/16$. On the other hand $\gamma^2 \leq 0$ and Lemma B.13 give $e_k^2 \geq d_{k-1}d_k \geq (3/32)^2$, and

$3/32 > 1/16$, so $e_k \geq 3/32 > 0$. At a centre the switches are reverse, the upper bounds $d \leq j/2$ give $e_k \leq X_k$, and now the logarithm is $-\log q_k < 0$ and $2a/(a+p) + 2a/(a+n) < 2$ by the strict minimum, so $X_k < 1/16$ and $e_k \leq -3/32$. □

Proof of Proposition B.16. Suppose $\gamma^2 \leq 0$. By Corollary D.3(c) every $d_n = y_n^{\mathsf{T}} S y_n$ is at least $3/32$. Let $h(v) = y_0^{\mathsf{T}} S v$, a nonzero linear form, and let l be the linear form with $h(u)l(v) - h(v)l(u) = c\det(u, v)$ for a constant $c > 0$. By Lemma D.4 with $y = y_0$, $h(y_n)h(y_m)\,(y_n^{\mathsf{T}} S y_m) > 0$ for all n, m; in particular $h(y_n) \neq 0$, and the projective coordinate $z_n = l(y_n)/h(y_n)$ is defined, with

$$z_{n+1} - z_n = \frac{c\,\det(y_n, y_{n+1})}{h(y_n)\,h(y_{n+1})}.$$

By Lemma B.10, at a turn $y_{n+1} = -y_n$ and the increment is 0. At a forward same-sign piece $\det(y_n, y_{n+1}) = 1/K_{n+1} > 0$ and $y_n^{\mathsf{T}} S y_{n+1} = e_{n+1} > 0$ (Lemma D.5), so $h(y_n)h(y_{n+1}) > 0$ and the increment is positive. At a centre $\det(y_n, y_{n+1}) = -1/K_{n+1} < 0$ and $e_{n+1} < 0$, using the strict minimum and the bounds $d \geq 3/32$ of Corollary D.3; so $h(y_n)h(y_{n+1}) < 0$ and the increment is again positive. A mixed period has a reverse switch, hence a centre (Corollary D.3(a)), so the increments over one period sum to a positive number. But $y_N = \pm y_0$ and z depends only on the line of the vector, so the sum is $z_N - z_0 = 0$. □

For an entirely forward period every piece is a forward same-sign piece and the same telescoping gives $\gamma^2 > 0$.

Appendix E. Class M1: every forward run short

A period of class M1 is mixed, every maximal run of reverse switches has length two and every maximal run of forward switches has length one (Definition 3.1). Its cyclic sign word is therefore $(--+)^m$: every reverse run is a pair of reverse switches around a centre piece, and consecutive pairs are separated by single forward switches. This section proves Theorem 4.3.

The theorem uses neither the compact domain nor a lower bound on the number of pieces. Its proof compares two ways of counting how far the switch vectors y_k of Definition B.9 turn over one period, measured in the positive definite form S. At a turn the switch vector changes sign, so the total turning is concentrated at the m centres. A whole-period winding identity fixes the total turning at exactly $(m+1)\pi/3$, because the frame of a competitive period turns by exactly $\pi/3$. A local estimate at each centre, which combines the stationarity identity, the Gram identity and the competitive cap on the Casimir γ^2, shows that each centre turns the switch vectors by more than $2\pi/5$. The average $(m+1)\pi/(3m)$ is at most $2\pi/5$ as soon as $m \geq 5$, and the word condition leaves $m = 2$ as the only other case. With two centres the recurrence of the switch vectors closes up only in a rigid configuration whose cost can be computed exactly, and it exceeds A_{oct}.

E.1. **The sign word.**

Lemma E.1. *Let a vertex representation with N pieces belong to class* M1*. Then $N = 3m$ for an integer $m \geq 1$ with $m \equiv 2 \pmod 3$, and there is $k_0 \in \mathbb{Z}$ such that for every $j \in \mathbb{Z}$ the switches $k_0 + 3j$ and $k_0 + 3j + 1$ are reverse and the switch $k_0 + 3j + 2$ is forward.*

Proof. A mixed representation has a forward switch followed by a reverse one: otherwise every switch after the first forward one would be forward, which the periodicity $\varepsilon_{k+N} = \varepsilon_k$ of the signs forbids. Start at such a pair. The reverse run that begins there has length two, and the forward run that follows it has length one, so the signs continue as $-, -, +, -, -, +, \ldots$ with period three. Hence $N = 3m$ and the word is $(--+)^m$. The signs of one period add up to $m - 2m = -m$, and Lemma B.6 gives $-m \equiv 1 \pmod 3$. □

We fix k_0 and m as in the lemma and call the piece $c_j = k_0 + 3j + 1$, which lies between the two reverse switches of the j-th run, the j-th *centre*. The pieces $c_j + 1$ and $c_j + 2$ are turns: one of their two switches is forward and the other reverse. At a turn the outgoing switch vector is the negative of the incoming one, so the three switch vectors that follow the centre c_j are z_j, $-z_j$, z_j with $z_j = y_{c_j}$. Between two consecutive centres the switch vectors therefore do not turn, and the whole turning of the sequence (y_k) is concentrated at the centres.

E.2. **The data at a centre.** Fix a centre c, with active margin $a = a_c$ and neighbouring active margins $p = a_{c-1}$ and $n = a_{c+1}$. In this section we write

$$r = \frac{a}{a+p}, \qquad t = \frac{a}{a+n}, \qquad \rho = \frac{a^2}{a^2+4}, \tag{E.1}$$
$$u = 8d_{c-1}, \qquad v = 8d_c, \qquad E = 16e_c, \qquad U(s) = 1 - s + s^2.$$

Here d_{c-1} and d_c are the switch norms of the two reverse switches at the ends of the centre and e_c is the cross pairing on the centre. The letter ρ is local to this section.

Lemma E.2. *Let P be a normal vertex period with the sign word of Lemma E.1, and let γ^2 be its Casimir. At every centre the numbers* (E.1) *satisfy*

(i) $0 < r < \frac{1}{2}$, $0 < t < \frac{1}{2}$ *and* $0 < \rho < rt$;

(ii) $\frac{3}{4} \le u \le U(r)$ *and* $\frac{3}{4} \le v \le U(t)$;

(iii) *the stationarity identity*

$$E = 1 + r + t - \rho + \log\frac{\rho}{rt} - \frac{1-u}{r} - \frac{1-v}{t}; \tag{E.2}$$

(iv) *the Gram identity* $16(1-\rho)^2\gamma^2 = 4uv - E^2$.

Proof. (i) The centre is the middle piece of a reverse run of length two, so its active margin is strictly smaller than both neighbouring ones (Proposition D.2). This gives $r, t < 1/2$. The factor of the centre is $q_c = (a^2+4)/((a+p)(a+n))$, and $q_c > 1$ is the inequality $(a+p)(a+n) < a^2+4$, which is $\rho < rt$.

(ii) Both neighbours of the centre are turns. The switch norms at the two ends of a turn are equal and at least $3/32$ (Lemma B.12); this gives $u, v \ge 3/4$. Both switches of the centre are reverse, so $d_{c-1} \le j_{c-1}/2$ and $d_c \le j_c/2$ by Lemma B.12, and $4j_{c-1} = (a^2+ap+p^2)/(a+p)^2 = U(r)$, and likewise $4j_c = U(t)$.

(iii) is the stationarity identity of Lemma B.11 with both incident signs equal to -1, written in the variables (E.1).

(iv) is the Gram identity $d_{c-1}d_c - e_c^2 = \gamma^2/(a^2+4)^2$ at a centre (Lemma B.13), multiplied by 256, since $1 - \rho = 4/(a^2+4)$. □

Since the period is mixed, $\gamma^2 > 0$ (Proposition B.16), and the switch norms are positive. The form S, whose determinant is γ^2, is then positive definite, and the number

$$\cos\varphi_c = \frac{e_c}{\sqrt{d_{c-1}d_c}} = \frac{E}{2\sqrt{uv}}$$

is the cosine of the angle $\varphi_c \in (0, \pi)$ between y_{c-1} and y_c in the inner product S; by (iv) its absolute value is less than one. The determinant $\det(y_{c-1}, y_c) = -1/(a^2+4)$ is negative at every centre, so all the centres turn the switch vectors in the same sense. We call φ_c the *centre angle*.

E.3. **The angle gap.** A centre with a large angle cosine needs a large pairing E. The stationarity identity (E.2) allows a large E only when $\log(\rho/(rt))$ is not too negative, that is, when the centre margin is not too small compared with its neighbours. A large centre margin makes the factor $16(1-\rho)^2$ in the Gram identity (iv) small, and then (iv) forces the Casimir γ^2 up, against the competitive cap of Corollary B.15. The next proposition makes the conflict quantitative.

Proposition E.3. *Let $\gamma^2, r, t, \rho, u, v, E$ be real numbers satisfying* (i)–(iv) *of Lemma E.2, and suppose that $\gamma^2 \le 5/27$. Then $E < \frac{3}{5}\sqrt{uv}$; that is, $\cos\varphi < 3/10$ for $\cos\varphi = E/(2\sqrt{uv})$.*

The proof, in Appendix E.7, reduces the seven variables to two in three steps. Write $E = (1-\rho+\log\rho) + \Psi(r,u) + \Psi(t,v)$ with $\Psi(r,u) = r - \log r - (1-u)/r$, and let $\omega(x) = -x - \log(1-x) = \sum_{n\ge2} x^n/n$ for $0 \le x < 1$.

(a) One reverse switch at a time: for $s = \sqrt{4u-3}$ one has $u = U((1-s)/2)$, $2r \le 1-s$, and $\Psi(r,u) \le \omega(s) + \log 2$, with equality when $r = (1-s)/2$, that is, when the switch bound $u \le U(r)$ is attained.

(b) The two switches together: define $S \ge 0$ by $S^2 = s^2 + s'^2 + s^2 s'^2/3$, where $s' = \sqrt{4v-3}$. Then $uv = 3(3+S^2)/16$, and $\omega(s) + \omega(s') \le \omega(S)$ because ω is superadditive in x^2.

(c) The centre: $z = 4\rho/(1-S)$ lies in $(0,1]$, $\rho = (1-S)z/4$, and $E \le (1-S)(1-z/4) + \log z$.

If $\cos\varphi \ge 3/10$, then $E^2 \ge \frac{9}{25}uv$, and the symmetric minimum proved in Lemma E.13 gives $16K_*(1-\rho)^2 < 4uv - E^2$ for the cap $K_* = 5/27$, which contradicts (iv). Figure 8 shows the region of (S,z) that survives the reductions and the least Casimir compatible with it.

Corollary E.4. *At every centre of a competitive normal vertex period of class* M1*, $\varphi_c > 2\pi/5$.*

Proof. The Casimir and the first switch norm are positive, so the competitive cap in Corollary B.15 gives $\gamma^2 < 5/27$. By Lemma E.2 and Proposition E.3, $\cos\varphi_c < 3/10$. Since $\cos(2\pi/5) = (\sqrt5 - 1)/4 > 3/10$, which is $\sqrt5 > 11/5$, and the arccosine decreases, $\varphi_c > 2\pi/5$. □

E.4. **The winding identity.**

Proposition E.5. *Let P be a competitive normal vertex period with the sign word of Lemma E.1, $N = 3m$, and positive Casimir. Then*

$$\sum_{j=0}^{m-1} \varphi_{c_j} = \frac{(m+1)\pi}{3}, \qquad \varphi_c = \arccos\frac{E_c}{2\sqrt{u_c v_c}},$$

where E_c, u_c, v_c are the numbers (E.1) *at the centre c.*

The identity says that the switch vectors, which turn only at the centres, turn in total by the angle that the frame of the period prescribes. The

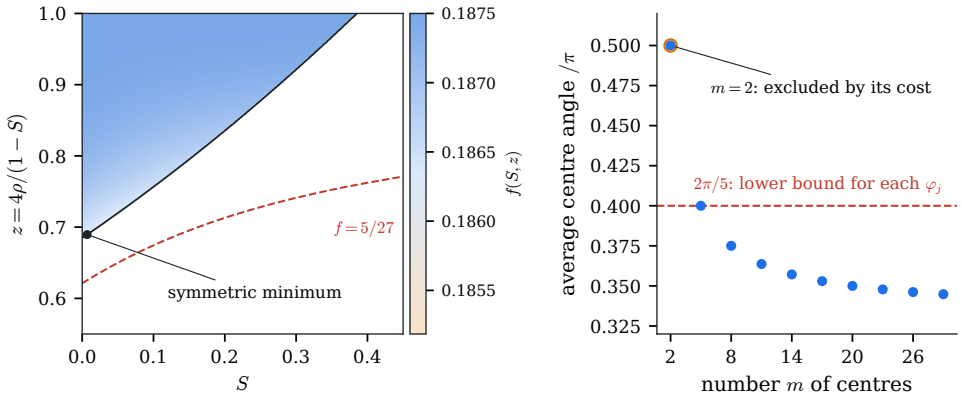


FIGURE 8. Left: the pairing constraint $E_{\max} \geq \frac{3}{5}\sqrt{uv}$ in the reduced (S, z) variables, with $E_{\max}$ the envelope in (c). The colour shows the Gram lower bound $(4uv - E_{\max}^2)/(16(1-\rho)^2)$. Its minimum occurs on the boundary at $S = 0$ (Lemma E.12); the dashed level $5/27$ lies outside the feasible region. Right: the average centre angle $(m+1)\pi/(3m)$ forced by winding, for $m \equiv 2 \pmod 3$, against the lower bound $2\pi/5$ for every centre. They meet at $m = 5$; only $m = 2$ lies above the line.

proof expresses the switch vectors through the physical frame and follows a continuous angle along the period.

Switch vectors in the physical frame. Let $V(t) = g(t)X(t)$ be the tangent frame. The switch vector at the switch n is, up to a sign and a positive factor, $KV(\tau_n)b$, where $K = R^{i_0}C(-X(0))$ is a fixed matrix with $\det K = -2\sqrt{3}$, C is the triangular change of coordinates, and b is one of two base vectors $b_{\text{in}}(i)$ and $b_{\text{out}}(i) = Rb_{\text{in}}(i)$ that depend only on the label i of the piece. Factor $S = Q^{\mathsf{T}}Q$ with $\det Q < 0$, so that angles for S become Euclidean angles after Q.

A continuous angle. The polar number $z(g) = (g_{00} + g_{11})/2 + i(g_{10} - g_{01})/2$ of a matrix of determinant one has modulus at least one, and for every vector $w \neq 0$ the quotient $gw/(z(g)w)$ lies in the right half-plane, since $\operatorname{Re}(gw \cdot \overline{z(g)w}) = (|gw|^2 + |w|^2)/2$. Let $\Phi(t)$ be a continuous angle of $z(g(t))$. For $w \neq 0$ let $\kappa_t(w)$ be the angle from w to $QKg(t)X(t)w$ obtained by adding $\Phi(t)$ to principal arguments that never leave the slit plane; it depends continuously on t.

Three facts. At a centre c the centre angle is $\varphi_c = \pi/3 + \kappa_{\tau_c}(b_{\text{out}}) - \kappa_{\tau_c}(b_{\text{in}})$. Along a piece the base vector that is being followed is an eigenvector of the tangent factor of the piece, so $g(t)X(t)b$ keeps its direction and $\kappa_t(b)$ is constant; between consecutive centres the outgoing base vector of one is, up to sign, the incoming base vector of the next. Across the seam, $\kappa_T(w) = \kappa_0(Rw) + \pi/3$, because $g(0) = I$, $g(T) = R$ and $\Phi(T) - \Phi(0) = \pi/3$.

The frame turns by exactly $\pi/3$. The endpoint conditions fix $\Phi(T)-\Phi(0)$ only modulo 2π. For a competitive period the cost is at most $A_{\mathrm{oct}}<\pi$, and the winding theorem behind the reverse comparison (Proposition A.9) shows that the reflected first row $(g_{00},-g_{01})$ of the frame turns by exactly $\pi/3$. The real part of the product of that row with the conjugate of $z(g)$ is $(g_{00}^2+g_{01}^2+1)/2>0$, so the two angles stay within $\pi/2$ of each other, and $\Phi(T)-\Phi(0)=\pi/3$.

Telescoping. The centre angles are periodic in j with period m, so we may take the m centres of one period in a window that starts at the seam. Summing the centre formula over them, the term $\kappa(b_{\mathrm{out}})$ of each centre cancels the term $\kappa(b_{\mathrm{in}})$ of the next, and the last link crosses the seam, where it gains $\pi/3$. What remains is $m\cdot\pi/3+\pi/3$.

E.5. **Five or more centres.**

Proof of Theorem 4.3 for $m\geq 5$. By Corollary E.4 every centre angle exceeds $2\pi/5$, so the sum in Proposition E.5 exceeds $2m\pi/5$. For $m\geq 5$, $2m/5\geq(m+1)/3$, which contradicts Proposition E.5. Since $m\equiv 2 \pmod 3$, the only other case is $m=2$. □

E.6. **Two centres.** For $m=2$, the average centre angle $\pi/2>2\pi/5$ is consistent with the local angle bound. We therefore use the switch-vector recurrence and the cost identity. For the centre c_j put $K_j=a_{c_j}^2+4$, write $b_j=a_{c_j+1}$ and $b'_j=a_{c_j+2}$ for the active margins of the two turns that follow it, and

$$D_j=(a_{c_j}+b_j)^2+(b'_j+a_{c_{j+1}})^2-(b_j+b'_j)^2=s_{c_j}^2+s_{c_j+2}^2-s_{c_j+1}^2.$$

Lemma E.6. *In a normal vertex period with the sign word of Lemma E.1, the vectors $z_j=y_{c_j}$ satisfy*

$$K_{j+1}z_{j+1}=-K_jz_{j-1}+D_jz_j,\qquad \det(z_{j-1},z_j)=-1/K_j,$$

and the cyclic product of the period is $\pm I$, so that $z_{j+m}=\pm z_j$ with one sign for all j.

Proof. Work in triangular coordinates and in the frame of the centre $c=c_j$, that is, multiply all vectors by the inverse of the prefix product up to c. Put $A_\pm(q)=T_0(q)R^{\mp 1}$, so that

$$A_+(q)=\begin{pmatrix}q & -q\\ 1/q & 0\end{pmatrix},\qquad A_-(q)=\begin{pmatrix}0 & q\\ -1/q & 1/q\end{pmatrix},$$

and write $K=K_j$ and $s_k=a_k+a_{k+1}$. The factors of the centre and of the two turns after it are $K/(s_{c-1}s_c)$, s_{c+1}/s_c and s_{c+1}/s_{c+2}. In this frame $z_{j-1}=-(1/s_{c-1},0)^{\mathsf T}$ and $z_j=(0,s_{c-1}/K)^{\mathsf T}$, which gives the determinant, and

$$\begin{aligned}K_{j+1}z_{j+1}&=s_{c+2}\,A_-\Big(\frac{K}{s_{c-1}s_c}\Big)A_+\Big(\frac{s_{c+1}}{s_c}\Big)A_-\Big(\frac{s_{c+1}}{s_{c+2}}\Big)\begin{pmatrix}0\\1\end{pmatrix}=\begin{pmatrix}K/s_{c-1}\\ (s_{c-1}/K)\,D_j\end{pmatrix}\\&=-Kz_{j-1}+D_jz_j.\end{aligned}$$

Multiplying back by the prefix product preserves the recurrence and the determinant. The cyclic product of the period is $\pm I$ by the tangent closure (Lemma B.4). □

Lemma E.7. *For every normal vertex period,*

$$\frac{2\mathcal{C}}{\sqrt{12}} = 8\sum_{k=0}^{N-1}(\varepsilon_{k-1}+\varepsilon_k)e_k - 2\sum_{k=0}^{N-1}\frac{\varepsilon_{k-1}+\varepsilon_k}{a_k^2+4}.$$

Proof. Multiply the stationarity identity of Lemma B.11 at the piece k by a_k and sum over a period. The switch norms and the cross pairings are formed from one weight w_k per piece, $d_k = (w_k + w_{k+1})/(2s_k^2)$ and $e_k = w_k/(2(a_k^2+4))$. Regrouped by switches, the left-hand sides add up to $\sum_k(\varepsilon_{k-1}+\varepsilon_k)w_k(1 - a_k^2/(a_k^2+4)) = 8\sum_k(\varepsilon_{k-1}+\varepsilon_k)e_k$. On the right-hand side, the one-arc cost formula of Lemma B.1, with the factors written in the active margins, gives $8\mathcal{C}/\sqrt{12} = \sum_k \varepsilon_k G(a_k, a_{k+1})$ for the signed action

$$G(a,b) = 3ab - 4 - (a^2+b^2)\log(a+b) + \tfrac{1}{2}a^2\log(a^2+4) + \tfrac{1}{2}b^2\log(b^2+4),$$

and the stationarity identity involves the partial derivatives of G. The Euler-type identity $aG_a(a,b)+bG_b(a,b) = 2G(a,b)+16/(a^2+4)+16/(b^2+4)$ turns the sum of the right-hand sides into $2\mathcal{C}/\sqrt{12}+2\sum_k(\varepsilon_{k-1}+\varepsilon_k)/(a_k^2+4)$. □

In the word $(--+)^m$, $\varepsilon_{k-1}+\varepsilon_k$ is -2 at a centre and 0 at a turn, so the lemma reads $\mathcal{C}/\sqrt{12} = 2\sum_j 1/K_j - 8\sum_j e_{c_j}$.

Proposition E.8. *A competitive normal vertex period with $N = 6$ does not have the sign word $(--+)^2$.*

Proof. The two recurrences of Lemma E.6 read $K_1 z_1 = -K_0 z_{-1} + D_0 z_0$ and $K_0 z_2 = -K_1 z_0 + D_1 z_1$, with $z_{-1} = \pm z_1$ and $z_2 = \pm z_0$, and z_0, z_1 are independent. The sign $+$ would give $(K_0 + K_1)z_1 = D_0 z_0$, which is impossible. So $z_{-1} = -z_1$, and the recurrences give $K_0 = K_1$ and $D_0 = D_1 = 0$. Hence both centres have the same active margin a, $K = a^2+4$, and $e_{c_0} = z_{-1}^{\mathsf{T}} S z_0 = -e_{c_1}$. Lemma E.7 gives the exact cost

$$\mathcal{C} = \sqrt{12}\cdot\frac{4}{K} = \frac{8\sqrt{3}}{a^2+4}.$$

It remains to bound a from above. The equation $D_j = 0$ is $(b_j - a)(b'_j - a) = 2a^2$, and the turn margins exceed a by Proposition D.2. By Lemma E.14, $b_j b'_j \geq (1+\sqrt{2})^2 a^2$ and $(a+b_j)(a+b'_j) = 2b_j b'_j$. The two centre factors exceed one, that is, $(a+b'_1)(a+b_0) < K$ and $(a+b'_0)(a+b_1) < K$. Multiplying, $K^2 > 4\,b_0 b'_0\, b_1 b'_1 \geq 4(1+\sqrt{2})^4 a^4$, so $K > 2(1+\sqrt{2})^2 a^2$ and $a^2 < 4/(5+4\sqrt{2})$. Consequently

$$\mathcal{C} > \frac{8\sqrt{3}}{4 + 4/(5+4\sqrt{2})} = \sqrt{3}(2\sqrt{2}-1).$$

This lower bound exceeds A_{oct}. Indeed, with $r=\sqrt{2}$, the midpoint inequality for $x\mapsto 1/x$ on $[1,r]$ gives $\log 2 > 4(r-1)/(r+1) = 12-8r$, and hence

$$(2r-1)\Big(\frac{2r-1}{2}-\delta_{\mathrm{oct}}\Big) = \log 2 + 2r - \frac{7}{2} > \frac{17}{2} - 6r = \frac{(3-2r)^2}{2} > 0.$$

Since $A_{\mathrm{oct}} = 2\sqrt{3}\,\delta_{\mathrm{oct}}$, the period is not competitive. □

Proof of Theorem 4.3. By Lemma E.1 the word is $(--+)^m$ with $m\equiv 2 \pmod 3$. The period is mixed, so its Casimir is positive (Proposition B.16), and the competitive cap applies. If $m\ge 5$ the angle count above is contradictory, and if $m=2$, Proposition E.8 applies. □

E.7. **Estimates for class M1.** This subsection proves the angle gap, Proposition E.3, and the arithmetic of the two-centre case. Throughout, $K_*=5/27$ is the competitive cap of Corollary B.15, $U(s)=1-s+s^2$, and $\omega(x)=-x-\log(1-x)$ for $0\le x<1$.

E.7.1. *Logarithm bounds.*

Lemma E.9. (1) *For $0<x\le 1$, $\frac{1}{2}(x-x^{-1})\le \log x$.*
(2) *If $a,b,c\ge 0$, $c<1$ and $a^2+b^2\le c^2$, then $\omega(a)+\omega(b)\le\omega(c)$.*

Proof. (1) is $y\le \sinh y$ for $y=-\log x\ge 0$. (2) Write $\omega(x)=\sum_{n\ge 0} x^{n+2}/(n+2)$. Since $a,b\le c$, termwise $a^{n+2}+b^{n+2}\le (a^2+b^2)c^n\le c^{n+2}$. □

E.7.2. *One reverse switch.* The reverse-switch term $\Psi(r,u)=r-\log r-(1-u)/r$ of the stationarity identity increases with u and, at fixed u, is largest when the switch bound $u\le U(r)$ is attained.

Lemma E.10. *Let $0<r<1/2$ and $3/4\le u\le U(r)$, and put $s=\sqrt{4u-3}$. Then $0\le s<1$, $2r\le 1-s$, and*

$$r-\log r-\frac{1-u}{r}\le \omega(s)+\log 2.$$

Proof. Put $r'=(1-s)/2$, so that $U(r')=(3+s^2)/4=u$. Since U decreases on $[0,1/2]$ and $U(r')=u\le U(r)$, we have $r\le r'$. The identity

$$2r'-1-\Big(r-\tfrac{1}{2}\big(\tfrac{r}{r'}-\tfrac{r'}{r}\big)-\frac{1-U(r')}{r}\Big)=\frac{(r'-r)^2(1-2r')}{2rr'}$$

and $1-2r'=s\ge 0$ give $r-\frac{1}{2}(r/r'-r'/r)-(1-u)/r\le 2r'-1=-s$. By Lemma E.9(1) with $x=r/r'\le 1$, $-\log(r/r')\le -\frac{1}{2}(r/r'-r'/r)$, so $r-\log r-(1-u)/r\le -s-\log r'$. Finally $-\log r'=-\log(1-s)+\log 2$. □

E.7.3. *Reduction to two variables.*

Lemma E.11. *Let γ^2,r,t,ρ,u,v,E satisfy* (i)–(iv) *of Lemma E.2, with $E>0$. Let $s=\sqrt{4u-3}$, $s'=\sqrt{4v-3}$, $S=\sqrt{s^2+s'^2+s^2s'^2/3}$ and $z=4\rho/(1-S)$. Then $0\le S<1$, $0<z\le 1$,*

$$uv=\frac{3(3+S^2)}{16},\qquad \rho=\frac{(1-S)z}{4},$$

$$E \le (1-S)\Big(1-\frac{z}{4}\Big) + \log z.$$

Proof. The identity (E.2) splits as $E = 1-\rho+\log\rho+\Psi(r,u)+\Psi(t,v)$, and Lemma E.10 bounds the two last terms by $\omega(s)+\log 2$ and $\omega(s')+\log 2$, with $2r \le 1-s$ and $2t \le 1-s'$. Then $4\rho < 4rt \le (1-s)(1-s')$. The bounds just obtained give

$$0 < E \le 1-\rho-s-s'+\log\frac{4\rho}{(1-s)(1-s')} < 1-\rho-s-s',$$

so $s+s' < 1$. From $u = (3+s^2)/4$ and $v = (3+s'^2)/4$ we obtain $uv = 3(3+S^2)/16$. Next,

$$(s+s'-ss')^2 - S^2 = \tfrac{2}{3}s^2s'^2 + 2ss'(1-s-s') \ge 0,$$

so $S \le s+s'-ss'$ and $(1-s)(1-s') \le 1-S$; in particular $S<1$ and $4\rho < 1-S$, so $0<z<1$. By Lemma E.9(2), $\omega(s)+\omega(s') \le \omega(S)$, and therefore

$$E \le 1-\rho+\log(4\rho)+\omega(S) = 1-S-\rho+\log\frac{4\rho}{1-S} = (1-S)\Big(1-\frac{z}{4}\Big)+\log z.$$

□

E.7.4. *The symmetric minimum.* The next lemma locates the minimum before estimating its value.

Lemma E.12. *Let $0<c<1$, $0 \le S < 1$, $0 < \rho \le (1-S)/4$, and put*

$$Q(S) = \frac{3(3+S^2)}{16}, \qquad F(S,\rho) = 1-S-\rho+\log\frac{4\rho}{1-S}.$$

If $c\sqrt{Q(S)} \le E \le F(S,\rho)$, then

$$\frac{4Q(S)-E^2}{16(1-\rho)^2} \ge \frac{9(4-c^2)}{256(1-\rho_0)^2},$$

where $\rho_0 \in (0,1/4)$ is the unique solution of $1-\rho_0+\log(4\rho_0) = 3c/4$.

Proof. Write $z = 4\rho/(1-S)$ and $f(S,\rho) = (4Q(S)-F(S,\rho)^2)/(16(1-\rho)^2)$. First fix S. On the region $F \ge 0$, the function f increases with ρ. To see this, use

$$\log z \le \frac{2(z-1)}{z+1}, \qquad H = 1-S-\rho+\frac{2(z-1)}{z+1} \ge F \ge 0.$$

The difference $\log z - 2(z-1)/(z+1)$ vanishes at one and has derivative $(z-1)^2/[z(z+1)^2] \ge 0$, which proves the logarithm inequality. The sign of f_ρ is the sign of $4Q - F^2 - F(1-\rho)^2/\rho$, which decreases when $F \ge 0$ is raised to H. At H it equals

$$\frac{(1-z)(1-z+S(1+z))}{4z(1-S)(1+z)^2}\Big((1-S)^2z^2 + (S^2+14S+1)z + 16\Big) \ge 0.$$

There is therefore a unique $\rho_c(S)$ with $F(S,\rho_c(S)) = c\sqrt{Q(S)}$, and lowering ρ to this root lowers f. Here $F_\rho = 1/\rho - 1 > 0$, F tends to $-\infty$ at zero, and

its value at $(1-S)/4$ is $3(1-S)/4$. The root exists for all S between zero and the actual value: its endpoint condition is $3(1-S)/4 \geq c\sqrt{Q(S)}$, with a decreasing left side and an increasing right side.

On this root curve,

$$f(S, \rho_c(S)) = \frac{(4-c^2)Q(S)}{16(1-\rho_c(S))^2}.$$

Implicit differentiation of $F = c\sqrt{Q}$ gives

$$\frac{\mathrm{d}}{\mathrm{d}S} \log \frac{Q(S)}{(1-\rho_c(S))^2} = \frac{2S}{(3+S^2)(1-\rho_c(S))^2}(1 - z + \rho_c(S)\log z) \geq 0.$$

Indeed, $\log z \geq (z-1)/z$ and $\rho_c(S)/z = (1-S)/4$, so the last bracket is at least $(1-z)(3+S)/4$. Lowering S to zero thus lowers the bound once more. At zero the root is ρ_0, $Q(0) = 9/16$, and the stated inequality follows, since $0 < E \leq F$ gives $(4Q - E^2)/(16(1-\rho)^2) \geq f(S, \rho)$. □

Lemma E.13. *Let $S \geq 0$, $0 < z \leq 1$ and $E > 0$ satisfy $E \leq (1-S)(1 - z/4) + \log z$ and $E^2 \geq \frac{9}{25}Q$, where $Q = 3(3+S^2)/16$. Then*

$$16K_*\Big(1 - \frac{(1-S)z}{4}\Big)^2 < 4Q - E^2.$$

Proof. The positive upper bound for E forces $S < 1$. Apply Lemma E.12 with $c = 3/5$ and $\rho = (1-S)z/4$. It remains to bound the symmetric root. Put $x_0 = 4\rho_0$. Then

$$\frac{9}{20} = 1 - \frac{x_0}{4} + \log x_0 \leq 1 - \frac{x_0}{4} + \frac{2(x_0-1)}{x_0+1},$$

so $5x_0^2 - 46x_0 + 29 \leq 0$, and hence $x_0 \geq (23 - 8\sqrt{6})/5$. Consequently

$$\frac{4Q - E^2}{16(1-\rho)^2} \geq \frac{273}{16(131 - 16\sqrt{6})} > \frac{273}{1472} > \frac{5}{27} = K_*.$$

Here $\sqrt{6} > 39/16$, since $6 \cdot 16^2 > 39^2$, gives the first strict comparison; the last follows from $273 \cdot 27 - 5 \cdot 1472 = 11 > 0$. □

Proof of Proposition E.3. Suppose $E \geq \frac{3}{5}\sqrt{uv}$. This gives $E > 0$ and $E^2 \geq 9uv/25$, so Lemma E.11 puts (S, z, E) in the hypotheses of Lemma E.13, whose conclusion, with $Q = uv$ and $(1-S)z/4 = \rho$, reads $16K_*(1-\rho)^2 < 4uv - E^2 = 16(1-\rho)^2\gamma^2$. This contradicts $\gamma^2 \leq K_*$. □

E.7.5. *The two-centre product bound.*

Lemma E.14. *If $0 < a < b$, $a < c$ and $(b-a)(c-a) = 2a^2$, then $bc \geq (1+\sqrt{2})^2 a^2$ and $(a+b)(a+c) = 2bc$.*

Proof. With $x = b/a$ and $y = c/a$, $(x-1)(y-1) = 2$, so $xy = x + y + 1$ and, by the inequality between the arithmetic and geometric means, $x + y - 2 \geq 2\sqrt{(x-1)(y-1)} = 2\sqrt{2}$. Hence $xy \geq (1+\sqrt{2})^2$. Expanding, $(a+b)(a+c) = a^2 + a(b+c) + bc$, and $bc - a(b+c) - a^2 = (b-a)(c-a) - 2a^2 = 0$. □

Appendix F. Class F: every switch forward

The smoothed octagon's period is entirely forward: its word $\mathbf{e}_2\mathbf{e}_0\mathbf{e}_1\mathbf{e}_2$ advances the label by one at each of its three switches, and by one more across the twisted seam. This section shows that no other entirely forward period can be competitive once it has six or more pieces (Theorem 4.4).

The proof compares two bounds for the Casimir γ^2 of the normal multiplier (Definition B.9). An upper bound follows from a total cost of at most A_{oct}: the frame must turn through $\pi/3$ between $g(0) = I$ and $g(T) = R$, and the cost per unit of turning increases with γ^2 (Proposition F.5). A local formula expresses γ^2 through the switch norms on either side of a piece and their cross pairing (Lemma B.13). We evaluate it at a piece of smallest active margin. In an all-forward period the number of pieces satisfies $N \equiv 1 \pmod 3$, so a period with $N \geq 6$ has at least seven pieces, and its cost is a sum of seven or more edge terms that depend only on consecutive active margins. At the octagon's margins, each edge contributes a quarter of A_{oct}. A seven-edge period within this budget must therefore have a strictly smaller minimum margin. The switching conditions at the five pieces around that small margin (stationarity of the cost, and the requirement that the unused vertex not be preferable at the two outer switches) then hold the two switch norms close to their least possible value $3/32$. The Gram identity then forces γ^2 above the competitive bound. The comparison retains the positive logarithmic correction in the central pairing; the resulting vertex bounds lie above the spectral cap (Figure 9).

We first express the cost through consecutive active margins, then extract the switching conditions at a minimum. The scalar exclusion reduces those conditions to three residual signs and a lower bound for the Gram determinant; their proofs occupy Appendix F.6.

F.1. **The count and the cost of a forward period.**

Lemma F.1. *If every switch of a normal vertex period is forward, then* $N \equiv 1 \pmod 3$. *In particular* $N \geq 6$ *implies* $N \geq 7$.

Proof. The label advances by one at each of the N switches of a period, so $i_N = i_0 + N$, while the twist gives $i_N = i_0 + 1$ in $\mathbb{Z}/3$ (Lemma B.6). □

At a forward switch from piece $k-1$ to piece k the state has the margins a_{k-1} and a_k for the labels i_{k-1} and i_k, and the relation $p_0p_1 + p_1p_2 + p_2p_0 = 4$ (Lemma A.6) determines the third margin, $(4 - a_{k-1}a_k)/(a_{k-1} + a_k)$. Its positivity and the one-arc formula (Lemma B.1) on piece k give the following.

Lemma F.2. *In a normal vertex period with every switch forward, for every* k,

$$a_{k-1}a_k < 4, \qquad q_k = \frac{(a_{k-1} + a_k)(a_k + a_{k+1})}{a_k^2 + 4} > 1,$$

and the cost is

$$\mathcal{C} = \frac{\sqrt{3}}{4} \sum_{j=0}^{N-1} G(a_j, a_{j+1}), \tag{F.1}$$

where

$$G(a,b) = 3ab - 4 - (a^2+b^2)\log(a+b) + \tfrac{1}{2}a^2\log(a^2+4) + \tfrac{1}{2}b^2\log(b^2+4).$$

So in a forward period the durations are determined by the active margins, and (F.1) expresses the cost through the margins alone. The function G is the signed action of Lemma B.3 with all signs positive. The arc costs of Lemma B.1 differ from the terms of (F.1) by a telescoping sum, which vanishes over a period. For the octagon, $N = 4$ and all four active margins equal a with $a^2 = 4/(2\sqrt{2}-1)$; then $(a^2+4)/(4a^2) = 1/\sqrt{2}$, so $G(a,a) = 3a^2 - 4 + a^2\log\frac{a^2+4}{4a^2} = 2\delta_{\mathrm{oct}}$ and the four edges cost $\sqrt{3}\,G(a,a) = A_{\mathrm{oct}}$.

Appendix F.6.1 collects the properties of G that we use: G is symmetric, increasing in each argument as long as $ab \le 8$, has positive mixed derivative $\partial_a\partial_b G = 2(a^2+ab+b^2)/(a+b)^2$, and satisfies $G(a,a) \ge 0$ when $a^2 \ge 4/3$.

F.2. **The window at a minimum.** Fix a piece m whose active margin $a = a_m$ is the least of all active margins of the period, and write

$$(p, b, a, c, d) = (a_{m-2}, a_{m-1}, a_m, a_{m+1}, a_{m+2})$$

for the active margins of the five pieces around it; all are at least a. Lowering every active margin outside this window to a lowers each edge term of (F.1), by monotonicity of G, and keeps the two edges $G(a,p)$ and $G(d,a)$ that close the window.

Lemma F.3 (The seven-entry bound)**.** *If $N \ge 7$, there is an integer $n \ge 0$ with*

$$G(a,p) + G(p,b) + G(b,a) + G(a,c) + G(c,d) + G(d,a) + (n+1)\,G(a,a) \le \sum_{j=0}^{N-1} G(a_j, a_{j+1}) = \frac{4}{\sqrt{3}}\,\mathcal{C}.$$

The window is described by dimensionless quantities. Put

$$\nu = \frac{4}{a^2}, \qquad z = \frac{b}{a}, \qquad w = \frac{c}{a}, \qquad \hat{p} = \frac{p}{a}, \qquad \hat{d} = \frac{d}{a},$$
$$\hat{s}_- = 1+z, \qquad \hat{s}_+ = 1+w, \qquad \hat{K} = 1+\nu, \qquad \hat{K}_- = z^2+\nu, \qquad \hat{K}_+ = w^2+\nu,$$

so that $\hat{s}_- = s_{m-1}/a$, $\hat{s}_+ = s_m/a$, and $\hat{K}, \hat{K}_-, \hat{K}_+$ are K_m, K_{m-1}, K_{m+1} divided by a^2. These letters are local to this section and to Appendix F.6. The factors of the three central pieces are

$$h := q_{m-1} = \frac{(\hat{p}+z)\hat{s}_-}{\hat{K}_-}, \qquad q := q_m = \frac{\hat{s}_-\hat{s}_+}{\hat{K}}, \qquad k := q_{m+1} = \frac{(w+\hat{d})\hat{s}_+}{\hat{K}_+}.$$

Minimality gives $\hat{p}, z, w, \hat{d} \geq 1$, and the products $pb, ab, ac, cd < 4$ of Lemma F.2 read $\hat{p}z < \nu$, $z < \nu$, $w < \nu$, $w\hat{d} < \nu$. Together these give

$$\frac{\hat{s}_-^2}{\hat{K}_-} \leq h \leq \frac{\hat{s}_-}{z}, \qquad \frac{\hat{s}_+^2}{\hat{K}_+} \leq k \leq \frac{\hat{s}_+}{w}. \tag{F.2}$$

F.3. **The normal multiplier at the window.** The costate enters through three numbers: eight times the switch norms on either side of the central piece, and sixteen times the cross pairing on it (Definition B.9),

$$u := 8d_{m-1}, \qquad v := 8d_m, \qquad E := 16e_m.$$

(The letter u does not denote the control here.) At a forward switch the switch norm is at least $j/2 \geq 3/32$ (Lemma B.12), so $u, v \geq 3/4$; the proof will show that u and v exceed $3/4$ by a uniformly small amount.

Three facts about (u, v, E) come from the Pontryagin conditions. First, stationarity (Lemma B.11) at the piece m gives one linear relation, and at the pieces $m \mp 1$, after the coordinates of the multiplier are transported across the pieces m and $m+1$ (Definition B.9), two more. Second, at the forward switch from piece $m-2$ into piece $m-1$, the Hamiltonians of the two active vertices coincide. The third vertex must have no larger Hamiltonian (Lemma A.15): its switching function is $\chi = -4\sqrt{3}\, G_L/(p+b)^2 \leq 0$ for an affine expression G_L in the entries of the multiplier. Read in the frame of the piece $m-1$, the inequality $G_L \geq 0$ is a lower bound for u in terms of the exterior factor h and the ratio z; the switch from $m+1$ into $m+2$ gives the same kind of bound for v. Third, the Gram identity (Lemma B.13) at the doubly forward piece m expresses the Casimir.

Proposition F.4 (The window data)**.** *In the setting above, put*

$$\begin{aligned}
A &= 1 + z + w - \frac{1}{\hat{s}_-} - \frac{1}{\hat{s}_+} + \frac{1}{\hat{K}} - \log q, \\
B &= \frac{\hat{K}_- h}{\hat{s}_-} + 1 - \frac{z^2 \hat{s}_-}{\hat{K}_- h} - \frac{z^2}{\hat{s}_-} + \frac{z^3}{\hat{K}_-} - z \log h, \\
\eta &= \frac{\hat{K}(h\hat{s}_- - z)}{\hat{K}_-}, \qquad n_1 = \frac{\hat{s}_-^2 (\nu - z)(h-1)}{\hat{K}_-}, \\
n_2 &= \frac{\hat{K}^2}{\hat{s}_- \hat{K}_-}(q - 1 + (h-1)(q\hat{s}_- - 1)),
\end{aligned}$$

and let $F_x(r) = 1 + x^2 r - xr \log r/(r-1)$ *for* $r > 1$. *Then:*

(1) (central row) $\hat{s}_- u + \hat{s}_+ v - E = A$*;*

(2) (left row) $n_1 u + n_2 v = \eta A - B$*, and the right row is obtained by exchanging* $(z, \hat{p}, h, u)$ *with* $(w, \hat{d}, k, v)$*;*

(3) (exterior floors) $u \geq F_{z/\hat{s}_-}(h)$ *and* $v \geq F_{w/\hat{s}_+}(k)$*;*

(4) (Gram identity) $\gamma^2 = \hat{K}^2(4uv - E^2)/(16\nu^2)$*, and* $\gamma^2 = \det \Lambda_1(0)/12$.

In physical units the central row reads $(a+b)u+(a+c)v-aE=\frac{1}{2}\partial_{a_m}\sum_j G(a_j,a_{j+1})$: it is the stationarity of the cost (F.1) in the central margin. For $x\ge 1/2$ the floor $F_x(r)$ increases in r and tends to $1-x+x^2\ge 3/4$ as $r\to 1$. The floor for u is the missing-control inequality: once the stationarity of piece $m-1$ is used, G_L is a positive multiple of $u-F_{z/\hat{s}_-}(h)$.

F.4. **The competitive cap.** Proposition B.14 bounds the cost below by a function of the Casimir. Case F uses the following sharper bound; since $(\kappa+\frac{3}{2})(2\kappa+3)-12\kappa=2(\kappa-\frac{3}{2})^2\ge 0$, it implies $\mathcal{C}\ge 4\pi\kappa/(2\kappa+3)$.

Proposition F.5 (Spectral cost–Casimir bound)**.** *Let P be a normal vertex period with every switch forward. Then $\det\Lambda_1(0)>0$, and with $\kappa=\sqrt{\det\Lambda_1(0)}$,*

$$\mathcal{C}\ge\frac{\pi}{3}\Big(\kappa+\frac{3}{2}\Big).$$

Proof. The Casimir is positive by the projective argument of Proposition B.16: at each doubly forward piece the projective coordinate of the switch vectors increases strictly, so over a period the increments cannot add up to zero. With one positive switch norm this makes $\Lambda_1(0)$ negative elliptic, $\Lambda_1(0)=-\kappa\,hJh^{-1}$ for some $h\in\mathrm{SL}_2(\mathbb{R})$. Put $B(t)=h^{-1}g(t)$; then $B'=BX$ and $B(T)=B(0)R$. Write $\rho=-\frac{1}{2}\operatorname{tr}(JX)$ and $\rho_R=-\frac{1}{2}\operatorname{tr}(JBXB^{-1})$, so that the cost density is 3ρ, and let $z_B=\frac{1}{2}(B_{00}+B_{11})+\frac{i}{2}(B_{10}-B_{01})$ be the rotation coordinate of B. Conservation, $\Lambda_1(t)=g(t)^{-1}\Lambda_1(0)g(t)$, gives $\operatorname{tr}(\Lambda_1X)=2\kappa\rho_R$, and the vertex Hamiltonians being nonpositive gives $\operatorname{tr}((\Lambda_1+\frac{3}{2}J)X)\le 0$, that is, $2\kappa\rho_R\le 3\rho$. The polar angle θ of z_B changes at the rate $\theta'=(\rho+\rho_R)/(2|z_B|^2)>0$ and turns through at least $\pi/3$, since $B(T)=B(0)R$. To prove the pointwise inequality

$$(\rho+\rho_R)^2\le 4|z_B|^2\rho\rho_R, \tag{F.3}$$

use the singular value decomposition $B=U\operatorname{diag}(\ell,\ell^{-1})V$ with U,V rotations and $\ell\ge 1$. Rotations preserve ρ and $|z_B|$, so replace X by $VXV^{-1}=\left(\begin{smallmatrix}a & -\rho-d\\ \rho-d & -a\end{smallmatrix}\right)$. Since $\det X=1$, $\rho^2-d^2=1+a^2$. With $c=(\ell^2+\ell^{-2})/2$ and $s=(\ell^2-\ell^{-2})/2$, one has $\rho_R=c\rho+sd$, $4|z_B|^2=2(c+1)$ and $c^2-s^2=1$. Consequently

$$4|z_B|^2\rho\rho_R-(\rho+\rho_R)^2=s^2(\rho^2-d^2)=s^2(1+a^2)\ge 0.$$

Thus

$$0<\theta'=\frac{\rho+\rho_R}{2|z_B|^2}\le\frac{2\rho\rho_R}{\rho+\rho_R}\le\frac{6\rho}{2\kappa+3}. \tag{F.4}$$

Integrating, $\pi/3\le 2\mathcal{C}/(2\kappa+3)$. □

Once negative ellipticity is known, this pointwise argument does not use the switch signs. In particular, on every subinterval $[s,t]\subseteq[0,T]$ it gives

$$\Big(\kappa+\frac{3}{2}\Big)\big(\theta(t)-\theta(s)\big)\le-\frac{3}{2}\int_s^t\operatorname{tr}(JX(r))\,\mathrm{d}r. \tag{F.5}$$

For a competitive period the total angle is exactly $\pi/3$: it is positive, congruent to $\pi/3$ modulo 2π, and at most $A_{\mathrm{oct}}/(\kappa+3/2) < 2\pi/3$. In the twisted extension the cost density and θ' are T-periodic: conjugation by R preserves $\mathrm{tr}(JX)$, while $B(t+T) = B(t)R$ multiplies z_B by $e^{i\pi/3}$. Splitting an interval at period endpoints therefore extends (F.5) to intervals in $\mathbb{R}$. The interval form will allow a bridge to be charged its exact cost while the complement is charged by its angle (Equation (G.14)).

Corollary F.6 (Competitive cap)**.** *A competitive normal vertex period with every switch forward has*

$$\gamma^2 \le \Gamma_{\mathrm{oct}} := \frac{1}{12}\left(\frac{3A_{\mathrm{oct}}}{\pi} - \frac{3}{2}\right)^2 < \frac{9193}{50000}.$$

Proof. The exact bound follows from Proposition F.5 and $\mathcal{C} \le A_{\mathrm{oct}}$. By Lemma C.5,

$$\frac{3A_{\mathrm{oct}}}{\pi} - \frac{3}{2} < \frac{101}{68}, \qquad \Gamma_{\mathrm{oct}} < \frac{1}{12}\left(\frac{101}{68}\right)^2 < \frac{9193}{50000}.$$

This is also the cap used for mixed periods in Lemma G.1. □

F.5. **The scalar exclusion.** From now on suppose $N \ge 6$, so $N \ge 7$ by Lemma F.1. The window data of Lemma F.3 and Proposition F.4, the bounds (F.2) and the cap $\gamma^2 < 9193/50000$ are finitely many real equations and inequalities. We show that they are inconsistent. The data are symmetric under the exchange of the two sides, which exchanges the left and right rows and the two floors, so we may assume $z \le w$. The steps follow; Appendix F.6 proves the scalar inequalities.

Step 1: the minimum is not small. The first restriction does not use the cost.

Lemma F.7 (Small forward minimum)**.** $\nu < 3$*, that is,* $a^2 > 4/3$*.*

If $\nu \ge 3$, the left row and the floors force the residual $\eta A - B - n_1 F_{z/\hat{s}_-}(h) - n_2 V$ to be nonnegative, where V is the floor for v evaluated at the smallest factor that minimality allows. Appendix F.6.7 shows that this residual is negative: lowering the larger neighbour w does not decrease it, which leaves three faces, and on each face the residual is a function of h with explicit value and slope at $h = 1$ and a controlled second derivative.

Step 2: the minimum is below the octagon's scale. Since $a^2 > 4/3$, $G(a,a) \ge 0$, so the seven-entry bound holds with $n = 0$. If $a^2 \ge 20/11$, each of the seven edges is at least $G(a_0, a_0)$ with $a_0^2 = 20/11$, and $G(a_0,a_0) = \frac{20}{11}(\frac{4}{5} - \log\frac{5}{4}) \ge \frac{23}{22}$, so the cost would exceed $161\sqrt{3}/88 > 29\sqrt{3}/16 > A_{\mathrm{oct}}$. The last comparison follows from the midpoint logarithm bound in Appendix F.6.3.

Lemma F.8. $11/5 < \nu < 3$.

Step 3: the exterior budget. The rectangle inequality $G(x,y) \geq G(x,a) + G(a,y) - G(a,a)$ for $x, y \geq a$, applied to the edges (p,b) and (c,d), turns the seven-entry bound into

$$\frac{\sqrt{3}}{\nu}\big(2g_\nu(\hat{p}) + 2g_\nu(z) + 2g_\nu(w) + 2g_\nu(\hat{d}) - g_\nu(1)\big) \leq A_{\mathrm{oct}}, \qquad g_\nu(x) = \frac{G(a,ax)}{a^2}.$$

On $[1,\nu]$ the function g_ν is increasing and concave, so for a given sum $\hat{p} + z + w + \hat{d}$ the left side is smallest when the excess over four is carried by one ratio. The joint concavity argument of Appendix F.6.3 then gives:

Lemma F.9 (Exterior budget)**.** *$\hat{p} + z + w + \hat{d} < (5\nu + 1)/3$. Consequently $z + w < 5(\nu - 1)/3$ and $h \leq \hat{s}_-(5\nu - 2 - 3w)/(3\hat{K}_-)$.*

The bound on h follows from $h\hat{K}_-/\hat{s}_- = \hat{p} + z$ and $k\hat{K}_+/\hat{s}_+ = \hat{d} + w \geq \hat{s}_+$.

Step 4: the larger neighbour. The floors and the left row give $\eta A - B = n_1 u + n_2 v \geq n_1 F_{z/\hat{s}_-}(h) + n_2 F_{w/\hat{s}_+}(\hat{s}_+^2/\hat{K}_+)$, since $n_2 \geq n_1 \geq 0$, the floor increases in the factor and $k \geq \hat{s}_+^2/\hat{K}_+$. The difference of the two sides is a residual $\mathcal{E}_N$, and $\mathcal{E}_N < 0$ whenever $w \geq 4/3$ (Appendix F.6.5).

Lemma F.10 (Neighbour bound)**.** *If $z \leq w$, then $w < 4/3$.*

Step 5: the switch norms are nearly minimal. Write $u = \frac{3}{4} + \delta u$ and $v = \frac{3}{4} + \delta v$. The floors become $\delta u \geq L(z,h) \geq 0$ and $\delta v \geq L(w, \hat{s}_+^2/\hat{K}_+) \geq 0$ for an explicit function L (Appendix F.6.4), and the left row becomes the centered identity

$$n_1\,\delta u + n_2\,\delta v = r, \qquad r := \eta A - B - \tfrac{3}{4}(n_1 + n_2), \tag{F.6}$$

with $n_2 > n_1 \geq 0$. If $\delta u + \delta v \geq 1/32$, then (F.6), written as $r = n_1(\delta u + \delta v) + (n_2 - n_1)\delta v$, and the floor for δv give

$$\mathcal{E}_\Sigma := r - \frac{n_1}{32} - (n_2 - n_1)\,L\Big(w, \frac{\hat{s}_+^2}{\hat{K}_+}\Big) \geq 0;$$

if $\delta v \geq 1/64$, they give $\mathcal{E}_v := r - n_1 L(z,h) - n_2/64 \geq 0$. Neither residual decreases when w is lowered to z. On the resulting symmetric domain, $\mathcal{E}_v < 0$ by Lemma F.14; the sum residual is negative by Lemma F.15.

Lemma F.11 (Norm bounds)**.** *If $z \leq w$, then $\delta u, \delta v \geq 0$, $\delta u + \delta v < 1/32$ and $\delta v < 1/64$.*

Step 6: the Casimir is too large. The central row and the identity $\frac{3}{4}(\hat{s}_- + \hat{s}_+) - A = e_0 - \varphi$, where

$$e_0 = 1 - \frac{\hat{K}}{8} + \frac{1}{\hat{K}}, \quad \varphi = f\Big(\frac{\hat{s}_-}{2}\Big) + f\Big(\frac{\hat{s}_+}{2}\Big) + f\Big(\frac{4}{\hat{K}}\Big), \quad f(r) = \frac{r - r^{-1}}{2} - \log r,$$

give $E = e_0 - \varphi + \hat{s}_-\delta u + \hat{s}_+\delta v$, with $0 \leq \varphi < e_0$ by the integral bound in Appendix F.6.6. Substituted into the Gram identity, the norm bounds give:

Lemma F.12 (Conditional Casimir bound)**.** *Let* $11/5 \le \nu < 3$, $1 \le z \le w \le 4/3$, $z + w \le 5(\nu - 1)/3$, $\delta u, \delta v \ge 0$, $\delta u + \delta v \le 1/32$, $\delta v \le 1/64$ *and* $E = e_0 - \varphi + \hat{s}_- \delta u + \hat{s}_+ \delta v$. *Then*

$$\frac{\hat{K}^2(4(\frac{3}{4} + \delta u)(\frac{3}{4} + \delta v) - E^2)}{16\nu^2} > \frac{9193}{50000}.$$

Proof of Theorem 4.4. Suppose P is competitive and all-forward, with $N \ge 6$. By Lemma F.1 $N \ge 7$; take the window at a global minimum of the active margins. Its data satisfy Lemma F.3 and Proposition F.4 and Corollary F.6; after exchanging the two sides if necessary, $z \le w$. Lemmas F.8 to F.11 give the hypotheses of Lemma F.12, which together with the Gram identity gives $\gamma^2 > 9193/50000$, contradicting Corollary F.6. □

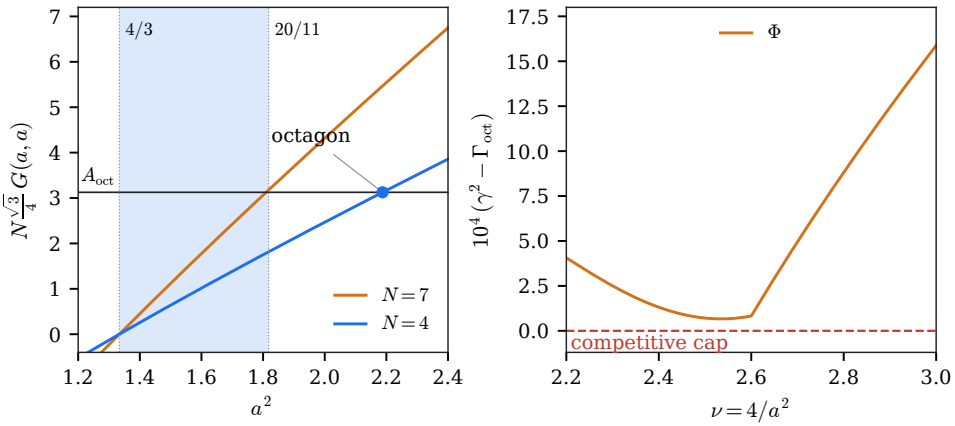


FIGURE 9. The two sides of the case F contradiction. Left: the cost $N\frac{\sqrt{3}}{4}G(a,a)$ of N equal edges against a^2, for $N = 4$ and $N = 7$. Four edges at the octagon's margin cost exactly A_{oct} (dot). Seven edges cost more than A_{oct} once $a^2 \ge 20/11$ (Lemma F.8), and $a^2 > 4/3$ by Lemma F.7, so the least active margin of a long competitive forward period lies in the shaded band. Right: on that band, under the bounds of Lemmas F.9 to F.11, the proof uses the scalar curve Φ of Appendix F.6.6. This curve bounds the Casimir at the two vertices on the norm-sum edge; the other two vertices are estimated separately in the proof. The curve is strictly convex up to the shape ceiling $\nu = 13/5$ and increases thereafter. Its minimum lies above the competitive cap Γ_{oct} (dashed), defined in Corollary F.6. The vertical scale is $10^4(\gamma^2 - \Gamma_{\text{oct}})$.

F.6. **Estimates for class F.** This subsection proves the scalar inequalities of Appendix F, in its notation: $\nu = 4/a^2$, the ratios $z, w, \hat{p}, \hat{d} \ge 1$, $\hat{s}_\pm$, $\hat{K}$,

$\hat{K}_{\pm}$, the factors h, q, k, the row coefficients A, B, η, n_1, n_2 and the floor $F_x(r)$ of Proposition F.4. Throughout,

$$f(r) = \frac{r - r^{-1}}{2} - \log r, \qquad \beta(y) = \frac{y^2 + y + 1}{(1+y)^2} = 1 - \frac{1}{1+y} + \frac{1}{(1+y)^2},$$

so that $\beta(w)$ is the value $1 - x + x^2$ of the floor F_x at $r = 1$ for $x = w/(1+w)$.

F.6.1. *The edge function.* Let G be the edge function of (F.1). Its partial derivatives are

$$\partial_b G(a,b) = 3a - 2b\log(a+b) - \frac{a^2+b^2}{a+b} + b\log(b^2+4) + \frac{b^3}{b^2+4},$$
$$\partial_a\partial_b G(a,b) = \frac{2(a^2+ab+b^2)}{(a+b)^2}, \qquad \partial_a\partial_b^2 G(a,b) = \frac{2a(b-a)}{(a+b)^3}.$$

(1) (Monotonicity) The bound $\log y \geq 1 - 1/y$ at $y = (b^2+4)/(a+b)^2$ gives $\partial_b G(a,b) \geq a(a+2b)(b^2 - ab + 8)/\big((a+b)(b^2+4)\big)$, which is positive when $ab \leq 8$. So G increases in each argument while the product of its arguments is at most 8. All products of consecutive active margins are below 4.

(2) (Rectangle inequality) Since $\partial_b G(x,\cdot) - \partial_b G(a,\cdot) \geq 0$ for $x \geq a$, integrating in the second variable gives $G(x,y) \geq G(x,a) + G(a,y) - G(a,a)$ for $x, y \geq a > 0$.

(3) (Concavity) If $a^2 \geq 4/3$, then $G(a,\cdot)$ is concave on $[a, 4/a]$. Since $\partial_a\partial_b^2 G \geq 0$ for $a \leq b$, it suffices to check $\partial_b^2 G < 0$ at $a = b$ when $4/3 \leq b^2 \leq 4$, and at $a = 4/b$. With $\varsigma = 4b^2/(b^2+4) \in [1,2]$ one has $\partial_b^2 G(b,b) = -\log\varsigma - \frac{3}{2} + \frac{5}{4}\varsigma - \frac{1}{8}\varsigma^2$, which increases in ς (its derivative is $(\varsigma-1)(4-\varsigma)/(4\varsigma)$) up to the value $\frac{1}{2} - \log 2 < 0$ at $\varsigma = 2$; and $\partial_b^2 G(4/b, b) = \log\varrho + 1 - \varrho < 0$ with $\varrho = b^2/(b^2+4)$.

(4) (Scaling) $G(ax, ay) = a^2 g_\nu(x,y)$ with $g_\nu(x,y) = 3xy - \nu - (x^2+y^2)\log(x+y) + \frac{1}{2}x^2\log(x^2+\nu) + \frac{1}{2}y^2\log(y^2+\nu)$; we write $g_\nu(x) = g_\nu(1,x)$. In particular $g_\nu(1) = 3 - \nu - \log\frac{4}{1+\nu} \geq \nu(3-\nu)/(1+\nu) \geq 0$ for $0 < \nu \leq 3$, by $\log y \geq 1 - 1/y$ again; so $G(a,a) \geq 0$ when $a^2 \geq 4/3$.

F.6.2. *Logarithm bounds.* The sum and neighbour comparisons use the following elementary logarithm bounds, each proved from the sign of a derivative. For $v \geq 1$ and $\tau = (v-1)/(v+1)$,

$$2\tau + \tfrac{2}{3}\tau^3 \leq \log v \leq 2\tau + \frac{2\tau^3}{3(1-\tau^2)},$$

and for $h \geq 1$, $0 \leq y < 1$,

$$\frac{(h-1)^3}{3h(h+1)} \leq f(h) \leq \frac{(h-1)^3}{6}, \qquad \log(1-y) \leq -y - \frac{y^2}{2} - \frac{y^3}{3}.$$

The lower bound for f holds because the difference vanishes at $h = 1$ and has derivative $(h-1)^4/(6h^2(h+1)^2)$; this Padé bound is used for the sum

and neighbour residuals below and, in the form $\log v \le \frac{v-v^{-1}}{2} - \frac{(v-1)^3}{3v(v+1)}$, for other logarithms. The upper bound for $\log v$ at $v = 5/4$ gives $\log \frac{5}{4} < \frac{9}{40}$.

F.6.3. *The threshold and the budget. The threshold.* With $a_0^2 = 20/11$ and $\nu_0 = 11/5$, $g_{\nu_0}(1) = 3 - \frac{11}{5} - \log \frac{20}{16} = \frac{4}{5} - \log \frac{5}{4}$, so $G(a_0, a_0) = \frac{20}{11}(\frac{4}{5} - \log \frac{5}{4}) > \frac{20}{11} \cdot \frac{23}{40} = \frac{23}{22}$. The midpoint inequality for $1/t$ on $[1, \sqrt{2}]$ gives $\log 2 > 12 - 8\sqrt{2}$, and hence

$$A_{\mathrm{oct}} < \frac{8\sqrt{3}(\sqrt{2}-1)}{2\sqrt{2}-1} < \frac{29\sqrt{3}}{16}.$$

The second comparison is $70\sqrt{2} < 99$, whose square is $9800 < 9801$. Since $\frac{7\sqrt{3}}{4}\frac{23}{22} = 161\sqrt{3}/88 > 29\sqrt{3}/16$, the threshold cost exceeds A_{oct}, which proves Lemma F.8.

Lemma F.13 (Joint concavity of the exterior cost)**.** *The function $\mathcal{F}(\nu, x) = 5g_\nu(1) + 2g_\nu(x)$ is strictly concave on $1 \le x \le \nu \le 3$.*

Proof. Put $D = x^2 + \nu$ and

$$\mathcal{E}(\nu, x) = -\partial_x^2 g_\nu(x) - \frac{\nu^2}{2D^2}.$$

We first show $\mathcal{E} > 0$. Differentiation gives

$$\partial_x^2 g_\nu(x) = -\log \frac{(1+x)^2}{D} + \frac{2(x+2)}{(x+1)^2} - \frac{\nu(x^2+3\nu)}{D^2}, \qquad \partial_\nu \mathcal{E} = \frac{\nu(2x^2-\nu)}{D^3}.$$

At $\nu = x$, writing $y = 1/(x+1)$,

$$\mathcal{E}(x,x) = -\log(1-y) - y - \frac{y^2}{2} = \int_0^y \frac{t^2}{1-t}\,\mathrm{d}t > 0.$$

For larger ν, substitution of $\nu = x^2 t$ gives

$$\mathcal{E}(\nu, x) - \mathcal{E}(x, x) = \int_{1/x}^{\nu/x^2} \frac{t(2-t)}{(1+t)^3}\,\mathrm{d}t.$$

The lower endpoint is at most one and the upper endpoint at most three. If the upper endpoint is at most two, the integrand is nonnegative. Otherwise the integral is at least the integral over $[1, 3]$, which is positive: with $k(t) = t/(1+t)^3$, pair the two points $2 \pm z$ to get

$$\frac{(2-z)z}{(3-z)^3} - \frac{(2+z)z}{(3+z)^3} = z\big(k(2-z) - k(2+z)\big) > 0 \qquad (0 < z \le 1),$$

since k decreases on $[1, 3]$. Thus $\partial_x^2 g_\nu(x) < -\nu^2/(2D^2)$ everywhere.

The other entries of the Hessian of $\mathcal{F}$ are

$$\mathcal{F}_{\nu\nu} = -\frac{6}{(1+\nu)^2} - \frac{x^2}{D^2} < 0, \qquad \mathcal{F}_{\nu x} = \frac{2\nu x}{D^2}.$$

Its determinant is therefore greater than

$$\frac{3\nu^2}{D^2}\left(\frac{2}{(1+\nu)^2} - \frac{x^2}{D^2}\right) > 0.$$

For the last sign use $D^2 \geq 4\nu x^2$ and $(1+\nu)^2 < 8\nu$ on $1 \leq \nu \leq 3$. The Hessian is negative definite, proving strict concavity. □

The budget. The rectangle inequality gives the seven-entry bound in Step 3. For an increasing concave function ϕ on $[1, M]$ and four points with sum at least σ, where $1 < \sigma - 3 \leq M$,

$$\phi(x_1) + \phi(x_2) + \phi(x_3) + \phi(x_4) \geq 3\phi(1) + \phi(\sigma - 3).$$

Indeed, first lower the points until their sum is σ. Each then lies in $[1, \sigma - 3]$; summing the lower chord bound on that interval proves the claim. If $\hat{p} + z + w + \hat{d} \geq (5\nu + 1)/3$, apply it to $\phi = g_\nu$ and $x_* = (5\nu - 8)/3$ to obtain

$$A_{\mathrm{oct}} \geq \frac{\sqrt{3}}{\nu}(5g_\nu(1) + 2g_\nu(x_*)) = \frac{\sqrt{3}}{\nu}\mathcal{F}(\nu, x_*).$$

The line (ν, x_*) lies in $1 \leq x \leq \nu \leq 3$. By Lemma F.13, $\mathcal{F}(\nu, x_*)$ lies above the chord joining its values at $\nu = 11/5$ and $\nu = 3$. After division by $\nu > 0$, that chord is a weighted average of the two endpoint ratios. It therefore suffices to bound both endpoint ratios below by $29/16$.

At $\nu = 11/5$, $x_* = 1$ and the threshold estimate gives

$$\frac{\mathcal{F}(11/5, 1)}{11/5} > \frac{161}{88} > \frac{29}{16}.$$

At $\nu = 3$, $x_* = 7/3$, $g_3(1) = 0$, and

$$g_3(7/3) = 4 - \log\frac{5}{3} - \frac{49}{18}\log\frac{25}{19}.$$

Use $\log v < (v-1)/\sqrt{v}$ for $v > 1$, which is $t < \sinh t$ at $t = \frac{1}{2}\log v$. Then

$$g_3(7/3) > 4 - \frac{2}{\sqrt{15}} - \frac{49}{15\sqrt{19}} > 4 - \frac{17}{32} - \frac{3}{4} = \frac{87}{32}.$$

The two rational comparisons follow by squaring: $64^2 < 15 \cdot 17^2$ and $196^2 < 19 \cdot 45^2$. Hence $\mathcal{F}(3, 7/3)/3 > 29/16$ as well. The exterior cost thus exceeds $29\sqrt{3}/16 > A_{\mathrm{oct}}$, a contradiction. This proves Lemma F.9.

F.6.4. *The centered system.* The following identities hold for $\nu, z, w, h > 0$, $h \neq 1$; each is an equality of rational expressions in the variables and the logarithms.

(1) (Floors) $F_{z/\hat{s}_-}(h) = \frac{3}{4} + L(z, h)$ with

$$L(z,h) = \frac{(z-1)(z-1+2z(h-1))}{4\hat{s}_-^2} + \frac{zh}{\hat{s}_-} \cdot \frac{f(h)}{h-1},$$

and $L(z, h) \geq 0$ for $z \geq 1$, $h > 1$.

(2) (Central row) $\frac{3}{4}(\hat{s}_- + \hat{s}_+) - A = e_0 - \varphi$.

(3) (Left row) With $\zeta = z - 1$ and $H = h - 1$, the quantity $r = \eta A - B - \frac{3}{4}(n_1 + n_2)$ of (F.6) is given by

$$\begin{aligned} 8\hat{s}_-\hat{K}_- r = {} & (3-\nu)\zeta(3-\nu+2\zeta) \\ & + H\big((3-\nu)^2(\zeta^2+4\zeta+2) + 8(3-\nu)\zeta z\big) + 2H\zeta^2(8+4\zeta-\zeta^2) \\ & + \frac{4\hat{s}_- H^2 z(3-\nu+4\zeta+\zeta^2)}{h} \\ & + 8\hat{s}_-\hat{K}(1+\hat{s}_- H)\,\varphi - 8\hat{s}_-\hat{K}_-\, z f(h). \end{aligned}$$

It depends on w only through φ.

(4) (Coefficients) $\hat{s}_-\hat{K}_-(n_2 - n_1) = \hat{K}(\hat{s}_-\hat{s}_+ - \hat{K}) + H\big(\hat{s}_-^4 - \hat{K}^2 + \hat{K}\hat{s}_-^2(\hat{s}_+ - \hat{s}_-)\big)$, which is positive for $\nu < 3$, $1 \le z \le w$, $h \ge 1$, because $\hat{s}_-\hat{s}_+ \ge 4 > \hat{K}$ and $\hat{s}_-^4 \ge 16 > \hat{K}^2$.

Two monotonicity facts complete the setup. The floor increases in the factor: for $x \ge 1/2$, $\partial_r F_x(r) = x^2 - x\,(r - 1 - \log r)/(r-1)^2 \ge x^2 - x/2 \ge 0$, since $r - 1 - \log r \le (r-1)^2/2$. And in the variable w, with ν, z, h fixed,

$$n_2(w) - n_2(z) = \eta\,(w - z), \qquad A(w) - A(z) \le (w - z)\,\beta(w) \qquad (z \le w),$$

because $\partial_w A = \beta(w)$ and $\beta'(y) = (y-1)/(1+y)^3 \ge 0$ for $y \ge 1$.

F.6.5. *The three residual signs. Lowering the larger neighbour.* The residuals of Appendix F depend on w through A, n_2, φ and the opposite floor. For $\mathcal{E}_v = r - n_1 L(z,h) - n_2/64$ the increments above give $\mathcal{E}_v(w) - \mathcal{E}_v(z) \le \eta(w-z)\big(\beta(w) - \frac{49}{64}\big) \le 0$, since $\beta(w) \le \beta(4/3) = 37/49 < 49/64$. For $\mathcal{E}_\Sigma = r - n_1/32 - (n_2 - n_1)L(w, \hat{s}_+^2/\hat{K}_+)$ put $V(w) = \frac{3}{4} + L(w, \hat{s}_+^2/\hat{K}_+)$, the floor at the least admissible factor; then $V(w) \ge \beta(w)$, and V is nondecreasing in w (the coefficients of L increase in w, $hf(h)/(h-1)$ increases in h, and $\hat{s}_+^2/\hat{K}_+$ increases in w), so

$$\mathcal{E}_\Sigma(w) - \mathcal{E}_\Sigma(z) \le \eta(w-z)\big(\beta(w) - V(w)\big) - (n_2(z) - n_1)\big(V(w) - V(z)\big) \le 0.$$

The neighbour residual is lowered in the same way to $w = \max(z, 4/3)$. After lowering, $w = z$ for $\mathcal{E}_\Sigma$ and $\mathcal{E}_v$, and the factor ranges over the symmetric domain

$$1 \le z \le \tfrac{4}{3}, \qquad 1 + \tfrac{6}{5}z \le \nu < 3, \qquad \frac{(1+z)^2}{z^2+\nu} \le h \le \frac{(1+z)(5\nu - 2 - 3z)}{3(z^2+\nu)}; \tag{F.7}$$

the lower bound for ν is exactly the condition that the interval for h be nonempty.

Concavity in the factor. As functions of h, with the other variables fixed, the three residuals have the form $c_0 + c_1 h + c_2 h^2 + c_{-1}/h + (d_0 + d_1 h)\log h$, with second derivative $(2c_2 h^3 + 2c_{-1} + d_1 h^2 - d_0 h)/h^3$. For $\mathcal{E}_v$ and $\mathcal{E}_N$ it equals $-zP(h)/(\hat{K}_- h^3)$ with

$$P(h) = (\nu - z)h^2(2zh - \hat{s}_-) + \hat{K}_- h - 2z\hat{s}_-.$$

For $h \geq 1$, $P'(h) = 2(\nu - z)h(3zh - \hat{s}_-) + \hat{K}_- > 0$, and at the least factor $H_0 = \hat{s}_-^2/\hat{K}_-$,

$$P(H_0) = -\frac{\hat{s}_-(2z + 1 - \nu)}{\hat{K}_-^3}\, Q(\nu, z),$$

$$Q(\nu, z) = (1 - z)\nu^2 - z(z^3 + 7z^2 + 5z + 3)\nu + z^2(z^4 + 2z^3 + 6z^2 + 5z + 2).$$

On the domains of these residuals, $1 \leq z \leq 5/3$, $\nu < 3 \leq 2z + 1$ and $\nu \geq 1 + 6z/5 > z + 1$. Since Q decreases in ν, it suffices to evaluate it at $z + 1$:

$$Q(z + 1, z) = z^3\big(z^2(z + 1) - 2z - 8\big) - 7z^2 - 2z + 1 < 0.$$

Indeed, $z^2 < 3$ gives $z^2(z + 1) - 2z - 8 < z - 5 < 0$, and $7z^2 + 2z - 1 > 0$ for $z \geq 1$. So $P \geq 0$ on $[H_0, \infty)$, and $\mathcal{E}_v, \mathcal{E}_N$ are concave in h there. A concave function on an interval lies below its tangent at an endpoint, and the tangent is affine, so the residual is negative on the interval as soon as the tangent is negative at both endpoints. This leaves two values for each residual: the value at one endpoint and the tangent's value at the other. The sum residual is treated in the variable $\tau = (h - 1)/h$ instead (below).

Lemma F.14 (The individual norm residual)**.** *On the symmetric domain* (F.7)*,* $\mathcal{E}_v < 0$.

Proof. Put $a = (z - 1)/(z + 1)$, $b = 1 - a$, and write the least and greatest admissible factors as

$$h_0 = \frac{(1 + z)^2}{z^2 + \nu} = 1 + t =: \ell, \qquad h_+ = \frac{(1 + z)(5\nu - 2 - 3z)}{3(z^2 + \nu)}.$$

Then $0 \leq a \leq 1/7$, $0 < t \leq 1/4$, and

$$z = \frac{1 + a}{b}, \qquad \nu = \frac{4}{b^2\ell} - z^2, \qquad \xi := \frac{\hat{K}}{\hat{K}_-} = 1 - a\ell \geq \frac{23}{28},$$

$$h_+ - \ell = \frac{2[1 - a^2 - (4 + a^2)t]}{3b}.$$

Thus $t \leq t_*(a) := (1 - a^2)/(4 + a^2) \leq 1/4$. Define the value and the other endpoint of its tangent by

$$\mathcal{V}(a, t) = \mathcal{E}_v(\ell), \qquad \mathcal{T}(a, t) = \mathcal{V}(a, t) + (h_+ - \ell)\, \partial_h \mathcal{E}_v(\ell).$$

The product $n_1 F_{z/\hat{s}_-}(h)$ has a removable singularity at $h = 1$, since n_1 contains $h - 1$. After cancellation both endpoint functions extend smoothly to the rectangle $0 \leq a \leq 1/7$, $0 \leq t \leq 1/4$. The original residual is concave in h, as proved above. It is therefore enough to show $\mathcal{V} < 0$ and $\mathcal{T} < 0$ on the actual domain.

Convexity of the value. The logarithmic part of $\mathcal{V}$ is

$$c \log \ell + \frac{(b + 2t)\xi}{b} \log \xi, \qquad c = \frac{a(a^2 + 3)\ell + 2(1 - a^2) - (1 + 3a^2)\ell^2}{b^2}.$$

The rational part has third t-derivative $3(1+a)^3/b^2 > 0$. The third derivative of $c\log\ell$ is

$$\frac{4(1-a^2)-2(1+3a^2)\ell^2-a(a^2+3)\ell}{b^2\ell^3} > 0.$$

Its numerator decreases in both variables and is positive at $a = 1/7$, $\ell = 5/4$. The last logarithmic term has third derivative $a^2[a(b+2t)+6\xi]/(b\xi^2) \geq 0$. Moreover

$$\mathcal{V}_{tt}(a,0) = \frac{a}{16b^2}\left[2+a-33a^2+64b(-\log b-a)\right] \geq 0,$$

since $-\log b \geq a$ and $2 - 33/49 > 0$. Hence $\mathcal{V}$ is convex in t. Its lower edge is negative for $a > 0$:

$$\mathcal{V}(a,0) = b\log b + \frac{a(31-16a)}{32} \leq \frac{a(16a^2-1)}{32} < 0,$$

using $\log(1-a) \leq -a - a^2/2$. At $a = 0$ this edge value is zero.

We show that the upper edge is convex in $y = z - 1 \in [0, 1/3]$. Put $S = y + 2$, $D = 2 - ty$, so $\xi = D/S$ and $q = \ell S/D$. Differentiation of the logarithmic terms, with their common terms combined, gives

$$\begin{aligned}\mathcal{V}_{yy} = 2t^3 + \frac{49}{32}t^2 - \frac{\ell(16t^2+15t-1)}{8S^3} - \frac{12\ell}{S^4} \\ - 2t\ell\log\ell - \frac{4\ell}{S^3}\log q + \frac{4\ell(3\ell+t^2S)}{DS^3}.\end{aligned}$$

The inequalities $\log\ell \leq t$, $\log q \leq q - 1$ therefore give

$$\mathcal{V}_{yy} \geq -\frac{15t^2}{32} - \frac{\ell(16t^2+15t-1)}{8S^3} - \frac{12\ell}{S^4} + \frac{4\ell[5\ell - S(1+2t-t^2)]}{DS^3}.$$

At $t = 1/4$ the right side equals

$$\frac{5}{S^4}\left[\frac{3(S-1)}{8} + \frac{y(19-43y)}{8(8-y)} - \frac{15S}{128} - \frac{3S^4}{512}\right] > 0.$$

Indeed, the second term in brackets is nonnegative, while $S^4 \leq 16 + 52y$ follows by integrating $4S^3 \leq 4(7/3)^3 < 52$. The remaining terms are at least $3(1-y)/64 > 0$.

It remains to check the two ends of this convex upper edge. Set $L_0 = \log(5/4)$. The reciprocal integral gives $1/5 < L_0 < 9/40$, the latter by its trapezoid bound. At $y = 0$,

$$\mathcal{V}(0,1/4) = \frac{56L_0 - 13}{128} < 0.$$

At $y = 1/3$, write $\mathcal{V} = R + C_L\log\ell + C_M\log\xi$. Direct substitution gives

$$R = -\frac{13}{84^2}, \qquad C_L = \frac{8}{3} - \frac{15}{14} - \frac{65}{144} \in (0, 5/4), \qquad C_M = \frac{1}{4} - \frac{1}{48} + \frac{15}{14},$$
$$15C_M - 17C_L = \frac{4}{63}, \qquad \ell\xi = \frac{115}{112}, \quad \xi = \frac{23}{28}.$$

Combine the logarithms before bounding them. Their sum is less than

$$C_L\left[\frac{3}{112}-\frac{2}{15}\left(\frac{5}{28}+\frac{25}{2\cdot 28^2}\right)\right]=\frac{C_L}{1176}<\frac{5}{4\cdot 1176}.$$

Here we used $\log(115/112)<3/112$ and $\log(23/28)<-5/28-25/(2\cdot 28^2)$. As $84^2=6\cdot 1176$, this proves $\mathcal{V}<(-13/6+5/4)/1176<0$ at the second corner. Convexity in y gives a strictly negative upper edge. Convexity in t, together with the lower edge, now proves $\mathcal{V}<0$ throughout the rectangle with $t>0$.

Monotonicity of the other tangent endpoint. We prove $\mathcal{T}_t\ge 0$ on the rectangle. Its logarithmic part is

$$c\log\ell+\frac{\xi U}{3b^2}\log\xi,\qquad U=b(7+a)-kt,\quad k=2(5+3a+2a^2),$$

where $c=(p_2\ell^2-p_1\ell+p_0)/(3b^3)$ and

$$\begin{aligned}p_2&=(1+3a^2)(5+3a+2a^2),\\ p_1&=10(1+3a^2)+a\big[8(4+a^2)-3b(a^2+3)\big],\\ p_0&=6b(1-a^2)+40a.\end{aligned}$$

All three coefficients are positive; for p_1 use $b\le 1$. The rational part has third derivative

$$-\frac{(1+a)^3(13+3a+4a^2)}{b^3}-\frac{5(1-16a^2)}{8b^2\ell^4}<-\frac{13(1+a)^3}{b^3}.$$

The logarithmic term $c\log\ell$ contributes at most $(2p_2+p_1+2p_0)/(3b^3)$, since $\ell\ge 1$. It is absorbed by this negative term:

$$13(1+a)^3-\frac{2p_2+p_1+2p_0}{3}=\frac{7+20a+a^2(56+4a-15a^2)}{3}>0.$$

The third derivative of the last logarithmic term is

$$\frac{a^2(aU-3k\xi)}{3b^2\xi^2}\le 0,$$

since $aU\le 7a\le 1$, $k\ge 10$ and $\xi\ge 23/28>3/4$. Thus $\mathcal{T}_{ttt}<0$, so $\mathcal{T}_t$ is concave in t.

On its lower edge, substitute $-\log b\ge a+a^2/2$ in the exact derivative to obtain

$$96b^2\mathcal{T}_t(a,0)\ge a\left[23+11a-2a^2(81+106a+40a^2)\right]\ge 0.$$

The bracket is positive: $81+106a+40a^2<100$ on $a\le 1/7$, so it exceeds $23-200/49>0$. Consequently this edge is strictly positive for $a>0$, and zero at $a=0$.

On the upper edge let $W(a)=\mathcal{T}_t(a,1/4)$. Its coefficient of $\log\xi$ is $(7a^3+11a^2+2a-10)/(3b^2)<0$, since $7a^3+11a^2+2a\le 26/49<1$. Replacing $\log\xi=\log(1-5a/4)$ by $-5a/4$ therefore gives a lower bound $W(a)\ge\mathcal{G}(a)$.

After two derivatives in b, its rational terms combine into three reciprocal kernels:

$$\mathcal{G}_{bb} = -\frac{113}{48} + C_2(3/b - 1/b^2)'' + C_3(6/b - 1/b^3)'' + D_0(-1/b)'',$$
$$C_2 = \frac{92}{3}L_0 - \frac{173}{48}, \qquad C_3 = \frac{133}{48} - \frac{20}{3}L_0, \qquad D_0 = \frac{28}{3}L_0 - \frac{113}{320}.$$

The bounds $1/5 < L_0 < 1/4$ make all three weights positive. The kernels are concave:

$$(3/b{-}1/b^2)'' = -\frac{6(1-b)}{b^4}, \quad (6/b{-}1/b^3)'' = -\frac{12(1-b^2)}{b^5}, \quad (-1/b)'' = -\frac{2}{b^3}.$$

Thus $\mathcal{G}_{bb} < 0$ on $6/7 \le b \le 1$. At $a = 0$, $\mathcal{G}$ has value $5L_0/6 - 49/480 > 0$. At $a = 1/7$, the rational part of W exceeds $-4/5$, its L_0 coefficient exceeds $1/5$, and minus its $\log \xi$ coefficient exceeds $30/7$. Thus

$$\mathcal{G}(1/7) > \frac{1}{25} + \frac{75}{98} - \frac{4}{5} = \frac{13}{2450} > 0.$$

Both endpoints are positive, so $W \ge \mathcal{G} > 0$ throughout the upper edge. Concavity in t now gives $\mathcal{T}_t \ge 0$ on the whole rectangle.

Finally, at $t = t_*(a)$ one has $h_+ = \ell$ and hence $\mathcal{T}(a, t_*) = \mathcal{V}(a, t_*) < 0$. Therefore $\mathcal{T}(a, t) \le \mathcal{T}(a, t_*) < 0$ on the actual domain. The two negative endpoint values of the affine tangent prove $\mathcal{E}_v < 0$. □

Lemma F.15 (The sum norm residual)**.** *On the symmetric domain* (F.7)*, the residual* $\mathcal{E}_\Sigma = r - n_1/32 - (n_2 - n_1)L(z, \hat{s}_-^2/\hat{K}_-)$ *is negative.*

Proof. Put $a = (z-1)/(z+1)$ and $t = (1+z)^2/(z^2+\nu) - 1$. Then $0 \le a \le 1/7$ and $0 < t \le 1/4$. The sum residual has the form

$$\mathcal{E}_\Sigma = A_0 + A_1 H + z\left(\frac{DH^2}{2h} - f(h)\right), \qquad H = h - 1, \qquad D = a + t + at,$$

where A_0, A_1 are independent of h. The reciprocal lower bound for $f(h)$ gives, with $\tau = H/h$,

$$\frac{\mathcal{E}_\Sigma}{H} \le \Psi(\tau) := A_0(1/\tau - 1) + A_1 + z\left(\frac{D\tau}{2} - \frac{\tau^2}{3(2-\tau)}\right).$$

We first prove $A_0 \le 0$, and then bound a fixed affine tangent to Ψ.

The constant coefficient of the sum residual. The sign $A_0 \le 0$ in the sum residual has a direct derivative proof. Write $a = (z-1)/(z+1)$, $b = 1 - a$, $h_0 = 1 + t$ and $\beta = 1 - ah_0$. The coefficient A_0 is the value $\mathcal{E}_\Sigma(1)$ of the continuously extended row expression. Since $n_1(1) = 0$, it does not depend on the chosen norm threshold. Substitution gives

$$A_0 = \beta \log \beta + \frac{a\beta(ah_0 + 3t + 1)}{bt} \log h_0 + \frac{a(a + t + at)}{2b}((1+a)^2 h_0 - 4).$$

The quotient $\log h_0/t$ takes its continuous value 1 at $t = 0$. At fixed t, three derivatives in a give

$$\partial_a^3 A_0 = \frac{h_0^3}{\beta^2} - 3h_0^2 + \frac{12(1+2t)(t-\log h_0)}{b^4} < 0.$$

Indeed $0 \le t - \log(1+t) \le t^2/2$, and on $0 \le a \le 1/7$, $0 \le t \le 1/4$,

$$\frac{h_0}{\beta^2} \le \frac{5/4}{(23/28)^2} < 2, \qquad \frac{6(1+2t)t^2}{b^4 h_0^2} \le \frac{9}{25(6/7)^4} < 1.$$

To determine the initial slope and curvature, set $L = \log h_0$. For $t > 0$ the exact values are

$$\partial_a A_0(0,t) = \frac{t^3 - 5t^2 + (6t+2)L - 2t}{2t},$$
$$\partial_a^2 A_0(0,t) = \frac{5t^3 - t^2 - 2t + (2-6t^2)L}{t}.$$

The convexity of the reciprocal kernel gives $L \le t - t^2/(2h_0)$. Both coefficients of L above are positive, so

$$\partial_a A_0(0,t) \le \frac{t^2(t-1)}{2h_0} < 0, \qquad \partial_a^2 A_0(0,t) \le \frac{2t(t^2-t-1)}{h_0} < 0.$$

At $t = 0$ these derivatives have value zero by continuity. Thus $\partial_a^2 A_0$ decreases from a nonpositive initial value, and $\partial_a A_0$ does likewise. Since $A_0(0,t) = 0$, we obtain $A_0 \le 0$, with strict inequality when $a > 0$.

A fixed tangent for the sum residual. Write

$$a = \frac{z-1}{z+1}, \qquad b = 1-a, \qquad k = \frac{t}{1+t},$$
$$r = 1-k, \qquad c = a+k, \qquad \bar{c} = 1-c.$$

Thus $0 \le a \le 1/7$, $0 \le k \le 1/5$, $z = (1+a)/b$ and $\beta = \bar{c}/r$. The actual shape domain is contained in this rectangle. The tangent to Ψ at $\tau = 1/5$ is

$$W(a,k,\tau) = A_1 + (9-25\tau)A_0 + z\left[\left(\frac{c}{2r} - \frac{19}{243}\right)\tau + \frac{2}{243}\right].$$

Since $A_0 \le 0$, one has

$$\Psi''(\tau) = \frac{2A_0}{\tau^3} - \frac{8z}{3(2-\tau)^3} < 0.$$

Concavity therefore gives $\Psi(\tau) \le W(a,k,\tau)$. The following rational terms come from the left row:

$$A_R = 1 + 2z - b + \frac{b^2}{4\bar{c}}, \qquad B_1 = \frac{2r - (1+a) + (1+a)^2/(2r)}{b},$$
$$N_1 = \frac{2(b-2k)}{b^2 r}, \qquad N_d = \frac{2c(2-c)}{br}, \qquad F_R = 1 + \frac{(1+a)^2}{4r},$$
$$R_0 = \frac{ac((1+a)^2 + 4k - 4)}{2br^2},$$
$$R_1 = \frac{2\beta}{b} A_R - B_1 - \frac{49}{32} N_1 - N_d F_R.$$

Here $49/32 = 3/2 + 1/32$ includes the norm threshold. Pair the two logarithms using

$$\ell(k) = \int_0^1 \frac{\mathrm{d}v}{1 - kv}, \qquad J(a,k) = \int_0^1 \frac{v\,\mathrm{d}v}{(1-kv)(1-(a+k)v)}.$$

Their elementary reciprocal bounds are

$$\ell(k) \le \ell_+ := 1 + \frac{k}{2} + \frac{k^2}{3r}, \qquad J(a,k) \ge J_0 := \frac{1}{2} + \frac{a+2k}{3}.$$

The first follows by bounding the geometric remainder under the integral; the second follows by retaining its constant and linear terms. For $m = 9 - 25\tau$, the exact logarithmic part of W is

$$\frac{c}{br}\left\{[bc + 2a(1+m\bar{c})]\ell(k) - a\bar{c}(mb+2)J(a,k)\right\}.$$

On $0 \le \tau \le 2/5$, one has $m \ge -1$, $1 + m\bar{c} \ge c \ge 0$ and $mb + 2 \ge 1 + a > 0$. The signs therefore give the rational upper bound

$$U(a,k,\tau) = R_1 + mR_0 + z\left[\left(\frac{c}{2r} - \frac{19}{243}\right)\tau + \frac{2}{243}\right]$$
$$+ \frac{c}{br}\left\{[bc + 2a(1+m\bar{c})]\ell_+ - a\bar{c}(mb+2)J_0\right\} \ \ge W(a,k,\tau).$$

All denominators in this expression are positive on the enlarged rectangle.

The interval for the factor. In these coordinates the actual shape domain satisfies

$$ab < k \le \frac{1-a^2}{5}, \qquad k \le \tau \le \tau_c := \frac{2 - 2a^2 - (7+3a)k}{5 - 3a - 2a^2 - 10k}.$$

The denominator $D_c = 5 - 3a - 2a^2 - 10k$ is $3b + 2(1 - a^2 - 5k) > 0$. The inequality $\tau_c \le 2/5$ is equivalent to $2ab \le 5(1+a)k$, which follows from $k > ab$. There is also a useful linear ceiling

$$\tau_c \le T(k) := \frac{8}{15} - \frac{5}{3}k.$$

Indeed, writing $N_c = 2 - 2a^2 - (7+3a)k$ and $d = 1 - 5k \geq 0$ gives

$$\begin{aligned} T(k)D_c - N_c &= \frac{2}{15}\left(5d^2 - a(12+5a)d + 12a^2\right) \\ &\geq \frac{2}{15}\left(5\left(d - \frac{13}{10}a\right)^2 + \frac{71}{20}a^2\right) \geq 0. \end{aligned}$$

Thus it suffices to prove negativity of the affine function U at $\tau = k$ and at $\tau = \min(2/5, T(k))$. The two upper ceilings meet at $k = 2/25$.

The lower sum endpoint. Set $X(a,k) = U(a,k,k)$. We prove $X < 0$ on the whole rectangle. First define

$$\begin{aligned} \mathcal{F}(a) &= \frac{1+a}{6b}\left[(1+30a)(1+a)^2 + 15ab^2\right], \\ Q(a) &= \frac{43}{48} + a\left[\frac{39}{2} - \frac{177}{16b} - \frac{15(1+b)}{8b^2}\right] + a^2\left[\frac{37}{3} + \frac{a}{3} - \frac{16a^2}{3}\right]. \end{aligned}$$

Differentiating the displayed row expression gives

$$X_{kkk} = \frac{50a(1-2a)}{b} + \frac{6Q(a)}{r^4} + \frac{6(1-5k)\mathcal{F}(a)}{r^5} > 0.$$

Indeed $\mathcal{F} > 0$, and $b \geq 6/7$ gives

$$\frac{177}{16b} \leq \frac{413}{32} < 13, \qquad \frac{15(1+b)}{8b^2} \leq \frac{455}{96} < \frac{19}{4}.$$

The first bracket in Q is consequently greater than $7/4$; the second is greater than 12, so $Q \geq 43/48 > 0$. The remaining factors are nonnegative since $a \leq 1/7$ and $k \leq 1/5$.

To determine the initial curvature, let $I(a) = X_{kk}(a,0)$. A second differentiation in a gives

$$I''(a) = \frac{4}{3}\left[41 - 12a(11+8a)\right] + \frac{8(35b-9)}{3b^4} + \frac{6-b}{4b^4} > 0.$$

Each summand is positive: $a < 1/6$ gives $12a(11+8a) < 74/3 < 41$, and $6/7 \leq b \leq 1$ handles the others. Since $I(0) = 17/8 > 2$ and $I'(0) = -199/24 > -9$, its tangent gives

$$X_{kk}(a,0) = I(a) \geq I(0) + aI'(0) > 2 - 9a \geq \frac{5}{7} > 0.$$

Together with $X_{kkk} > 0$, this proves that X is strictly convex in k.

It remains to check its two k edges. At $k = 0$ the row simplifies to

$$bX(a,0) = -\frac{1}{16} + \frac{2(1+a)}{243} + \frac{a^2}{2} - \frac{a^3}{6}\left[(2-a)(11-18a) + 4a\right].$$

The final bracket is positive. Moreover $2(1+a)/243 < 1/72$ and $a^2/2 < 1/72$, whence $bX(a,0) < -1/16 + 1/36 < 0$.

For the upper edge put $Y(a) = X(a,1/5)$ and

$$H(b) = 20b^2 - \frac{393}{16}b - \frac{2}{3} + \frac{119}{16b^3} - \frac{45}{16b^4}.$$

The exact identities

$$H(b) - Y''(a) = \frac{101}{4320b^3} > 0, \qquad H''(b) = 40 + \frac{357b - 225}{4b^6} > 0$$

reduce its curvature to two scalar values: $H(6/7) < -2/5$ and $H(1) < -2/5$. Convexity of H therefore gives $Y'' < -2/5$ throughout the interval. At the midpoint $m = 1/14$, direct substitution in the displayed row gives

$$Y(m) + \frac{5}{4}Y'(m)^2 < -\frac{1}{900}.$$

The quadratic tangent bound now yields

$$Y(a) \le Y(m) + Y'(m)(a - m) - \frac{(a - m)^2}{5} \le Y(m) + \frac{5}{4}Y'(m)^2 < -\frac{1}{900}.$$

Both k edges are negative. Convexity in k gives $X(a, k) < 0$ everywhere on the rectangle, proving negativity of the lower sum endpoint.

The upper endpoint of the sum residual. Put $k_c = 2/25$ and define

$$\begin{aligned} H(a, k) &= U(a, k, 2/5) < 0 \qquad && (0 \le k \le k_c), \\ G(a, k) &= U(a, k, T(k)) < 0 \qquad && (k_c \le k \le 1/5). \end{aligned} \tag{F.8}$$

All the estimates below hold on the independent rectangle $0 \le a \le 1/7$, $0 \le k \le 1/5$ wherever the indicated functions are used; no enlargement of the physical cost interval is needed.

The useful derivative structure of U comes from its original row. Put

$$m = 9 - 25\tau, \qquad \ell_+ = 1 + \frac{k}{2} + \frac{k^2}{3r}, \qquad \mu = \frac{19}{243}, \qquad \eta_0 = \frac{2}{243}.$$

Its simple-pole weight in b is the following displayed row expression:

$$\begin{aligned} \mathcal{A}(k, \tau) = {} & \frac{2k - 1 - 1/16}{r} + \frac{2mk(1 + k)}{r^2} + \frac{\tau(1 + k)}{r} + 2(\eta_0 - \mu\tau) \\ & + \frac{2(1 + k)(1 - mk)}{r}\ell_+ + \frac{2k(1 + k)}{r}\left(\frac{5}{6} + \frac{2k}{3}\right). \end{aligned} \tag{F.9}$$

The double-pole weight is $-15k/(8r)$. For example, the rational row contributes $(2k - 1 - 1/16)/(rb) - 15k/(8rb^2)$; the remaining terms in (F.9) are, respectively, the mR_0 term, the transport term, and the two paired logarithmic moments. Thus these are the poles of the actual residual majorant.

The nonconstant polynomial quotient in a also has a short form:

$$U = q_0 + q_1 a + q_2 a^2 + q_3 a^3 + q_4 a^4 + \frac{\mathcal{A}(k, \tau)}{b} - \frac{15k}{8rb^2}, \tag{F.10}$$

where $q_0 = U(0,k,\tau) - \mathcal{A}(k,\tau) + 15k/(8r)$ is independent of a and

$$\begin{aligned} q_4 &= \frac{m}{3r}, \qquad q_i = -\frac{T_i}{6r^2} \quad (i=1,2,3),\\ T_3 &= m(8k^2-7k+2)+7-4k,\\ T_2 &= 2mk(5k^2-4k+3)+3+19k-15k^2,\\ T_1 &= mk^2(4k^2-k+10)+6+10k+11k^2-18k^3+3r\tau. \end{aligned}$$

To obtain these identities, divide the row only in b and use $r\ell_+ = 1-k/2-k^2/6$; the logarithmic approximations are not expanded into a shape-box polynomial.

Concavity of the high-k function. Write $\mathcal{A}_G(k) = \mathcal{A}(k,T(k))$. On $4/5 \le r \le 23/25$ define the intrinsic pole and central-derivative functions

$$\begin{aligned} P(r) &= r^4\mathcal{A}_G''(1-r),\\ D(r) &= r^4 G_{akk}(0,1-r),\\ C(r) &= 2P(r) - \frac{45}{2}r - r^4 G_{aakk}(0,1-r). \end{aligned}$$

Differentiating the displayed row and quotient gives

$$\begin{aligned} P'' &= \frac{200}{3}r^2(44-25r) > 0,\\ D'' &= r^2\left(\frac{5000}{3}r - 164\right) > 0,\\ C'' &= \frac{10000}{3}r^2 > 2000. \end{aligned}$$

The endpoint substitutions in the same row give

$$\max\{P(4/5),P(23/25)\} < 75, \qquad \max\{D(4/5),D(23/25)\} < -7.$$

Convexity extends both bounds across the interval. At $v = 6/7$, the row gives

$$C(v) - \frac{C'(v)^2}{4000} > 149.$$

The quadratic lower bound from $C'' > 2000$ therefore proves $C > 149$.

The resulting exact mixed-derivative identity is

$$\begin{aligned} r^4 G_{akk} = D + \frac{896}{9}ra^3 + (112-457k)a^2 - Ca\\ + P(b^{-2}-1) - \frac{15}{2}r(b^{-3}-1). \end{aligned}$$

On $0 \le a \le 1/7$,

$$b^{-2} - 1 \le \frac{8}{3}a, \qquad b^{-3} - 1 \ge 3a.$$

For the first inequality, multiplication by b^2 leaves the positive factor $a(2-13a+8a^2)/3$; the second follows by multiplying by b^3. Using $a^3 \le a/49$, $a^2 \le a/7$, $112-457k<76$ and $15r/2 \ge 6$, we obtain

$$r^4 G_{akk} < -7 + \left(\frac{100}{49} + \frac{76}{7} - 149 + 200 - 18\right) a \le -7 + 46a \le -\frac{3}{7} < 0.$$

At the central shape, direct cancellation in the row gives

$$G_{kk}(0,k) = -\frac{40k^4 + 80k^2(3-2k) + (1-k)}{120(1-k)^4} < 0.$$

Hence $G_{kk}(a,k) \le G_{kk}(0,k) < 0$: G is concave in k.

The tangent at $k = 1/6$. Let $k_f = 1/6$, $F(a) = G(a, k_f)$, and

$$B_\ell(a) = F(a) - \frac{13}{150} G_k(a, k_f).$$

The factor $13/150 = k_f - k_c$. Along $\tau = T(k)$, the values at k_f are

$$r_f = \frac{5}{6}, \qquad \tau_f = \frac{23}{90}, \qquad m_f = \frac{47}{18}, \qquad m' = \frac{125}{3}, \qquad \tau' = -\frac{5}{3}.$$

The explicit weights in (F.9)–(F.10) give the small scalar bounds

$$\mathcal{A}_G(k_f) > \frac{37}{12},$$

$$q_4(k_f) > 0, \qquad 3q_3(k_f) > -7, \qquad 2q_2(k_f) > -4, \qquad q_1(k_f) > -\frac{9}{4},$$

$$\mathcal{A}'_G(k_f) < \frac{33}{2},$$

$$4q'_4(k_f) < 72, \qquad 3q'_3(k_f) < -36, \qquad 2q'_2(k_f) < -37, \qquad q'_1(k_f) < -\frac{23}{2}.$$

The q_i comparisons follow from the displayed products: for $i = 1, 2, 3$,

$$q'_i = -\frac{rT'_i + 2T_i}{6r^3}, \qquad q'_4 = \frac{m'r + m}{3r^2},$$

evaluated at the displayed values above. All q_i and their primes in these comparisons are restricted to $\tau = T(k)$.

The double pole of F is $-3/(8b^2)$. It follows that

$$\begin{aligned} b^3 F' &> -7a^2b^3 - 4ab^3 - \frac{9}{4}b^3 + \frac{37}{12}b - \frac{3}{4} \\ &= \frac{1}{12} - \frac{a}{3} - \frac{7}{4}a^2 + a^3\left(\frac{45}{4} - 17a + 7a^2\right) > 0. \end{aligned}$$

The first three terms decrease to zero at $a = 1/7$, and the cubic factor is positive. Similarly, the k derivative of the double-pole weight at k_f gives the negative pole $-27/(5b^3)$ in G_{ak}. The same coefficient bounds yield

$$\begin{aligned} G_{ak}(a, k_f) &< 72a^3 - 36a^2 - 37a - \frac{23}{2} + \frac{33}{2b^2} - \frac{27}{5b^3} \\ &\le -\frac{2}{5} - \frac{46}{5}a + a^2(72a - 36) < 0. \end{aligned}$$

Consequently $B'_\ell = F' - (13/150)G_{ak} > 0$ as well. At their common upper shape endpoint, substitution in the original row gives the two fixed scalar comparisons

$$G(1/7, 1/6) < -\frac{1}{900}, \qquad G(1/7, 1/6) - \frac{13}{150} G_k(1/7, 1/6) < 0. \tag{F.11}$$

Monotonicity therefore proves $F < -1/900$ and $B_\ell < 0$ throughout the shape interval. The k_f tangent to the concave function G is negative at both ends of $[k_c, k_f]$, proving $G < 0$ there.

A coarse curvature budget on the short right interval. Put $a_* = 1/7$. The original row has only simple and double poles at $r = 0$, and its polynomial quotient in k has degree three, with leading coefficient $125a(2a-1)/(9b)$. Define its actual Laurent weights

$$f = \Big[r^2 G(a_*, 1-r)\Big]_{r=0}, \qquad e = \left[\frac{d}{dr}\{r^2 G(a_*, 1-r)\}\right]_{r=0},$$

where the bracketed rational functions have removable singularities. Set $\alpha_L = -2e$, $\beta_L = 6f$ and $\kappa_L = G_{kk}(a_*, 0) + \alpha_L - \beta_L$. Elementary partial fractions give

$$Z(k) := G_{kk}(a_*, k) = -\frac{625}{63}k + \kappa_L - \frac{\alpha_L}{r^3} + \frac{\beta_L}{r^4}.$$

The following bounds suffice:

$$\kappa_L > \frac{21}{2}, \qquad \alpha_L < 35, \qquad \beta_L > \frac{45}{2}. \tag{F.12}$$

These comparisons follow by direct evaluation of the row. Set $p = 1 + a$, $M = 112/3$, $N = 125/3$ and

$$A_p = 1 + 2a - (1 + 2M)a^2, \qquad B_p = -b + 2a(M + Na).$$

Its double-pole weight is

$$f(a) = \frac{p^3}{2}\left(\frac{Ma}{b} - 1\right) + \frac{pA_p}{3b}, \qquad f(1/7) > \frac{15}{4}.$$

For the simple pole, the rational row and transport terms give

$$e_{\mathrm{rat}}(a) = \frac{p}{b^2}\left(\frac{119}{240} - 3a + \frac{17}{30}a^2 + a^3\right) > 0,$$

since $119/240 > 3/7$. The remaining original-row residues are

$$e_m(a) = -\frac{ap\{Np^2 + M(a+5)\}}{2b},$$
$$e_\ell(a) = \frac{(5p-2)A_p + 2pB_p}{6b},$$
$$e_J(a) = \frac{a^2 p(Mb+2)(7+2a)}{6b}.$$

At a_* these satisfy $e_m > -47/2$, $e_\ell > 44/9$ and $e_J > 10/9$. Their sum with $e_{\rm rat} > 0$ gives $e > -47/2 + 6 = -35/2$. Finally the constant polynomial term in Z gives

$$\kappa_L = \frac{2\{1250/343 - 427/49 + 90 - 3\}}{108/7} > \frac{2 \cdot 569}{108} > \frac{21}{2},$$

using $1250/343 > 3$ and $427/49 = 61/7$. This proves (F.12) from the row itself.

For $1/6 \le k \le 1/5$ we have $4/5 \le r \le 5/6$. The function $(45/2 - 35r)/r^4$ decreases there, since its derivative is $(105r - 90)/r^5 < 0$. Hence

$$Z(k) > -2 + \frac{21}{2} - \frac{20}{3}\left(\frac{6}{5}\right)^4 > -6.$$

Since $G_{akk} < 0$, also $G_{kk}(a,k) \ge Z(k) > -6$. We already know $G(a,1/6) < -1/900$, while the lower-endpoint proof gives $G(a,1/5) = Y(a) < -1/900$. Convexity of $G + 3k^2$ now gives

$$G(a,k) < -\frac{1}{900} + 3(k - 1/6)(1/5 - k) \le -\frac{1}{900} + \frac{1}{1200} = -\frac{1}{3600} < 0.$$

This proves the whole high-k assertion in (F.8).

The low-k function. In $H = U(a,k,2/5)$ the fixed-tangent multiplier is $m = -1$. Differentiating its original row in k gives

$$H_k = 2A_2 k + A_1 + \frac{E}{r^2} - \frac{2F_p}{r^3}, \qquad H_{kk} = 2A_2 + \frac{2E}{r^3} - \frac{6F_p}{r^4},$$

where

$$A_2 = -\frac{4a^2 + 5a + 1}{6b}, \quad A_1 = -\frac{10a^3 + 15a^2 + 10a + 5}{6b}, \quad F_p = \frac{(1+a)^3}{6b},$$

and the simple-pole weight is exposed by the positive difference

$$E - \frac{15}{2}F_p = \frac{1+a}{240b^2}\{43 - 220a + 12a^2 + 20a^3(15 - 14a - 4a^2)\} > 0.$$

Indeed $43 - 220a \ge 81/7 > 0$, and the last bracket is positive. For $k \le k_c$, $r \ge 23/25$. Therefore

$$H_{kk} > 2A_2 + \frac{(15r - 6)F_p}{r^4} \ge -\frac{44}{63} + \frac{39}{5}\frac{1}{6} = \frac{379}{630} > 0.$$

At $k = 0$,

$$H_k > A_1 + \frac{11}{2}F_p = \frac{1 + 13a + 3a^2 - 9a^3}{12b} > 0.$$

Thus H increases in k, and $T(k_c) = 2/5$ gives $H(a,k) \le H(a,k_c) = G(a,k_c) < 0$. Both assertions in (F.8) follow. Together with the negative lower τ endpoint and affine dependence on τ, this proves $U < 0$ on the actual forward window and hence $\mathcal{E}_\Sigma < 0$ for the norm-sum threshold $1/32$. □

The neighbour residual. The residual in Step 4 is

$$\mathcal{E}_N = \eta A - B - n_1 F_{z/\hat{s}_-}(h) - n_2 F_{w/\hat{s}_+}(\hat{s}_+^2/\hat{K}_+).$$

For $w \geq 4/3$, lowering w to $\max(z, 4/3)$ leaves two faces. On the symmetric face $w = z \in [4/3, 5/3]$, the factor budget gives $\nu \geq 1 + 6z/5$. We first prove that the value at the least factor is negative.

Lemma F.16 (The symmetric neighbour value)**.** *Let $4/3 \leq z \leq 5/3$, $1 + 6z/5 \leq \nu < 3$ and $w = z$. The value of $\mathcal{E}_N$ at $h_0 = (1+z)^2/(z^2+\nu)$ is strictly negative.*

Proof. Put $a = (z-1)/(z+1)$, $b = 1-a$, $p = 1+a$ and $h_0 = 1 + t =: \ell$. The physical domain is contained in

$$\frac{1}{7} \leq a \leq \frac{1}{4}, \qquad \frac{1}{8} \leq t \leq \frac{1}{4}.$$

Indeed, $\nu < 3$ gives $t > 2(z-1)/(z^2+3) \geq 6/43 > 1/8$; the rational function increases on $4/3 \leq z \leq 5/3$. For the upper bound, $\nu \geq 1 + 6z/5$ and $5(z^2 + 1 + 6z/5) - 4(z+1)^2 = (z-1)^2 \geq 0$ give $h_0 \leq 5/4$. We prove the sign on this whole rectangle.

Writing $\beta = 1 - a\ell$, the original row at h_0 has the exact form

$$V(a,t) := \mathcal{E}_N(h_0) = R + C_L \log \ell + C_M \log \beta,$$

where

$$\begin{aligned} R &= \frac{p^3(1+2a-a^2)}{2b^2}\ell^3 - \frac{ap(13+4a-a^2)}{2b^2}\ell^2 \\ &\quad - \frac{a^4 - 4a^3 - 14a^2 - 8a + 9}{2b^2}\ell + \frac{4p}{b}, \\ C_L &= -\frac{A}{b^2}\ell^2 + \frac{B}{b^2}\ell + \frac{(2-a)p^2}{b} + \frac{ap}{t}, \\ C_M &= \frac{(b+2t)\beta}{b}, \qquad A = 1 + 2a + 6a^2 - a^4, \quad B = a^4 - a^2 + 6a + 2. \end{aligned}$$

Both logarithmic weights are positive. For C_L, use

$$B - \frac{5}{4}A = \frac{9}{4}a^4 - \frac{17}{2}a^2 + \frac{7}{2}a + \frac{3}{4} > \frac{7}{32} > 0;$$

the other terms are positive, and $\ell \leq 5/4$.

Use the two integral upper bounds

$$S(t) = \frac{t}{6}\left(1 + \frac{8}{2+t} + \frac{1}{1+t}\right) \geq \log(1+t),$$

$$P(y) = -y - \frac{y^2}{2} - \frac{y^3}{3} \geq \log(1-y).$$

For the first, $(S - \log(1+t))' = t^4/[6(1+t)^2(2+t)^2] \geq 0$; the second integrates the positive geometric-series remainder. Thus the rational majorant

$$\mathcal{B}(a,t) = R + C_L S(t) + C_M P(a\ell)$$

satisfies $V \le \mathcal{B}$. Its apparent singularity at $t = 0$ is removable, and we use that extension below.

Decrease in the shape. Introduce the positive moment weights

$$\begin{aligned} Q &= \ell^3(2t^2 + 2t + 1), \\ N_1 &= t^2(4\ell^4 + 10\ell^3 + 19\ell^2 + 12\ell + 1), \\ N_2 &= t\ell(4\ell^4 + 6\ell^3 + 10\ell^2 + \ell - 3), \\ R_2 &= \frac{4t^3(5t+6)}{3(t+2)}, \\ R_1 &= \frac{t^2\{16 + 24t + 11t^2 + t^3(18 + 15t + 2t^2)\}}{3\ell(t+2)}. \end{aligned}$$

Differentiating the displayed row gives

$$\mathcal{B}_a = \frac{4}{3}\ell^4 a^3 - Qa^2 - \frac{N_2 a}{3(t+2)} - \frac{N_1}{6(t+2)} + \frac{2R_2}{b^3} + \frac{R_1}{b^2}.$$

Since $a \le 1/4$, combine its first two terms. Also $1/[a(1-a)^n]$ decreases for $a \le 1/(n+1)$. Its maximum for $n = 2, 3$ is therefore at $a = 1/7$. In particular $2/(ab^3) < 23$ and $1/(ab^2) < 10$, so division by a gives

$$\frac{\mathcal{B}_a}{a} \le -\frac{Q - \ell^4/3}{7} - \frac{N_2}{3(t+2)} - \frac{2N_1}{3(t+2)} + 23R_2 + 10R_1.$$

The binomial lower bounds for $\ell^j = (1+t)^j$ imply

$$\begin{aligned} Q - \ell^4/3 &\ge \frac{2}{3} + \frac{11}{3}t + 9t^2, \\ N_1 &\ge 46t^2 + 96t^3 + 73t^4, \\ N_2 &\ge 18t + 73t^2 + 107t^3 + 74t^4. \end{aligned}$$

Furthermore $R_2 \le 13t^3/3$, since $7t \le 2$. For $R_1 \le 11t^2/4$, multiply by $3\ell(t+2)/t^2$. The resulting difference is at least

$$\frac{1}{2} + \frac{3}{4}t - \frac{33}{4}t^2 > 0,$$

using $18 + 15t + 2t^2 \le 22$. This concave quadratic has positive endpoint values $1/2$ and $11/64$. Now $1/[3(t+2)] \ge 4/27$, and the scalar comparisons

$$\begin{aligned} 10\frac{11}{4} - \frac{4}{27}(73 + 92) - \frac{9}{7} &< 2, \\ 23\frac{13}{3} - \frac{4}{27}(107 + 192) < 56, \qquad -\frac{4}{27}(74 + 146) &< -32 \end{aligned}$$

give

$$\frac{\mathcal{B}_a}{a} \le -\frac{2}{21} - \frac{67}{21}t + 2t^2 + 56t^3 - 32t^4 \le -\frac{2}{21} - \frac{67}{21}t + 14t^2 < 0.$$

The last comparison uses $14-(2+56t-32t^2)=4(1-4t)(3-2t)\geq 0$. The last quadratic is convex on $[0,1/4]$, with negative endpoint values $-2/21$ and $-1/56$. Hence $\mathcal{B}$ decreases in a.

Convexity on the remaining edge. Let $u(t)=\mathcal{B}(1/7,t)$. The polynomial part of $\mathcal{B}$ as a rational function of t has degree five. The coefficients of t^5,t^4,t^3 are respectively

$$p_5=\frac{2a^4}{3b}>0,\qquad p_4=\frac{a^3(-a^2+9a+1)}{3b}>0.$$

The cubic coefficient is determined by the positive-product identity

$$6b^2\left(p_3-\frac{5}{6}\right)=a^4(22-42a+8a^2)+13a^3+(19a^2+23a-3)>0;$$

the last bracket is at least $33/49$. Thus $p_3>5/6$. The two poles arise from Simpson's reciprocal kernel. Their weights at $t=-2$ and $t=-1$ are

$$C_2=\frac{4p(-a^3+2a^2+13a+2)}{3b^2},\qquad -\frac{p}{3b}.$$

At $a=1/7$, $6C_2<49$, because $2/49+13/7+2<63/16$ and $8p/b^2=112/9$. All polynomial terms of degree at most two disappear after three derivatives. We consequently obtain

$$u'''>5-\frac{49}{(t+2)^4}+\frac{8}{3(1+t)^4}>\frac{31}{16}>0.$$

The initial curvature, computed in the same displayed row, is

$$u''(0)=-\frac{4}{9}+\frac{5}{3\cdot 7^4}>-\frac{4}{9}.$$

Integrating on $[0,1/8]$ gives

$$u''(1/8)>-\frac{4}{9}+\frac{1}{8}\left(\frac{31}{16}+\frac{8}{3}\left(\frac{8}{9}\right)^4\right)>0.$$

Thus u is convex on $[1/8,1/4]$. The two fixed scalar substitutions in the displayed rational definition of $\mathcal{B}$ give

$$\mathcal{B}(1/7,1/8)<0,\qquad \mathcal{B}(1/7,1/4)<0.$$

It follows that $V(a,t)\leq\mathcal{B}(a,t)\leq u(t)<0$ on the full rectangle, as required. □

Lemma F.17 (The symmetric neighbour residual)**.** *Suppose $w=z$, $4/3\leq z\leq 5/3$ and $1+6z/5\leq\nu<3$. Then $\mathcal{E}_N(h)<0$ for every $h\geq h_0=(1+z)^2/(z^2+\nu)$.*

Proof. Use a,b,p,t,ℓ,β from Lemma F.16. The actual domain has $1/7\leq a\leq 1/4$ and $t>1/8$. The symmetric exterior budget also gives

$$t\leq\frac{1-a^2}{4+a^2}\leq\frac{48}{197},$$

since this rational function decreases in a. We first prove that the partial derivative of $\mathcal{E}_N$ in h, holding z, ν fixed, is negative at h_0. Differentiate the original row before substituting $h = h_0$ and changing to (a, t). The resulting partial slope is

$$D(a,t) = R_D + C_D \log \ell + \frac{2\beta}{b} \log \beta,$$

where

$$\begin{aligned} R_D &= \frac{p(-a^4+5a^2+6a+2)}{2b^2}\ell^2 - \frac{-a^4+9a^2+10a+2}{b^2}\ell \\ &\quad - \frac{a^3-5a^2-13a+1}{2b^2} + \frac{p}{2\ell}, \\ C_D &= -\frac{A}{b^2}t + \frac{P_D}{b^2} + \frac{K_D}{t}, \\ P_D &= 2a^4-2a^3-9a^2+4a+1, \qquad K_D = \frac{p(1+2a-a^2)}{b}. \end{aligned}$$

The A here is the positive weight from the value proof. Both logarithmic weights are positive: $K_D > 0$, and

$$P_D - \frac{A}{4} = \frac{9}{4}a^4 - 2a^3 - \frac{21}{2}a^2 + \frac{7}{2}a + \frac{3}{4} > \frac{1}{16}.$$

Using the same Simpson and geometric upper bounds S, P as before, define

$$\mathcal{D}(a,t) = R_D + C_D S(t) + \frac{2\beta}{b} P(a\ell).$$

Use the continuous extension at $t = 0$ for the derivative comparisons below. Then $D \le \mathcal{D}$. We shall prove that this rational majorant is negative on the physical domain.

A derivative maximum principle in the shape. The derivative in a has the exact reciprocal-kernel form

$$\mathcal{D}_a = -3P_3a^2 - 2P_2a - P_1 + \frac{2R_2}{b^3} + \frac{R_1}{b^2},$$

with the following positive row weights:

$$\begin{aligned} P_3 &= \frac{\ell^2(4\ell^2+3)}{6}, \\ P_2 &= \frac{\ell(4\ell^4+6\ell^3+10\ell^2-7\ell-7)}{6(t+2)}, \\ P_1 &= \frac{t(4\ell^5+10\ell^4+16\ell^3-8\ell^2+\ell+3)}{6\ell(t+2)}, \\ R_2 &= \frac{2t^2(16t^2+35t+18)}{3\ell(t+2)}, \\ R_1 &= \frac{t(10-25t-49t^2+15t^4+2t^5)}{3\ell(t+2)}. \end{aligned}$$

Positivity follows by expanding $\ell = 1 + t$ into positive binomial terms; for R_1 use $10 - 25t - 49t^2 \geq 11/16$. In particular

$$\mathcal{D}_{aaaa} = \frac{120R_2}{b^6} + \frac{24R_1}{b^5} > 0 \qquad (t > 0).$$

The same binomial bounds and $0 \leq t \leq 1/4$ give

$$\begin{aligned} P_3 \geq L_3 &:= \frac{7}{6} + \frac{11}{3}t + \frac{9}{2}t^2, \\ P_2 \geq L_2 &:= \frac{2}{27}(6 + 53t + 99t^2), \\ P_1 \geq L_1 &:= \frac{8}{135}(26t + 93t^2), \\ R_2 \leq U_2 &:= \frac{20}{3}t^2, \qquad R_1 \leq U_1 := \frac{5}{3}t - 6t^2. \end{aligned}$$

For the last two comparisons, multiplication by the positive denominators leaves, respectively, $t^2(4 - 10t - 12t^2)$ and $t^2(4 - 18t^2 - 15t^3 - 2t^4)$, both nonnegative.

At the two shape endpoints we use the quadratic bounds

$$\begin{aligned} Q_a(t) &= -3a^2L_3 - 2aL_2 - L_1 + \frac{2U_2}{(1-a)^3} + \frac{U_1}{(1-a)^2}, \\ Q_*(t) &= -\frac{6}{7}L_3 - 2L_2 + 6\left(\frac{7}{6}\right)^4 U_2 + 2\left(\frac{7}{6}\right)^3 U_1. \end{aligned}$$

Thus $\mathcal{D}_a(a,t) \leq Q_a(t)$ at $a = 1/7, 1/4$, and $\mathcal{D}_{aa}(1/7,t) \leq Q_*(t)$. Each of $Q_{1/7}, Q_{1/4}, Q_*$ is convex. The displayed products give

$$Q_{1/7}(0), Q_{1/7}(1/4), Q_{1/4}(0), Q_{1/4}(1/4), Q_*(0), Q_*(1/4) < 0.$$

Convexity extends these signs over $0 \leq t \leq 1/4$. For fixed $t > 0$, set $f(a) = \mathcal{D}_a(a,t)$. Its derivative $f' = \mathcal{D}_{aa}$ is convex, and is negative at $a = 1/7$. Once such a convex function becomes positive it stays positive to the right, by its positive secant slope from $1/7$. Thus f can only decrease and then increase, so its maximum occurs at one of the two shape endpoints. Both endpoint values are negative. We have proved $\mathcal{D}_a < 0$ throughout the rectangle.

Convexity on the remaining edge. Set $d(t) = \mathcal{D}(1/7, t)$. Its two Simpson-pole weights are read directly from the displayed original row. At $a = 1/7$ put

$$A_e = \frac{8}{3}\left(\frac{2A + P_D}{b^2} - \frac{K_D}{2}\right) < 13, \qquad B_e = p - \frac{A + P_D}{3b^2} + \frac{K_D}{3} > 0.$$

Then

$$d''(t) = \Pi(t) - \frac{2A_e}{(t+2)^3} + \frac{B_e}{(1+t)^3},$$

where Π is a polynomial of degree two. Its row derivatives give

$$\Pi(0) > 4, \qquad \Pi'(0) = \frac{5}{7^3} > 0, \qquad \Pi'' = \frac{8}{3 \cdot 7^3} > 0.$$

Consequently $\Pi > 4$ for $t \geq 0$, and $d'' > 4 - 26/8 = 3/4 > 0$. The two scalar substitutions in the original rational definition give

$$d(1/8) < 0, \qquad d(48/197) < 0.$$

Convexity proves $d < 0$ between them. Shape monotonicity then gives $D(a,t) \leq \mathcal{D}(a,t) \leq d(t) < 0$ on the physical symmetric face.

Finally, the established concavity in h yields

$$\mathcal{E}_N(h) \leq \mathcal{E}_N(h_0) + (h - h_0)\,\partial_h \mathcal{E}_N(h_0) \leq \mathcal{E}_N(h_0) < 0$$

for every $h \geq h_0$, by Lemma F.16. □

Lemma F.18 (The asymmetric neighbour residual)**.** *Extend $\mathcal{E}_N$ continuously to $h = 1$. If $w = 4/3$ and*

$$\frac{12}{5} \leq \nu \leq 3, \qquad 1 \leq z \leq \frac{4}{3}, \qquad 1 \leq h \leq \frac{3}{2},$$

then $\mathcal{E}_N(h) < 0$.

Proof. Write $K = 1 + \nu$, $s = 1 + z$, $x = h - 1$, $T = hs - z = 1 + sx$ and

$$F = F_{4/7}\left(\frac{49}{16 + 9\nu}\right), \qquad U = \frac{\mathcal{E}_N}{\eta}, \qquad \eta = \frac{KT}{z^2 + \nu} > 0.$$

The opposite floor F decreases in ν. At $\nu = 3$, the reciprocal Simpson bound gives

$$F \geq \frac{59}{43} - \frac{14}{3}\log\frac{49}{43} > \frac{59}{43} - \frac{14}{3 \cdot 43}\left(1 + \frac{86}{23} + \frac{43}{49}\right) > \frac{16}{21}.$$

The last comparison reduces to $86 \cdot 98 < 367 \cdot 23$.

Convexity in the neighbour shape. We differentiate at fixed h, ν. Dividing the original row by η makes the coefficient of $\log q$ equal to -1, and that of F equal to $-7/3 + Kh/(sT)$. Its second derivative is

$$\begin{aligned} U_{zz} = {} & \frac{2}{K}\left(h - h^{-1} - \log h\right) + \frac{1}{s^2} + \frac{2hK(F-1)}{s^3} \\ & + \frac{2h}{KT^3}\left\{\nu(Kx^3 + x^2 - x + \log h) - K^2 F x^3\right\}. \end{aligned}$$

The coefficient of F here is $2hK(s^{-3} - x^3T^{-3}) > 0$, since $T = 1 + sx > sx$. We may therefore replace F by $16/21$. The braces then become

$$\begin{aligned} & \nu(x^2 - x + \log h) - \frac{K(16 - 5\nu)}{21}x^3 \\ & \qquad \geq x^2\left(\frac{\nu}{2} - \frac{K(16 - 5\nu)}{21}x\right) \geq x^2\left(\frac{6}{5} - \frac{8}{21}\right) \geq 0, \end{aligned}$$

using $\log(1 + x) \geq x - x^2/2$ and $0 \leq x \leq 1/2$. Using $\log h \leq x$ in the remaining terms gives

$$\begin{aligned} U_{zz} & \geq \frac{s - 10K/21}{s^3} + x\left(\frac{2}{hK} - \frac{10K}{21s^3}\right) \\ & \geq \frac{s - 10K/21}{s^3} + \frac{2}{21}x > 0. \end{aligned}$$

Indeed, $s \geq 2$, $K \leq 4$ and $h \leq 3/2$ give $s - 10K/21 \geq 2/21$, $2/(hK) \geq 1/3$ and $10K/(21s^3) \leq 5/21$. It remains to prove $U < 0$ at $z = 1$ and $z = 4/3$.

The unit-shape boundary. At $z = 1$ put $q = 14/(3K) \in [7/6, 70/51]$. The continuous original row at $h = 1$ is

$$\mathcal{E}_N(1) = (q-1)\left\{\frac{7(1-F)}{3q} + \frac{3}{7}\right\} - \log q < (q-1)\left\{\frac{5}{9q} + \frac{3}{7} - \frac{2}{q+1}\right\} < 0.$$

For the last sign, multiplication of the braces by $63q(q+1)$ gives the convex quadratic $27q^2 - 64q + 35$, negative at both endpoints of the larger interval $[7/6, 7/5]$. Differentiation of the original row, keeping ν, z fixed, gives

$$\partial_h \mathcal{E}_N = 2A - \frac{K}{2} - \frac{2}{Kh^2} + \frac{1}{h} - \left(1 - \frac{2}{K}\right)(1 + 2h - 2\log h) - \left(\frac{14}{3} - \frac{K}{2}\right)F.$$

Here $A = 101/42 + 1/K - \log q$. Since $h - \log h \geq 1$ and $h^{-1} - 2/(Kh^2) \leq K/8$, completing a square and using $F > 16/21 = 3/4 + 1/84$ bounds this derivative above by

$$R(q) = \frac{12q}{7} - \frac{4(q-1)}{q+1} - \frac{71}{42} - \frac{2q-1}{36q}.$$

Here $R'' = 16/(q+1)^3 + 1/(18q^3) > 0$, while $R(7/6) < 0$ and $R(70/51) < 0$ by direct substitution. Thus $\mathcal{E}_N$ decreases for $h \geq 1$, proving this boundary.

The symmetric boundary and the scale parameter. Now fix $z = w = 4/3$. In this paragraph $s = 7/3$, $T = (7h-4)/3$, and $K \in [17/5, 4]$ varies at fixed h. Collecting powers of K in the original normalized row gives

$$U = A_h K + B_h + \frac{C_h}{K} + \log K + \left(dK - \frac{7}{3}\right)F(K), \qquad d = \frac{9h}{7(7h-4)},$$

where A_h, B_h are independent of K and

$$C_h = \frac{7\{16h^3 + 41h^2 - 64h + 16 - 4h(7h-1)\log h\}}{9h(7h-4)}.$$

The bound $\log h \leq (h - h^{-1})/2$ shows

$$C_h - \frac{7}{3} \geq \frac{14(h-1)(h^2 + 12h - 7)}{9h(7h-4)} \geq 0.$$

For the derivatives of the opposite floor write $t = (42 - 9K)/49 \in [6/49, 57/245] \subset [0, 1/4]$ and $a = 4/7$. Its reciprocal integral is

$$F(K) = 1 + \frac{a^2}{1-t} - a\int_0^1 \frac{\mathrm{d}r}{1 - tr}.$$

All its positive-order derivatives in t are positive: for $j \geq 1$,

$$\frac{1}{j!}\frac{\mathrm{d}^j F}{\mathrm{d}t^j} = \frac{a^2}{(1-t)^{j+1}} - a\int_0^1 \frac{r^j\,\mathrm{d}r}{(1-tr)^{j+1}} > 0,$$

because the integral is at most $1/((j+1)(1-t)^{j+1})$ and $a > 1/(j+1)$. Thus the first two derivatives are bounded above by their values at $t = 1/4$. Apply weighted Jensen to their negative integral terms; the probability densities $2r$ and $3r^2$ have means $2/3$ and $3/4$. As $t_K = -9/49$, this gives

$$0 < -F_K \le \frac{9}{49}\left\{\frac{16}{49(3/4)^2} - \frac{2}{7(5/6)^2}\right\} < \frac{1}{30},$$
$$0 < F_{KK} \le \left(\frac{9}{49}\right)^2\left\{\frac{32}{49(3/4)^3} - \frac{8}{21(13/16)^3}\right\} < \frac{1}{32}.$$

Moreover, $27/91 \le d \le 3/7$, so $1 < dK < 7/3$ and $0 < 7/3 - dK < 4/3$. Consequently

$$(U - \log K)_{KK} = \frac{2C_h}{K^3} - \left(\frac{7}{3} - dK\right)F_{KK} + 2dF_K > \frac{7}{96} - \frac{1}{24} - \frac{1}{35} > 0.$$

The two scale endpoints. We prove the bounds

$$U(17/5, h) < -\frac{1}{288}, \qquad U(4, h) < -\frac{1}{192} \qquad (1 \le h \le 3/2).$$

Put $\alpha = z/s = 4/7$, $H = s^2/(z^2+\nu)$ and define

$$S(r) = \frac{r-1}{6}\left(1 + \frac{8}{1+r} + \frac{1}{r}\right), \qquad \rho = \frac{q-1}{q+1}, \qquad L_q = 2\rho\left(1 + \frac{\rho^2}{3}\right),$$

and

$$F_\nu^- = \beta(z) + \left(\alpha^2 - \frac{\alpha}{2}\right)(H-1) + \frac{\alpha(H-1)^2}{3(H+1)}.$$

Here S is the reciprocal Simpson bound for the logarithm, and $F_\nu^- = 1+\alpha^2 H - \alpha H S(H)/(H-1)$ is the resulting lower bound for the opposite floor. Let $\mathcal{B}_\nu(h)$ be the rational function obtained from the original normalized row U by replacing $\log h$, $\log q$ and F by $S(h)$, L_q and F_ν^-, respectively. The coefficient of $\log h$ is

$$\frac{z(\nu - z)}{K} + \frac{z\nu s}{KT} > 0,$$

that of $\log q$ is -1, and that of F is $dK - 7/3 < 0$. Thus $U < \mathcal{B}_\nu$; the strictness follows from the cubic lower bound for $\log q$ because $q > 1$.

The derivative of this majorant has three reciprocal poles:

$$\mathcal{B}_\nu'(h) = -a_\nu - \frac{sB_\nu}{T^2} - \frac{c_\nu}{(h+1)^2} + \frac{d_\nu}{h^2},$$

where the row residues are

$$a_\nu = \frac{z(\nu - z)}{K}\left(\alpha - \frac{1}{6}\right),$$
$$B_\nu = \frac{z\{K^2 F_\nu^- - \nu^2 + \nu(z^2 + 3z + 1)\}}{Ks^2} + \frac{z\nu s}{K} S(\alpha),$$
$$c_\nu = \frac{8z^2}{3K}\left(1 - \frac{\nu}{1+2z}\right), \qquad d_\nu = \frac{5z^2 + 6z - \nu}{6K}.$$

The value $S(\alpha)$ here is evaluated by its rational formula. For an anchor v the exact reciprocal identity gives

$$\begin{aligned}&\mathcal{B}_\nu(h)-\mathcal{B}_\nu(v)-(h-v)\mathcal{B}'_\nu(v)\\&\qquad=(h-v)^2\left\{\frac{s^2B_\nu}{T(v)^2T(h)}+\frac{c_\nu}{(v+1)^2(h+1)}-\frac{d_\nu}{v^2h}\right\}.\end{aligned}$$

For $1\le h\le 3/2$ we have $h/T(h)\le 1$ and $h/(h+1)\le 3/5$.

At $\nu=12/5$ the residues satisfy $0<B_\nu<1/5$, $0<c_\nu<1/2$ and $d_\nu>7/10$. With $v=7/5$, multiplying the braces by h bounds them above by

$$\frac{245}{841}+\frac{5}{96}-\frac{5}{14}<0.$$

At $\nu=3$ they satisfy $0<B_\nu<5/36$, $0<c_\nu<2/9$ and $d_\nu=125/216$. With $v=7/6$ the corresponding bound is

$$\frac{49}{125}+\frac{24}{845}-\frac{125}{294}<0.$$

To check the residue bounds, use $F^-_\nu>3/4$ and the upper bounds $F^-_{12/5}<4/5$, $F^-_3<49/64$ in the displayed formulas. The upper bounds follow from $\log H\ge 2(H-1)/(H+1)$:

$$F^-_\nu\le F\le J(H):=1+\alpha^2H-\frac{2\alpha H}{H+1}.$$

The function J increases for $H\ge 1$. At the first endpoint $H<4/3$ and $J(4/3)<4/5$; at the second $H=49/43<8/7$ and $J(8/7)<49/64$. Substitution proves the stated bounds.

The negative remainder kernels put $\mathcal{B}_\nu$ below its tangent at the respective anchors on the whole factor interval. The two fixed rational substitutions in the original majorant give

$$\begin{aligned}\mathcal{B}'_{12/5}(7/5)>0,&\qquad \mathcal{B}_{12/5}(7/5)+\frac{1}{10}\mathcal{B}'_{12/5}(7/5)<-\frac{1}{288},\\ \mathcal{B}'_3(7/6)<0,&\qquad \mathcal{B}_3(7/6)-\frac{1}{6}\mathcal{B}'_3(7/6)<-\frac{1}{192}.\end{aligned}$$

These are the maxima of the two affine tangent bounds, proving both endpoint comparisons.

The logarithmic chord. The endpoint comparisons just proved give, for $\lambda=(K-17/5)/(3/5)\in[0,1]$,

$$\begin{aligned}U(K,h)&\le-\frac{1-\lambda}{288}-\frac{\lambda}{192}+\log K-(1-\lambda)\log\frac{17}{5}-\lambda\log 4\\&\le-\frac{1-\lambda}{288}-\frac{\lambda}{192}+\frac{9}{578}\lambda(1-\lambda)\\&\le-\frac{9(\lambda-4/9)^2+2/9}{576}<0.\end{aligned}$$

For the second line use $-(\log K)'' \le 25/289$ and the quadratic chord bound. For the last line use $9/578 < 1/64$ and complete the square. This proves the second shape boundary, and hence the whole rectangle. □

For the actual asymmetric neighbour face the budget gives $\nu \ge 9/5 + 3z/5 \ge 12/5$. Its lower factor endpoint $h_0 = s^2/(z^2+\nu)$ is at least 1, since $1 + 2z \ge \nu$. Its upper factor endpoint is $h_c = s(5\nu-6)/(3(z^2+\nu)) \le 3/2$, since

$$9(z^2+\nu) - 2s(5\nu-6) = 9(z-1)^2 + (3-\nu)(10z+1) \ge 0.$$

Thus Lemma F.18 proves the remaining neighbour sign. Together with Lemma F.17, it completes the proof of Lemma F.10.

F.6.6. *The Casimir lower bound.* Assume the hypotheses of Lemma F.12. The integral

$$f(r) = \int_1^r \frac{(s-1)^2}{2s^2}\,\mathrm{d}s \le \frac{(r-1)^3}{6} \qquad (r \ge 1)$$

and the fact that the three arguments in φ lie in $[1, 5/4]$ give $0 \le \varphi \le 3(1/4)^3/6 < 3/4 \le e_0$. In particular the central pairing E is positive throughout the norm polygon. Put $s_1 = \hat{s}_-$, $s_2 = \hat{s}_+$ and $c = 1/32$. For fixed shape the Gram numerator is

$$Q(x,y) = 4\Big(\frac{3}{4}+x\Big)\Big(\frac{3}{4}+y\Big) - \big(e_0 - \varphi + s_1 x + s_2 y\big)^2.$$

Its Hessian has negative diagonal entries and determinant $16(s_1 s_2 - 1) > 0$. Thus Q is strictly concave. Its minimum on $x, y \ge 0$, $x + y \le c$, $y \le c/2$ occurs at one of

$$(0,0), \quad (0,c/2), \quad (c,0), \quad (c/2,c/2).$$

We keep the positive term φ when estimating the last two vertices.

The two vertices below the sum edge. At $(x,y) = (0,T)$, where $T = 0$ or $1/64$, dropping φ and using $s_2 \le 7/3$ give

$$\gamma^2 \ge \frac{(1+\nu)^2}{16\nu^2}\left(\frac{9}{4} + 3T - \Big(e_0 + \frac{7}{3}T\Big)^2\right).$$

These two lower curves increase with ν. To see this, set $t = 1/(1+\nu)$ and

$$E_0 = 1 + t - \frac{1}{8t}, \qquad H_0 = 2 + \frac{1-2t}{8t^2}, \qquad d = \frac{7}{3}T.$$

The derivative in t has the sign of $9/4 + 3T - (E_0 + d)(H_0 + d)$. On $1/4 \le t \le 5/16$, $E_0 + H_0 \ge 11/4$, and

$$E_0 H_0 - \frac{9}{4} = \frac{(2t+1)(4t-1)^3}{64t^3} \ge 0. \tag{F.13}$$

For $T > 0$ the cross term $d(E_0 + H_0)$ exceeds $3T$; for $T = 0$ the right side is positive except at the endpoint $t = 1/4$. Their minima are therefore their values at $\nu = 11/5$, both greater than $9193/50000$.

Equalizing the two shapes on the sum edge. At either remaining vertex $x + y = c$, $0 \le y \le c/2$. Drop only $f(4/\hat{K})$ from φ. For fixed $S = s_1 + s_2$, the resulting upper bound on E has derivative in s_1

$$c - 2y - \frac{1}{2}f'(s_1/2) + \frac{1}{2}f'(s_2/2) \ge 0,$$

since $s_1 \le s_2$ and f' increases. Hence its maximum occurs at $s_1 = s_2 = s = S/2$:

$$E \le e_0 - 2f(s/2) + cs.$$

This bound increases in s, because $c - f'(s/2) \ge 1/32 - 1/98 > 0$ on $1 \le s/2 \le 7/6$. The shape interval and the budget therefore allow

$$s_*(\nu) = \min\left\{\frac{7}{3}, \frac{5\nu + 1}{6}\right\}.$$

On the sum edge the norm product is

$$4\Big(\frac{3}{4} + c - y\Big)\Big(\frac{3}{4} + y\Big) = P + 4y(c - y) \ge P, \qquad P = \frac{75}{32}.$$

The reciprocal integral bound gives

$$2f(1 + x) \ge L(x) := \frac{2x^3}{3(1 + x)(2 + x)}.$$

It also follows by differentiation: the derivative of $2f(1 + x) - L(x)$ is $x^4/[3(1 + x)^2(2 + x)^2]$. Define

$$\begin{aligned} \Phi(\nu) &= B(\nu)(P - M(\nu)^2), \qquad B(\nu) = \frac{(1 + \nu)^2}{16\nu^2}, \\ M(\nu) &= e_0(\nu) + \frac{s_*(\nu)}{32} - L\Big(\frac{s_*(\nu)}{2} - 1\Big). \end{aligned} \tag{F.14}$$

The upper pairing bound M is positive, as the estimates below show. Thus both sum-edge vertices have Casimir at least Φ.

Convexity before the shape ceiling. On $11/5 \le \nu \le 13/5$, set

$$x = \frac{5\nu - 11}{12}, \qquad \mathcal{L}(\nu) = L(x), \qquad M_0 = e_0 + \frac{5\nu + 1}{192}, \qquad M = M_0 - \mathcal{L}.$$

Here $0 \le x \le 1/6$. The useful derivatives of the reciprocal kernel are

$$\begin{aligned} L_{xx} &= \frac{4}{3}\left(\frac{8}{(2 + x)^3} - \frac{1}{(1 + x)^3}\right) \ge 0, \\ L_{xxx} &= 4\left(\frac{1}{(1 + x)^4} - \frac{8}{(2 + x)^4}\right) > 0. \end{aligned}$$

For the second sign, use $(2 + x)/(1 + x) \ge 13/7$ and $(13/7)^4 > 8$. Consequently $\mathcal{L}, \mathcal{L}', \mathcal{L}''$ are nonnegative, $\mathcal{L}''$ increases, and $\mathcal{L}' \le (\nu - 11/5)\mathcal{L}''$, since

$\mathcal{L}'(11/5) = 0$. Endpoint substitution, using the monotonicity of B, $-B'$, $-M_0'$ and M, gives

$$\frac{7}{60} < B < \frac{2}{15}, \qquad 0 < -B' < \frac{1}{24},$$
$$0 < -M_0' < \frac{1}{5}, \qquad \frac{8}{9} < M < 1,$$
$$0 \le \mathcal{L}' < \frac{1}{100}.$$

For example $M(11/5) = 39/40$, while $M(13/5) > 8/9$. Also $B'', M_0'' > 0$.

Separate the curve into its base and its positive correction:

$$\Phi = \Phi_0 + C, \qquad \Phi_0 = B(P - M_0^2), \qquad C = B(2M_0\mathcal{L} - \mathcal{L}^2).$$

The base is strictly convex. Indeed, write

$$A = \frac{19}{192}, \quad U = \frac{25}{32}, \quad V = \frac{361}{192}, \quad D = P - UV > 0, \quad F = V^2 - P > 0.$$

Then $(1+\nu)M_0/\nu = -A\nu + U + V/\nu$, and differentiation gives

$$\Phi_0'' = \frac{1}{8}\left(-A^2 + \frac{2D}{\nu^3} - \frac{3F}{\nu^4}\right), \qquad \Phi_0''' = \frac{3(2F - D\nu)}{4\nu^5} > 0.$$

The last sign follows from $2F > (13/5)D$. Direct substitution gives $\Phi_0''(11/5) > 0$ and $\Phi_0''(5/2) > 1/768$.

The correction is convex as well. Its exact second derivative is

$$C'' = 2BM\mathcal{L}'' - 2B(\mathcal{L}')^2 + 4(BM_0' + B'M)\mathcal{L}'$$
$$+ 2BM_0''\mathcal{L} + 4B'M_0'\mathcal{L} + B''(2M_0\mathcal{L} - \mathcal{L}^2).$$

The last three terms are nonnegative. Put $d = \nu - 11/5 \le 2/5$ and use $\mathcal{L}' \le d\mathcal{L}''$ in the others. Their sum is at least

$$\mathcal{L}''\big(2BM - 4(B|M_0'| + |B'|M)d - 2B\mathcal{L}'d\big) \ge 0,$$

because the expression in parentheses exceeds

$$2\frac{7}{60}\frac{8}{9} - 4\left(\frac{2}{15}\frac{1}{5} + \frac{1}{24}\right)\frac{2}{5} - 2\frac{2}{15}\frac{1}{100}\frac{2}{5} > 0.$$

Thus Φ is strictly convex on this branch.

Controlling its minimum. On $5/2 \le \nu \le 13/5$, the stronger endpoint bounds

$$B < \frac{1}{8}, \qquad -B' < \frac{1}{32}, \qquad -M_0' < \frac{3}{16}, \qquad \mathcal{L}'' > \frac{3}{100}$$

give, in the same derivative identity,

$$\Phi'' > \frac{1}{768} + 2\frac{7}{60}\frac{8}{9}\frac{3}{100} - 4\left(\frac{1}{8}\frac{3}{16} + \frac{1}{32}\right)\frac{1}{100} - 2\frac{1}{8}\left(\frac{1}{100}\right)^2 > \frac{1}{192}.$$

At $v = 5/2$, substitution in (F.14) gives

$$\Phi'(v) < 0, \qquad \Phi(v) - 96\Phi'(v)^2 > \frac{9193}{50000}.$$

Strict convexity implies $\Phi(\nu) \geq \Phi(v)$ to the left of v. To its right, integrating the curvature bound twice and completing the square give

$$\Phi(\nu) \geq \Phi(v) + \Phi'(v)(\nu - v) + \frac{(\nu - v)^2}{384} \geq \Phi(v) - 96\Phi'(v)^2 > \frac{9193}{50000}.$$

After the shape ceiling. For $13/5 \leq \nu \leq 3$, one has $M = e_0 + d$ with $d = 7/96 - 1/819 > 1/16$. It is positive. In the coordinate $t = 1/(1+\nu)$ the derivative of Φ has the sign of $P - (E_0 + d)(H_0 + d)$, with E_0, H_0 as above. By (F.13),

$$(E_0 + d)(H_0 + d) > \frac{9}{4} + \frac{1}{16}\frac{11}{4} > \frac{75}{32} = P.$$

Hence Φ increases with ν on this branch. The two definitions agree at $13/5$, where $L(1/6) = 1/819$, so the strict bound from the first branch applies here too. All four norm vertices therefore have Casimir greater than $9193/50000$.

F.6.7. *The small forward minimum.* Suppose $\nu \geq 3$ and, after the exchange of sides, $z \leq w$. The left row and the floors give $\mathcal{E}(h) \geq 0$ for the residual

$$\mathcal{E}(h) = \eta A - B - n_1 F_{z/\hat{s}_-}(h) - n_2 V, \qquad V = F_{w/\hat{s}_+}\Big(\max\Big\{1, \frac{\hat{s}_+^2}{\hat{K}_+}\Big\}\Big),$$

the floor for v at the least factor that minimality allows. Lowering w does not decrease $\mathcal{E}$, by the increments of Appendix F.6.4 and $V \geq \beta(w)$, as long as $q \geq 1$. Lower w until either $q = 1$ or $w = z$. Three faces remain:

(1) $q = 1$, that is, $w = \hat{K}/\hat{s}_- - 1$ (possible when $\hat{s}_-^2 \leq \hat{K}$);

(2) $w = z$ with $\hat{s}_-^2 \geq \hat{K}$ and $\hat{s}_-^2 \leq \hat{K}_-$;

(3) $w = z$ with $\hat{s}_-^2 \geq \hat{K}$ and $\hat{s}_-^2 > \hat{K}_-$, the hard face.

On each face $\mathcal{E}$ has the form in h of Appendix F.6.5, with second derivative $-zP(h)/(\hat{K}_- h^3)$ and the same P. Since $P(1) = -z\hat{K}_- d$ with $d = \hat{s}_-^2/\hat{K}_- - 1$ and $P'(h) \geq \hat{K}_-$ for $h \geq 1$, integrating twice from $h = 1$ with $\int_1^h (h - r)r^{-3}\,\mathrm{d}r = (h-1)^2/(2h)$ and $\int_1^h (h-r)(r-1)r^{-3}\,\mathrm{d}r = f(h)$ gives

$$\mathcal{E}(h) \leq \mathcal{E}(1) + (h-1)\,\mathcal{E}'(1) + \frac{z^2 d\,(h-1)^2}{2h} - z\,f(h). \tag{F.15}$$

The value and slope at $h = 1$ are explicit: with $\psi(q) = (q-1)/\hat{s}_+ + \frac{w}{\hat{s}_+}(1 - q^{-1}) - \log q$,

$$\mathcal{E}(1) = \frac{\hat{K}}{\hat{K}_-}\,\psi(q) - \frac{\hat{K}^2(q-1)}{\hat{s}_-\hat{K}_-}(V - \beta(w)),$$

$$\mathcal{E}'(1) = \frac{\hat{s}_-\hat{K}}{\hat{K}_-}\Big(\psi(q) + \frac{z(q-1)(q-w)}{\hat{s}_-\hat{s}_+ q}\Big) - \frac{\hat{K}^2(\hat{s}_- q - 1)}{\hat{s}_-\hat{K}_-}(V - \beta(w)).$$

Since $\psi'(q) = (q-1)(q-w)/(\hat{s}_+ q^2)$, ψ first decreases and then increases on $[1, \infty)$.

(1) On the face $q = 1$: $\psi(1) = 0$, and $V = \beta(w)$ because $\hat{s}_+^2 \le \hat{K}_+$ there; so $\mathcal{E}(1) = \mathcal{E}'(1) = 0$, $d \le 0$, and (F.15) gives $\mathcal{E}(h) \le -zf(h) < 0$.

(2) On the second face: $V = \beta(z)$, $d \le 0$, and $1 \le q \le z$, so $\psi(q) \le 0$ and the extra slope term is nonpositive; again $\mathcal{E}(h) \le -zf(h) < 0$.

(3) On the hard face, $\psi(q) \le 0$ still holds; by the shape of ψ it suffices to check the largest admissible q. If $z \le 3$, then $\nu \ge 3$ gives $q \le \hat{s}_-^2/4$, and $\psi(\hat{s}_-^2/4) = \frac{\hat{s}_-}{4} + 1 - \frac{2}{\hat{s}_-} - \frac{4}{\hat{s}_-^2} + \frac{4}{\hat{s}_-^3} - 2\log\frac{\hat{s}_-}{2}$ vanishes at $\hat{s}_- = 2$ and has derivative $(\hat{s}_- - 6)(\hat{s}_- - 2)^2(\hat{s}_- + 2)/(4\hat{s}_-^4) \le 0$ for $2 \le \hat{s}_- \le 4$. If $z \ge 3$, then $\nu \ge z$ gives $q \le \hat{s}_-$, and $\psi(\hat{s}_-) = x + x^2 - \log\hat{s}_-$ with $x = z/\hat{s}_-$ decreases in $\hat{s}_-$ (its derivative is $-(\hat{s}_- - 1)(\hat{s}_- - 2)/\hat{s}_-^3$) from the value $\frac{21}{16} - \log 4 < 0$ at $\hat{s}_- = 4$. The opposite floor exceeds $\beta(z)$ by at least $\Delta_0 = x(x - \frac{1}{2})d + xd^2/(3(d+2))$, by the Padé bound, and $f(h) \ge (h-1)^3/(3h(h+1))$. Inserting these into (F.15) gives a rational upper bound $\mathcal{B}$ for $\mathcal{E}(h)$. Put $\mu = (2z+1-\nu)/\hat{s}_- \in (0,1)$, $\lambda = 1-\mu$ and $v = \hat{s}_-(h-1)/h$, so that $\mu \le v \le 1$ and $\hat{K}_- = \hat{s}_-(\hat{s}_- - \mu)$, $\hat{K} = \hat{s}_-(2-\mu)$, $q = \hat{s}_-/(2-\mu)$, $d = \mu/(\hat{s}_- - \mu)$. After the two weakenings $\Delta_0 \ge z\mu(3(z-1)+\mu)/(6\hat{s}_-^2(\hat{s}_- - \mu))$ and $(h-1)^3/(3h(h+1)) \ge (h-1)^3/(6h^2)$, an exact simplification gives $\mathcal{B} \le -zh\,P_5/(6\hat{s}_-\hat{K}_-^2)$ with the cubic

$$\begin{aligned} P_5 = {} & (z+\lambda)^2 v^3 - 3z\mu\hat{s}_-(z+\lambda)v^2 \\ & + \big[(1+\lambda)\mu\hat{s}_- z(3(z-1)+\mu) - 6(z+\lambda)(z-\lambda)(1-z\lambda)\big]v \\ & + (1+\lambda)\mu\hat{s}_-(3(z-1)+\mu)(z-\lambda). \end{aligned}$$

It remains to show $P_5 > 0$.

The cubic P_5. We prove its positivity directly by the inequality between the arithmetic and geometric means. Write $s = 1 + z$, $\ell = z + \lambda$ and

$$P_5 = Av^3 - Bv^2 + Cv + D,$$

where

$$\begin{aligned} & A = \ell^2, \qquad B = 3z\mu s\ell, \\ & C = (2-\mu)\mu s z\big(3(z-1)+\mu\big) - 6\ell(z-\lambda)(1-z\lambda), \\ & D = (2-\mu)\mu s\big(3(z-1)+\mu\big)(z-\lambda). \end{aligned}$$

Here $0 < \mu < 1$, $\lambda = 1-\mu$, $0 < v \le 1$, and $\nu \ge 3$ is equivalent to $z \ge z_0 = (2+\mu)/(2-\mu) > 1$. In particular $A, B, D > 0$.

First we establish the bound $C > (4-3\mu)\mu^2 s^2$. The inequalities $z \ge z_0$ give

$$3(z-1)+\mu \ge \frac{\mu(8-\mu)}{2-\mu}, \qquad 1 - z\lambda \le \frac{\mu^2}{2-\mu}.$$

Therefore

$$C - (4-3\mu)\mu^2 s^2 \ge \frac{\mu^2}{2-\mu}\,T(z),$$

where the remaining expression factors into positive terms:

$$\begin{aligned} T(z) &= (2-\mu)\big(sz(8-\mu)-(4-3\mu)s^2\big)-6(z^2-\lambda^2) \\ &= 4\mu(2-\mu)+2(1-\mu^2)(z^2-1)+5\mu(2-\mu)(z-1)>0. \end{aligned}$$

This proves the bound for C.

Since $0<v\le 1$, AM–GM now gives

$$\frac{P_5}{v^2}=Av+\frac{C}{v}+\frac{D}{v^2}-B\ge 2\sqrt{AC}+D-B.$$

By the strict bound for C, the right side is greater than $\mu s\ell\,\mathcal{R}$, where

$$\mathcal{R}=2\sqrt{4-3\mu}+\frac{D}{\mu s\ell}-3z.$$

Use the exact factorization

$$\frac{D}{\mu s\ell}-3z+4-2\mu=\frac{\lambda(\mu+z-2)(\mu+3z-5)}{\ell}$$

and the square-root identity

$$\sqrt{4-3\mu}-(2-\mu)=\frac{\mu\lambda}{\sqrt{4-3\mu}+2-\mu}\ge\frac{\mu\lambda}{4}.$$

The denominator in this quotient is at most four. These bounds give

$$\mathcal{R}\ge\lambda\left(\frac{\mu}{2}+\frac{(\mu+z-2)(\mu+3z-5)}{\ell}\right).$$

If $\mu\ge 1/6$, complete the square:

$$(\mu+z-2)(\mu+3z-5)=3\left(\mu+z-2+\frac{1-2\mu}{6}\right)^2-\frac{(1-2\mu)^2}{12}\ge-\frac{1}{12}.$$

As $\ell\ge 1$, this implies $\mathcal{R}\ge\lambda(\mu/2-1/12)\ge 0$ and hence $P_5>0$. The same conclusion holds for $\mu<1/6$ whenever the product is nonnegative.

In the remaining case the two roots of the product give $(5-\mu)/3<z<2-\mu$, so $3/2<z<2$, $s<3$ and $z\lambda>1$. The second term in C is then positive. Moreover,

$$(2-\mu)(3(z-1)+\mu)>\frac{9}{4},\qquad \frac{9}{4}z\mu s<\frac{9}{4}.$$

It follows that $C>\frac{9}{4}z^2\mu^2s^2$, whence $2\sqrt{AC}>3z\mu s\ell=B$. Again AM–GM gives $P_5>0$. Thus $\mathcal{E}(h)<0$ on every face, a contradiction; therefore $\nu<3$.

Appendix G. Class M2: longer forward runs

A period of class M2 has switches of both signs, every maximal reverse run has exactly two switches, and some maximal forward run has at least two (Definition 3.1). Unlike the classes R, F and M1, such a period has no rigid sign word: the forward runs may have any lengths, the number of reverse runs is arbitrary, and the factors are unrelated. The exclusion uses a window of at most seven consecutive pieces around a smallest active margin. Expressing the period's normal multiplier U in this window gives a finite system of scalar necessary conditions. We prove that this system is inconsistent. The whole period enters only through three inputs: the compact domain (Theorem A.12), the cap on the Casimir that competitiveness imposes, and a lower bound on the Casimir obtained from the winding of the switch vectors over the period.

We prove Theorem 4.5.

The proof has five steps.

(1) A global minimum of the active margins occurs at the centre of a reverse run (Proposition G.3).

(2) Of the two turns beside that centre, choose the one with the smaller active margin; after reflecting the period and reversing time if necessary, it follows the centre (Lemma G.5).

(3) The switch after the chosen turn is forward or reverse. If it is reverse, the forward run through the turn has a single switch (it is short), and Appendix G.3 excludes the window (Theorem G.33). If it is forward, the run is nonshort, and the switch after the next piece is again forward or reverse (Propositions G.6 and G.7).

(4) A comparison that moves one margin to a boundary value turns the window into one scalar residual N that must be negative (Lemma G.8), together with a short list of necessary bounds (Lemma G.10).

(5) A reverse successor contradicts $N < 0$ directly (Theorem G.12). A forward successor leaves a small region of parameters on which the Casimir of the period exceeds the competitive cap (Theorem G.14).

The first subsection carries out steps (1), (2), (4), (5) and the nonshort half of (3). The polynomial inequalities are collected in Appendix G.2. Throughout, γ^2 is the Casimir of Definition B.9, and the switch norms and turn secants are those of Lemma B.12.

G.1. **The minimum and the nonshort continuations.**

G.1.1. *Three inputs from the whole period.* Class M2 requires a sharper cap than Corollary B.15. We use the spectral form of the cost–Casimir inequality, whose pointwise proof (F.4) applies whenever the primary costate is negative elliptic.

Lemma G.1 (Casimir bounds)**.** *Let P be a normal vertex period with cost $\mathcal{C}$ and primary costate Λ_1.*

(a) *If* $\det \Lambda_1(0) > 0$ *and* $\Lambda_1(0)_{10} < 0$, *then* $\mathcal{C} \geq \frac{\pi}{3}(\kappa + \frac{3}{2})$ *with* $\kappa = \sqrt{\det \Lambda_1(0)}$.
(b) *If* P *is competitive, then* $\gamma^2 < 9193/50000$.
(c) *If* P *is competitive and mixed, then* $\gamma^2 > 27/256$.

Sketch. (a) Write $\Lambda_1(0) = -\kappa\, hJh^{-1}$ and $B = h^{-1}g$. The maximum condition gives pointwise $2\kappa m_s \leq 3m_b$, where $m_b = -\frac{1}{2}\operatorname{tr}(JX)$ is a third of the cost density and $m_s = -\frac{1}{2}\operatorname{tr}(JBXB^{-1})$. The argument θ of $(B_{00} + B_{11}) + i(B_{10} - B_{01})$ has $\theta' = (m_b + m_s)/(z+1)$ with $z = \frac{1}{2}\|B\|_F^2 \geq 1$, and $g(T) = R$ forces $\Delta\theta \geq \pi/3$. Since $|z_B|^2 = (z+1)/2$, (F.3) is $(m_b+m_s)^2 \leq 2(z+1)m_b m_s$. Hence $\theta' \leq 2m_b m_s/(m_b + m_s) \leq 6m_b/(2\kappa + 3)$, and integration gives (a). (b) Class R is impossible (Theorem 4.1), so some switch is forward and has positive norm; when $\gamma^2 > 0$, $\Lambda_1(0)$ is then negative elliptic (Proposition B.16), and (a) with $\mathcal{C} \leq A_{\text{oct}}$ gives $\kappa \leq 3A_{\text{oct}}/\pi - 3/2$. The cost comparison of Lemma C.5 and the calculation in Corollary F.6 then give $\gamma^2 < 9193/50000$. (c) If $\gamma^2 \leq 27/256$, the Gram identity (Lemma B.13) with $d_\pm \geq 3/32$ puts the angle between consecutive switch vectors below $\pi/3$ at a forward piece, and above $2\pi/3$ at a reverse centre, by the angle gap $\cos\varphi < 3/10$ of Appendix E; a turn contributes nothing. The increments then add up to more than $\pi/3$, while the winding of the switch vectors over one period is exactly $\pi/3$ (Appendix E). □

At a reverse centre (the middle piece of a reverse run of two) the pairing and the Casimir are explicit functions of four quantities. With active margin a, turn neighbours p, n and incident norms $u_{\text{i}}, u_{\text{o}}$ (eight times the switch norms, which are turn secants), put $r = a/(a+p)$, $t = a/(a+n)$ and $\rho = a^2/(a^2+4)$. Stationarity at the centre and the Gram identity read

$$\gamma^2 = \frac{4u_{\text{i}}u_{\text{o}} - E^2}{16(1-\rho)^2}, \qquad E = 1 + r + t - \rho + \log\frac{\rho}{rt} - \frac{1-u_{\text{i}}}{r} - \frac{1-u_{\text{o}}}{t}. \tag{G.1}$$

The strict centre minimum gives $r, t < \frac{1}{2}$, and the positive reverse factor gives $\rho < rt$.

Lemma G.2 (Scale of a reverse centre)**.** *In a competitive mixed normal vertex period, every reverse centre has active margin* a *with* $a^2 < 12/17$.

Proof. The bound is $\rho < 3/20$. Lowering both incident norms to their common floor $3/4$, and then raising both ratios r, t to $\frac{1}{2}$, only decreases the right side of (G.1), and gives $\gamma^2 > G(\rho) = (9/4 - B(\rho)^2)/(16(1-\rho)^2)$ with $B(\rho) = 1 - \rho + \log(4\rho)$. The function G increases on $[3/20, 1/4)$ and $G(3/20) > 23/125 > 9193/50000$, which contradicts Lemma G.1(b) when $\rho \geq 3/20$; and $\rho < rt < 1/4$ always. Figure 10(a) shows G. □

G.1.2. *The minimum is a reverse centre.*

Proposition G.3 (The minimum is a reverse centre)**.** *Let* P *be a competitive normal vertex period of class* M2 *with at least six pieces. There is an index*

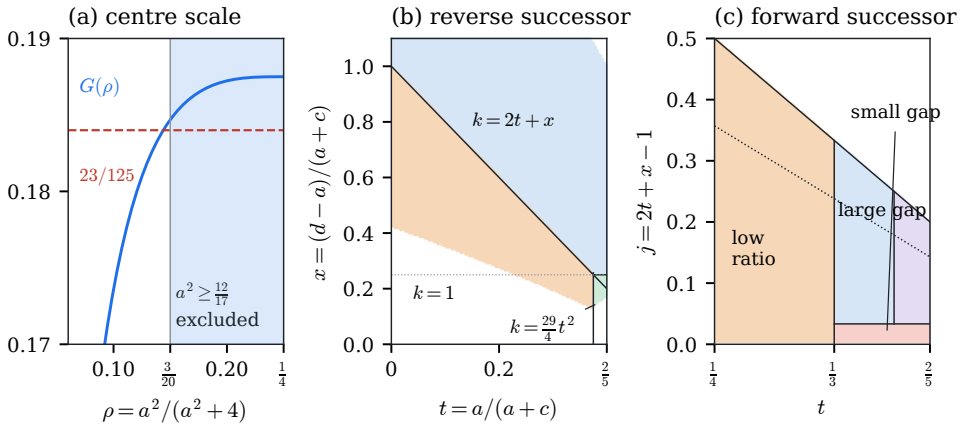


FIGURE 10. Parameter bounds for class M2. (a) The lower bound $G(\rho) = (9/4 - B(\rho)^2)/(16(1-\rho)^2)$ for the Casimir at a reverse centre, with $\rho = a^2/(a^2+4)$. It exceeds the cap 23/125 (dashed) on $[3/20, 1/4)$, which proves $a^2 < 12/17$ (Lemma G.2). (b) The parameters (t, x) of the scalar reverse exclusion (Proposition G.11), below $x = 1.1$, by the face to which k is lowered: the unit shape $k = 1$, the order face $k = 2t + x$, and the scale strip $k = 29t^2/4$, which lies in $3/8 < t < 2/5$, $x \leq 1/4$. Points above $x = 1$ belong to the order face. (c) The parameters (t, j) of a forward successor, in $1/4 < t < 2/5$, $0 < j < 1 - 2t$: the low ratio $t \leq 1/3$, the two large-gap regions $j \geq 1/30$, split at $t = 3/8$, and the small-gap region, where the Casimir exceeds the cap. The dotted line is $j = \frac{5}{7}(1 - 2t)$.

m with $a_m \leq a_k$ for every k such that the switches $m-1$ and m are reverse and the switches $m-2$ and $m+1$ are forward.

Let a_m be a smallest active margin of the period; the active margins are periodic, so a minimum exists. A minimum cannot be at a turn: at three pieces with labels i, j, i the active margins are strictly log-concave, $a_k a_{k+2} < a_{k+1}^2$ (Lemma D.1), so the middle one is not the smallest of the three. If both switches at m are reverse, m is the centre of a reverse run. The remaining case, both switches forward, is excluded by the following theorem, which uses only stationarity and the floors that the two missing controls impose.

Proposition G.4 (No small forward minimum)**.** *Let P be a normal vertex period with at least three pieces, and let m be an index such that the switches $m-2, m-1, m, m+1$ are forward, $a_m \leq a_j$ for $m-2 \leq j \leq m+2$, and $a_m^2 \leq 4/3$. This is impossible.*

The proof writes the multiplier in the frame of the central piece, eliminates one coordinate with stationarity at the centre, and keeps the stationarity row at the smaller neighbour with the two exterior floors $v \geq F_x(q)$ at the forward switches. The remaining face is a one-variable residual with an exactly controlled second derivative; two integrations and the Padé bound $\frac{1}{2}(h-h^{-1})-\log h \geq (h-1)^3/(3h(h+1))$ leave a rational expression, negative by an elementary cubic comparison. No Casimir bound is used.

Proof of Proposition G.3. Take a global minimum m. If exactly one of the switches $m-1, m$ is reverse, the pieces $m-1, m, m+1$ have labels i, j, i, and Lemma D.1 contradicts minimality. If both are forward, the strict centre minimum of the neighbouring reverse runs (Proposition D.2) forces the switches $m-2$ and $m+1$ to be forward as well: otherwise the centre of the adjacent reverse run would have a smaller active margin. The period is mixed, so it has a reverse centre; by Lemma G.2 its active margin squared is below $12/17$, hence $a_m^2 < 12/17 < 4/3$, and Proposition G.4 gives a contradiction. If both switches are reverse, they form a maximal reverse run, of length two in class M2, and m is its centre. □

G.1.3. *Orientation by reflection.* Let $S = \mathrm{diag}(1,-1)$. Reflecting the plane and reversing time is a symmetry of the control problem: it maps a normal vertex period to another one with the same cost, and reverses the order of the pieces.

Lemma G.5 (Reflection)**.** *For every normal vertex period P there is a normal vertex period P' with the same number of pieces and the same cost, whose state stays in $\mathfrak{h}^{\star\star}$ when that of P does, with labels $i'_k = c - i_{-1-k}$ for a constant $c \in \mathbb{Z}/3$ and active margins $a'_k = a_{-1-k}$. Its switch k is forward if and only if the switch $-2-k$ of P is forward, and P' is of class* M2 *when P is.*

The construction is $u'(t) = \varsigma u(T-t)$, with ς the permutation of the vertices that reflects the labels, $g'(t) = SR^{-1}g(T-t)S$, $X'(t) = -SX(T-t)S$, and $\Lambda'(t) = -S\Lambda(T-t)S$ for both costates. Since $SJS = -J$, the cost density is preserved, and $\mathfrak{h}^{\star\star}$ is symmetric under $x \mapsto -x$. Reading the signs in reverse order does not negate them: a reverse switch of P is a reverse switch of P'.

By Proposition G.3 and Lemma G.5 we may assume that the global minimum $a = a_m$ is a reverse centre and that the turn after it is the smaller one: $a_{m+1} \leq a_{m-1}$. (At a tie either choice works.) After relabeling the vertices by a power of R, which changes neither the cost nor the active margins, the pieces $m-2, \dots, m+3$ have labels and active margins

$$0(p), \quad 1(b), \quad 0(a), \quad 2(c), \quad 0(d), \quad 1(r) \tag{G.2}$$

when the switch $m+2$ is forward. These are consecutive pieces of the period, continued through the twisted seam when necessary; they need not be distinct from other pieces named later, and they are not a closed competitor.

Symbol	Meaning (in Appendix G and Appendix G.2)
p, b, a, c, d, r	active margins of six consecutive pieces (G.2); a is a global minimum, and $c \le b$
$K_a,\ Q,\ K_r$	a^2+4, d^2+4, r^2+4
S, T, W, Z, Y	$a+b$, $a+c$, $c+d$, $d+r$, W^2-T^2 (here T is not the terminal time)
u, v, w	eight times the norms of the switches $b \to a$; $a \to c$ and $c \to d$; $d \to r$
h_a, h_d	the diagonal entries of the multiplier in the frames of the pieces a and d
$U(x),\ g_B(x)$	$1-x+x^2$ and $(B^2+Bx+x^2)/(B+x)^2 = 1-Bx/(B+x)^2$
$\mathrm{Sec}(A, B, P)$	the average of g_B over the interval between A and P
$F_x(q)$	$1+x^2q-xq\log q/(q-1)$, with $F_x(1)=U(x)$
t, x, k, q	a/T, $(d-a)/T$, K_a/T^2, WZ/Q (normalized coordinates)
j	$2t+x-1$, the gap of a forward successor

TABLE 2. Local notation of class M2. The letters p, r, t, u, x have their usual meaning elsewhere in the paper.

G.1.4. *The nonshort window.* Table 2 lists the local notation. All quantities in the window come from the one normal multiplier of the period. The factors of the same-sign pieces follow from the one-arc map (Lemma B.1) and the margins $(a_k, a_{k+1}, (4-a_ka_{k+1})/(a_k+a_{k+1}))$ of the state at a switch: the factor of a piece with incident signs $++$, $--$, $-+$, $+-$ is $s_{k-1}s_k/K_k$, $K_k/(s_{k-1}s_k)$, s_k/s_{k-1}, s_{k-1}/s_k respectively. In the window this gives the central factor $K_a/(ST) > 1$, the factor $q = WZ/Q > 1$ of the piece d, and $d > a$ from the turn c.

Both switches at a turn have the same norm, the turn secant (Lemma B.12): $u = \mathrm{Sec}(a, b, p)$ and $v = \mathrm{Sec}(a, c, d)$, where

$$\mathrm{Sec}(A, B, P) = 1 + \frac{B^2}{(A+B)(P+B)} - \frac{B}{P-A}\log\frac{P+B}{A+B}.$$

In the frame of the central piece the multiplier is

$$U_a = \begin{pmatrix} h_a & -K_a^2 v/(8S^2) \\ S^2u/8 & -h_a \end{pmatrix}, \qquad \gamma^2 = \det U_a = \frac{uvK_a^2}{64} - h_a^2, \tag{G.3}$$

and stationarity at the centre (Lemma B.11) fixes the diagonal:

$$h_a = \frac{K_a}{16a}(Su + Tv - C(b, a, c)),$$
$$C(p, a, n) = p + a + n - \frac{a^2}{p+a} - \frac{a^2}{a+n} + \frac{a^3}{a^2+4} - a\log\frac{(p+a)(a+n)}{a^2+4}. \tag{G.4}$$

Here $C(p, a, n)$ is four times the partial derivative of the normalized cost $\mathcal{C}/\sqrt{12}$ in the active margin of a same-sign piece. Transport across the centre

and the turn to the frame of the piece d gives

$$h_d = h_a + \frac{vY}{8}, \qquad w = \frac{uK_a^2 + 16h_aY + vY^2}{Q^2}, \tag{G.5}$$

and stationarity at d, the first piece of the forward run after the turn, is the root equation

$$R_d := C(c,d,r) - Wv - Zw + \frac{16\,d\,h_d}{Q} = 0. \tag{G.6}$$

Proposition G.6 (The nonshort window)**.** *Let P be competitive, with state in $\mathfrak{h}^{\star\star}$, of class* M2 *and with at least six pieces; let m be as in Proposition G.3 with $a_{m+1} \le a_{m-1}$, and let the switch $m{+}2$ be forward. Then $(p,b,a,c,d,r) = (a_{m-2},\dots,a_{m+3})$, with u, v, h_a, w as above, satisfy:*

(i) the order $a < b$, $a < c \le b$, $a < p$, $a < d$, $a \le r$;

(ii) strict star at the five switches, $pb, ab, ac, cd, dr < 4$; the factors $ST < K_a$ and $Q < WZ$; three-arc log-concavity $ad < c^2$;

(iii) the compact domain: $a \ge 2/15$, $cd + \frac{2}{15}(c+d) \le 4$, $dr + \frac{2}{15}(d+r) \le 4$, $d, r \le 8$, and $a+c$, $c+d$, $d+r \ge \frac{1}{2}$; and the scale bound $a^2 < 12/17$;

(iv) the turn secants $u = \mathrm{Sec}(a,b,p)$, $v = \mathrm{Sec}(a,c,d)$; the switching inequalities $g_b(p) \le u \le g_b(a)$, $g_c(d) \le v \le g_c(a)$, $w \ge g_d(r)$; the missing-control floors $v \ge F_{d/W}(q)$ and $w \ge F_{d/Z}(q)$;

(v) the identities (G.4), (G.5) and (G.6);

(vi) $27/256 < uvK_a^2/64 - h_a^2 < 9193/50000$.

Sketch. Each item is read off the costate cycle at the switches of the window: (i) from minimality, the choice $c \le b$ and Proposition D.2; (ii) from the margin transports (Lemma B.1); (iii) from Theorem A.12 at the switching times (every margin is at least $2/15$, and $y \le 9/2$ on $\mathfrak{h}^{\star\star}$ gives the pair sums and $d, r \le 8$) and Lemma G.2; (iv) from Lemma B.12 and, for the floors, from $H_i \le 0$ for the two controls missing at a forward switch, written through stationarity, an identity algebraic in x that holds for every x; (v) from exact transports; (vi) from Lemma G.1. □

G.1.5. *The successor of the forward piece.* The window does not yet contain a contradiction: the root equation (G.6) has solutions compatible with (i)–(vi). The next actual switch, after the piece r, supplies one more condition. Finiteness of the representation guarantees that it exists, and equal neighbouring labels have been merged, so it is forward or reverse. Let

$$t_{\mathrm{next}} = \frac{Q^2 v - 16h_dZ^2 + wZ^4}{K_r^2}$$

be the norm of that switch when it is forward, read in the frame of the piece r.

Proposition G.7 (The continuation alternative)**.** *In the situation of Proposition G.6:*

(a) if the switch $m+3$ is forward, to a piece with active margin s, then $s \geq a$, $rs < 4$, $K_r < Z(r+s)$, $t_{\text{next}} \geq g_r(s) \geq \frac{3}{4}$, and both missing-control floors hold at the piece r;

(b) if the switch $m+3$ is reverse, to a piece with active margin A, then $a \leq A < d$, $Ad < r^2$, $w = \mathrm{Sec}(A, r, d)$ and $w \leq g_r(A)$.

In (a) the inequality for t_{next} is the forward switching inequality at the piece r, whatever the margin s and the length of the forward run. In (b) the piece r is a turn, both its switches have the norm $\mathrm{Sec}(A, r, d)$, $A \geq a$ by minimality, and $Ad < r^2$ is three-arc log-concavity. Since g_r decreases on $(0, r)$ and $A < r$, extending the average from $[A, d]$ to $[a, d]$ also gives $w \leq \mathrm{Sec}(a, r, d)$; we use (b) as stated.

G.1.6. *The boundary comparison and the residual.* The root equation (G.6) is affine in the incident norm u, with slope

$$\partial_u R_d = \frac{K_a}{aQ^2}\big(dSQ - Z(aK_a + SY)\big) < 0, \tag{G.7}$$

negative because $Y > 0$, $Z > Q/W$, and $dSQ - Z(aK_a + SY) < (Q/W)\{a(ST - K_a) + Sc(a-d)\} < 0$. The reverse switch $b \to a$ caps u by the ceiling $U(a/S) = g_b(a)$, so the actual data satisfy $R_d(U(a/S)) \leq R_d(u) = 0$. Stationarity at d thus imposes an upper bound on the incident norm. On the necessary domain, we will show that stationarity instead requires the norm to exceed this ceiling.

Before testing this, we remove the margins b and p. Keep a, c, d, r and the actual v fixed and raise b to $b_* = K_a/T - a$, the value at which the central factor $K_a/(ST)$ equals one. Write $G(b) = R_d(U(a/(a+b)); b)$ for the residual at the ceiling, with h_a given by central stationarity at the margin b.

Lemma G.8 (Boundary comparison)**.** *For the window of Proposition G.6, $G(b_*) < 0$; the root u_* of $R_d(\,\cdot\,; b_*) = 0$ satisfies $u_* < U(aT/K_a)$; and the outgoing norm w_* of that boundary root satisfies $w_* < w$.*

Proof. Along $[b, b_*]$ the derivative is $G'(b) = \partial_u R_d \cdot a(b-a)/(a+b)^3 < 0$, and at the actual margin $G(b) = \partial_u R_d\,(U(a/S) - u) \leq 0$, so $G(b_*) < 0$. Along the curve of roots $u(b)$, the outgoing norm has $dw/db = dK_a^3 G(b)/(aQ^3(\partial_u R_d)^2) < 0$, so w decreases from its actual value. □

This is a scalar comparison: the point b_* need not belong to a trajectory, and no switching inequality, floor or Casimir bound is transported to it.

At b_* the logarithm of the centre disappears ($C(b_*, a, c) = 4/T + 4c/K_a$), and homogeneity lets us normalize by $T = a + c$. In the coordinates $t = a/T$, $x = (d-a)/T$, $k = K_a/T^2$, $q = WZ/Q$, write $W = 1 + x$, $p = t + x$ and $B = k + 2tx + x^2$ for the normalized values; the margins are recovered from (t, x, k, q), since $a^2(k - t^2) = 4t^2$. The positive multiple

$$N(t, x, k, q, v, \log q) = t\,B\,q\,W \cdot \frac{G(b_*)}{T}$$

is a polynomial in t, x, k, q, v and $L = \log q$, affine in k, in v and in L; Appendix G.2.3 gives its coefficients. The normalized incoming secant is $v = \mathrm{Sec}(t, 1-t, t+x) = 1 + (1-t)^2/W - (1-t)\log W/x$.

Lemma G.9 (Shape monotonicity)**.** *If $0 < t < 1$, $x > 0$, $q \geq 1$, $v \leq U(t)$ and $L < q - 1$, then $N(t, x, k, q, v, L)$ is nondecreasing in k.*

The coefficient of k is at least $q\,t\,x(1-t)\{(q-1)(2x+1) + tx\} > 0$ (Appendix G.2.3). Since k moves with a at fixed t, the lemma lets every later argument lower k to the smallest value that the necessary conditions allow, and prove $N > 0$ there.

Lemma G.10 (Necessary conditions)**.** *Every window of Proposition G.6 satisfies*

(i) $0 < t < \frac{1}{2}$, $x > 0$, $k > 1$, $1 < q < W/p$, $v = \mathrm{Sec}(t, 1-t, t+x)$, $1 - (1-t)p/W^2 \leq v \leq U(t)$ *and* $F_{p/W}(q) \leq v$*;*

(ii) $N(t, x, k, q, v, \log q) < 0$*;*

(iii) $t < 2/5$*;*

(iv) $a < 4/5$*, that is,* $k > 29t^2/4$*;*

(v) $d^2 + 2cd > 4$*, that is,* $k < 2t + 2x + x^2$*;*

(vi) $3d < 7c$*, that is,* $x < \frac{7}{3} - \frac{10}{3}t$*;*

(vii) if the successor is reverse, $d \leq b_*$*, that is,* $k \geq 2t + x$*.*

Item (i) restates the window ($k > 1$ is $ST < K_a$ with $S \geq T$, and $q < W/p$ is $dr < 4$); the others are proved in Appendix G.2.4.

G.1.7. *The reverse successor.*

Proposition G.11 (The scalar reverse exclusion)**.** *Let $0 < t < 2/5$, $x > 0$, $\max\{1, 2t + x, 29t^2/4\} \leq k < 2t + 2x + x^2$, $x < \frac{7}{3} - \frac{10}{3}t$, $1 < q < (1+x)/(t+x)$, $v = \mathrm{Sec}(t, 1-t, t+x) \leq U(t)$ and $F_{(t+x)/(1+x)}(q) \leq v$. Then $N(t, x, k, q, v, \log q) > 0$.*

Proof. By Lemma G.9 it suffices to find a shape $k' \leq k$ allowed by the hypotheses with $N(k') > 0$. There are three exhaustive cases (Figure 10(b)).

(1) If $t \leq 3/8$ and $x \leq 1 - 2t$, lower k to 1, where $N(1) > 0$ (Lemma G.21).

(2) If $2t + x \geq 1$ and $x > 1/4$, lower k to the order face $k = 2t + x$. The floor and the swapped secant order give $q \leq (1+x)/(2t+x)$. On this face N is a positive multiple of a function $\alpha + \beta q + \gamma/q - p\log q$, concave in q, so it suffices to check $q = 1$ and the upper end, where it is positive (Lemma G.22).

(3) Otherwise $3/8 < t < 2/5$ and $x \leq 1/4$; lower k to the scale face $k = 29t^2/4$, where $N > 0$ (Lemma G.23). □

On each face the logarithms are first replaced by rational bounds with the correct signs (v by a larger $\bar{v}$, $\log q$ by $L_+(q) \geq \log q$, the floor by a smaller F_ℓ), which lowers N; the result is a cubic in $y = q - 1$ whose positivity for

$y \geq 0$ follows from six polynomial signs in (t, x). Appendix G.2.5 gives the reduction and proves the signs.

Theorem G.12 (Reverse successor)**.** *No window of Proposition G.6 has a reverse successor.*

Proof. By Proposition G.7(b) and Lemma G.10, the normalized window satisfies every hypothesis of Proposition G.11, which gives $N > 0$, against Lemma G.10(ii). □

G.1.8. *The forward successor.* When the successor is forward the order bound (vii) fails, and the residual alone does not suffice. Two facts replace it. First, the scalar reverse exclusion applies to every window with $k \geq 2t + x$, without any assumption on the successor, so every window has the opposite order

$$k < 2t + x, \qquad \text{equivalently } d > b_* \text{ and } j = 2t + x - 1 > 0. \tag{G.8}$$

Second, the forward successor gives a lower bound for the next norm at the boundary root. Both the actual point and the boundary point satisfy the root equation (G.6), which makes h_d an affine function of w; on that line the next norm is a decreasing affine function of w, and $w_* < w$, so the next norm $t_{\mathrm{next},*}$ at the boundary is at least $t_{\mathrm{next}} \geq \frac{3}{4}$. As a function of the incident norm the next norm is affine, and its value at the root differs from its value $\mathcal{T}_U$ at the ceiling by a multiple of N. For multipliers $\alpha > 0$ and $M \geq 0$ the combined residual

$$\begin{aligned} \mathcal{H} &= \alpha N + C_*(\tfrac{3}{4} - t_{\mathrm{next},*}) + M(F_{p/W}(q) - v) \\ &= (\alpha + 1 - Je^2)N + C_*(\tfrac{3}{4} - \mathcal{T}_U) + M(F_{p/W}(q) - v), \end{aligned} \tag{G.9}$$

with $e = q/W$, $J = t + x(x+2)$ and an explicit $C_* > 0$, is therefore negative at every window with a forward successor. It is affine in v, in $\log q$ and in the floor value, and a concave quadratic in k.

The unit face of the low-scale argument gives $t > 1/4$ (Lemma G.24); with the derivative floor $v \geq g_c(d)$ this gives $x < 1$ and $4t + x < 2$, and a quantitative form of the swapped secant order gives $q < W/(1 + 2j)$. The remaining parameters are cut down in three steps (Figure 10(c)).

Proposition G.13 (The forward successor in three steps)**.** *Every window of Proposition G.6 with a forward successor satisfies*

(a) $t > 1/3$*;*

(b) $j = 2t + x - 1 < 1/30$*;*

and then the Casimir of the period is at least 9193/50000*, against Lemma G.1(b).*

Outline. (a) If $t \leq 1/3$, take $\alpha = 6$ and $M = 0$. The coefficients of v and of $\log q$ in $\mathcal{H}$ are negative, so raising v to a rational majorant $\hat{v}$ and $\log q$ to $L_+(q)$ keeps $\mathcal{H}$ negative. The result increases in k, so it is negative at $k = 1$ as well; but at $k = 1$ it is positive (Lemma G.25).

(b) Suppose $j \geq 1/30$. Then $1/3 < t < 2/5$ by (a), and the derivative floor in integral form gives $j < \frac{5}{7}(1-2t)$. For $1/3 \leq t \leq 3/8$ take $\alpha = \frac{7}{2} - 10(q-1)$ and $M = 8j + 8(q-1) - \frac{1}{6}$, both positive here; after the substitutions of v, $\log q$ and the floor by rational bounds, the residual increases in k and is positive at $k = 1$. For $3/8 \leq t < 2/5$ take $\alpha = 4$ and $M = 1$; the residual increases in k and is positive at the scale face $k = 29t^2/4$ (Lemma G.26).

(c) Now $1/3 < t < 2/5$, $0 < j < 1/30$ and $1 \leq k < 31/30$. With $N < 0$, a rational comparison in q gives $\chi = pq/W > 4/7$ (Lemma G.27). On the stationarity line of the piece d, the diagonal h_d is an affine function $H_d(w)$ of the outgoing norm, and the Casimir is $\Gamma(w) = (4B^2vw - H_d(w)^2)/(16(k-t^2)^2)$ in normalized form. For $w \geq 3/4$ one has $\Gamma'(w) < 0$ (Lemma G.28). The actual w lies between $3/4$ and the central upper envelope w_U, the transported norm with u at its ceiling and $b = b_*$, so $\gamma^2 = \Gamma(w) > \Gamma(w_U)$. At w_U the event factor q is eliminated with the floor (Lemma G.29), and the remaining function of (t, x, k, v) exceeds $9193/50000$ (Lemma G.30). □

Theorem G.14 (Forward successor)**.** *No window of Proposition G.6 has a forward successor.*

This is Proposition G.13. The obstruction in its last step is the cost itself: the turn data force the Casimir of the period above $9193/50000$, and by Lemma G.1(a) a period with such a Casimir costs more than A_{oct}.

Remark G.15 (Small scale). The same methods exclude every window with $a \leq \frac{1}{2}$, whatever the sign of the successor: then $k \geq 17t^2$, and N can be lowered to $k = 1$ or to $k = 2t + \frac{1}{2}$. The proof above needs only the unit face of that argument (Lemma G.24), which gives $t > 1/4$.

G.1.9. *Proof of Theorem 4.5.* Let P be as in Theorem 4.5, and take m from Proposition G.3. If $a_{m+1} > a_{m-1}$, replace P by its reflection (Lemma G.5): it is again competitive, of class M2, with state in $\mathfrak{h}^{\star\star}$ and the same number of pieces; the minimum moves to $-1-m$, the four switches around it keep their signs in reverse order, and the smaller turn now follows the centre. If the switch $m+2$ after the turn is reverse, the seven pieces $m-2, \ldots, m+4$ form a smaller-short window, excluded by Theorem G.33. If it is forward, Proposition G.6 gives a nonshort window, and Proposition G.7 gives a forward or a reverse successor; Theorem G.14 and Theorem G.12 exclude them. □

G.2. **Estimates for class M2: the nonshort cases.** This subsection proves the scalar inequalities behind Appendix G, in the notation of Table 2. Every step concerns explicit functions of at most five real variables. The logarithm estimates follow from one reciprocal integral, using Jensen's inequality and pointwise bounds on the integrand. They give bounds for the secants and the duration. The remaining face arguments reduce polynomial signs, one variable at a time, to inequalities on intervals. Several of these polynomials are long; they are determined by the displayed definitions, and we state the identities and bounds on which each sign rests.

G.2.1. *Secants and floors.* For $B > 0$ the function $g_B(x) = 1 - Bx/(B+x)^2$ decreases on $(0, B)$, increases on (B, ∞), and has minimum $g_B(B) = 3/4$; also $g_B(x) = U(B/(B+x)) = U(x/(B+x))$. The turn secant $\mathrm{Sec}(A, B, P) = (P-A)^{-1}\int_A^P g_B$ has the antiderivative $s - B\log(B+s) - B^2/(B+s)$.

Lemma G.16 (Secants)**.** *(a) If $0 < a \le c < d$, then $\mathrm{Sec}(a, c, d) < \mathrm{Sec}(a, d, c)$.*

(b) If $0 < c \le d$, then $z \mapsto \mathrm{Sec}(z, d, c)$ strictly decreases on $(0, c]$.

(c) For $P, C > 0$ and $q > 1$, $F_{P/(P+C)}(q) = \mathrm{Sec}((P+C)/q - P,\, P,\, C)$. In particular $F_{d/W}(q) = \mathrm{Sec}(B_0, d, c)$ with $B_0 = (4-dr)/(d+r)$, the third margin at the switch $d \to r$.

(d) If $ad < c^2$, then $\mathrm{Sec}(a, c, d) < \frac{1}{2}(g_c(a) + g_c(d))$.

(e) If $0 < A < P$, $B > 0$ and $B/(A+B) \ge 3/5$, then $\mathrm{Sec}(A, B, P) > \mu_* := (2\sqrt{130} - 4)/25$.

Proof. (a) $\mathrm{Sec}(a, d, c) = \int_0^1 g_1(u)\,d\theta$ and $\mathrm{Sec}(a, c, d) = \int_0^1 g_1(v)\,d\theta$ with $u = (a + \theta(c-a))/d$ and $v = (a + \theta(d-a))/c$. Here $v > u$, $uv < 1$, and $g_1(u) - g_1(v) = (v-u)(1-uv)/((1+u)^2(1+v)^2) > 0$. (b) The derivative is $(\mathrm{Sec}(z, d, c) - g_d(z))/(c-z)$, and the average of the decreasing function g_d over $[z, c]$ is below its value at z. (c) Substitute $(P+C)/q - P$ in the closed form of the secant; $W/q - d = Q/Z - d = B_0$. (d) For $a \le x \le d$,

$$\frac{(d-x)g_c(a) + (x-a)g_c(d)}{d-a} - g_c(x) \\ = \frac{c(x-a)(d-x)\left[c^2(a+d+2c) + (c^2-ad)x\right]}{(a+c)^2(d+c)^2(x+c)^2} \ge 0;$$

integrate over $[a, d]$. (e) With $\eta = B/(A+B)$ and $q = (P+B)/(A+B) > 1$, the secant is $1 + \eta^2/q - \eta\log q/(q-1)$. The upper bound of Lemma G.17(b) gives

$$\mathrm{Sec}(A, B, P) > 1 - \frac{\eta}{6} + \frac{a}{q} - \frac{b}{q+1}, \qquad a = \eta\left(\eta - \frac{1}{6}\right), \quad b = \frac{4\eta}{3}.$$

For $a, b > 0$, completing a square gives

$$\frac{a}{q} - \frac{b}{q+1} + (\sqrt{b} - \sqrt{a})^2 = \frac{\left(\sqrt{a}(q+1) - \sqrt{b}\,q\right)^2}{q(q+1)} \ge 0.$$

Thus the secant exceeds

$$m(\eta) = 1 - \eta^2 - \frac{4\eta}{3} + \frac{2\eta}{3}\sqrt{12\eta - 2}.$$

This lower bound increases on $1/2 \le \eta \le 1$. Indeed, with $w = \sqrt{12\eta - 2} - 2$, it is $3/4 + w^2(4 - w^2)/144$, and $0 \le w^2 < 2$ on that interval. For $\eta \ge 3/5$ we obtain $\mathrm{Sec}(A, B, P) > m(3/5) = \mu_*$. □

G.2.2. *Rational bounds for the logarithms.*

Lemma G.17 (Logarithms)**.** *Let $0 < t < 1$, $x > 0$, $q > 1$, $c = 1 - t$, $W = 1+x$, $X = x+2$, $p = t+x$, $g = 1-cp/W^2$, and let $v = 1+c^2/W - c\log W/x$ be the normalized incoming secant.*

(a) $\log(1+x) > 2x/X + 2x^3/(3X^3)$*; hence* $v < \bar{v} = 1 + c^2/W - (2c/X)(1 + x^2/(3X^2))$.

(b) $\log q < L_+(q) = (q-1)(q^2+10q+1)/(6q(q+1))$*; hence* $F_{p/W}(q) > F_\ell = 1 + p^2q/W^2 - (p/W)(q^2+10q+1)/(6(q+1))$.

(c) With $\tau = x/X$, $\log(1+x) > L_J(x) := 2\tau\big(1 + \tau^2/[3(1-3\tau^2/5)]\big)$*; hence* $v < v_G = g + (cx/W^2)(T_G(x) - t)$ *with* $T_G = (90 - 19x^2)/(12(x^2+10x+15))$.

(d) $\log(1+x) > 3x(x+2)/(x^2+6x+6)$*; hence* $v < V = 1 + c^2/W - 3c(x+2)/(x^2+6x+6)$.

(e) $\bar{v} - g = ((1-t)x/W^2)(T(x) - t)$ *with* $T(x) = (-2x^3 - x^2 + 10x + 12)/(3(x+2)^3)$*; so* $v \geq g$ *forces* $t < T(x)$*. The function T decreases, and* $T(1) = 19/81 < 1/4$.

Proof. All the logarithm bounds follow from the reciprocal integral

$$\log\frac{1+\tau}{1-\tau} = 2\tau I(\tau), \qquad I(\tau) = \int_0^1 \frac{\mathrm{d}s}{1-\tau^2 s^2} \qquad (0 < \tau < 1).$$

For (a), the integrand exceeds $1+\tau^2 s^2$. For (b), it is less than $1+\tau^2 s^2/(1-\tau^2)$ for $0 < s < 1$. Integration gives $I(\tau) < 1+\tau^2/[3(1-\tau^2)]$; at $\tau = (q-1)/(q+1)$ this is the displayed upper bound.

For (c), write $I(\tau) = 1 + \tau^2 \int_0^1 s^2/(1-\tau^2 s^2)\,\mathrm{d}s$. Apply strict Jensen to $y \mapsto (1-\tau^2 y)^{-1}$ with probability density $3s^2$. The mean of s^2 is $3/5$, giving $I(\tau) > 1 + \tau^2/[3(1-3\tau^2/5)]$. As $c > 0$, this implies $v < v_J := 1 + c^2/W - cL_J(x)/x$. The required ceiling follows from

$$v_G - v_J = \frac{cx^6}{12W^2(x+2)(x^2+10x+10)(x^2+10x+15)} > 0.$$

For (d), Jensen with uniform measure gives $I(\tau) > 1/(1-\tau^2/3)$. The resulting logarithm bound also has the direct verification

$$\frac{\mathrm{d}}{\mathrm{d}x}\left(\log(1+x) - \frac{3x(x+2)}{x^2+6x+6}\right) = \frac{x^4}{(1+x)(x^2+6x+6)^2} > 0;$$

the difference vanishes at $x = 0$. Part (e) follows by subtraction and $T'(x) = -(11x^2+24x+16)/(3(x+2)^4)$. □

G.2.3. *The residual.* Put $c = 1-t$, $p = t+x$, $W = 1+x$, $Y = x(x+2)$, $B = k + 2tx + x^2$, $Z = qB/W$ (the normalized $d + r$), and

$$H_0 = \frac{k}{t}\,(v - 1 + t^2) + t(2-t), \qquad w_U = \frac{k^2 - kt + t^2 + H_0 Y + vY^2}{B^2},$$

the central diagonal and the transported outgoing norm with the incident norm at the boundary ceiling $U(t/k)$. The root function at the ceiling, divided by T, is

$$R = c + Z - \frac{p^2}{W} - \frac{p^2}{Z} + \frac{p^3}{B} - p\log q - Wv - Zw_U + \frac{p(H_0 + 2vY)}{B},$$

and $N = tBqW\,R$. Clearing denominators cancels every negative power, and N is a polynomial, affine in k, v and $L = \log q$:

- the coefficient of v is $N_v = -xq\,I$ with $I = k + tx + tW(k - 2t) + (q - 1)\big(kX + txX^2\big)$, nonpositive when $k \geq 2t$;
- the coefficient of L is $N_L = -tpqWB \leq 0$;
- the coefficient of k is

$$A = q\{x[(1 - v)(1 + tW) - t(1 - t^2)] \\ + (q - 1)[t^2(1 + 2x - x^2) + 2tx^2 + (1 - v)Y] - tpWL\}.$$

In A the coefficient of v is $-qx[(q - 1)(x + 2) + 1 + tW] < 0$ and that of L is $-qtpW < 0$. Inserting $v \leq U(t) = 1 - t + t^2$ and $L < q - 1$ and collecting gives $A > q\,t\,x(1 - t)[(q - 1)(2x + 1) + tx] > 0$, which is Lemma G.9.

Lemma G.18 (Rational relaxation)**.** *If $k \geq 2t$, $q > 1$ and $F_{p/W}(q) \leq v$, then*

$$N(v, \log q) \;\geq\; N(\bar v, L_+) - (q - 1)(\bar v - F_\ell),$$

and $P_k := 6(q + 1)W^2X^3\{N(\bar v, L_+) - (q - 1)(\bar v - F_\ell)\}$ is a polynomial in t, x, k and $y = q - 1$, cubic in y.

Proof. $N_v \leq 0$, $N_L \leq 0$, $v < \bar v$ and $\log q < L_+$, while $F_\ell < F_{p/W}(q) \leq v < \bar v$ makes the subtracted term nonnegative. □

G.2.4. *The necessary bounds.* We prove Lemma G.10(iii)–(vii).

The scale. At the centre, (G.1) holds with $u_{\mathrm{i}} = u$, $u_{\mathrm{o}} = v$, $r = a/S$ and $t = a/T$. Lowering both norms to a floor μ and raising both ratios to a cap ζ gives $\gamma^2 \geq (4\mu^2 - B(\rho)^2)/(16(1 - \rho)^2)$ with $B(\rho) = 1 - \rho + \log\rho + 2\psi_\mu(\zeta)$, $\psi_\mu(\zeta) = \zeta - \log\zeta - (1 - \mu)/\zeta$. The cleared comparison $\varphi(\rho) = 4\mu^2 - B(\rho)^2 - 16\,\Gamma_{\max}(1 - \rho)^2$, with $\Gamma_{\max} = 9193/50000$, has derivative $2(1 - \rho)(16\Gamma_{\max} - B/\rho)$ and increases, so its positivity at the left end of an interval of ρ excludes that interval by Lemma G.1(b). With the floor $3/4$ and the cap $1/2$ this gives $a^2 < 2/3$, that is $\rho < 1/7$. When $t \leq 2/5$, both ratios are at most $2/5$ and both turns have middle-margin ratio $B/(A + B) \geq 3/5$, so Lemma G.16(e) gives the intrinsic floor $\mu = \mu_*$. For completeness, put $\rho_0 = 4/29$, $\zeta = 2/5$ and

$$B_* = 1 - \rho_0 + 2\zeta - \frac{2(1 - \mu_*)}{\zeta} - \frac{2(\zeta^2 - \rho_0)}{\zeta^2 + \rho_0}.$$

The elementary bound $\log y < y - 1$ and $\mu_* > 3/4$ give $B(\rho_0) > 0$. The symmetric reciprocal bound for $\log(\zeta^2/\rho_0)$ then gives $B(\rho_0) < B_*$, and a

squaring in $\sqrt{130}$ gives

$$4\mu_*^2 - B_*^2 - 16\Gamma_{\max}(1-\rho_0)^2 > 0.$$

Moreover B increases, and the same logarithm bound, together with $\mu_* < U(\zeta)$, gives $B(3/20) < 16\Gamma_{\max}\rho_0$. Thus φ increases throughout $[\rho_0, 3/20]$, whose left endpoint is already positive. This excludes that interval and proves $a < 4/5$. In normalized form $a^2(k-t^2) = 4t^2$, so $a^2 < 2/3$ is $k > 7t^2$, and $a < 4/5$ is $k > 29t^2/4$.

Lemma G.19 (The centre gap)**.** *Every window of Proposition G.6 has $d^2 + 2cd > 4$, or equivalently $k < 2t + 2x + x^2$ in normalized coordinates.*

Proof. First raise the incident norm u to its ceiling at the actual margin b, and then raise b to $b_* = K_a/T - a$. The outgoing norm increases in u. For the second comparison put $S = a + b$ and $S_* = a + b_* = K_a/T$. Since $2a < S < S_*$, the diagonal at the upper incident norm satisfies

$$\begin{aligned} h_U(S_*) - h_U(S) &= \frac{K_a}{16}\left\{2a\left(\frac{1}{S_*} - \frac{1}{S}\right) - \log\frac{S}{S_*}\right\} \\ &\geq \frac{K_a}{16}\left(1 - \frac{S}{S_*}\right)\left(1 - \frac{2a}{S}\right) > 0. \end{aligned}$$

Here we used $\log y \leq y - 1$. Also $U(a/S_*) > U(a/S)$. The transport has positive coefficients in the incident norm and diagonal, so the actual outgoing norm satisfies $w < w_U(b_*)$.

Now use the normalized coordinates of Appendix G.2.3, and put $c = 1 - t$, $X = x + 2$, $Y = xX$, $p = t + x$, $W = 1 + x$ and $B = k + 2tx + x^2$. The strict three-arc inequality gives $tX < 1$. By Lemma G.16(d), the actual incoming secant is below

$$V_{\mathrm{ch}} = \frac{1}{2}\big(U(t) + g_c(p)\big), \qquad U(t) - V_{\mathrm{ch}} = \frac{cx(1 - tX)}{2W^2} > 0.$$

The transported ceiling increases in this norm:

$$\partial_v w_U = \frac{Y(k/t + Y)}{B^2} > 0.$$

Suppose that the asserted gap fails, so $k \geq k_0 := 2t + Y$. Set $r_0 = B/W - p$ and clear the natural norm denominator:

$$D(k, v) = B^2\big(g_p(r_0) - w_U(v)\big), \qquad g_p(r_0) = 1 - \frac{pW}{B} + \frac{p^2W^2}{B^2}.$$

The k^2 terms cancel. Thus D is affine in k, and direct differentiation of the transport gives

$$D_k = cx(1 + 2x) + \frac{Y}{t}\big(U(t) - v\big).$$

In particular, $D(k, V_{\mathrm{ch}})$ increases in k. At $k = k_0$ one has $B = 2pW$ and $r_0 = p$. Substitution gives the positive boundary identity

$$D(k_0, V_{\mathrm{ch}}) = \frac{cx^2}{2tW^2}\big\{xX^2(1 + t^2X) + tx^3(3x + 7) + 2t(x^2 - x + 1)\big\} > 0,$$

since $x^2 - x + 1 = (x - 1/2)^2 + 3/4 > 0$. It follows that $w_U(v) < w_U(V_{\mathrm{ch}}) < g_p(r_0)$.

But $r_0 - p = (k - k_0)/W \geq 0$, and the actual successor margin in these coordinates is $\hat{r} = qB/W - p > r_0$. The function g_p increases on $[p, \infty)$, so its switching inequality gives $w \geq g_p(\hat{r}) \geq g_p(r_0)$. This contradicts $w < w_U(v) < g_p(r_0)$ and proves the gap. □

Lemma G.20 (The ratio $t < 2/5$)**.** *Every window of Proposition G.6 has $t < 2/5$.*

Proof. Suppose $t \geq 2/5$. Normalize the margins by a, and write

$$C = \frac{c}{a}, \qquad D = \frac{d}{a}, \qquad K = \frac{a^2 + 4}{a^2}, \qquad v = \mathrm{Sec}(1, C, D).$$

The scale bound above gives $K > 7$. Also $1 < C \leq 3/2$ and $D < C^2$. By Lemma G.17(e), $t < T(x)$; since T decreases and $T(1/3) < 2/5$, we have $x < 1/3$ and hence $D = 1 + x/t < 11/6$. The centre gap of Lemma G.19 gives $D^2 + 2CD > K - 1 > 6$. As $C \leq 3/2$, this also implies $D > 4/3$. The incoming norm satisfies

$$\frac{3}{4} < v \leq g_C(1), \qquad v \geq g_C(D), \qquad F_{D/(C+D)}(q) \leq v, \qquad 1 < q < \frac{C + D}{D}.$$

The outgoing norm comparison. Put

$$W = C + D, \quad Q = D^2 + K - 1, \quad Y = (D - 1)(D + 2C + 1),$$

$$A = K^2 - K(1+C) + (1+C)^2, \qquad H = K(1+C)(v-1) + \frac{K}{1+C} + 1 + 2C.$$

Raising the incident norm to its ceiling and then raising b to b_*, as in the proof of Lemma G.19, gives $w < w_U$, where $w_U = (A + HY + vY^2)/Q^2$. Thus its switching floor gives

$$\Delta := Q^2 - DWQ/q + D^2W^2/q^2 - A - HY - vY^2 < 0.$$

Indeed, this is $Q^2\{g_D(qQ/W - D) - w_U\}$. The scale comparison is affine, with

$$\Delta_K = \frac{C(D-1)(C+2D-1)}{C+1} + (C+1)Y\big(g_C(1) - v\big) + DW(1 - 1/q) > 0.$$

We may therefore lower K to 7, preserving $\Delta < 0$.

The middle margin. Keep D, q fixed and raise C to $3/2$, always using the actual secant $v(C) = \mathrm{Sec}(1, C, D)$. Along this path $D < C^2$, so $v \le g_C(1)$. While $C \le D$, homogeneity gives

$$Cv_C = \frac{g_C(1) - v + D(v - g_C(D))}{D-1}.$$

The inequality $v \ge g_C(D)$ persists: at a first zero of its difference, this identity gives $v_C > 0$, whereas $\partial_C g_C(D) \le 0$. Hence $v_C \ge 0$ up to $C = D$. For $C \ge D$ the same sign follows pointwise from $\partial_C g_C(s) = s(C-s)/(C+s)^3 \ge 0$ on $[1, D]$.

The total C-derivative of Δ has negative coefficients of v and v_C. Its q-dependent part is $-DQ/q + 2D^2W/q^2$. This is convex in $1/q$, with maximum at $q = 1$, since $D(D^2 + 2CD - 6) > 0$. Lowering v to $3/4$ and v_C to zero therefore gives

$$\frac{\mathrm{d}\Delta}{\mathrm{d}C} \le -\frac{(D-1)Z(C,D)}{4(C+1)^2},$$

where, with $H_- = (C-3)/4 + 7/(C+1)$,

$$\frac{Z(C,D)}{4(C+1)^2} = 3Y + 2H_- + H_-'(D + 2C + 1) \\ - D^2 - (2C+1)D - 2C + 5.$$

This derivative comparison has a simple sign. Namely,

$$Z_D = 4(C+1)^2\left(4D + 4C - \frac{3}{4} - \frac{7}{(C+1)^2}\right) > 0.$$

At $D = 4/3$, $Z_{CC} = -40C - 2/9 < 0$, and substitution at $C = 1, 3/2$ gives positive values. Concavity then proves $Z > 0$ throughout the required region. Raising C to $3/2$ consequently decreases Δ.

When $D \ge 3/2$, the incoming floor also survives this change. Its secant interval has endpoints $A_q = (C+D)/q - D$ and C. Both increase with C, and g_D decreases on $[0, D]$. Thus $\mathrm{Sec}(A_q, D, C) = F_{D/(C+D)}(q)$ decreases, while v increases. We now have $C = 3/2$ and still $\Delta < 0$.

The boundary with $D \le 3/2$. Here $v \le g_{3/2}(1) = 19/25$. The transported norm increases in v, so it is enough to use that ceiling. With $Q = D^2 + 6$ and $Y = (D-1)(D+4)$, its cleared gap below $3/4$ is

$$F(D) = \frac{3}{4}Q^2 - \frac{151}{4} - \frac{13}{5}Y - \frac{19}{25}Y^2.$$

On $4/3 \le D \le 3/2$ this function is concave. Indeed,

$$F'' = \frac{3}{2}(4D^2 + 2Q) - \frac{26}{5} - \frac{38}{25}\big((2D+3)^2 + 2Y\big) < 0,$$

since $2D \le 3$, $Q \le 33/4$, $2D + 3 \ge 17/3$ and $Y > 0$. Both endpoint values of F are positive, so $w_U < 3/4$ on this interval. Since $g_D(qQ/W - D) \ge 3/4$, this gives $\Delta > 0$, a contradiction.

A variance ceiling on the other boundary. It remains to consider $3/2 \le D < 11/6$. Set $u = 2D - 3 \in [0, 2/3]$. For $f(D) = \mathrm{Sec}(1, 3/2, D)$, differentiation under the integral gives $f''' < 0$, because

$$g_C'''(s) = \frac{6C(s - 3C)}{(C+s)^5} < 0 \qquad (1 \le s \le D < 3C).$$

If $e_0 = f(3/2) - 3/4$, then $f'(3/2) = -2e_0$ and $f''(3/2) = 8e_0$. Taylor's bound and the positive variance integral give

$$f(D) \le \frac{3}{4} + e_0(1 - 2(D - 3/2) + 4(D - 3/2)^2),$$

$$e_0 = \frac{1}{4(C-1)} \int_1^C \frac{(C-s)^2}{(C+s)^2}\,\mathrm{d}s < \frac{(C-1)^2}{12(C+1)^2} = \frac{1}{300} \qquad (C = 3/2).$$

Consequently

$$v < V(u) := \frac{3}{4} + \frac{u^2 - u + 1}{300}.$$

Write $z = 1 - 1/q \in (0, 1)$ and $\varepsilon = 4(V - 3/4)$. The retained incoming floor is $\mathrm{Sec}(C - Wz, D, C) \le v < V$. The function g_D is strictly convex on $[0, C]$; Jensen's inequality for this secant therefore gives

$$J(u, z) := \varepsilon(2 - z)^2 - \left(z + \frac{2u}{u+6}\right)^2 > 0.$$

The quadratic minimum. Raise v to V, which lowers Δ, and denote the resulting comparison by $G(u, z)$. Here

$$D = \frac{u+3}{2}, \quad W = \frac{u+6}{2}, \quad Y = (D-1)(D+4), \quad M = Y(35/2 + Y) > 0.$$

Subtracting the incoming-floor slack with its variance multiplier gives

$$P(u, z) := G(u, z) - \frac{M}{4}J(u, z) = a_2(u)z^2 + b_2(u)z + c_2(u),$$

where

$$a_2 = D^2W^2 + \frac{M}{4}(1-\varepsilon),$$
$$b_2 = -DW(Y-2) + M\left(\varepsilon + \frac{u}{u+6}\right),$$
$$c_2 = G(u,0) - M\left(\varepsilon - \frac{u^2}{(u+6)^2}\right).$$

We show that its minimum $h = c_2 - b_2^2/(4a_2)$ is positive by convexity. All derivatives in the rest of the proof are with respect to u.

First $a_2 > 0$ and $0 < a_2'' < a_2'$. To see the derivative comparison in the displayed products, use $Y' = W$, $Y'' = 1/2$ and

$$M' < 28W, \qquad M'' < 12W, \qquad M''' = 3W,$$

which follow from $Y < 5$ and $3 \le W \le 10/3$. Together with $\varepsilon \le 1/75$, $\varepsilon' \le 1/225$ and $\varepsilon'' = 2/75$, they give

$$a_2''' = \frac{3}{2}(D+W) + \frac{1}{4}\left(M'''(1-\varepsilon) - 3M''\varepsilon' - 3M'\varepsilon''\right) > 0.$$

The endpoint comparisons $a_2''(0) > 0$ and $a_2''(2/3) < a_2'(0)$ now give $0 < a_2'' < a_2'$ throughout the interval. Put $\ell = a_2'(0)/a_2(0) > 1$. The logarithmic derivative $r = a_2'/a_2$ satisfies $r' < r - r^2$; comparison at the level ℓ therefore gives $r \le \ell$.

Next $b_2 < 0$ and $b_2'' > 0$. For an explicit check of the curvature, put $r_0 = 3(5/2)^3(7 - 5/2) > 0$. Division by W in the formula for b_2 gives

$$b_2'''' = \frac{3}{2}\varepsilon + 12W\varepsilon' + \frac{4}{25}M'' + \frac{3r_0}{2W^5} > \frac{8W^2 - 4W + 35 + 4Y}{25} > 0,$$

using $\varepsilon' \ge -1/75$. Now $b_2'''(0) > 0$ and $b_2''(0) > 0$ make b_2'' positive and increasing. Since $b_2'(2/3) < 0$, the function b_2 decreases from $b_2(0) < 0$. For the last coefficient, the same division gives

$$c_2'''' = -\frac{3}{2}W^2 + \frac{13}{4}W + \frac{3}{10} + \frac{3r_0(2W-15)}{2W^6} < 0.$$

The quadratic part is negative and decreasing for $W \ge 3$, and $2W \le 20/3 < 15$. Thus c_2'' is concave; its two endpoint values are positive, so $c_2'' > 0$.

It remains to compare the quotient curvature. Set $E = b_2' - b_2 a_2'/a_2$. Then

$$E' = b_2'' - E\frac{a_2'}{a_2} - b_2\frac{a_2''}{a_2}.$$

Substitution gives $-3 < E(0) < 3$ and $E(2/3) < 3$. At $E = -3$ the derivative is positive. At $E = 3$ it is again positive, since $b_2''(0) > 3\ell$ by direct substitution. Thus E cannot cross below -3, and an upward crossing of 3 could never return below it, contrary to the right endpoint. Hence $|E| \le 3$.

It follows that

$$\left(\frac{b_2^2}{a_2}\right)'' = \frac{2}{a_2}\left(E^2 + b_2 b_2'' - \frac{b_2^2 a_2''}{2a_2}\right) < \frac{2}{a_2}(9 + b_2(0)b_2''(0)) < 0,$$

where the last endpoint comparison again follows by substitution. Therefore $h'' > 0$. At the midpoint of $[0, 2/3]$, substitution gives

$$h'(1/3) < 0, \qquad h(1/3) + \frac{1}{3}h'(1/3) > 0.$$

The tangent at $1/3$ is consequently positive on the whole interval. Thus $h > 0$ and

$$P = a_2\left(z + \frac{b_2}{2a_2}\right)^2 + h > 0.$$

Since $G = P + (M/4)J > 0$, the original outgoing norm comparison is positive as well. This contradicts $\Delta < 0$ and proves the lemma. □

The duration $3d < 7c$. Since $v = \mathrm{Sec}(a, c, d) \le g_c(a)$ and $g_c \ge g_c(a)$ on $[0, a]$, extending the interval to $[0, d]$ raises the average: $\mathrm{Sec}(0, c, d) > v \ge g_c(d)$. With $h = d/c$ this reads

$$\log(1 + h) < \frac{h(1 + 2h)}{(1 + h)^2}.$$

The uniform Jensen bound of Lemma G.17(d) gives $\log(1 + h) > 3h(h + 2)/(h^2 + 6h + 6)$. Comparing the two positive rational expressions yields $h^2 - h - 3 < 0$, so $h < (1 + \sqrt{13})/2 < 7/3$.

The reverse order $d \le b_*$. Suppose $d > b_* \ge c$. By Lemma G.16(a),(c), $\mathrm{Sec}(a, d, c) > \mathrm{Sec}(a, c, d) \ge \mathrm{Sec}(B_0, d, c)$, and Lemma G.16(b) forces $B_0 > a$, that is $(a + d)(a + r) < K_a$; with $d > b_*$ this gives $r < c < d$. The reverse successor gives $w \le \mathrm{Sec}(a, r, d)$, and the outgoing floor is $\mathrm{Sec}(A_0, d, r)$ with $A_0 = (4 - cd)/(c + d)$. The same two steps force $A_0 > a$, but $d > b_*$ is equivalent to $A_0 < a$. In normalized form $b_*/T = k - t$ and $d/T = t + x$, so $d \le b_*$ is $k \ge 2t + x$.

G.2.5. *The reverse successor.*

Lemma G.21 (Unit shape)**.** *If* $0 < t \le 3/8$, $0 < x \le 1 - 2t$, $2t + 2x + x^2 > 1$, $q > 1$, $v = \mathrm{Sec}(t, 1 - t, t + x)$ *and* $F_{p/W}(q) \le v$, *then* $N(t, x, 1, q, v, \log q) > 0$.

Proof. Use the reciprocal-mean ceiling V of Lemma G.17(d), and put $\mathcal{D} = x^2 + 6x + 6$ and $r = 1 - 2t - x$. The shape inequalities give

$$\frac{1}{10} < x < 1, \qquad \max\{0, \tfrac{1}{4} - x\} \le r \le \min\{xW, 1 - x\}.$$

For fixed t, x, write $q = 1 + xz$ and define the cubic $f(z) = C + Az + Bz^2 + Gz^3$ by

$$x^2 f(z) = 24(q + 1)W^2\mathcal{D}\left\{N(t, x, 1, q, V, L_+) - (q - 1)(V - F_\ell)\right\}.$$

The constant coefficient has the positive factorization

$$C = 6W(W+r)^2\{xW(x^2+2x+2) + r(x^3+4x^2+8x+6)\}.$$

The quantities that control the cubic are its value at one and three related combinations:

$$P = f(1), \quad S = A + 2C - G, \quad K = 3P - S, \quad H = B + 4G - C/9.$$

We first establish

$$G > 0, \quad P > 0, \quad K > 0, \quad H \geq 0, \quad C + 8S > 0, \quad S + 32P > 0. \tag{G.10}$$

The order gap reduces these inequalities to boundary estimates. The functions C, P, S, K increase in r. For P, write

$$P = xW\{Wa_0 + a_1 r + W^2 a_2 r^2 + W^2 a_3 r^3\}.$$

Here a_1 has positive coefficients, $a_2 \geq 0$, and direct expansion gives

$$2a_2 + 3xWa_3 = 9x^6 + 84x^5 + 246x^4 + 261x^3 + 46x^2 + 24(1-x) \geq 0.$$

If $a_3 < 0$, use $r \leq xW$ in the derivative; if $a_3 \geq 0$, all its terms are nonnegative. For S, its linear and quadratic coefficients in r are at least $72W$ and 72, and its cubic coefficient is at least $-12xW^2$. Since $Wr \leq 1$, this gives

$$S_r \geq 72W - 36x > 0.$$

The same power bounds, $x^n \leq x^m$ for $n \geq m$ and $0 \leq x \leq 1$, give

$$K_r \geq -72 + 516x + 2928x^2 + 6352x^3 + 8316x^4 + 4412x^5 > 0;$$

the last inequality already follows from $x > 1/10$ and its first three terms. Consequently the inequalities for $P, C + 8S, S + 32P, K$ reduce to $r = \max\{0, 1/4 - x\}$.

On $r = 0$, divide P and $S + 32P$ by xW^2, and K by x^2W^2. The resulting polynomials increase for $x \geq 1/4$ and are positive at $1/4$. For the remaining combination, dropping higher powers gives

$$C + 8S \geq 2xW^2(983x^3 + 1005x^2 + 258x - 138) > 0.$$

Indeed, its negative terms of degrees five through eight are at least $-552x^4$; combining these with $411x^4 + 1124x^3$ leaves at least $983x^3$. The last cubic increases and is positive at $1/4$. On the cap edge $r = 1/4 - x$, the polynomial $-64P/(xW)$ has positive nonconstant coefficients and is negative at $1/4$. The polynomials $32(C + 8S)$ and $64(S + 32P)$ are concave on $[0, 1/4]$ and positive at both endpoints. For the former, the positive terms of its second derivative sum to less than 40000, whereas its negative constant term is -189864; for the latter, every term of the second derivative is negative. Finally $64K$ is concave on $[1/10, 1/4]$ and positive at both endpoints: its second derivative is at most $41856 - 910872x < 0$. This proves the four lower-boundary inequalities.

For H, expansion in r gives $H_{rr} \leq 0$ on $0 \leq r \leq xW$. To check the sign, the quadratic coefficient is at most a negative quadratic in x. If the cubic coefficient is positive, bound r by xW. The resulting upper bound is negative

on $[0,1/2]$ using $x^2 \le x/2$, $x^3 \le x/4$; on $[1/2,1]$, use $x^2 \le x$ and $x^4 \ge x^3/2$ instead. It therefore suffices to check the endpoints of the gap interval. On $r=0$, the only terms requiring attention in $H/(xW^2)$ are a positive multiple of

$$320 + 132x - 428x^2 = 320(1-x) + 24x + 428x(1-x) \ge 0.$$

On $r = 1/4 - x$, the polynomial $192H$ increases on $[1/10, 1/4]$ and is positive at $1/10$. Its derivative is positive because the negative terms of degree at least three have total magnitude at most $160000x$, whereas its positive linear term is $227864x$. On $r = xW$, the bracket after removing the positive factor is at least $200 - 192x > 0$. On $r = 1-x$, this edge is active only for $x > 2/5$, and

$$H = \frac{8}{3}(72x^6 + 408x^5 + 912x^4 + 810x^3 + 140x^2 - 91x - 6) > 0.$$

The last sign follows from $x^3 \ge 2x^2/5$ and $x^2 \ge 2x/5$. Thus $H \ge 0$ throughout the domain.

It remains to check G. Write $G = xF(r)$, with $F(r) = a + br + cr^2 + er^3$ and $Q(r) = a + br + cr^2$. For $x \le 1/3$, the coefficients satisfy $a \ge 36W^2$, $b \ge -80W$, $c \ge -16$, $e \ge 0$; hence

$$F(r) \ge W^2(36 - 80x - 16x^2) \ge 7W^2 > 0.$$

For $x \ge 1/3$, put $R = 1 - x$. Expansion gives $a > 0$ and

$$Q(R) \ge -18 + 303x^2 > 0, \qquad F(R) = 4x(12x^4 + 59x^3 + 107x^2 + 60x - 6) > 0.$$

On $[1/3, 1/2]$, one has $c \le 0$ and $e \ge 0$, so the concavity of Q proves $F > 0$. On $[1/2, 1]$, one has $b \ge 0$. If $e \ge 0$, interpolate Q using $R^2Q(r) = (R^2 - r^2)a + r^2Q(R) + bRr(R-r) > 0$. If $e < 0$, use instead

$$R^2F(r) = (R^2 - r^2)a + r^2F(R) + bRr(R-r) - eR^2r^2(R-r) > 0.$$

The endpoint values cover $r = 0$ as well. This proves (G.10).

We now recover positivity of the cubic. If $S < 0$, the last two inequalities imply $4CP - S^2 > 0$. Completing a square gives

$$4Cf(z) = \{2C(1-z) + Sz\}^2 + (4CP - S^2)z^2 + 4CGz(1-z)^2 > 0.$$

At $z = 0$, use $f(0) = C > 0$. If $S \ge 0$, use the following identities in the three consecutive ranges:

$$\begin{aligned}
f(z) &= (C + Gz)(1-z)^2 + Sz(1-z) + Pz^2, && 0 \le z \le 1,\\
f(z) &= C(z-1)^2 + z\{(3-2z)P + (z-1)K + G(z-1)^2\}, && 1 \le z \le \tfrac{3}{2},\\
f(z) &= \tfrac{C}{9}(z-3)^2 + z\{K/2 + (z - \tfrac{3}{2})H + G(z-2)^2\}, && z \ge \tfrac{3}{2}.
\end{aligned}$$

Every term has the required sign, and $C, P, K > 0$ give strictness at the endpoints. Finally, Lemma G.18, with V as the incoming ceiling, transfers this positivity to the exact secant and logarithm. □

Lemma G.22 (Order face)**.** *If* $0 < t < 2/5$, $x > 1/4$, $2t + x \ge 1$, $x < \frac{7}{3} - \frac{10}{3}t$, $v = \mathrm{Sec}(t, 1-t, t+x)$ *and* $1 < q \le (1+x)/(2t+x)$, *then* $N(t, x, 2t + x, q, v, \log q) > 0$.

Proof. Put $k = 2t + x$. Then $B = Wk$, $Z = qk$, $r = Z - p$, and with t, x, k, v fixed, $R'(q) = k(g_p(r) - w_U)$ and $R''(q) = k^2 p(r-p)/(r+p)^3 \le 0$, since $t \le r \le c \le p$ on the interval. So R is concave, and it suffices to check $q = 1$ and $q = W/k$.

At $q = 1$, $N = x\{(tW^2 + p)(1 - v) - 2t(1 - t)p\}$. Raise v to the Jensen ceiling V of Lemma G.17(d), which lowers this value. With $\mathcal{D} = x^2 + 6x + 6$, completing a square gives

$$N(t, x, 2t + x, 1, V, 0) = \frac{(1-t)x^3}{W\mathcal{D}^2}\left\{\left(\mathcal{D}t - 3x - 4\right)^2 + 2W^3\right\} > 0.$$

At $q = W/k$ replace $v, \log q$ by $\bar{v}, L_+$, which lowers R. With $h = 1 - 2t$, $s = k - 1$ and $x = h + s$ the domain is $\frac{1}{5} < h < 1$, $0 \le s < \frac{4}{3}$, $h + s > \frac{1}{4}$, and

$$\bar{R}(W/k) = \frac{P(h,s)}{24tk^2W^2(W+k)X^3}, \qquad P = \sum_{j=0}^{7} C_j(h)s^j,$$

$$C_0 = h^3(1+h)^2F(h), \qquad F = -16+24h+120h^2+140h^3+78h^4+23h^5+3h^6,$$

$$C_1 = h^3(1+h)(8+396h+1020h^2+1080h^3+597h^4+179h^5+24h^6),$$

and $C_2, \dots, C_7$ are polynomials in h whose only negative monomials are $-70h^2$ in C_5, $-108h^2$ in C_6, and $-4h$, $-20h^2$ in C_7. These are absorbed: $184hs^5 - 70h^2s^5 \ge 0$, and $316hs^4 - 108h^2s^6 - (4h + 20h^2)s^7 \ge hs^4(316 - 108 \cdot \frac{16}{9} - 24 \cdot \frac{64}{27}) \ge 0$, the term $316hs^4$ being part of C_4s^4. So $P \ge C_0 + C_1s$. For $h \ge \frac{1}{4}$, the factorization

$$F(h) = 8(4h-1)(5h^2+5h+2) + h^3(78h + 23h^2 - 20) + 3h^6$$

is positive: the first two terms are nonnegative and the last is positive. Indeed, $78h + 23h^2 - 20 \ge \frac{39}{2} + 1 - 20 > 0$.

For $\frac{1}{5} \le h \le \frac{1}{4}$, the order gap gives $s > \frac{1}{4} - h$. Since $C_1 > 0$, it suffices to bound $C_0 + C_1(\frac{1}{4} - h) = \frac{h^3(1+h)}{4}Q(h)$, where

$$Q(h) = -56 + 396h + 12h^2 - 1960h^3 - 2851h^4 - 1805h^5 - 588h^6 - 84h^7.$$

The upper bound $h \le \frac{1}{4}$ gives $2851h \le 713$, $1805h^2 \le 113$ and $588h^3 + 84h^4 < 10$. Thus

$$Q(h) \ge -56 + 396h + 12h^2 - 2796h^3 = 4h^3 + 16(\tfrac{1}{4} - h)(175h^2 + 43h - 14) > 0,$$

because $175h^2 + 43h - 14 \ge 7 + \frac{43}{5} - 14 > 0$. □

Lemma G.23 (Scale strip)**.** *If $\frac{3}{8} \le t \le \frac{2}{5}$, $0 < x \le \frac{1}{4}$, $\frac{29}{4}t^2 < 2t + 2x + x^2$, $q > 1$, $v = \mathrm{Sec}(t, 1-t, t+x)$ and $F_{p/W}(q) \le v$, then $N(t, x, \frac{29}{4}t^2, q, v, \log q) > 0$.*

Proof. Use the reciprocal-mean ceiling V and the normalization of Lemma G.21, with $\mathcal{D} = x^2 + 6x + 6$. Write the cleared residual on $k = 29t^2/4$ as $\sum_{j=0}^{3} C_j(t, x)y^j$, where $y = q - 1$, and put $t_0 = 3/8$. Each coefficient increases with t on $3/8 \le t \le 2/5$, $0 \le x \le 1/4$. Indeed, write

$\partial_t C_0 = x^2 D_0$, $\partial_t C_1 = x D_1$, and $\partial_t C_2 = D_2$, $\partial_t C_3 = D_3$. The elementary bounds

$$\frac{9}{64} \le t^2 \le \frac{4}{25}, \qquad \frac{27}{512} \le t^3 \le \frac{8}{125}$$

make every coefficient of D_0, D_1 nonnegative. For D_2, the constant coefficient is at least 120, the next three are nonnegative, and the coefficients of degrees four through eight are at least -2400. For D_3, the corresponding bounds are 160, four nonnegative coefficients, and coefficients of degrees five through eight at least -400. Since $\sum_{i=0}^{4} x^i \le 4/3$,

$$D_2 \ge 120 - 2400x^4 \sum_{i=0}^{4} x^i > 100, \qquad D_3 \ge 160 - 400x^5 \sum_{i=0}^{3} x^i > 150.$$

Thus the cleared residual is nondecreasing in t for every $y \ge 0$.

The function $29t^2/4 - 2t$ increases on the strip, so its defining inequality gives

$$2x + x^2 > \frac{29}{4} t_0^2 - 2t_0 = \frac{69}{256} > \frac{1}{4}.$$

Hence (t_0, x) satisfies the unit-shape inequalities. The polynomial part of Lemma G.21 gives strict positivity there at $k = 1$. Moreover, $V < U(t_0)$ follows from

$$U(t_0) - V = \frac{(1 - t_0)x}{2W\mathcal{D}} \{ Wx(x + 4) + \mathcal{D}(1 - 2t_0 - x) \} > 0.$$

Shape monotonicity therefore raises k to $29t_0^2/4 = 261/256 > 1$. Coefficient monotonicity transports this positive value to the original t. Apply Lemma G.18 at that point, where the control floor holds, to obtain $N > 0$. □

Lemma G.24 (Low-scale unit face)**.** *If* $0 < t \le \frac{1}{4}$, $\frac{1}{2} < x \le \frac{7}{3}$, $x \ge 1 - 2t$, $t \le T(x)$, $q > 1$, $v = \mathrm{Sec}(t, 1 - t, t + x)$ *and* $F_{p/W}(q) \le v$, *then* $N(t, x, 1, q, v, \log q) > 0$.

Proof. With $I_0 = 1 + tx + tW(1 - 2t)$, $I_1 = X + txX^2$ and $\mathcal{A} = xq(I_0 + I_1 y) = -N_v$, one has $N > N(\bar{v}, L_+) - \mathcal{A}(\bar{v} - F_\ell)$, and $Q = 6(q + 1)W^2 \{ N(\bar{v}, L_+) - \mathcal{A}(\bar{v} - F_\ell) \}$ is a quartic $P_0 + P_1 y + \cdots + P_4 y^4$. It is positive for $y \ge 0$ when $P_0 > 0$, $P_1 + 2P_0 > 0$, $P_2 + 48P_0 > 0$, $P_3 + 4P_2 > 0$, $P_3 - P_0 > 0$ and $P_4 - P_0 > 0$; the extreme cases reduce to $1 - 2y - \frac{1}{4}y^2 + y^3 + y^4 = \frac{4}{81} + \frac{5}{27}(y - \frac{2}{3}) + (\frac{29}{12} + \frac{7}{3}y + y^2)(y - \frac{2}{3})^2$ and $1 - 2y - 48y^2 + 192y^3 = \frac{2}{9} - 2(y - \frac{1}{6}) + (16 + 192y)(y - \frac{1}{6})^2$, both positive. The first four signs factor as $4(1 - t)x^2 F_0/X^3$, $2(1 - t)xF_1/X^3$, F_2/X^3, F_3/X^3 with polynomials $F_i(t, x)$, and $-\partial_t F_i$ is smallest at $t = \frac{1}{4}$ and positive there (in the variables $1 - 4t$ and $2x - 1$ its coefficients are nonnegative after absorbing a few negative terms), so $F_i(t, x) \ge F_i(\frac{1}{4}, x) + (\frac{1}{4} - t)(-\partial_t F_i(\frac{1}{4}, x))$. For $\frac{1}{2} \le x \le 1$ the values $F_i(\frac{1}{4}, x)$ are positive one-variable polynomials. For $1 \le x \le \frac{7}{3}$, where $t < T(x) < \frac{1}{4}$, the tangent value at $t = T(x)$ is a positive multiple of an explicit polynomial $h_i(x)$, positive on $[1, \frac{7}{3}]$ after charging its few negative high-degree terms to positive lower ones. The comparisons $P_3 > P_0$ and $P_4 > P_0$ are proved in the same way. □

G.2.6. *The forward successor.* In this subsection write $o = 2t + x$, $j = o - 1$, $h = 1-2t$, $y = q-1$, $e = q/W$, $J = t+Y$, $D = eJ-p$ and $E = e^2B+1-2pe$. The next norm at the ceiling is $\mathcal{T}_U = (v - e^2(H_0 + 2vY) + e^4B^2w_U)/E^2$, and the combined residual (G.9) has $C_* = tW^2DE^2/e$; it is a concave quadratic in k with leading coefficient $-tW^2De^3/4$.

Lemma G.25 (Low ratio)**.** *Let $\frac{1}{4} \le t \le \frac{1}{3}$, $1 - 2t \le x \le 1$, $1 \le k \le o$, $1 \le q \le W/o$, and let $\widehat{G} = \mathcal{H}(\hat{v}, L_+)$, with $\alpha = 6$, $M = 0$, $\hat{v} = v_G + \rho(v_G - F_\ell)$ and $\rho = \frac{3}{4}(q+o-1)$. Then $\widehat{G} > 0$. Consequently every window with a forward successor has $t > 1/3$.*

Proof. At fixed (t, x, q) the coefficients of v and $\log q$ in $\mathcal{H}$ are negative: $\mathcal{H}_v < -\frac{17}{12}xq(I_0 + I_1y)$, from $Je^2 \le \frac{10}{3}$ and $WD/x \le \frac{9}{4}$, and $\mathcal{H}_{\log q} = (7 - Je^2)N_L < 0$. So a window with a forward successor and $t \le \frac{1}{3}$, which lies in the domain (with $q < W/(1 + 2j) \le W/o$), has $\widehat{G} \le \mathcal{H}(v, \log q) < 0$.

The three-term secant majorant $v_3 = 1 + c^2/W - (2c/X)(1 + x^2/(3X^2) + x^4/(5X^4))$ satisfies $v_3 - v_G = cx^6(31x^2+82x+46)/(60W^2X^5(x^2+10x+15)) > 0$, and $v_3 + \rho(v_3 - F_\ell) < U(t)$ (using $F_\ell - g = \eta(\eta - \frac{1}{2})y + \eta y^2/(3(y+2)) \ge 0$ with $\eta = p/W \ge \frac{1}{2}$, and $\rho < 2$); so $\hat{v} < U(t)$. Also $\mathcal{H}_{kv} < 0$ and $\mathcal{H}_{k,\log q} < 0$, so $\widehat{G}_k > \mathcal{H}_k(U(t), y)$, which decreases in k. At $k = o$,

$$\mathcal{H}_k(U(t), y)\big|_{k=o} = \frac{tq}{2W^2}(d_0 + d_1y + d_2y^2 + d_3y^3),$$

with $d_0 = xc\{1 - 2t + x(-2t^2 + 14t - 1) + 24tx^2 + 12tx^3\} > 0$, $d_3 = xWJ > 0$, $d_2 < 0$ (every coefficient in x is negative for $\frac{1}{4} \le t \le \frac{1}{3}$), and $2d_1 + d_2 > 0$; as $y \le \frac{1}{2}$, $d_1 + d_2y \ge d_1 + \frac{1}{2}d_2 > 0$. Hence $\widehat{G}_k > 0$, and it suffices to show $\widehat{G} > 0$ at $k = 1$.

For $k = 1$, put $t = \frac{1-h}{2}$, $x = h + j$ and $q = 1 + y$ on $\frac{1}{3} \le h \le \frac{1}{2}$, $j \ge 0$, $0 \le y \le h$, a region that contains the domain, the polynomial $\Pi = 192q(q + 1)W^3(x^2 + 10x + 15)\,\widehat{G}(t, x, q, 1)$ equals $\sum_i \tilde{C}_i(h, y)j^i$. Its part of degree at most two in j is $h^3w^3(\mathcal{A}_0 + \beta\mathcal{A}_1 + w\beta^2\mathcal{A}_2)$ with $w = 1 + h$ and $\beta = j/(hw)$; each $\mathcal{A}_i$ increases in h, its h-derivative being a polynomial with nonnegative coefficients after factoring, and at $h = \frac{1}{3}$ the quadratic in β exceeds 4 by three square identities. The coefficients $\tilde{C}_3, \ldots, \tilde{C}_{10}$ are nonnegative from their factored forms. Hence $\Pi > 4h^3w^3 + 700j^3 + 3900j^4 > 0$. □

Lemma G.26 (Large gap)**.** *Every window with a forward successor has $j < 1/30$.*

Proof. Suppose $j \ge \frac{1}{30}$. By Lemma G.25, $t > \frac{1}{3}$; Lemma G.17(c) with $v \ge g$ gives $t < T_G(x)$, hence $x < \frac{3}{5}$ and $7x < 12h$, that is, $j < \frac{5}{7}h$.

For $\frac{1}{3} \le t \le \frac{3}{8}$, take $\alpha = \frac{7}{2} - 10y \ge \frac{11}{16}$ and $M = 8j + 8y - \frac{1}{6} \ge \frac{1}{10}$. The v-coefficient of $\mathcal{H}$ is negative, so $v \mapsto V_L$ (a rational majorant of v), $\log q \mapsto L_+$ and $F \mapsto F_\ell$ keep $\mathcal{H} < 0$. The resulting residual increases in k on $[1, o]$: it is quadratic in k, its shape derivative at the midpoint of two

shapes is affine in v, $\log q$ and k with nonpositive slopes, and at the extreme values it equals $q^2S/(12W)$ with a quartic S in y that is concave in y and positive at both ends. At $k = 1$ a positive multiple S of the residual, a polynomial in (h, j, y), increases in h at fixed (j, y): a cubic Taylor bound from $\partial_h^4 S \geq 500000$ and two derivative signs at $h = \frac{1}{4}$ give $\partial_h S > 0$. So it suffices to check the three faces $h = \frac{1}{4}$, $j = \frac{5}{7}h$ and the largest y, where $S > 0$ by termwise monomial bounds after a shift of the variable.

For $\frac{3}{8} \leq t < \frac{2}{5}$, take $\alpha = 4$, $M = 1$ and $v \mapsto v_G$. The residual increases in k on $(\frac{29}{4}t^2, o)$, by the same midpoint argument with endpoint value $2tq^2(d'_0 + d'_1y + d'_2y^2 + d'_3y^3)$, a positive cubic. At $k = \frac{29}{4}t^2$ a positive multiple S satisfies $-\partial_h^2 S > 0$, and $-\partial_h S > 0$ at $h = \frac{1}{5}$, which reduces the sign to $h = \frac{1}{4}$; there S is concave in y, and along the remaining line a ninth-derivative bound and exact values at the ends give $S \geq \frac{13}{4}$. In both cases $\mathcal{H} > 0$, a contradiction. □

Lemma G.27 (The ratio χ)**.** *If $\frac{1}{3} \leq t \leq \frac{19}{50}$, $0 \leq j \leq \frac{1}{30}$, $k \geq 1$, $q > 1$, $v = \mathrm{Sec}(t, 1-t, t+x) \leq U(t)$ and $N(t, x, k, q, v, \log q) < 0$, then $\chi = pq/W > \frac{4}{7}$.*

Proof. Lower k to 1 by Lemma G.9, and raise v to the secant ceiling

$$V_2 = 1 - \frac{cp}{W^2} + \frac{cx}{W^2}\left(\frac{1}{2} - \frac{x}{3} + \frac{x^2}{12} - t\right).$$

Both operations lower N. We prove that the resulting function of q is concave and positive at the two ends of the interval in question.

Suppose $1 < q \leq q_\chi = 4W/(7p)$. Since $j \geq 0$, one has $q_\chi \leq 8/7$. Also $V_2 \geq 3/4$: its first two terms equal $3/4 + j^2/(4W^2)$, and

$$\frac{1}{2} - \frac{x}{3} + \frac{x^2}{12} - t = \frac{h - 2j}{6} + \frac{x^2}{12} > 0.$$

Write the lowered residual as $N_1(q) = A_2q^2 + A_1q - tp^2W^2 - Cq\log q$, where $C = tpBW > 0$. Its second derivative is $2A_2 - C/q$. The coefficient of v in A_2 is $-Y - tY^2 < 0$, so it suffices to check $7C - 16A_2 > 0$ at $v = 3/4$. At that value, $7C - 16A_2 = \sum_{i=0}^4 b_i(h)j^i$, where

$$b_0 = -\tfrac{h+1}{4}(16h^4+23h^3-9h^2-7h+1), \qquad b_1 = -\tfrac{h+1}{4}(87h^3+12h^2-55h-12),$$

$$b_2 = -\tfrac{1}{2}(66h^3 + 33h^2 - 58h - 33), \qquad b_3 = \tfrac{67}{4}(1 - h^2), \qquad b_4 = \tfrac{3}{2}(1 - h),$$

These coefficients are positive for $1/5 \leq h \leq 1/3$. For example, using $h^{m+1} \leq h^m/3$, the polynomial in b_0 is at most $1 - 185h/27 < 0$. The polynomials in b_1 and b_2 are at most $-124h/3 - 12$ and $-119h/3 - 33$, respectively. Thus N_1 is concave on $[1, 8/7]$.

At $q = 1$, write $48W^2N_1(1) = (h + 1)^2R(h, j)$. Expansion and absorption of the negative terms give

$$R(h, j) \geq 2h^3 - 6hj^2 - 10j^3 - 5j^4 - j^5 \geq 2h^3 - 6hj^2 - 16j^3 > 0.$$

The last inequality follows from $h^3 \geq h/25$, $j^2 \leq 1/900$ and $j^3 \leq 1/27000$.

At $q = q_\chi$, replace $\log q$ by its upper bound $L_+(q)$. The resulting residual is

$$\frac{P(h,j)}{1176(h+2j+1)^2(15h+22j+15)}, \qquad P(h,j) = \sum_{i=0}^{8} a_i(h) j^i.$$

The constant and linear coefficients increase with h on $[6/25, 1/3]$. Indeed, after discarding positive terms, their derivatives are bounded below by

$$a_0'(h) \geq 21140h^3 + 2055h^2 - 414h - 35 > 0,$$
$$a_1'(h) \geq 174325h^4 + 84328h^3 + 8877h^2 - 4932h - 941 > 0.$$

These signs follow by successively using $h^{m+1} \geq h^m/5$ in the first line and $h^{m+1} \geq (6/25)h^m$ in the second. Their values at $h = 6/25$ give $a_0 \geq 18$ and $a_1 \geq -400$. Bounding the remaining powers of h by those of $1/3$ gives

$$a_2 \geq -700, \qquad a_3, a_4 \geq 0, \qquad a_8 \geq 0,$$
$$a_5 \geq -11000, \qquad a_6 \geq -19000, \qquad a_7 \geq -6000.$$

Consequently,

$$P(h,j) \geq 18 - 400j - 700j^2 - 36000j^5 \geq 18 - 450j \geq 3,$$

where $j^2 \leq j/30$ and $j^5 \leq j/30^4$ absorb the tail. Both endpoint values are therefore positive. Concavity gives $N_1(q) > 0$ throughout $[1, q_\chi]$, contradicting $N < 0$. □

Lemma G.28 (Casimir slope)**.** *Let $\frac{1}{3} \leq t \leq \frac{2}{5}$, $0 < j < \frac{1}{4}$, $k \geq 1$, $q > 1$, $\chi \geq \frac{4}{7}$, $v \geq \frac{3}{4}$, $F_{p/W}(q) \leq v$, and let $r = qB/W - p$ satisfy $t \leq r \leq p$. With $H_d(w) = BL' + \frac{e}{p}B^2(w-1) + p(1-\chi)/e$ and $L' = \log q + (Wv - c + p^2/W)/p$, one has $2v - \frac{e}{p}H_d(w) < 0$ for every $w \geq \frac{3}{4}$; so $\Gamma(w) = (4B^2vw - H_d(w)^2)/(16(k-t^2)^2)$ decreases on $[\frac{3}{4}, \infty)$.*

Proof. $\Gamma'(w) = \frac{B^2}{8(k-t^2)^2}(2v - \frac{e}{p}H_d(w))$ and H_d increases in w, so it suffices to take $w = \frac{3}{4}$. With $A = Z/p = 1 + r/p$ the hypotheses give $\frac{15}{11} \leq A < 2$ and $A/\chi = B/p^2 > 2$, and $j < \frac{1}{4}$ gives $k - t^2 - p^2 > \frac{7}{144}$. The floor bounds $\log q$ below in terms of v, and $2v - \frac{e}{p}H_d(\frac{3}{4})$ is then bounded above by a convex quadratic in one ratio, negative at both ends of its range. □

Lemma G.29 (Eliminating the event factor)**.** *Let $\frac{1}{3} \leq t \leq \frac{2}{5}$, $0 \leq j \leq \frac{1}{30}$, $k \geq 1$, $v \leq \min\{1, V\}$, $\eta = p/W$, $\alpha_U = B(w_U - 1)/(pW)$ and $\beta_U = pW/B$. Then for $q > 1$,*

$$\log q + \alpha_U(q-1) + \beta_U(1/q - 1) < F_\eta(q) - U(\eta).$$

Proof. The comparison concerns the change from the boundary value $q = 1$. Put $y = q - 1$. By Lemma G.17(b), the difference between the right and left sides exceeds

$$\frac{y}{6(y+1)(y+2)}(C_0 + C_1 y + C_2 y^2)$$

with $C_0 = 12\eta^2 - 6\eta - 12\alpha_U + 12\beta_U - 12$, $C_1 = 18\eta^2 - 7\eta - 18\alpha_U + 6\beta_U - 12$ and $C_2 = 6\eta^2 - \eta - 6\alpha_U - 1$. Each coefficient increases in k: indeed, $\partial_k \alpha_U < \partial_k \beta_U = -pW/B^2 < 0$. Each decreases in v, since $\partial_v \alpha_U = Y(k/t + Y)/(BpW) > 0$. It therefore suffices to take $k = 1$ and $v = V$.

The boundary gap. Write $\alpha = \alpha_U|_{k=1,v=V}$ and $\beta = \beta_U|_{k=1}$, and set

$$\rho = \eta^2 - \frac{\eta}{2} - \alpha + \beta - 1, \qquad L = 12\beta - 6 - 2\eta, \qquad \varepsilon = \eta - \frac{1}{2}.$$

Thus $C_0 = 12\rho$: ρ measures the gap left at the boundary. The geometric ratios satisfy

$$\frac{1}{2} \le \eta \le \frac{31}{60}, \qquad \frac{3}{5} \le \beta \le \frac{7}{10},$$

and hence $\varepsilon \ge 0$ and $0 < L \le 7/5$. For the lower bound on β, put $h = 1 - 2t$ and use

$$\beta - \frac{1+h}{2} = \frac{j\{2 + (h+j)(1-h)\}}{2B} \ge 0.$$

For the upper bound, clear the denominator and note that

$$7B - 10pW = 2 - 5h^2 - 3h - 8hj - 3j^2 - 8j > 0$$

on $1/5 \le h \le 1/3$, $0 \le j \le 1/30$. The bounds on η follow from $\eta - 1/2 = j/(2W)$.

We claim that the boundary gap controls the displacement:

$$L^2 < 10(12 + L)\rho. \tag{G.11}$$

To prove this, put $D = x^2 + 6x + 6$ and clear the positive denominator:

$$P(h, j) = 8tpW^3B^2D\{10(12 + L)\rho - L^2\}, \qquad t = \frac{1-h}{2}, \quad x = h + j.$$

The expression P is a polynomial. It is concave in the short gap j. Indeed, its first five coefficients in P_{jj} are nonpositive, using $h^{m+1} \le h^m/3$, and its last four terms give

$$P_{jj} \le -100000j^5 + 400000j^6 + 200000j^7 + 30000j^8 \le 0.$$

The last inequality uses $j^6 \le j^5/30$ and $j^7, j^8 \le j^6$. It remains to examine the two ends of the gap interval.

At $j = 0$ one has $P(h, 0) = 2(h + 1)^5 g(h)$, where

$$g(h) = 30h^7 + 211h^6 + 604h^5 + 559h^4 - 50h^3 - 139h^2 + 66h - 6.$$

For $h \ge 1/5$,

$$g''(h) \ge 12000h^3 + 6700h^2 - 300h - 280 \ge 0.$$

Since $g(1/5) > 2$ and $g'(1/5) > 27$, the polynomial g increases and remains positive.

At the other end, put $E(h) = P(h, 1/30)$. For $1/5 \le h \le 1/4$, absorption of the higher powers using $h^{m+2} \le h^2/4^m$ gives

$$E'(h) \le 100 - 180h - 8000h^3 \le 0.$$

For $h \geq 1/4$, the lower powers alone give positive curvature:

$$\begin{aligned} E''(h) &\geq -200 - 10000h - 26000h^2 + 100000h^3 + 600000h^4 + 1200000h^5 \\ &\geq 12000h^2 > 0. \end{aligned}$$

The supporting tangent at $h = 2/7$ is positive on this branch, because

$$E(2/7) > 1, \qquad 0 < E'(2/7) < 12, \qquad h - 2/7 \geq -1/28.$$

Convexity therefore gives $E(h) \geq 4/7$ for $h \geq 1/4$. On the left branch, monotonicity gives $E(h) \geq E(1/4) > 0$. Thus both endpoint values of P are positive. Concavity in j proves $P(h, j) > 0$, which establishes (G.11).

Completing the square. In these variables the quadratic is

$$C_0 + C_1 y + C_2 y^2 = 12\rho + (18\rho - L)y + Ay^2, \qquad A = \frac{5 - L}{2} + \varepsilon + 6\rho.$$

The gap inequality gives $\rho > 0$, and hence $A > 0$. If $18\rho - L \geq 0$, all three terms are nonnegative and the constant term is positive. Otherwise $L - 18\rho > 0$, and

$$48\rho A - (18\rho - L)^2 = \{10(12 + L)\rho - L^2\} + 2\rho(L - 18\rho) + 48\varepsilon\rho > 0.$$

Multiplying the quadratic by $4A$ now gives the sum of this positive quantity and the square $(2Ay + 18\rho - L)^2$. This proves the required strict inequality. □

Lemma G.30 (The envelope Casimir)**.** *On* $\frac{1}{3} \leq t \leq \frac{2}{5}$, $0 \leq j \leq \frac{1}{30}$, $1 \leq k \leq \frac{31}{30}$, $\frac{3}{4} \leq v \leq V$,

$$\mathcal{G} = \frac{B^2\{4vw_U - (\phi(1) + v - g)^2\}}{16(k - t^2)^2} > \frac{3}{16} - \frac{(1 - 2t)^2}{32},$$

where $\phi(1) = \alpha_U + \beta_U + (Wv - c + p^2/W)/p - p^2/B$.

Proof. The numerator $\mathcal{N}(v) = B^2\{4vw_U - (\phi(1) + v - g)^2\}$ is a quadratic in v. The numerator of w_U has v-derivative $P_v = Y(k/t + Y)$, and $D_1(v) = B(\phi(1) + v - g)$ has $D_1' = B(1 + W/p) + P_v/(pW)$, so the arithmetic–geometric mean inequality and $B > p^2$ give $D_1'^2 > 4P_v$ and $\mathcal{N}'' = 8P_v - 2D_1'^2 < 0$.

We next show $\mathcal{N}'(3/4) < 0$ by bounding the transported norm from below. Put $w_0 = w_U|_{v=3/4}$ and $f(x) = 12tB^2(w_0 - 2/3)$, keeping t, k fixed. Direct differentiation of the transport gives

$$f(0) = C, \qquad f'(0) = -2D, \qquad f''(x) = 2E + 24t(9 - 8t)x + 12tx^2,$$

where

$$\begin{aligned} C &= 4t(k^2 - 3kt + 3t^2), \\ D &= 4kt^2 + 3k + 12t^3 - 24t^2, \\ E &= k(12t^2 - 16t - 3) - 44t^3 + 24t^2 + 36t. \end{aligned}$$

On $1/3 \leq t \leq 2/5$, $1 \leq k \leq 31/30$, C increases in k and $\partial_t C|_{k=1} = 4(3t - 1)^2$, so $C \geq 4/9$. The function D increases in k and decreases in t, while E

decreases in k and increases in t. Evaluation at $(t,k) = (1/3, 31/30)$ gives $D < 7/5$ and $E > 5$. Hence, for every $x \geq 0$,

$$f(x) \geq \frac{4}{9} - \frac{14}{5}x + 5x^2 > 0.$$

The last sign follows by AM–GM, since $(4/9)5 > (7/5)^2$. In particular $w_0 > 2/3$ throughout the required domain.

Write $\eta = p/W$, $\beta = pW/B$ and $m = P_v/B^2$. The geometric ratios satisfy

$$\frac{1}{2} \leq \eta \leq \frac{19}{37}, \qquad \frac{4}{7} \leq \beta \leq \frac{3}{4}, \qquad 0 \leq m \leq 2.$$

Indeed, $p = c + j$, $W = 2c + j$ and $c \geq 3/5$ give the first bounds. For the lower bound on β, clear the positive denominator:

$$7pW - 4B = c(14c - 8) - 4(k - 1) + j(13c + 3j) > 0.$$

The right side is smallest at $c = 3/5$, $k = 31/30$, $j = 0$, where it is positive. For the upper bound, use $x \leq 11/30$ in

$$4pW - 3B = 4t + (4 - 2t)x + x^2 - 3k \leq \frac{8}{5} + \frac{10}{3}\frac{11}{30} + \left(\frac{11}{30}\right)^2 - 3 < 0.$$

To bound m, put $\theta = Y/B$ and $\kappa = k/(tB)$. Then $3Y - 2B = x^2 + (6 - 4t)x - 2k < 0$, again using $x \leq 11/30$, so $\theta < 2/3$. Also $2t + x \geq 1$, $x \geq 1 - 2t$ give

$$\frac{tB}{k} = t + \frac{tx(2t + x)}{k} \geq t + \frac{30}{31}t(1 - 2t) > \frac{3}{7}.$$

The middle expression is concave in t, so its minimum is at an endpoint. Thus $\kappa < 7/3$ and $m = \theta(\kappa + \theta) < 2$.

For these fixed ratios put

$$d(w) = \frac{w - 1}{\beta} + \beta(1 - \eta) - \frac{1}{4\eta} + \frac{3}{4} + 2\eta - \eta^2,$$
$$\Omega(w) = 4w + 3m - 2d(w)(1 + 1/\eta + m/\beta).$$

The original envelope definitions give $\mathcal{N}'(3/4)/B^2 = \Omega(w_0)$. Since $\Omega'(w) = 4 - 2(1 + 1/\eta + m/\beta)/\beta < 0$, we may lower w_0 to $2/3$. The function $d(2/3) - \beta$ increases in η: its derivative is $-\beta + 1/(4\eta^2) + 2 - 2\eta > 0$ even on $1/2 \leq \eta \leq 2/3$. At $\eta = 1/2$ it is $1 - \beta/2 - 1/(3\beta)$. Its difference from the value $11/84$ at $\beta = 4/7$ is

$$(\beta - 4/7)\left(\frac{7}{12\beta} - \frac{1}{2}\right) \geq 0,$$

so it exceeds $1/8$. Consequently

$$\frac{\mathcal{N}'(3/4)}{B^2} < \frac{8}{3} - 2(\beta + \tfrac{1}{8})(1 + 1/\eta) + m\left(1 - \frac{1}{4\beta}\right)$$
$$\leq \frac{8}{3} - 2\left(\frac{4}{7} + \frac{1}{8}\right)\frac{56}{19} + \frac{4}{3} = -\frac{2}{19} < 0.$$

Together with concavity this proves that $\mathcal{N}$ decreases on $[3/4, V]$. It remains to bound the envelope at $v = V$. At the balanced limit $t = 1/2$, $x = 0$, $v = 3/4$, the envelope has value $3/16$. We measure its decrease by the order imbalance $h = 1 - 2t$ and prove

$$\frac{\mathcal{N}(V)}{16(k-t^2)^2} > \frac{3}{16} - \frac{h^2}{32}.$$

Put $u = k - 1$, so $0 \le u \le 1/2$, and write $t = (1-h)/2$, $x = h + j$. With $D = x^2 + 6x + 6$, clear the positive denominator and set

$$P(h, j, u) = t^2 p^2 W^4 D^2 \left\{ \mathcal{N}(V) - 16(k-t^2)^2 \left(\frac{3}{16} - \frac{h^2}{32} \right) \right\}.$$

This is a quadratic in the shape increment u. Write it as $P_0 + uP_1 + u^2 P_2$, where the P_i are polynomials in (h, j), and put $P_m = P_1 + P_2/2$. The two shape polynomials P_1 and P_m are nonnegative for $j \ge 0$ and $0 \le h \le 1/3$. For their coefficients in j, first factor out powers of h, $1 + h$ and $1 - h$. In each remaining factor, keep the constant, linear and quadratic terms in h, and absorb each unfavorable higher power using

$$h^n \le \frac{h^2}{3^{n-2}} \qquad (n \ge 3).$$

The resulting quadratic lower bounds are nonnegative on $[0, 1/3]$: discard nonnegative quadratic terms, replace each negative quadratic term using $h^2 \le h/3$, and check the resulting linear expression at the two endpoints. Thus

$$P = P_0 + u\{(1-2u)P_1 + 2uP_m\} \ge P_0.$$

On the unit face, the terms of degree at least three in j are nonnegative by the same power estimate. The first three coefficients give

$$P_0 \ge \frac{h^2(h+1)^6}{512} g(h) + \frac{h(h+1)^5}{128} f(h) j + \frac{(h+1)^4}{256} b(h) j^2,$$

where $b(h)$ is $256/(h+1)^4$ times the coefficient of j^2 in P_0, and

$$\begin{aligned} g(h) &= h^{10} - 18h^9 - 223h^8 - 940h^7 - 1425h^6 + 54h^5 \\ &\qquad + 1607h^4 + 224h^3 - 1044h^2 - 216h + 180, \\ f(h) &= h^{11} - 76h^{10} - 747h^9 - 2942h^8 - 4457h^7 - 238h^6 \\ &\qquad + 4727h^5 + 1622h^4 - 2312h^3 - 858h^2 + 324h + 36. \end{aligned}$$

These two coefficients are controlled by concavity. For $0 \le h \le 1/3$, charging the higher powers to h gives

$$g'''(h) \ge 1344 + 10000h > 0, \qquad g''(1/3) < 0.$$

Thus $g'' < 0$ throughout the interval, and the endpoint values $g(0) = 180$, $g(1/3) > 16$ give $g \ge 16$. Similarly,

$$f''(h) \le -1716 + 3200h < 0, \qquad f(0) = 36, \qquad f(1/3) > -1,$$

so $f \geq -1$. The quadratic coefficient satisfies, by the higher-power estimate above,

$$b(h) \geq 144 + 1116h - 5200h^2 \geq -64.$$

Finally, on $1/5 \leq h \leq 1/3$, $(1+h)^6 \geq 8/3$, $(1+h)^5 \leq 8$ and $(1+h)^4 \leq 4$. Consequently

$$P_0 \geq \frac{h^2}{12} - \frac{hj}{16} - j^2 > \frac{h^2}{24} > 0,$$

where the last comparison uses $j \leq 1/30 \leq h/6$. This proves the endpoint bound and hence the asserted envelope inequality. □

The small gap. By Lemma G.27, the hypotheses of Lemma G.28 hold, so $\gamma^2 = \Gamma(w) > \Gamma(w_U)$. At w_U the diagonal divided by B is $\phi(q) = \log q + \alpha_U q + \beta_U/q + (Wv - c + p^2/W)/p - p^2/B$, and Lemma G.29 with the floor gives $0 < \phi(q) \leq \phi(1) + F_\eta(q) - U(\eta) \leq \phi(1) + v - g$; positivity comes from Lemma G.28, and $U(\eta) = g$. Since $0 \leq 1 - 2t \leq 1/3$,

$$\Gamma(w_U) \geq \mathcal{G} > \frac{3}{16} - \frac{1}{288} > \frac{9193}{50000}$$

by Lemma G.30, which is the last step of Proposition G.13.

G.3. **The smaller-short window.** This subsection completes the exclusion of class M2. By Appendix G, a competitive normal period of class M2 with at least six pieces has a global minimum of its active margins at the centre of a reverse run (Proposition G.3). Of the two turns adjacent to that centre we choose the one with the smaller active margin, and after reflecting and reversing time if necessary (Lemma G.5) it follows the centre. If the forward run that starts at this turn has at least two switches, Appendix G excludes the period. The remaining case is a forward run with a single switch: a short bridge between two reverse runs. Its data are fixed by six real numbers, and no free continuation of the period is left to compare with. The proof therefore works with a fixed list of necessary conditions on those numbers and shows that the list is inconsistent.

The argument has seven steps, which we call gates. Gates S0 to S3 confine the shape of the bridge to a small box, gate S4 gives a lower bound for the Casimir, gate S5 shows that the bridge is sheared in a definite direction, and gate S6 shows that a bridge sheared in that direction costs more, together with the rest of the period, than the octagon.

G.3.1. *The window.* Index the pieces of the period so that piece m is the global minimum. The seven pieces $m-2, \dots, m+4$ have active margins

$$p_b,\ b,\ a,\ C,\ B,\ A,\ P_f,$$

and the switches between them and after the last one have signs $+, -, -, +, -, -, +$. Thus a is the centre of a reverse run with the two turns b and C; the turn C, the chosen smaller one, is joined by the single forward switch $C \to B$ to the turn B; and A is the centre of the next reverse run, with far neighbour P_f. At a turn the two incident switch norms are

equal (Lemma B.12). Hence the three switches $a \to C$, $C \to B$, $B \to A$ of the bridge share one norm, and we write u for eight times it. Similarly v is eight times the norm of the switch $b \to a$ and w eight times that of $A \to P_f$. The turn secant of Lemma B.12 reads, in this normalization,

$$\begin{aligned} \operatorname{Sec}(A_0, B_0, P_0) &= 1 + \frac{B_0^2}{(A_0 + B_0)(P_0 + B_0)} - \frac{B_0}{P_0 - A_0} \log \frac{P_0 + B_0}{A_0 + B_0} \\ &= \frac{1}{P_0 - A_0} \int_{A_0}^{P_0} g_{B_0}(s)\, \mathrm{d}s, \end{aligned} \tag{G.12}$$

with $g_{B_0}(s) = 1 - B_0 s/(B_0 + s)^2$, and with the continuous value $g_{B_0}(A_0)$ at $A_0 = P_0$. The identity

$$g_{B_0}(s) = \frac{3}{4} + \frac{(s - B_0)^2}{4(s + B_0)^2} \tag{G.13}$$

shows that every secant lies in $[3/4, 1]$. We also write $\mathcal{U}(s) = 1 - s + s^2$, so that $g_{B_0}(A_0) = \mathcal{U}(A_0/(A_0 + B_0))$; this is four times the quantity j_k of Lemma B.12, and the switching inequalities of that lemma become $u \geq g$ at a forward switch and $u \leq g$ at a reverse one.

Definition G.31 (Smaller-short window)**.** A *smaller-short window* consists of real numbers p_b, b, a, C, B, A, P_f (margins), u, v, w (norms), E, E_L (centre pairings) and Γ (the Casimir) such that, with $\rho = a^2/(a^2 + 4)$:

(i) (order) $0 < a < b$, $a < C \leq b$, $a \leq A < C$, $a \leq p_b$, $A < B$ and $A < P_f$;

(ii) (strict star at the six switches) $p_b b$, ab, aC, CB, BA and AP_f are less than 4;

(iii) (reverse centres) $(a+b)(a+C) < a^2 + 4$ and $(B+A)(A+P_f) < A^2 + 4$;

(iv) (scales) $2/15 \leq a$, $a^2 < 12/17$ and $A^2 < 12/17$;

(v) (turn secants) $u = \operatorname{Sec}(a, C, B) = \operatorname{Sec}(C, B, A)$ and $v = \operatorname{Sec}(a, b, p_b)$;

(vi) (switching inequalities) $u \leq g_C(a)$, $g_B(C) \leq u$, $u \leq g_B(A)$, $v \leq g_b(a)$, $g_b(p_b) \leq v$, $w \leq g_{P_f}(A)$ and $3/4 \leq w$;

(vii) (centre stationarity) $E = \mathcal{E}(a, b, C, v, u)$ and $E_L = \mathcal{E}(A, B, P_f, u, w)$, where

$$\mathcal{E}(a_0, p, n, u_i, u_o) = 1 + r' + t' - \rho_0 + \log \frac{\rho_0}{r' t'} - \frac{1 - u_i}{r'} - \frac{1 - u_o}{t'},$$

with $r' = a_0/(a_0 + p)$, $t' = a_0/(a_0 + n)$ and $\rho_0 = a_0^2/(a_0^2 + 4)$;

(viii) (Gram determinants and Casimir bounds) $\Gamma = (4uv - E^2)/(16(1-\rho)^2)$, $\Gamma = (4uw - E_L^2)/(16(1-\rho_A)^2)$ with $\rho_A = A^2/(A^2 + 4)$, and $27/256 < \Gamma < 9193/50000$;

(ix) (same-lift transport) $w = \lambda^2 v + d^2 u - d\lambda E$ and $E_L = 2du - \lambda E$, where $\lambda = (a^2 + 4)/(A^2 + 4)$ and $d = \lambda \rho D$ with the bridge diagonal D of Table 3;

(x) (window defect) the inequality (G.14) below, with $0 < \mathcal{B} < \mathcal{A}$.

The conditions are the facts of Appendices B and G written at the seven pieces. The orders come from the minimum structure (Theorem 4.2) and the choice of the smaller turn; the products in (ii) and (iii) from the strict star condition and the reverse factors (Lemma B.1); the scales from the compact domain (Theorem A.12) and the reverse-centre scale bound; and (v) to (ix) from Lemmas B.11 to B.13 applied to the one normal multiplier of the period. The Casimir bounds are those of Appendix G.

TABLE 3. The normalized variables of a smaller-short window. Lengths are divided by the central margin a.

Symbol	Definition
$x,\ \beta,\ \alpha$	C/a, B/a, A/a (ratio, bridge, outer centre)
$\delta,\ \eta$	$\beta/x - 1 = B/C - 1$ (displacement); $(\alpha - 1)/\delta$
$\rho,\ r,\ t$	$a^2/(a^2+4)$; $1/(1+x) = a/(a+C)$; $a/(a+b)$
$q,\ r_A$	$\rho_A = A^2/(A^2+4)$ (outer scale); $\alpha/(\alpha+\beta) = A/(A+B)$
q_c	$\alpha^2/(x(x+2)+\alpha^2)$, the value of q at $\rho = r^2$
$\tau,\ \tau_{\max}$	$(A+P_f)/A$ (outer duration); r_A/q
$\lambda,\ D$	$(a^2+4)/(A^2+4)$; $\alpha^2 + 2\alpha\beta + 1 + 2x - 2\beta x$ (bridge diagonal)
$u,\ v,\ w$	bridge, central and outer switch norms (times 8)
$E,\ E_L$	pairings at the centres a and A
$\Gamma,\ \kappa$	the Casimir γ^2; $\sqrt{12\Gamma}$

Proposition G.32 (Extraction of the window)**.** *Let P be a competitive normal period of class* M2 *in the compact domain with at least six pieces, let m be a piece of least active margin, and suppose that the switches after pieces $m-2, m-1, m, m+1, m+2$ have the signs $+,-,-,+,-$ and that $a_{m+1} \le a_{m-1}$. Then the active margins $a_{m-2}, \dots, a_{m+4}$ are the margins of a smaller-short window.*

The window defect imposes a global cost constraint on the local data. Let $\mathcal{C}_{\mathrm{br}}$ be the cost of the two bridge pieces, let θ_{br} be the angle swept over the bridge by the continuous polar angle of the costate frame, and let $\kappa = \sqrt{\det \Lambda_1} = \sqrt{12\Gamma}$. Over a competitive period the polar angle increases by exactly $\pi/3$, as shown after (F.5). That inequality bounds the cost of any subinterval below by $\kappa + 3/2$ times its polar-angle increment. Applying it to the intervals before and after the bridge, and retaining the exact cost on the bridge, gives the window defect below. If the bridge crosses the original seam, use the twisted extension and the period $[t_-, t_- + T]$ starting at its first switch; the bridge is $[t_-, t_+]$ and its complement is $[t_+, t_- + T]$. The extension of (F.5) applies there, with the same normal multiplier and the same total angle $\pi/3$. Thus

$$\mathcal{C}_{\mathrm{br}} + \Big(\kappa + \frac{3}{2}\Big)\Big(\frac{\pi}{3} - \theta_{\mathrm{br}}\Big) \le \mathcal{C}(T, X) \le A_{\mathrm{oct}}. \tag{G.14}$$

Both sides are explicit. Because the two turn secants agree, the logarithms in the two arc costs cancel and

$$\mathcal{C}_{\mathrm{br}} = \sqrt{3}\Big[\frac{\rho u}{1-\rho}(2\beta x - x - \alpha\beta) + \frac{\beta x - \alpha}{(1+x)(\alpha+\beta)}\Big], \tag{G.15}$$

by the rational identity $\frac{x(\beta-1)}{(1+x)(\beta+x)} + \frac{\beta(x-\alpha)}{(\alpha+\beta)(\beta+x)} = \frac{\beta x-\alpha}{(1+x)(\alpha+\beta)}$. The angle is $\theta_{\mathrm{br}} = \arctan(\sqrt{3}\mathcal{B}/\mathcal{A}) \in (0, \pi/3)$. We compute explicit formulas for the two traces $\mathcal{A}, \mathcal{B}$, so the defect is an inequality in the window data alone. Put

$$\begin{aligned} S_0 &= 1 + x, \qquad S_1 = \alpha + \beta, \qquad S_2 = x + \beta, \\ U_0 &= \frac{1-\rho}{\rho}, \qquad n_0 = \frac{U_0 - x}{S_0}, \qquad z_0 = \frac{U_0 - \alpha\beta}{S_1}, \end{aligned}$$

and define $m = (m_{ij})$ by the product

$$m = \begin{pmatrix} -n_0 - 2 & x+2 \\ n_0 & x \end{pmatrix} \begin{pmatrix} S_0/S_1 & 0 \\ (S_0^2 - S_2^2)/(S_0 S_1) & S_1/S_0 \end{pmatrix} \frac{1}{2U_0} \begin{pmatrix} -\alpha & \alpha + 2\beta \\ z_0 & z_0 + 2\beta \end{pmatrix}. \tag{G.16}$$

The three factors have determinants $-2U_0$, 1 and $-1/(2U_0)$, respectively, so $\det m = 1$. The scaled costate entries are

$$\begin{aligned} h_{11} &= E(r - \rho S_0) - u(1 - (\rho S_0)^2) + v(1 - r^2), \\ h_{12} &= u(1 + \rho S_0)^2 + v(1+r)^2 - E(1+r)(1+\rho S_0), \\ h_{21} &= -u(1 - \rho S_0)^2 - v(1-r)^2 - E(1-r)(1-\rho S_0). \end{aligned}$$

With $T_\kappa = 2\kappa(1-\rho)^2$, the required expressions are

$$\begin{aligned} \mathcal{A} &= T_\kappa(m_{00} + m_{11}) + h_{12}m_{11} - h_{11}(m_{01} + 3m_{10}) - 3h_{21}m_{00}, \\ \mathcal{B} &= h_{11}(m_{11} - m_{00}) + h_{12}m_{10} + h_{21}m_{01} + \frac{T_\kappa}{3}(3m_{10} - m_{01}). \end{aligned} \tag{G.17}$$

To obtain these formulas, use the two turn factors $q_C = (C+B)/(a+C)$ and $q_B = (C+B)/(A+B)$ from Lemma B.2, and transport the same normal multiplier through both pieces. Write X_-, X_+ for the states at the two ends of the bridge in triangular coordinates, with margin triples $(C, a, (4-aC)/(a+C))$ and $(A, B, (4-AB)/(A+B))$ in the label order of its first piece. The relative frame is $-X_- T_0(q_C) T_1(q_B) X_+$, since $X_+^{-1} = -X_+$; conjugation by $\left(\begin{smallmatrix} -1 & 2 \\ 1 & 0 \end{smallmatrix}\right)$ gives (G.16). Substitution of the two switch norms and the central pairing in that multiplier gives the three h_{ij} above, and multiplication gives (G.17).

For the costate frame $B_*(t) = h^{-1}g(t)$ of the spectral proof, write $z_*(t) = \frac{1}{2}(B_{*,00} + B_{*,11}) + \frac{i}{2}(B_{*,10} - B_{*,01})$. If t_-, t_+ are the ends of the bridge, the trace expressions satisfy

$$\mathcal{A} + i\sqrt{3}\mathcal{B} = 4T_\kappa z_*(t_+)\overline{z_*(t_-)} = T_\kappa \rho_W e^{i\theta_{\mathrm{br}}}, \qquad \rho_W = 4|z_*(t_+)||z_*(t_-)| \geq 4.$$

This identity follows by expanding the rotation coordinate of a product of two matrices; the real and imaginary parts are the two traces in (G.17). Since $0 < \theta_{\mathrm{br}} < \pi/3$, it gives $0 < \mathcal{B} < \mathcal{A}$ and also $\mathcal{A} \geq 2T_\kappa$ and $\mathcal{A} + \mathcal{B} \geq 4T_\kappa$.

The window defect therefore compares its explicit cost with its explicit angle. The gates show that no window satisfies this comparison.

Theorem G.33 (The smaller-short window does not exist)**.** *There is no smaller-short window. Consequently no period satisfies the hypotheses of Proposition G.32.*

The second sentence follows from the first by Proposition G.32. It is the case of a short forward run after the chosen smaller turn, and Appendix G uses it to finish class M2 (Theorem 4.5). The rest of this section proves the first sentence.

G.3.2. *Normalized data and the shape of the bridge.* The secant is invariant under scaling of its three arguments, so in the variables of Table 3

$$u = \mathrm{Sec}(1, x, \beta) = \mathrm{Sec}(\alpha, \beta, x). \tag{G.18}$$

The first secant averages g_x over $[1, \beta]$ and the second averages g_β over $[\alpha, x]$. Only the conditions of Definition G.31 are used below. We collect the first consequences.

Lemma G.34 (First consequences)**.** *Every smaller-short window satisfies:*

(a) $C \le B$*, so that* $1 \le \alpha < x \le \beta$*;*

(b) $u \le V(x) := \mathrm{Sec}(1, x, x) < 3/2 - \log 2$*;*

(c) $\beta \le \alpha x$*;*

(d) the outer centre is stationary: with $\tau = (A+P_f)/A$ *one has* $2 < \tau < \tau_{\max}$ *and*

$$E_L = (w-1)\tau + \frac{1}{\tau} + \frac{u-1}{r_A} + r_A + 1 - q + \log\frac{\tau q}{r_A}; \tag{G.19}$$

(e) $q \le q_c$ *and* $q < 1/7$*.*

Proof sketch. (a) Suppose $B < C$. The cross inequality for secants gives $\mathrm{Sec}(a, B, C) < \mathrm{Sec}(a, C, B) = u$. On the other hand, g_B decreases on $[a, A]$ and $g_B \ge g_B(A) \ge u$ there, so adding $[a, A]$ to the interval of $\mathrm{Sec}(A, B, C) = u$ gives $\mathrm{Sec}(a, B, C) \ge u$. (b) Split the integral of $\mathrm{Sec}(1, x, \beta)$ at x. On $[x, \beta]$ the integrand is at most $g_x(\beta) \le u$, the switching inequality at $C \to B$, so $u \le V(x)$; its exact upper limit follows from the normalized integral in Appendix G.4.1. (c) follows from the stronger order of Lemma G.35(a), proved by the integral comparison in Appendix G.4.2. (d) is the stationarity (vii) at A, rewritten: $\tau > 2$ is $A < P_f$, and $\tau < \tau_{\max}$ is the reverse factor at A. (e) $q \le q_c$ is equivalent to $\rho \le r^2$, which follows from $\rho < rt \le r^2$. The bound $q < 1/7$ is Lemma G.45, applied at the outer centre: if $q \ge 1/7$, the Gram determinant at A would be at least $9193/50000$. □

Equation (G.19) is the one place where the outer reverse run enters. We write it as $R_\tau(t, v) = 0$, where the *outer residual* $R_\tau(t, v)$ is the right side of

(G.19) minus E_L, with w and E_L given by the transport (ix) as functions of the opposite turn (t, v) and with E given by the stationarity at a:

$$E(t,v) = (u-1)(1+x) + \frac{v-1}{t} + 1 + \frac{1}{1+x} + t - \rho + \log\frac{\rho(1+x)}{t}. \quad \text{(G.20)}$$

The shape (x, α, β) is constrained by (G.18) alone. The gates use it through explicit inequalities between α and β at fixed x, which we call orders.

Lemma G.35 (Orders of the bridge)**.** *Let $1 < x$, $1 \le \alpha < x \le \beta$ and $\mathrm{Sec}(1, x, \beta) = \mathrm{Sec}(\alpha, \beta, x)$, and put $\delta = \beta/x - 1$.*

(a) $\alpha \ge \beta/x + k_s(x)(\beta - x)$ with $k_s(x) = 4(x+1)/(7x+1)$.

(b) $\alpha \le \beta/x + \frac{5}{4}(\beta - x)$, and $\alpha \le \beta/x + \frac{9}{8}(\beta - x)$ if $x \le 5$.

(c) If $x \le 13/6$ and $\delta \le 1/19$, then $\alpha \le 1 + 3\delta$.

(d) If $2 \le x \le 13/6$, then $\alpha \ge 1 + \frac{4x^2+11x+1}{7x+1}\,\delta + \frac{\delta^2}{1+4\delta}$.

In particular, at a window with $x \ge 2$ one has $\alpha \ge 1 + (1 + \frac{4x(x+1)}{7x+1})\delta \ge 1 + \frac{13}{5}\delta$.

The last inequality holds because $4x(x+1)/(7x+1)$ increases, with derivative $4(7x^2+2x+1)/(7x+1)^2$, and equals $8/5$ at $x = 2$. In the slope $\eta = (\alpha - 1)/\delta$, parts (a) and (c) say that $\eta \in [13/5, 3]$ on the box $2 \le x \le 13/6$, $0 < \delta \le 1/19$; Figure 11(b) shows the actual slopes.

G.3.3. *Comparison with boundary faces.* Gates S0 to S4 use a common comparison argument. Each assumes that the window lies outside a specified region and derives a scalar inequality at the actual window: $G \ge 0$ for a weighted outer residual G, or $\Gamma \ge G$ for a corrected Casimir G. It then moves the opposite turn, the scale and the outer duration, each time in a direction that keeps the inequality valid, until the point reaches one of two faces where the data are explicit functions of the shape (x, α, β) and the norm u. On these faces the inequality fails. The transformations are described below.

The opposite turn. The norm v and the ratio t of the turn before the centre are not controlled individually. They satisfy $v \le \mathcal{U}(t)$, the switching inequality at $b \to a$, and $t_0 := \rho(1+x) \le t \le r$, from the central reverse factor and the choice of the smaller turn. For a weight $m \ge 0$ put $G = R_\tau + m(w - 3/4)$. Then

$$\frac{\partial G}{\partial v} = \frac{\lambda}{t} B_{\tau+m}(t), \qquad \frac{\mathrm{d}}{\mathrm{d}t} G(t, \mathcal{U}(t)) = \frac{\lambda(2t-1)}{t} B_{\tau+m}(t),$$

with the linear factor $B_z(s) = 1 + z(\lambda s - \lambda\rho D)$. When $B_{\tau+m} > 0$ on $[t_0, t]$, G increases in v and $G(s, \mathcal{U}(s))$ decreases in s (since $s < 1/2$), so

$$G(t,v) \le G(t_0, \mathcal{U}(t_0)) \quad \text{(G.21)}$$

. Without weight, the positivity of B_τ for $\tau \le \tau_{\max}$ reduces to $\alpha(\alpha+\beta) + (1+x) - D = x(\beta - 1) + \beta(x - \alpha) > 0$; with a weight each gate checks it separately. At the point $(t_0, \mathcal{U}(t_0))$ the opposite turn has zero duration and the logarithm in (G.20) vanishes.

The scale. At $(t_0, \mathcal{U}(t_0))$ every quantity is rational in ρ. We reparametrize by the outer scale $q = \rho_A$, through $\rho = q/(\alpha^2 - (\alpha^2 - 1)q)$. Then

$$w_* = 1 - qA_1 + q^2 A_2, \qquad E_{L,*} = -E_0 + qA_3, \qquad E_0 = (1+x)(u-1) + \frac{1}{1+x}, \tag{G.22}$$

with A_1, A_2, A_3 affine in u and rational in x, α, β. At fixed $p = \tau q$ the residual becomes $R^*_{p/q} = \mathcal{C}(p) + q\mathcal{L}(p)$, where $\mathcal{C}(p) = E_0 + (u-1)/r_A + r_A + 1 - pA_1 + \log(p/r_A)$ and $\mathcal{L}(p) = pA_2 + 1/p - 1 - A_3$. So at fixed p every weighted residual is a quadratic polynomial in q. The actual scale satisfies $q \le q_c$ and $q < 1/7$ (Lemma G.34), and $q \le p/2$ because $\tau \ge 2$.

The faces. The scale is raised at fixed p up to $q_{\max} = \min\{q_c, 1/7, p/2\}$. If $q_{\max} = p/2$, the duration has reached $\tau = 2$, and the scale is raised further along $\tau = 2$. The end point lies on the *central face* $q = q_c \le 1/7$ or on the *outer face* $q = 1/7 \le q_c$. On the central face $\rho = r^2$, that is $(a+C)^2 = a^2 + 4$: the central reverse factor is one when both turns equal C, so the central piece has zero duration. On the outer face the outer centre attains the reverse-centre scale bound $A^2 = 2/3$. On each face all data are rational in (x, α, β, u), except for the duration $\tau \in [2, \tau_{\max}]$ and the term $\log(\tau q/r_A)$.

The duration. The duration enters only through $(w-1)\tau + 1/\tau + \log\tau$. The identity

$$R_\tau - R_2 - (\tau - 2)\Big(w - \frac{3}{4}\Big) = \log\frac{\tau}{2} - \frac{\tau}{4} + \frac{1}{\tau} \le 0 \qquad (\tau \ge 2) \tag{G.23}$$

bounds every R_τ with $2 \le \tau \le \tau_{\max}$ by $\max\{R_2,\, R_2 + (\tau_{\max} - 2)(w - 3/4)\}$. A variant with two tangent lines to the logarithm reduces the duration to three test values.

The incoming norm. The bridge norm u is the secant (G.18), a transcendental function of (x, β). On each face the tested quantity is monotone in u, and u is replaced by a rational ceiling or floor, for instance $V(x) \le \bar{V}(x) = \frac{3}{4} + \frac{2(x-1)^2}{(x+1)(35x+13)}$ (Appendix G.4.1). It remains to determine the sign of an explicit function of (x, α, β) on a region bounded by the orders. In some cases this function contains one logarithm. These are the *face signs*.

G.3.4. *Gates S0 to S3: the ratio and the displacement.* The ratio $x = C/a$ is the shape parameter of the central reverse run, and the displacement δ measures how far the bridge is from symmetric. Gates S0 to S3 place (x, δ) in the box $[2, 13/6] \times [0, 1/19]$ (Figure 11(a)).

Proposition G.36 (Gate S0)**.** *Every smaller-short window has $x < 12/5$, $D > 0$ and $E_L > 0$.*

Proof sketch. Suppose $x \ge 12/5$. The root gives $R_\tau = 0$ and $w \ge 3/4$, so the floor-weighted residual $G = R_\tau + \frac{1}{2}(w - \frac{3}{4})$ is nonnegative at the actual window. Large ratios force an asymmetric outer bridge: the equal secants and $x \ge 12/5$ give $\alpha/\beta < 3/4$ and then $D < \alpha(\alpha + \beta)$. The weighted

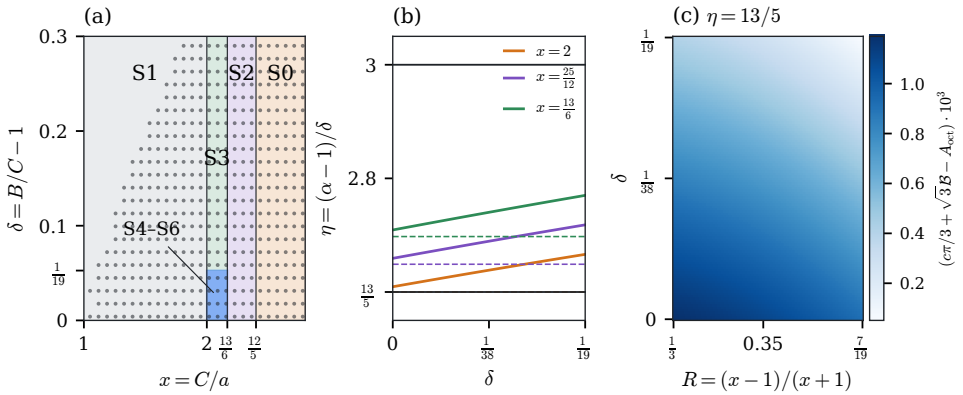


FIGURE 11. Parameter bounds from the gates. (a) The plane of the ratio $x = C/a$ and the displacement $\delta = B/C - 1$. The dots are computed points of the equal-secant locus (G.18) with $1 \le \alpha < x$. Gate S1 removes $x \le 2$, S0 removes $x \ge 12/5$, S2 removes $13/6 \le x < 12/5$ and S3 removes $\delta \ge 1/19$, leaving the box where S4 to S6 work. (b) On that box, the slope $\eta = (\alpha - 1)/\delta$ of the locus for $x = 2, 25/12, 13/6$ (solid), between the lower order $1 + 4x(x+1)/(7x+1)$ (dashed) and the upper order 3 of Lemma G.35. (c) The score gap $c\pi/3 + \sqrt{3}\,\mathfrak{B}(u_*; R, T, \delta) - A_{\mathrm{oct}}$ of gate S6 at $\eta = 13/5$: it is positive throughout the box.

envelope (G.21) applies (its factor stays positive for $x > 5/3$), so $G^*_\tau \ge 0$ at the actual scale, and the two shape facts make G^*_τ increase with u. At fixed p its value at the zero scale is negative, and an endpoint argument for the quadratic in q gives a nonnegative value at $q_{\max}$; the move along $\tau = 2$ keeps it nonnegative. On the face reached, u is raised to the ceiling $\hat{u}(x, \beta)$ (central face) or $\bar{u}(x, \beta)$ (outer face), and the duration is reduced to the three tests $F(2, \cdot)$, $F(8\tau_{\max}/9, \cdot)$, $F(\tau_{\max}, \cdot)$ of Appendix G.3.3. The three face signs of Proposition G.46 say that each test is negative on the region cut out by the orders. This contradiction gives $x < 12/5$. For $x \le 12/5$ the identities

$$D - \big[(1+2x)\alpha^2 - 2x^2\alpha + 1 + 2x\big] = 2(\beta - \alpha x)(\alpha - x),$$
$$(1+2x)\alpha^2 - 2x^2\alpha + 1 + 2x - D_0 = (\alpha - 1)\big[(1+2x)(\alpha+1) - 2x^2\big],$$

with $D_0 = 2 + 4x - 2x^2 \ge 2/25$, give $D \ge \alpha D_0 > 0$. Finally $E_L = \lambda(2\rho D u - E) > 0$, because E is at most its envelope value $(1+x)(u-1) + \frac{1}{1+x} + \rho(2x+1)$, which is less than $2\rho D u$. □

Proposition G.37 (Gate S1)**.** *Every smaller-short window has $x > 2$.*

Proof sketch. Suppose $1 < x \le 2$. In this range we combine the residual with the upper bound $\Gamma < 9193/50000$. Moving the opposite turn changes the Casimir $\Gamma(t, v)$ computed at the central reverse run, but the combination

$\Gamma - \frac{7}{96} R_\tau$ does not increase along the two moves of (G.21): in v it is concave with a nonpositive slope at the actual point, and along $v = \mathcal{U}(s)$ its derivative is the derivative of R_τ, which is negative, times a factor that stays below zero. Since $R_\tau = 0$ at the actual window,

$$\Gamma \geq \Gamma_* - \tfrac{7}{96} R^*_\tau$$

at the actual scale, where Γ_* is the Gram expression at $(t_0, \mathcal{U}(t_0))$. At fixed p the right side decreases in q, and along $\tau = 2$ the slope of R^*_2 in q exceeds one, so the right side does not increase when the point moves to a face. On the central face $q = q_c \leq 1/7$ and on the outer face $q = 1/7 \leq q_c$, the face signs $\Gamma_* - \frac{7}{96} R^*_\tau > 9193/50000$ for every admissible duration (Proposition G.47) contradict $\Gamma < 9193/50000$. □

Proposition G.38 (Gate S2)**.** *A smaller-short window with $x < 12/5$ has $x < 13/6$.*

Proof sketch. Suppose $13/6 \leq x < 12/5$. The weight is now variable, $m = 2 - 7\tau/(4\tau_{\max}) \in [1/4, 2)$, and the weighted residual $G_\tau = R_\tau + m(w - 3/4)$ is still nonnegative at the root. The envelope applies with this weight, the scale polynomial is convex in q with a negative value at the zero scale, and the $\tau = 2$ continuation reaches the face $q^* = \min\{q_c, 1/7\}$. After raising u to the ceiling $u_C(x, \beta)$ on the central face, and to $u_s(x, \beta)$ or $\bar{V}(x)$ on the outer face, the face signs $G_\tau < 0$ of Proposition G.48 give the contradiction. □

Proposition G.39 (Gate S3)**.** *A smaller-short window with $2 \leq x \leq 13/6$ has $\delta < 1/19$.*

Proof sketch. Suppose $\delta \geq 1/19$. The displacement is penalized by the weight $m = 4(4 - 3\tau/\tau_{\max})/(5\alpha - 1)$, with which $G_\tau = R_\tau + m(w - 3/4) \geq 0$ at the root. On this range $\beta < 2\alpha$, so $D = \alpha^2 + 1 + 2x + 2\beta(\alpha - x) > 5\alpha^2 - 4\alpha x + 1 + 2x \geq 6 - 2x > 0$. The same three moves reach the face $q^* = \min\{q_c, 1/7\}$. The duration is reduced to two tests, F_0 at $\tau = 2$ and F_1 at $\tau = \tau_{\max}$, the norm is raised to a ceiling, and on the central face a monotonicity in β moves the point to the curve $\beta = x + \frac{25x}{15x+36}(\alpha - 1)$. The face signs $F_0, F_1 < 0$ of Proposition G.49 give the contradiction. □

G.3.5. *Gate S4: the small box and a floor for the Casimir.* After gates S0 to S3 the shape lies in $2 < x < 13/6$, $0 \leq \delta < 1/19$. Gate S6 needs a two-sided control of α in terms of δ, and also a lower bound for κ, because the completion term in (G.14) is weighted by $\kappa + 3/2$.

Proposition G.40 (Gate S4)**.** *A smaller-short window with $2 \leq x \leq 13/6$ and $\delta \leq 1/19$ has $\alpha \leq 1 + 3\delta$ and $\Gamma > 1829/10000$.*

Proof sketch. The first inequality is Lemma G.35(c), proved by the integral comparison in Appendix G.4.2. For the second, the monotonicity used in gate S1, now with the constant $1/7$, gives $\Gamma \geq \Gamma_* - R^*_\tau/7$ at the actual scale.

The scale is moved at fixed p, and along $\tau = 2$, up to the central scale q_c, and the duration maximum (G.23) leaves the two functions

$$\Phi = \Gamma_*(q_c) - \tfrac{1}{7}R_2^*(q_c), \qquad \Psi = \Phi - \tfrac{1}{7}(\tau_{\max} - 2)(w_*(q_c) - \tfrac{3}{4}).$$

Both decrease in u, so the norm may be raised to its ceiling $\bar{u}(x, \beta)$ (Appendix G.4.1). The face sign $\min\{\Phi, \Psi\} > 1829/10000$ on the closed box $2 \le x \le 13/6$, $0 \le \delta \le 1/19$, $\frac{13}{5}\delta \le \alpha - 1 \le 3\delta$ (Proposition G.50) concludes. □

With Lemma G.35 the window now lies in the *small-shape domain*

$$2 \le x \le \tfrac{13}{6}, \qquad 0 \le \delta \le \tfrac{1}{19}, \qquad 1 + \tfrac{13}{5}\delta \le \alpha \le 1 + 3\delta, \tag{G.24}$$

that is $\alpha = 1 + \eta\delta$ with $13/5 \le \eta \le 3$ when $\delta > 0$, and $\alpha = 1$ when $\delta = 0$. No division by δ is used, so the symmetric boundary $\delta = 0$ is included.

G.3.6. *Gate S5: the root forces a negative shear.* The defect inequality compares the cost of the bridge with the angle it turns, and both are governed by one number. Let M be the relative frame of the bridge, from the central switch to the far switch. After a fixed rotation that depends only on x, its lower row does not depend on ρ, and $\det M = 1$ then determines the upper row up to one real parameter h, the *shear*. Put $R_x = (x-1)/(x+1)$, $T_s = (\beta - \alpha)/(\beta + \alpha)$, $\mathcal{U}_1 = \mathcal{U}(r) = (3 + R_x^2)/4$, $\mathcal{U}_2 = \mathcal{U}(r_A) = (3 + T_s^2)/4$ and the anisotropy $\nu = \mathcal{U}_1/\mathcal{U}_2 - 1 \ge 0$. Then

$$\sqrt{3}\,h = \frac{(\beta x - \alpha)\big((x+2)\alpha + (2x+1)\beta\big)}{(1+x)^2(\alpha^2 + \alpha\beta + \beta^2)} - 2\mathcal{U}_1\,(2\beta x - x - \alpha\beta)\,\frac{\rho}{1-\rho}. \tag{G.25}$$

This is a strictly decreasing function of the central scale ρ, and the bridge cost (G.15) is affine in it:

$$\mathcal{C}_{\mathrm{br}} = H_0 - J_0 h, \qquad J_0 = \frac{3u}{2\mathcal{U}_1}, \qquad H_0 = \frac{\sqrt{3}}{2}\Big[\Big(1 + \frac{u}{\mathcal{U}_1}\Big)R_x + \Big(1 + \frac{u}{\mathcal{U}_2}\Big)T_s\Big]. \tag{G.26}$$

Decreasing the shear increases the bridge cost. Gate S5 shows that the actual outer root is compatible only with a negative shear.

Proposition G.41 (Gate S5)**.** *A smaller-short window in the small-shape domain* (G.24) *has* $q > \frac{15}{16}q_c$, $\sqrt{3}hR_x + \frac{7}{2}\nu < 0$, $h < 0$ *and* $-hR_x > 2\nu$.

Proof sketch. Suppose $q \le q_b := \frac{15}{16}q_c$. With weight zero, (G.21) and (G.22) give $\mathcal{R}(u; p, q) := \mathcal{C}(p) + q\mathcal{L}(p) \ge 0$ at $p = \tau q$, where $2q \le p \le r_A$. The residual is affine and increasing in u (Appendix G.4.6). The integral ceiling gives $u < 19/25$ on the small-shape domain, so $\mathcal{R}(19/25; p, q) \ge 0$. At fixed q, however, the residual is concave in p/r_A on its allowed interval. Its tangent at $p = r_A$ has two endpoint values, both negative by the shape comparisons in Appendix G.4.6. This is a contradiction, so $q > q_b$. The ratio comparison in Appendix G.4.6 then gives $-\sqrt{3}hR_x > 7\nu/2$. Since $R_x > 0$ and $\nu \ge 0$, this implies $h < 0$ and $-hR_x > 2\nu$, using $7 > 4\sqrt{3}$. □

G.3.7. *Gate S6: the completion cost bound.*

Proposition G.42 (Gate S6)**.** *For a smaller-short window in the small-shape domain with* $\Gamma > 1829/10000$*,* $h < 0$ *and* $-hR_x > 2\nu$*,*

$$\mathcal{C}_{\mathrm{br}} + \Big(\kappa + \frac{3}{2}\Big)\Big(\frac{\pi}{3} - \theta_{\mathrm{br}}\Big) > A_{\mathrm{oct}}.$$

The proof lowers the left side in four steps to an explicit function of the shape.

Step 1: lowering κ*.* Since $12 \cdot \frac{1829}{10000} - (\frac{40}{27})^2 = \frac{23}{1822500} > 0$, the floor of gate S4 gives $\kappa > \kappa_0 = 40/27$. Since $\theta_{\mathrm{br}} < \pi/3$, we may replace $\kappa + 3/2$ by $c = \kappa_0 + 3/2 = 161/54$.

Step 2: the angle in terms of the shear. Let $\psi = \arctan(R_x/\sqrt{3}) + \arctan(T_s/\sqrt{3})$ be the angle of the fixed lower row of the rotated frame, let $K = \nu + 2 + 2\kappa_0/J_0$, and

$$\delta(h) = 2\arctan\frac{\sqrt{h^2+\nu^2} - h}{K + \sqrt{K^2 - \nu^2}}.$$

In the rotated basis, $\mathcal{A} + i\sqrt{3}\mathcal{B}$ is, up to a positive factor and the rotation by ψ, a quadratic polynomial in one unknown entry of the costate at the central switch. The condition that its values stay on one side of the line through the origin at angle $\psi + \delta$ is a discriminant inequality, which holds at $\delta = \delta(h)$. This gives $\theta_{\mathrm{br}} \le \psi + \delta(h)$ for the actual shear. With (G.26), the left side of the proposition is at least $H_0 + c(\pi/3 - \psi) + \tilde{F}(\delta(h))$, where $\tilde{F}(\delta) = -J_0\tilde{h}(\delta) - c\delta$ and $\tilde{h}$ is the inverse of $h \mapsto \delta(h)$.

Step 3: continuation to the central scale. The shear decreases in ρ, and $\rho < r^2$. Let $h_c < h < 0$ be the shear at the central scale $\rho = r^2$ and $\omega_c = \delta(h_c)$. Throughout $[\delta(h), \omega_c]$, the quantity $H = -\tilde{h}$ satisfies $\nu < HR_x/2$ and $H \le -h_c < R_x^2$. Together with the norm ceiling $u \le 3/4 + R_x^2/12$, these bounds make $\tilde{F}$ strictly decreasing, by the loss comparison in Appendix G.4.7. The left side is therefore at least the central score

$$\mathcal{F}_c = \mathcal{C}_c + c\Big(\frac{\pi}{3} - \psi - \omega_c\Big), \qquad \mathcal{C}_c = \sqrt{3}\Big[\frac{u(2\beta x - x - \alpha\beta)}{x(x+2)} + \frac{R_x + T_s}{2}\Big].$$

Here $\mathcal{C}_c$ is the bridge cost (G.15) at the central scale.

Step 4: minimizing the shape bound. The integral estimate for the additional angle in Appendix G.4.7 and a norm floor $u_*(R_x, \delta)$ obtained from the mean and variance of a reciprocal integral give

$$\mathcal{F}_c > \frac{c\pi}{3} + \sqrt{3}\,\mathfrak{B}(u_*; R_x, T_s, \delta),$$

with $\mathfrak{B}$ the function of Appendix G.4.7, retaining the exact fixed-row angles. It is concave in T_s, and T_s is monotone in η, so its minimum over $13/5 \le \eta \le 3$ is attained at an endpoint. The bound decreases with displacement, and at the largest displacement it is concave in R_x. Its minimum is therefore at one of four corners (Proposition G.51, Figure 11(c)), where a reciprocal-integral

bound gives $\mathfrak{B} > 9/4000$. The direct octagon comparison of Lemma C.5, with $c/3 = 161/162$, now gives

$$A_{\mathrm{oct}} < \frac{c\pi}{3} + \frac{9\sqrt{3}}{4000} < \mathcal{F}_c.$$

G.3.8. *Proof of Theorem G.33.* Let a smaller-short window be given. Gate S0 gives $x < 12/5$, gate S1 gives $x > 2$, and gate S2 then gives $x < 13/6$. Gate S3 gives $\delta < 1/19$, gate S4 gives $\alpha \le 1 + 3\delta$ and $\Gamma > 1829/10000$, and Lemma G.35 gives $\alpha \ge 1 + \frac{13}{5}\delta$. So the window lies in the small-shape domain (G.24). Gate S5 gives $h < 0$ and $-hR_x > 2\nu$, and gate S6 then contradicts the window defect (G.14). □

G.4. **Estimates for the smaller-short window.** This subsection gives the scalar inequalities behind Appendix G.3, organized by the gate that uses them. They are identities between rational functions, and inequalities proved by exhibiting a difference as a sum of terms of known sign, by the sign of a derivative on an interval, or by monotonicity or convexity in one variable at a time. The face signs for gates S0 to S4 in Appendix G.4.8 reduce, one variable at a time, to polynomials in one or two variables, mostly of degree at most eight. We give the reductions and final signs; every polynomial is specified by the rational expressions at the face where it is evaluated. Gates S5 and S6 use the direct concavity and derivative comparisons given below. Throughout, Sec, g_B and $\mathcal{U}$ are as in Appendix G.3.1, and the variables are those of Table 3.

G.4.1. *Secants and the incoming norm.* By (G.12), $(P_0 - A_0)\,\mathrm{Sec}(A_0, B_0, P_0)$ is the integral of g_{B_0} over $[A_0, P_0]$, hence additive over adjacent intervals, and by (G.13) every secant lies in $[3/4, 1]$. The function g_{B_0} decreases on $(0, B_0]$ and increases on $[B_0, \infty)$. For the bridge norm, (G.13) gives

$$u - \frac{3}{4} = \frac{1}{4(\beta - 1)} \int_1^\beta \frac{(s - x)^2}{(s + x)^2}\,\mathrm{d}s. \tag{G.27}$$

The central value. Normalize the integration variable by x and put $a = 1/x$. Then

$$V(x) = \frac{1}{1 - a} \int_a^1 g(s)\,\mathrm{d}s, \qquad g(s) = 1 - \frac{s}{(1 + s)^2}.$$

Since g strictly decreases on $(0, 1)$, adding $[0, a]$ strictly raises the average. Hence V increases with x and

$$V(x) < V_\infty := \int_0^1 g(s)\,\mathrm{d}s = \frac{3}{2} - \log 2, \qquad \lim_{x \to \infty} V(x) = V_\infty.$$

With $z = (x - 1)/(x + 1)$ one also has $2x/(1 + x) = 1 + z$ and

$$V(x) - \frac{3}{4} = \mathrm{Sec}(1, x, x) - \frac{3}{4} = \frac{1}{2} + \frac{z}{4} - \frac{(1 + z)\log(1 + z)}{2z}.$$

A lower bound for $\log(1+z)$ is therefore an upper bound for V. The weighted Jensen bound $L_J(z)$ of Lemma G.17(c) gives

$$V(x) - \frac{3}{4} < U(z) := \frac{1}{2} + \frac{z}{4} - \frac{(1+z)L_J(z)}{2z}.$$

This proves the ceiling used below,

$$V_G(x) = \frac{3}{4} + \frac{(x-1)^2(71x^2+48x+1)}{96x(x+1)(13x^2+14x+3)},$$

because direct subtraction gives the positive difference

$$V_G(x) - \frac{3}{4} - U(z) = \frac{(x-1)^6}{96x(3x+1)(13x^2+14x+3)(21x^2+18x+1)} > 0.$$

Finally

$$\bar{V}(x) - V_G(x) = \frac{(x-1)^3(11x^2+96x+13)}{96x(x+1)(35x+13)(13x^2+14x+3)} \geq 0$$

gives $V \leq \bar{V} = \frac{3}{4} + \frac{2(x-1)^2}{(x+1)(35x+13)}$.

Ceilings and floors. Splitting (G.27) at x, bounding the integral over $[1,x]$ by $(x-1)(\bar{V}-\frac{3}{4})$ or by a sharper rational bound, and bounding $(s-x)^2/(s+x)^2$ on $[x,\beta]$ by a quadratic or linear function of $s-x$, gives the rational ceilings of the bridge norm used by the gates:

$$\bar{u} = \tfrac{3}{4} + \tfrac{1}{\beta-1}\Big[\tfrac{2(x-1)^3}{(x+1)(35x+13)} + \tfrac{(\beta-x)^3}{48x^2}\Big], \qquad \hat{u} = \tfrac{3}{4} + \tfrac{(x-1)^2(\beta+7x-8)}{4(x+1)(35x+13)(\beta-1)},$$
$$u_\ell = \tfrac{3}{4} + \tfrac{(x-1)^2(\beta+15x-16)}{8(x+1)(35x+13)(\beta-1)}, \qquad u_C = \tfrac{3}{4} + \tfrac{680(x-1)^3+3(x+1)^2(\beta-x)}{9600(x+1)^2(\beta-1)},$$
$$u_s = \tfrac{3}{4} + \tfrac{1}{\beta-1}\Big[\tfrac{17(x-1)^3}{240(x+1)^2} + \tfrac{\beta-x}{1200}\Big], \qquad u_O = \tfrac{3}{4} + \tfrac{(x-1)^2(x+\beta-2)}{(\beta-1)(x+1)(35x+13)},$$
$$u_{CB} = \tfrac{3}{4} + \tfrac{(x-1)(V_G(x)-3/4)+(\beta-x)/5808}{\beta-1},$$

together with the floor $u_f = \frac{3}{4} + \frac{(x-1)^2}{28(x+1)^2}$ and the integral floor u_* of gate S6 (Appendix G.4.7). The ceiling $\bar{u}$ holds for all $1 < x \leq \beta$; each of the others holds on the shapes where its gate uses it.

G.4.2. *Orders and the comparison argument.* Normalize the shape by β: put $\ell = 1/x$, $p_1 = \alpha/\beta$, $q_1 = x/\beta$ and $\mathrm{avg}(p,q) = \mathrm{Sec}(p,1,q)$, the mean of $g(s) = 1 - s/(1+s)^2$ over $[p,q]$, with $\mathrm{avg}(p,p) = g(p)$ by continuity. The equal secants (G.18) read $\mathrm{avg}(\ell, 1/q_1) = \mathrm{avg}(p_1, q_1)$. An order compares p_1 with a *barrier* $P(z)$, a differentiable function with $P(1) = \ell$, through the comparison function

$$\Phi(z) = \mathrm{avg}(P(z), z) - \mathrm{avg}(\ell, 1/z), \qquad \Phi(1) = 0.$$

Because avg decreases in its left end, the sign of $\Phi(q_1)$ decides whether p_1 lies above or below $P(q_1)$.

Lemma G.43 (First contact)**.** *Let $a < 1$ and let Φ be continuous on $[a, 1]$ with $\Phi(1) = 0$. If Φ has a negative derivative at each of its zeros in $(a, 1]$, then $\Phi > 0$ on $(a, 1)$.*

Proof. Since $\Phi(1) = 0$ and $\Phi'(1) < 0$, $\Phi > 0$ on some interval $(1 - \varepsilon, 1)$. If $\Phi(z_0) \le 0$ for some $z_0 \in (a, 1)$, then Φ has a zero in $[z_0, 1 - \varepsilon]$; let z^* be the largest one. Then $\Phi > 0$ on $(z^*, 1)$ and $\Phi(z^*) = 0$, so $\Phi'(z^*) \ge 0$, a contradiction. □

At a zero of Φ both averages equal a common value u, and differentiation gives

$$\begin{aligned}\Phi'(z)\,(z - P(z))\,z\,(x - z)\\ = -P'(z)\big(g(P(z)) - u\big)z(x - z) - \big(u - g(z)\big)\big((2z - P(z))x - z^2\big).\end{aligned}$$

Thus the sign at a contact depends on the ratio of the two endpoint gaps. We first bound this ratio by comparing positive weights in its integral representation.

Lemma G.44 (Endpoint gaps)**.** *For $0 < p < q \le 1$, put $m = \operatorname{avg}(p, q)$. Then*

$$\frac{3 - 2p - q}{3 - p - 2q} + \frac{q - p}{6} < \frac{g(p) - m}{m - g(q)} < \frac{\log 2 - 1/2}{3/4 - \log 2}.$$

Proof. Write $d = q - p$ and

$$h(t) = -g'(p + dt) = \frac{1 - p - dt}{(1 + p + dt)^3}, \qquad 0 < t < 1.$$

Integration by parts gives $g(p) - m = d\int_0^1 (1 - t)h(t)\,\mathrm{d}t$ and $m - g(q) = d\int_0^1 th(t)\,\mathrm{d}t$. Their ratio is $(1 - \mu)/\mu$, where μ is the mean of t with weight h. Multiplying a positive weight by a decreasing function lowers this mean: its covariance with t is the integral of $(t - s)(a(t) - a(s))/2$ against two copies of the probability measure, and is nonpositive. Apply this to the linear weight $1 - p - dt$ and the strictly decreasing factor $(1 + p + dt)^{-3}$. The ratio for the linear weight is $(3(1 - p) - d)/(3(1 - p) - 2d)$, giving the first term of the lower bound.

More generally, put $b = 1 - p$. The quotient $h(t)/[(b - dt)(1 - \eta t)]$ decreases whenever $0 \le \eta \le 3d/(1 + p + 4d)$: its logarithmic derivative has the sign of $\eta(1 + p + 4dt) - 3d$, which is negative for $0 < t < 1$. The same covariance comparison gives

$$\frac{g(p) - m}{m - g(q)} > \mathcal{Q}_\eta(p, q) := \frac{\int_0^1 (1 - t)(b - dt)(1 - \eta t)\,\mathrm{d}t}{\int_0^1 t(b - dt)(1 - \eta t)\,\mathrm{d}t}. \tag{G.28}$$

These weights are positive on the interior, since $\eta < 3/4$. For the quantitative improvement take $\eta = d/2$, which is admissible because $1 + p + 4d < 6$. If μ_0, m_2 and σ^2 are the mean, second moment and variance for the linear weight,

direct integration gives

$$\frac{\sigma^2}{\mu_0^2} = \frac{1}{3} + \frac{d(6b-5d)}{6(3b-2d)^2} > \frac{1}{3},$$

because $0 < d \le b$. Tilting increases the gap ratio by

$$\frac{\eta\sigma^2}{\mu_0(\mu_0 - \eta m_2)} > \frac{\eta\sigma^2}{\mu_0^2} > \frac{\eta}{3} = \frac{d}{6}.$$

The denominator is positive, as it is a positive multiple of the first moment of the tilted weight. This proves the stated refinement.

For the upper bound compare h with $h_0(t) = (1-t)/(1+t)^3$. The quotient

$$\frac{h(t)}{h_0(t)} = \frac{1-p-dt}{1-t}\left(\frac{1+t}{1+p+dt}\right)^3$$

increases: the two factors have derivatives with numerators $1-q \ge 0$ and $1+2p-q > 0$, respectively. Thus h has a larger mean than h_0 and a smaller gap ratio. The latter ratio is exactly $(\log 2 - 1/2)/(3/4 - \log 2)$. Its numerator and denominator are positive integrals of $(1-t)h_0(t)$ and $th_0(t)$, respectively, so no numerical logarithm bound is needed. □

The global upper order. Take $k = 5/4$ and $P(z) = \ell + k(1-z)$. If $P(q_1) \ge q_1$, the upper bound $p_1 \le P(q_1)$ is automatic from $p_1 < q_1$. On the interval $P(z) < z$, put

$$a = 1-\ell, \qquad \theta = \frac{1-z}{a}, \qquad R_0 = \frac{3-2P(z)-z}{3-P(z)-2z}.$$

Here $0 < a < 1$ and $0 \le \theta < 4/9$, since $P(z) < z$. The following identity makes the contact sign positive:

$$\begin{aligned} &kR_0 z(x-z) - \big((2z-P(z))x - z^2\big) \\ &\quad = \frac{a}{4(1-a)(4+3\theta)}\Big[1 + 6\theta^2 + 7(3\theta-1)^2 \\ &\qquad + 6(1-a)\theta\big(2(4-9\theta) + 7\theta^2\big) + 6(1-a)^2\theta^2(4-7\theta)\Big] > 0. \end{aligned}$$

Every factor in the bracket is nonnegative, and the constant term is positive. By the lower gap bound of Lemma G.44, $\Phi' > 0$ at every zero, including $z = 1$. Apply Lemma G.43 to $-\Phi$: then $\Phi < 0$ to the left of one, and decreasing the left endpoint of the average gives $p_1 < P(q_1)$ if $q_1 < 1$. This proves the upper order $\alpha \le \beta/x + \frac{5}{4}(\beta - x)$ of Lemma G.35(b). If $q_1 = 1$, equality of the two averages forces $p_1 = \ell$, so the same bound holds.

The upper order for $x \le 5$. Now take $k = 9/8$ and $P(z) = \ell + k(1-z)$. Again only the interval $P(z) < z \le 1$ needs a comparison. Put

$$a = 1-\ell, \qquad 0 \le \theta = \frac{1-z}{a} < \frac{8}{17}, \qquad E = 1 + \frac{7}{8}\theta, \qquad D = 1 - \frac{17}{8}\theta > 0.$$

Let $R_0 = (3 - 2P(z) - z)/(3 - P(z) - 2z)$ and define the normalized contact expression

$$\mathcal{N}(\ell, \theta) = \frac{E(1-a)}{a}\left[kR_0 z(x-z) - ((2z - P(z))x - z^2)\right].$$

This expression does not decrease in ℓ. Indeed,

$$\partial_\ell \mathcal{N} = \frac{\theta}{32}\left[73\theta^2 - 186\theta + 80 + 2\ell\theta(40 - 73\theta)\right] \geq 0.$$

On $0 \leq \theta < 1/2$, the first quadratic decreases and is at least its value $21/4$ at $1/2$; also $40 - 73\theta > 7/2$. By the refined gap bound, the actual normalized contact sign is strictly greater than

$$\mathcal{N} + \frac{3}{16}aDE(1 - a\theta)(1 + \ell\theta) \geq \mathcal{N} + \frac{3}{16}aD(1 - \theta).$$

For $1 < x \leq 3$, use $\ell \geq 1/3$ and the identity

$$\mathcal{N}(1/3, \theta) = \frac{9}{4}\left(\theta - \frac{1}{3}\right)^2 + \frac{\theta(30 + 53\theta + 292\theta^2)}{576} > 0.$$

For $3 \leq x \leq 5$, use $a \geq 2/3$ and $\ell \geq 1/5$. The lower bound is then at least

$$\begin{aligned}\mathcal{N}(1/5, \theta) + \frac{1}{8}D(1 - \theta) &= \frac{27}{8}\left(\theta - \frac{1}{3}\right)^2 + \theta\left(\frac{5}{64} - \frac{13\theta}{160}\right) + \frac{73\theta^3}{200} \\ &\geq \frac{27}{8}\left(\theta - \frac{1}{3}\right)^2 + \frac{3\theta}{80} > 0,\end{aligned}$$

since $\theta < 1/2$. The contact derivative is therefore positive in both ranges. The same first-contact argument as for $k = 5/4$ gives $\alpha \leq \beta/x + \frac{9}{8}(\beta - x)$, including equality when $\beta = x$.

The upper order in the small box. We prove Lemma G.35(c). Here $q_1 = 1/(1 + \delta) \geq 19/20$, so it suffices to compare barriers on $19/20 \leq z \leq 1$. If $x \leq 16/9$, the upper order just proved gives $\alpha \leq 1 + (1 + 9x/8)\delta \leq 1 + 3\delta$.

For $16/9 \leq x \leq 2$, take $P(z) = \ell + 1 - z$, put $a = 1 - \ell$ and $\theta = (1 - z)/a$, and use (G.28) with $\eta = (z - P(z))/2$. We have $a \geq 7/16$ and $\theta \leq 4/35 < 1/8$. Direct integration of the tilted linear weight gives

$$\begin{aligned}&\mathcal{Q}_\eta(P(z), z)z(x - z) - ((2z - P(z))x - z^2) \\ &\quad = \frac{a^2(1 - 2\theta)}{(1 - a)(4a\theta^2 + 4\theta - a + 4)}[1 - 6\theta - 6\theta^2 - 2a(1 - a)\theta^3] > 0.\end{aligned}$$

The bracket exceeds $1 - 6/8 - 6/64 - 1/1024 > 0$, using $a(1 - a) \leq 1/4$. First contact therefore gives $\alpha \leq \beta/x + (\beta - x) = 1 + (1 + x)\delta \leq 1 + 3\delta$.

It remains to consider $2 \leq x \leq 13/6$, so $6/13 \leq \ell \leq 1/2$. Take $P(z) = \ell(3 - 2z)$ and, at a contact, put $p = P(z)$, $s = 1 - z$, $b = 1 - p$ and $d = z - p = b - s$. The maximal tilt $\eta = 3d/(1 + p + 4d)$ in (G.28) has the ratio

$$\mathcal{T}(b, s) := \mathcal{Q}_\eta(p, z) = \frac{N}{D}, \qquad \begin{aligned} N &= 9b^2 - 10bs - 5s^2 + 8b + 4s, \\ D &= 3b^2 - 2bs - 7s^2 + 4b + 8s. \end{aligned}$$

We compare $2\ell\mathcal{T}$ with $M = 1 + (z-p)/[z(1-\ell z)]$, since their difference has the sign of the contact derivative. On this rectangle $9/20 \le b \le 7/13$, $0 \le s \le 1/20$ and $p \le 11/20$. Also $d/b \ge 8/9$, so the linear-weight ratio is at least $19/11$ and $\mathcal{T} > 17/10$. The positive expressions above give

$$D > \frac{12}{5}, \qquad N_b < 18, \qquad D_b > \frac{13}{2}, \qquad \mathcal{T}_b = \frac{N_b - \mathcal{T}D_b}{D} < 3.$$

For the first inequality use $D \ge b(3b+4)$; for the last use $18-(17/10)(13/2) < 3(12/5)$. Consequently

$$\partial_p(p\mathcal{T}) = \mathcal{T} - p\mathcal{T}_b > \frac{1}{20}, \qquad M_\ell = \frac{(z-1)(z+3)}{z(1-\ell z)^2} \le 0.$$

Thus $2\ell\mathcal{T} - M$ increases in ℓ, and it suffices to take $\ell = 6/13$.

Along this edge $b = (7-12s)/13$. Write $k = 12/13$ for b_z. The two quadratics satisfy $N_{zz} = 18k^2 + 20k - 10 > 0$ and $D_{zz} = 6k^2 + 4k - 14 < 0$. Their endpoint values and the upper tangent to D at $z = 1$ give

$$N_z \ge N_z(19/20) > 16, \qquad D_z \le D_z(19/20) < \frac{1}{8}, \qquad D < \frac{31}{10}.$$

Moreover $\mathcal{T} < 5/2$, since $5D - 2N = b(4-3b) + s(10b + 32 - 25s) > 0$. It follows that

$$\mathcal{T}_z = \frac{N_z - \mathcal{T}D_z}{D} > \frac{16 - 5/16}{31/10} > 5.$$

On the same edge,

$$M_z = \frac{6(25z^2 - 36z + 39)}{z^2(13-6z)^2} < 4 :$$

its numerator is at most 168, while $z(13-6z) > 13/2$. Therefore $(2\ell\mathcal{T} - M)_z > 8/13 > 0$, and the smallest value is at $z = 19/20$. At that corner $s = 1/20$ and $d = 23/52$. The inequality $(23/52)^2 > 39/200$ gives $N - 2D = 3d^2 - 6s(2-s) > 0$, so $\mathcal{T} > 2$. Also $d < 4/9$ and $z(1-\ell z) > 8/15$, giving $M < 11/6 < 24/13 = 4\ell$. The contact sign is positive. First contact gives $p_1 \le \ell(3 - 2q_1)$, or $\alpha \le 3\beta/x - 2 = 1 + 3\delta$, as required.

Feasibility of a lower barrier. The lower-order proofs below use the following observation. Suppose $0 < z_m < 1$, $P(1) = \ell$, $P(z_m) = z_m$ and $P(z) < z$ for $z_m < z \le 1$. If its comparison function Φ is positive on $(z_m, 1)$, the actual equal-secant shape must have $q_1 > z_m$. Indeed, put

$$H(b) = \int_\ell^b g(s)\,\mathrm{d}s - (b-\ell)g(b), \qquad b \ge 1.$$

This decreases because $H'(b) = -(b-\ell)g'(b) \le 0$. At the actual shape, $\operatorname{avg}(\ell, 1/q_1) = \operatorname{avg}(p_1, q_1) > g(q_1)$, so $H(1/q_1) > 0$, using $g(1/q_1) = g(q_1)$. It follows that $\operatorname{avg}(\ell, 1/z) > g(z)$ for $q_1 \le z \le 1$. If $q_1 \le z_m$, this makes $\Phi(z_m) < 0$, contrary to continuity and $\Phi > 0$ on $(z_m, 1)$. Thus $q_1 > z_m$.

The strong lower order. We prove Lemma G.35(a), without a restriction on x. Put $a = 1 - \ell \in (0,1)$ and $k = 4(2-a)/(8-a) \in (0,1)$, and take $P(z) = \ell + k(1-z)$. For $p = P(z) < z$, use the relative endpoint deficits

$$b = 1 - p, \qquad r = \frac{1-z}{1-p}, \qquad b = \frac{a}{1+kr}, \qquad 0 < b < 1, \quad 0 \le r < 1.$$

Write $\mathcal{R}(b,r)$ for the endpoint-gap ratio on $[1-b, 1-br]$. At fixed r its integral weight, up to a constant factor, is

$$w_b(t) = \frac{v(t)}{(2 - bv(t))^3}, \qquad v(t) = 1 - (1-r)t.$$

The ratio w_{b_2}/w_{b_1} decreases in t when $b_2 > b_1$: its logarithmic derivative with respect to v is $6(b_2 - b_1)/[(2 - b_1 v)(2 - b_2 v)] > 0$. The covariance argument in Lemma G.44 therefore shows that $\mathcal{R}$ increases in b.

Simpson's bound for the reciprocal integral gives a convenient upper estimate. In the form

$$L(y) = \frac{y-1}{6}\left(1 + \frac{8}{1+y} + \frac{1}{y}\right), \qquad (L - \log)'(y) = \frac{(y-1)^4}{6y^2(1+y)^2},$$

it follows directly that $\log y < L(y)$ for $y > 1$. Substitute $y = (1+q)/(1+p)$ in

$$\operatorname{avg}(p,q) = 1 + \frac{1}{(1+p)(1+q)} - \frac{\log y}{q-p}.$$

The resulting lower mean m_L lies above $g(q)$: for $p = 1-b$, $q = 1-br$,

$$m_L - g(q) = \frac{b^2(1-r)\,[4 + 8r - br(5+r)]}{6(2-b)(2-br)^2(4-b-br)} > 0,$$

since the bracket is at least $4 + 3r - r^2 > 0$. The gap ratio decreases as the mean increases, so

$$\mathcal{R}(b,r) < \widehat{\mathcal{R}}(b,r) := \frac{2-br}{2-b}\,\frac{8 + 4r - b(1+5r)}{4 + 8r - br(5+r)}.$$

The slope satisfies $k - (1 - a/2) = a(2-a)/[2(8-a)] > 0$. Thus $b \le b_* := a/[1 + (1 - a/2)r]$, and comparison of the weights followed by the last bound gives $\mathcal{R}(b,r) < \widehat{\mathcal{R}}(b_*, r)$. Set $B = 1 + (1-r)(2/k - 1)/[1 + (2-a)r]$ and write

$$H_* = 4 + (12 - 7a)r + (8 - 5a)r^2,$$
$$Q = (1+r)\{4(1-a)(7r+1) + a(9r+11)\} + 2a(1-a)r(5-r).$$

Multiplication of the two fractions gives the positive-product identity

$$B - \widehat{\mathcal{R}}(b_*, r) = \frac{ar(1-r)Q}{2(2-a)(1+r)(1+(2-a)r)H_*} \ge 0.$$

Here $H_*, Q > 0$, so $\mathcal{R} < B$, also when $r = 0$ by the strict Simpson bound.

It remains to compare this upper bound with the contact threshold $M = 1 + (z-p)/[z(1-\ell z)]$. Put $h = k - a$ and

$$B_0 = 1 + \frac{(1-r)(2/k-1)}{1+(1+h)r}.$$

Since $k < 1$, we have $B \le B_0$, while substitution of $p = 1 - a/(1+kr)$ and $z = 1 - ar/(1+kr)$ gives

$$M - kB_0 = \frac{r\big[(1-r)(2-k(1+h)) + r(1-k)(1+h)(2+h)\big]}{(1+hr)(1+(1+h)r)} \ge 0.$$

Here $-1 < h < k < 1$ and $2 - k(1+h) > 2 - k(1+k) = (1-k)(k+2) > 0$. Consequently $k\mathcal{R} < kB \le kB_0 \le M$, so $\Phi' < 0$ at every zero, including $z = 1$. First contact gives $\Phi > 0$ between the crossing $P(z_m) = z_m$ and 1. The preceding feasibility argument, applied to this crossing, shows that the actual $q_1 > z_m$. Thus $p_1 > P(q_1)$ when $q_1 < 1$; when $q_1 = 1$, $p_1 = \ell$. Multiplying by β proves the claimed lower order.

The quadratic lower order. For Lemma G.35(d), retain $a = 1 - \ell$ and $k = 4(1+\ell)/(7+\ell)$, but curve the barrier:

$$P(z) = \ell + ks + \frac{\ell s^2}{1+3s}, \qquad s = 1 - z.$$

Here $6/13 \le \ell \le 1/2$, so

$$\frac{1}{2} \le a \le \frac{7}{13}, \qquad \frac{3}{4} < k \le \frac{4}{5}, \qquad \frac{1}{5} < k - a \le \frac{3}{10}.$$

For $p = P(z) < z$, put

$$K = k + \frac{\ell s}{1+3s}, \qquad j = -P'(z) = K + \frac{\ell s}{(1+3s)^2}, \qquad r = \frac{s}{1-p}.$$

Then $0 \le r < 1$, $b = 1 - p = a/(1+Kr)$ and $s = ar/(1+Kr)$. The condition $p < z$ gives $s < a/(1+k) < 1/3$, hence $k \le K < 9/10$ and $j \ge k > 3/4$. At a contact it suffices to prove $j\mathcal{R}(b,r) < M(K)$, where

$$M(v) = 1 + \frac{(1-r)(1+vr)}{(1+(v-a)r)(1+(v+\ell)r)}.$$

This is the contact threshold $1 + (z-p)/[z(1-\ell z)]$ when $v = K$.

First keep the positive gain in the preceding comparison. With the same B, B_0, H_*, Q as there, put $E = B - \widehat{\mathcal{R}}(b_*, r) \ge 0$. Since $K \ge k$, comparison of the weights still gives $\mathcal{R}(b,r) < B - E$. Write $D = M(k) - kB_0$. The preceding positive-product identity and the bounds on $k - a$ give

$$D \ge \frac{r\big[\frac{19}{20}(1-r) + \frac{1}{2}r\big]}{(1+3r/10)(1+13r/10)}.$$

Indeed its two numerator factors $2-k(1+k-a)$ and $(1-k)(1+k-a)(2+k-a)$ exceed $19/20$ and $1/2$. The remaining two gains satisfy

$$k(B_0 - B) \geq \frac{r(1-r)}{5(1+2r)^2}, \qquad jE \geq \frac{r(1-r)}{5(1+2r)^2}. \tag{G.29}$$

For the first, subtract the two fractions:

$$k(B_0 - B) = \frac{r(1-r)(2-k)(1-k)}{(1+(1+k-a)r)(1+(2-a)r)}.$$

Both denominator factors are at most $1+2r$ and $(2-k)(1-k) > 1/5$. For the second, the positive expressions for Q, H_* give

$$Q \geq 7(1+r)(1+3r/2), \qquad H_* \leq 4(1+2r)^2.$$

Together with $a/[2(2-a)] \geq 1/6$ and $2-a \leq 3/2$, these imply $E \geq 7r(1-r)/[24(1+2r)^2]$. Now use $j > 3/4$ and $7/32 > 1/5$.

Next bound the two costs of curving the barrier. Set $\eta = K - k$. The identities for s give

$$\eta = \frac{\ell a r}{1+(K+3a)r}, \qquad j - k = \frac{\ell a r\,[2+(2K+3a)r]}{(1+(K+3a)r)^2}.$$

Since $\ell a \leq 1/4$, $K+3a > 9/4$ and $2K+3a < 7/2$,

$$\eta \leq \frac{r}{4(1+9r/4)}, \qquad j - k \leq \frac{r(2+7r/2)}{4(1+9r/4)^2}. \tag{G.30}$$

Also $2/k - 1 < 8/5$ and $2-a > 7/5$, so $B \leq (13-r)/(5+7r)$. For the change in M, put

$$F(v) = \frac{1+vr}{(1+(v-a)r)(1+(v+\ell)r)}, \qquad k \leq v \leq K.$$

With $u = 1+(v-a)r \geq 1+r/5$, direct differentiation gives

$$0 \leq -F'(v) = \frac{r(u^2+2aru+ar^2)}{u^2(u+r)^2} \leq \frac{r}{u}.$$

For the last inequality compare the three numerator terms with those of $u(u+r)^2$, using $u \geq 1 > a$. Integration over $[k,K]$, followed by (G.30), yields

$$0 \leq M(k) - M(K) \leq \frac{(1-r)\eta r}{1+r/5} \leq \frac{r^2(1-r)}{4(1+9r/4)(1+r/5)}.$$

For $r > 0$, normalize by $S = (1+2r)^2/r$. The three denominator comparisons

$$(1+3r/10)(1+13r/10) \leq 1+2r,$$
$$1+2r \leq 1+9r/4,$$
$$(1+9r/5)(1+r/5) \geq 1+2r$$

give

$$SD \geq \left(\frac{19}{20} - \frac{9}{20}r\right)(1+2r),$$
$$S(j-k)B \leq \frac{(4+7r)(13-r)}{8(5+7r)} \leq \frac{13}{10} + \frac{r}{10} + \frac{3}{10}r(1-r),$$
$$S\big(M(k) - M(K)\big) \leq \frac{r(1-r)(1+9r/5)}{4}.$$

The second upper bound follows by division:

$$\frac{(4+7r)(13-r)}{8(5+7r)} = \frac{13}{10} + \frac{3r}{40} + \frac{7r(1-r)}{5(5+7r)}.$$

Combining these estimates with (G.29) gives

$$\begin{aligned} S\big(M(K) - j\mathcal{R}(b,r)\big) &> \left(\frac{19}{20} - \frac{9}{20}r\right)(1+2r) + \frac{2}{5}(1-r) \\ &\quad - \left(\frac{13}{10} + \frac{r}{10} + \frac{3}{10}r(1-r)\right) - \frac{r(1-r)(1+9r/5)}{4} \\ &= \frac{1+r^3}{20} + \frac{2}{5}r(1-r)^2 > 0. \end{aligned}$$

When $r = 0$, $j = k$, $M(K) = 2$ and $E = 0$, while the Simpson bound still gives $\mathcal{R} < B = 2/k$. Thus the contact derivative is negative in this case as well.

The function $P(z) - z$ decreases strictly and has a unique zero $z_m \in (0,1)$. First contact gives $\Phi > 0$ on $(z_m, 1)$; the same feasibility function H as above excludes $q_1 \leq z_m$. Hence $p_1 > P(q_1)$ for $q_1 < 1$, with equality for $q_1 = 1$. Finally, $s = \delta/(1+\delta)$ and multiplication by $\beta = x(1+\delta)$ transform the barrier into

$$\beta P(q_1) = 1 + (1+xk)\delta + \frac{\delta^2}{1+4\delta},$$

which is the order claimed in part (d).

G.4.3. *The outer scale bound.*

Lemma G.45 (Reverse-centre scale bound)**.** *At a reverse centre with neighbour ratios $0 < r, t < 1/2$, scale $0 < \rho < rt$, and norms $3/4 \leq u \leq \mathcal{U}(r)$, $3/4 \leq v \leq \mathcal{U}(t)$ and the stationarity $E = 1 + r + t - \rho + \log\frac{\rho}{rt} - \frac{1-u}{r} - \frac{1-v}{t}$, the inequality $(4uv - E^2)/(16(1-\rho)^2) < 9193/50000$ implies $\rho < 1/7$.*

Proof. Assume $\rho \geq 1/7$. First consider the pairing E_0 at $u = v = 3/4$. With $\sigma = rt$ and $\chi = r + t$, the inequalities $r, t < 1/2$ give $\chi \leq 1/2 + 2\sigma$, so

$$E_0 = 1 - \rho + \log\frac{\rho}{\sigma} + \chi\left(1 - \frac{1}{4\sigma}\right) \geq L(\sigma) := 1 - \rho + \log\frac{\rho}{\sigma} + 2\sigma - \frac{1}{8\sigma}.$$

Since $L'(\sigma) = (4\sigma - 1)^2/(8\sigma^2) \geq 0$ and $\sigma > \rho$, $E_0 \geq L(\rho) = 1 + \rho - 1/(8\rho) > 0$. The pairing increases with either norm, hence $E \geq E_0 > 0$.

The switch ceilings also give $E < -1 + 2r + 2t - rt \le 3r/2$, and symmetrically $E \le 3t/2$. For the first inequality use that $\log\rho - \rho$ increases for $\rho < 1$; for the second use $-1 + r/2 + 2t - rt = (1-2t)(r/2-1) \le 0$. Consequently

$$\partial_u(4uv - E^2) = 4v - \frac{2E}{r} \ge 4v - 3 \ge 0,$$

and the analogous derivative in v is nonnegative. Lowering both norms to $3/4$ therefore lowers the Gram quotient. At those norms, $\partial_r E_0 = (r-1/2)^2/r^2 \ge 0$, and likewise for t. Letting both ratios increase to $1/2$ raises the positive pairing and lowers the quotient again:

$$\frac{4uv - E^2}{16(1-\rho)^2} \ge G(\rho) := \frac{9/4 - B(\rho)^2}{16(1-\rho)^2}, \qquad B(\rho) = 1 - \rho + \log 4\rho.$$

On $[1/7, 1/4)$ the function B is positive, and G increases by the fixed-S monotonicity in the proof of Lemma E.12, at $S = 0$. To estimate the initial value, pair the reciprocal integral about its midpoint. With $m = 11/8$ and $h = 3/8$, strict Jensen gives

$$\log\frac{7}{4} = 2m\int_0^h \frac{\mathrm{d}t}{m^2 - t^2} > \frac{2mh}{m^2 - h^2/3} = \frac{33}{59}.$$

Here Jensen applies to the strictly convex function $s \mapsto 1/s$ and the nonconstant positive function $m^2 - t^2$, whose average is $m^2 - h^2/3$. Since $0 < B(1/7) < 6/7 - 33/59$,

$$G(1/7) > \frac{9/4 - (6/7 - 33/59)^2}{16(6/7)^2} > \frac{9193}{50000},$$

where the last step is an exact rational comparison. This contradicts the assumed Casimir cap. □

Applied at the outer centre A, with $r = r_A$, $t = 1/\tau$, $\rho = q$, norms u, w and pairing E_L, the lemma gives $q < 1/7$ (Lemma G.34(e)).

G.4.4. *The comparison transformations.* The transformations in Appendix G.3.3 use the following formulas.

The scale coefficients. With $\varepsilon = 1 - \alpha^{-2}$, $n = (1+x)/\alpha^2$, $g = D/\alpha^2$ and $E_0 = (1+x)(u-1) + 1/(1+x)$, the coefficients of (G.22) are

$$\begin{aligned} A_1 &= 2\varepsilon + n + gE_0, \\ A_2 &= \varepsilon^2 + \varepsilon n + n^2 + g^2 u + g\varepsilon E_0 - g(2(1+x)-1)/\alpha^2, \\ A_3 &= 2gu + \varepsilon E_0 - (2(1+x)-1)/\alpha^2. \end{aligned}$$

The two scales are related by $\rho/(1-\rho) = q/(\alpha^2(1-q))$, and $q = q_c$ exactly when $\rho = r^2$; also $q_c \le 1/7$ if and only if $6\alpha^2 \le x(x+2)$.

The duration tests. The identity (G.23) is the logarithm bound $\log y \le (y - 1/y)/2$ at $y = \tau/2$. For the three tests of gate S0, the logarithm $\log(\tau/\tau_{\max})$ is bounded by $\log y \le y - 1$ and by the tangent $\log y \le \frac{5}{4}y - \frac{11}{9}$ at $y = 4/5$ (using $\log\frac{5}{4} \ge \frac{2}{9}$), and $1/\tau$ by its chord; an affine function of τ is

then at most its larger endpoint value. This gives the values $\tau = 2$, $8\tau_{\max}/9$ and $\tau_{\max}$, with the logarithm replaced by $\frac{5}{2\tau_{\max}} - \frac{11}{9}$, $-\frac{1}{9}$ and 0 respectively.

G.4.5. *Gates S0 to S4.* Apart from the face signs of Appendix G.4.8, these gates use the following inequalities.

Gate S0. The positive diagonal is in the proof of Proposition G.36. For the positive pairing, the envelope bound for E and the replacements $D \geq D_0$ and $u \leq u_0 = \frac{3}{4} + \frac{(x-1)^2}{12(x+1)^2}$ give $E - 2\rho D u \leq \Lambda(\rho) := \big((1+x) - 2\rho D_0\big)u_0 - (1+x) + \frac{1}{1+x} + \rho(2x+1)$, affine in $\rho \in [0, (1+x)^{-2}]$, with

$$\Lambda(0) = -\frac{(x-1)(x+5)}{6(1+x)},$$
$$\Lambda\big((1+x)^{-2}\big) = -\frac{x^5 - 13x^4 + 6x^3 + 32x^2 + 37x + 9}{6(1+x)^4},$$

both negative on $1 < x \leq 12/5$. The shape bound $\alpha/\beta < 3/4$ for $x \geq 12/5$ compares two bounds for $u - 3/4$: if $\alpha \geq 3\beta/4$ then $u - \frac{3}{4} \leq \frac{(\beta-\alpha)^2}{4(\beta+\alpha)^2} \leq \frac{1}{196}$, while (G.27) and the tangent bound $1/y^2 \geq 1/c^2 - 2(y-c)/c^3$ give $u - \frac{3}{4} \geq \frac{4(x-1)^3}{(4x-3)(5x+3)^2}$, which increases in x and exceeds $1/196$ at $x = 12/5$.

Gate S1. The combination $\Gamma - \frac{7}{96}R_\tau$ is monotone along the envelope path because the ratio of the derivatives of Γ and R_τ along that path is at most $7/96$. Along $\tau = 2$ the slope of R_2^* in q exceeds one: after completing the square in Δq with $\Delta = D/\alpha^2$, $q\,\mathrm{d}R_2^*/\mathrm{d}q$ is bounded below by a quadratic in $a = \alpha^{-2} \in (0, 1]$ with nonpositive leading coefficient, and its values at $a = 0$ and $a = 1$ exceed $1/7 \geq q$ by explicit rational functions of $z = x - 1$ whose numerators have positive coefficients.

Gate S4. At the central scale put

$$S = 1+x, \quad V = x(x+2), \quad N = \alpha^2 + V, \quad R = \frac{x-1}{x+1}, \quad d = \frac{D}{N}, \quad C_* = \frac{S^3}{N}.$$

Here $q_c = \alpha^2/N$, $\rho(q_c) = S^{-2}$ and $\Gamma_c := \Gamma_*(q_c)$. The central row and Gram determinant have the compact form

$$16\Gamma_c = 3 - b_0^2 - 2Rb_0, \qquad b_0 = \frac{S^3(\mathcal{U}_1 - u)}{V}, \quad \mathcal{U}_1 = \frac{3+R^2}{4}.$$

Consequently $\partial_u \Gamma_c = S^3(b_0 + R)/(8V)$. For $u \geq 3/4$, $S^3/V = S(1 + 1/V) < 18/5$ and $R^2 < 1/7$ give $b_0 < 9/70 < 1/6$. Since $R < 3/8$,

$$\partial_u \Gamma_c < \frac{18}{5}\frac{1/6 + 3/8}{8} < \frac{1}{4}.$$

The transport formulas give

$$\partial_u w_* = d^2 - dC_*, \qquad \partial_u E_{L,*} = 2d - C_*,$$

so the exact residual at any duration satisfies

$$\partial_u R^*_\tau = \frac{1}{r_A} + C_*(1-\tau d) + \tau d^2 - 2d > 2.$$

To see the last sign, first note that $0 < D < S$. On this box $\alpha \geq 1$, $\beta \leq 130/57$ and $x-\alpha \leq 7/6$, so $D = \alpha^2+1+2x-2\beta(x-\alpha) \geq 116/171 > 0$. For the upper bound use $\beta \geq x$ and $\alpha \leq 22/19$: $D-S \leq \alpha^2+2\alpha x+x-2x^2 \leq \alpha^2+4\alpha-6 < 0$; the middle expression decreases in $x \geq 2$, and $(22/19+2)^2 < 10$. As $N \geq S^2$, it follows that $d < 1/S \leq 1/3$. Also $r_A \leq 11/30 < 3/8$ and $\tau d \leq \tau_{\max} d = g r_A < 1$; the last inequality is $\beta(2x-\alpha) > 1+2x$, proved directly in Appendix G.4.6. Thus the positive C_* term may be discarded and $\partial_u R^*_\tau > 1/r_A - 2d > 2$.

It follows that $\partial_u(\Gamma_c - R^*_\tau/7) < 0$ for every $2 \leq \tau \leq \tau_{\max}$ and every $u \geq 3/4$. For Φ this is the duration 2. For Ψ it is the duration $\tau_{\max}$: the difference between its affine duration bound and $R^*_{\tau_{\max}}$ is independent of u, by (G.23). Hence both S4 tests decrease in the norm throughout this range.

G.4.6. *Gate S5. Monotonicity in the norm.* At fixed shape, scale and duration the residual $\mathcal{R}(u;p,q)$ is affine in u, with

$$\partial_u \mathcal{R} = (1+x) + \frac{1}{r_A} - pg(1+x) + q\{p(g^2+g\varepsilon(1+x)) - 2g - \varepsilon(1+x)\}.$$

It is positive throughout the small-shape domain, for $0 \leq q < 1/2$ and $0 < p \leq r_A$. Indeed, $\alpha \leq 22/19 < 3/2$ and $\beta \geq x \geq 2$ imply

$$\beta(2x-\alpha) > x(2x-3/2) \geq 1+2x,$$

where the second difference is $(x-2)(2x+1/2) \geq 0$. Since $D = \alpha(\alpha+\beta) + 1 + 2x - \beta(2x-\alpha)$, this gives $g r_A < 1$. Now factor the derivative as

$$\partial_u \mathcal{R} = (1+x)(1-\varepsilon q)(1-pg) + \frac{1}{r_A} - qg(2-pg).$$

If $g \leq 0$, positivity is immediate. If $g > 0$, then $0 < pg \leq r_A g < 1$, so the first term is positive and the remaining terms are at least $(1-2q)/r_A > 0$. Here $0 \leq \varepsilon < 1$ because $\alpha \geq 1$.

The ceiling. The integral defining $V(x)$ admits the simpler global bound

$$V(x) < U(x) := \frac{3}{4} + \frac{(x-1)^2}{17x^2+26x+5} \qquad (x > 1). \tag{G.31}$$

To prove it, put $R = (x-1)/(x+1) \in (0,1)$ and

$$B(R) = \frac{R^2}{12+6R-R^2}, \qquad L(R) = \frac{2R}{1+R}\left(\frac{1}{2} + \frac{R}{4} - B(R)\right).$$

The formula for V in Appendix G.4.1 shows that $V < 3/4 + B$ is equivalent to $\log(1+R) > L(R)$. The difference vanishes at zero and has derivative

$$\frac{\mathrm{d}}{\mathrm{d}R}\big(\log(1+R) - L(R)\big) = \frac{R^4(8+12R-R^2)}{2(1+R)^2(12+6R-R^2)^2} > 0.$$

This proves (G.31).

On the small-shape domain, $R \le 7/19 < 3/8$. The function B increases in R, so

$$u \le V(x) < \frac{3}{4} + B(3/8) = \frac{3}{4} + \frac{3}{301} < \frac{19}{25}.$$

This constant ceiling leaves the shape variables independent in the duration comparison below.

Convexity of the scale quadratic. With $a = \alpha^{-2}$, $\varepsilon = 1 - a$, $\bar{H} = (1+x)/4 - 1/(1+x)$, $\mathcal{A} = \varepsilon^2 + \varepsilon(1+x)a + (1+x)^2a^2$ and $\bar{K} = \varepsilon\bar{H} + (2x+1)a$, the coefficient A_2 for $3/4 \le u < 4/5$ is at least $\mathcal{A} - \bar{K}^2/3$ for every value of g. To see this, put $k = (2x+1)a - \varepsilon E_0$. The bounds $3/4 \le u < 4/5$ give $-\bar{H} \le E_0 < 0$, so $0 < k \le \bar{K}$. Completing the square in $A_2 = \mathcal{A} + ug^2 - kg$ gives the claim.

The positivity of this lower bound is a comparison of two quadratic forms. Set $S = 1 + x$, $X = \varepsilon$, $Y = Sa$ and $b = 2 - 1/S$. Since $3 \le S \le 19/6 < 7/2$, $\bar{H} < b/2$ and $0 < b < \sqrt{3}$. Therefore $A_2 > 0$, as

$$\bar{K}^2 = (\bar{H}X + bY)^2 \le b^2(X/2 + Y)^2 \le b^2(X^2 + XY + Y^2) < 3\mathcal{A}.$$

The same argument proves $A_2 > 0$ throughout $2 \le x \le 5/2$, $\alpha \ge 1$, $3/4 \le u < 4/5$, with no restriction on β.

The duration tangent. For the rest of this gate put $u = 19/25$ and $\lambda_0 = 15/16$. We prove negativity on the larger rectangular shape domain

$$2 \le x \le 13/6, \qquad 1 \le \alpha \le 7/6, \qquad x \le \beta \le 20x/19.$$

It contains (G.24), since $\alpha \le 22/19 < 7/6$. Write $r = r_A$, $e = (1+x)(u-1) + 1/(1+x)$ and

$$X = e + \frac{u-1}{r} + r, \qquad q_b = \lambda_0 q_c.$$

The residual has the exact form

$$\mathcal{R}(p,q) = X + 1 - pA_1 + \log(p/r) + q(pA_2 + 1/p - 1 - A_3).$$

At fixed q, put $z = p/r$. On the allowed interval $2q/r \le z \le 1$,

$$\frac{\partial^2}{\partial z^2}\mathcal{R}(zr,q) = \frac{2q/r - z}{z^3} \le 0.$$

Its tangent at $z = 1$ is therefore an upper bound. The two endpoint values of this *affine tangent* are

$$\mathcal{J}(q) = X + 1 - rA_1 + q(rA_2 + 1/r - 1 - A_3),$$
$$\mathcal{T}(q) = X + q(4/r - 2A_1 - 1 - A_3) + 2q^2(A_2 - 1/r^2).$$

Consequently $\mathcal{R}(p,q) \le \max\{\mathcal{J}(q), \mathcal{T}(q)\}$. The norm $19/25$ is the common zero of these two tangent values at the central reference shape $(x, \alpha, \beta) = (2, 1, 2)$, $q = q_c$, when the norm is allowed to vary. When $q = 0$ the second value is the value of the affine tangent at zero; no logarithm is evaluated there.

Here $3/10 < r < 3/8$, $-1/2 < e \le -1/3$ and $0 < g \le 9/4$. For the last two bounds one may use $3/4 \le u < 7/9$ and $3 \le S = 1 + x < 16/5$. The diagonal D increases in α and decreases in β; evaluation at $\alpha = 1$, $\beta = 20x/19$ therefore gives $D \ge 2[1 + 2x - (20/19)x^2] > 0$, since $x^2 < 19/4$. For its upper bound set $\beta = x$ and then $x = 2$, giving $g \le 1 + 4/\alpha - 3/\alpha^2 \le 9/4$ for $\alpha < 6/5$. The identity

$$2A_1 + 1 + A_3 = 5 - 3\alpha^{-2} + 2g(u + e) + (1 - \alpha^{-2})e$$

and $u + e < 4/9$ give $2A_1 + 1 + A_3 < 59/12 < 2/r$. Since $A_2 > 0$,

$$\mathcal{T}'(q) \ge 2/r - (2A_1 + 1 + A_3) > 0 \qquad (0 \le q \le r/2).$$

Also $q_c < r/2$: the equivalent inequality $\alpha^2 + 2\alpha\beta < x(x + 2)$ follows already from $\alpha < 6/5$, $\beta < 7/3$ and $x \ge 2$. The function $\mathcal{J}$ is affine in q. At zero it is negative, since $A_1 \ge 7/8 + \alpha^{-2} > 3/2$ and

$$\mathcal{J}(0) < -\frac{1}{3} - \frac{16}{27} + \frac{3}{8} + 1 - \frac{9}{20} < 0.$$

It remains to prove $\mathcal{J}(q_b) < 0$ and $\mathcal{T}(q_b) < 0$.

Convexity in the outer shape. Both functions at q_b are convex in β. For $\mathcal{T}$ the exact second derivative is

$$\mathcal{T}_{\beta\beta} = \frac{2\alpha}{(\alpha + \beta)^3} + \frac{4q_b^2}{\alpha^4}(4u(x - \alpha)^2 - \alpha^2) > 0.$$

Here $4u(x-\alpha)^2 \ge 3(5/6)^2 > (7/6)^2 \ge \alpha^2$. For $\mathcal{J}$, g is affine and A_2 quadratic in β. Division shows that $\mathcal{J}$ is affine apart from one term $C/(\alpha + \beta)$, where

$$C = \alpha(1 - A_{1,*} + q_b A_{2,*});$$

the star means evaluation of these polynomial formulas at $\beta = -\alpha$. This residue is a rational function. The quadratic-form argument above gives $A_{2,*} > 0$, and $e \le -1/3$ gives

$$\alpha^2(1 - A_{1,*}) \ge \frac{-4\alpha^2 + 2\alpha(S - 1) - S + 5}{3} \ge \frac{2 - 4\alpha(\alpha - 1)}{3} > 0.$$

Thus $C > 0$. It suffices to prove the two signs on $\beta = x$ and $\beta = 20x/19$.

The initial values of $\mathcal{J}$. First set $\alpha = 1$, $S = x + 1$, $u = 19/25$ and $\lambda_0 = 15/16$. On the lower face $\beta = x$,

$$\mathcal{J}_c = -\frac{2x(x - 1)}{25S^3}\left[(x - 2)(6x^2 - 8x + 3) + \frac{38x^2 - 26x - 13}{16}\right].$$

Both positive factors increase on $2 \le x \le 13/6$; the derivative of the first has sign $3 - (x - 2)^2 > 0$. Thus $\mathcal{J}_c \le -29/900 < -1/80$.

For the upper face put $d = x/19$, $B = S + d$, $m = 2x - 1$ and $H = 2x^2 - 3x - 1$. With $e = -6S/25 + 1/S$ and $q = \lambda_0/S^2$, the exact identity in

the displacement is

$$B\mathcal{J} = S\mathcal{J}_c + dL + d^2Q,$$
$$L = em - \frac{12S}{25} + 1 + q\left[-1 + \frac{12S}{25} + 2uHm + m(m+2)\right],$$
$$Q = -\frac{6}{25} + q\left(\frac{6}{25} + um^2\right).$$

Only the broad bounds $L < 1$, $L' < 2$, $Q < 1$, $Q' < 1$ are needed. For Q, use $q \le 5/48$, $m \le 10/3$ and $u < 4/5$ to obtain

$$Q < \frac{5}{48}\left(\frac{1}{4} + \frac{80}{9}\right) < 1, \qquad Q' = \frac{2\lambda_0(3um - 6/25)}{S^3} < \frac{16}{27} < 1.$$

For L direct differentiation, retaining the factors, gives

$$L' = -\frac{6}{25}(4x+3) + \frac{3}{S^2} + \frac{\lambda_0}{S^3}\big[2 - \frac{12S}{25} + 2uH'mS + 4uH(2-x) + 6m + 4(2-x)\big].$$

Here $H > 0$, $H' < 6$, $m/S^2 \le 1/3$, and $(2+6m)/S^3$ decreases from $20/27$. Discarding the nonpositive terms gives $L' < -66/25 + 1/3 + 16/5 + 20/27 < 2$. Since $L(2) = 29/60$ and $x - 2 \le 1/6$, also $L < 1$. The two positive factors in $-S\mathcal{J}_c$ increase; the first is at least $4/225$ and the derivative of the second is at least $151/8$. Hence $(S\mathcal{J}_c)' < -151/450 < -1/3$, and consequently

$$(B\mathcal{J} + B/80)' < -\frac{1}{3} + \frac{2}{7} + \frac{1}{36} + \frac{1}{72} < 0.$$

The three positive bounds here follow from $(L + xL')/19 < (1+2x)/19 \le 16/57 < 2/7$, $(2xQ + x^2Q')/19^2 < (2x + x^2)/19^2 < 1/36$, and $B'/80 = 1/76 < 1/72$. At $x = 2$ the displacement identity and $Q(2) < 1/2$ give

$$B(\mathcal{J} + 1/80) < -\frac{29}{300} + \frac{2}{19}\frac{29}{60} + \frac{1}{2}\left(\frac{2}{19}\right)^2 + \frac{3 + 2/19}{80} < 0.$$

Thus the upper face has the same strict bound.

The endpoint at maximum duration. Put $u_* = 19/25$, $\lambda_0 = 15/16$, $d_0 = 1 - u_*$, $S = x + 1$, $V_0 = x(x+2)$, $N = \alpha^2 + V_0$ and $e = S(u_* - 1) + 1/S$. In this paragraph the norm is fixed at u_*. Write $\mathcal{J}$ for the maximum-duration endpoint of the exact duration tangent, with $q_b = \lambda_0\alpha^2/N$. We show that

$$\partial_\alpha \mathcal{J} < \frac{3}{40} \quad \text{on each of} \quad \beta = x,\ \beta = 20x/19, \qquad 1 \le \alpha \le 7/6.$$

Together with the previously proved $\mathcal{J}(1) < -1/80$, this gives $\mathcal{J}(\alpha) < -1/80 + 3(\alpha - 1)/40 \le 0$, with strict inequality. The convexity in β then gives the sign throughout the rectangle.

The following division keeps the dependence on α in four resolvents. Set

$$\begin{aligned} c_0 &= 1+2x-2\beta x, & L_0 &= d_0\beta^2+S-2+ec_0,\\ P &= L_0/\beta, & Q &= P+\beta(u_*-e),\\ Q_0 &= 2u_*c_0-e-2S+1, & D_4 &= S^2-S+1+c_0(u_*c_0-e-2S+1), \end{aligned}$$

and define the transport polynomial

$$\mathsf{R}(z)=d_0z^4+\beta(1-2u_*+e)z^3+(\beta^2+S-2+ec_0)z^2+\beta Q_0z+D_4.$$

Put

$$\begin{aligned} \mathsf{A} &= \frac{D_4}{\beta V_0}, & \mathsf{B} &= -\frac{\mathsf{R}(-\beta)}{\beta(\beta^2+V_0)},\\ \mathsf{L} &= \beta(e-u_*)-\mathsf{A}-\mathsf{B}, & \mathsf{M} &= \beta^2+S-2+ec_0-d_0V_0-\beta(\mathsf{A}+\mathsf{L}),\\ c_1 &= -P+\lambda_0\mathsf{A}, & c_2 &= Q+\lambda_0\mathsf{B},\\ l &= \lambda_0\mathsf{L}, & m &= \lambda_0\mathsf{M}. \end{aligned}$$

Division of R gives the exact identity

$$\mathcal{J}=(\lambda_0-1)d_0+\frac{c_1}{\alpha}+\frac{c_2}{\alpha+\beta}+\frac{l\alpha+m}{N}. \tag{G.32}$$

On $\beta=x$ the needed coefficient bounds follow from four one-variable comparisons. Differentiating c_1 gives

$$c_1'=-\frac{24x}{25}+\frac{261}{100}-\frac{201}{16(x+2)^2}+\frac{3}{16(x+1)^2}-\frac{51}{400x^2}<\frac{69}{100}-\frac{1809}{2500}+\frac{1}{48}<0.$$

Thus $c_1\ge c_1(13/6)=269/7410>1/36$. Similarly

$$c_2'=\frac{153x}{200}-\frac{221}{100}+\frac{143}{40(x+1)^2}+\frac{51}{400x^2}>-\frac{17}{25}+\frac{1287}{3610}+\frac{459}{16900}>-\frac{3}{10}.$$

Since $c_2(2)=-817/2400>-7/20$, this gives $c_2>-2/5$. The derivative of l satisfies

$$l'=\frac{9x}{40}-\frac{27}{80}+\frac{201}{16(x+2)^2}-\frac{153}{40(x+1)^2}\le\frac{3}{20}+\frac{201}{256}-\frac{1377}{3610}<\frac{3}{5}.$$

Its initial value $l(2)=27/64$ therefore gives $l<27/64+1/10<8/15$. Finally,

$$m''=\frac{81x}{40}-\frac{18}{5}+\frac{153}{20(x+1)^3}>0,\qquad m(2)=-\frac{9}{80},\quad m'(2)=-\frac{1}{20},$$

so its lower tangent gives $m\ge-29/240>-1/8$.

We next obtain the four bounds on $\beta=20x/19$. The first can be compared directly with the preceding face. At fixed x,

$$\partial_\beta c_1=\frac{B_2}{V_0}-\frac{B_{00}}{\beta^2V_0},\qquad B_2=\frac{x(261x-48)}{100},$$

$$B_{00}=\frac{x(192x^4+464x^3-61x^2-256x+102)}{400(x+1)}>0.$$

For the sign of B_{00} pair its first and third terms, and its second and fourth terms, using $x \geq 2$. Hence $\partial_\beta c_1$ increases in β. At $\beta = x$ it equals

$$R_0(x) = \frac{213}{100} - \frac{939}{160(x+2)} - \frac{1}{16(x+1)} + \frac{409}{800x} - \frac{51}{400x^2}.$$

Here $R_0' > 939/[160(25/6)^2] - 409/(800 \cdot 2^2) > 0$, and $R_0(2) = 8311/9600 > 6/7$. The previous lower bound also gives $c_1(x,x) > 1/28$. Consequently

$$c_1(x, 20x/19) > \frac{1}{28} + \frac{12}{133} > \frac{1}{8}.$$

For the remaining bounds let $\chi = 20/19$ and $D_\chi = 1+\chi^2$. The x-resolvents of the natural functions above have the forms

$$\begin{aligned} c_2 &= a_2 x^2 + b_2 x + d_2 - \frac{\tau}{D_\chi x + 2} + \frac{p}{x+1} - \frac{h}{x}, \\ l &= a_l x^2 + b_l x + d_l + \frac{\tau}{D_\chi x + 2} - \frac{B}{x+2}, \\ m &= a_m x^3 + b_m x^2 + c_m x + d_m + \frac{\tau_m}{D_\chi x + 2}. \end{aligned}$$

Only the following constants occur after differentiation:

$$\begin{aligned} a_2 &= \frac{d_0(\chi+2)(1+(1-\lambda_0)\chi^2)}{D_\chi}, & p &= \frac{(\chi+1)^2}{16\chi}, \\ h &= \frac{51}{400\chi}, & a_l &= \frac{9\chi(2\chi-1)}{40D_\chi}, \\ B &= \frac{3(4\chi+1)(76\chi-9)}{80\chi}, & a_m &= \frac{9\chi(\chi+2)}{40D_\chi}, \\ \tau_m &= \frac{2\chi\tau}{D_\chi}. \end{aligned}$$

The remaining residues have useful reciprocal forms. Put $\rho_\chi = \chi + \chi^{-1} = 2 + 1/380$ and $\zeta_\chi = \chi - \chi^{-1} = 39/380 < 1/8$. Then

$$\begin{gathered} b_l = -\frac{3\chi}{80}\left\{44 - \frac{76}{\rho_\chi} + \frac{12}{\rho_\chi^2} + \zeta_\chi\left(\frac{7}{\rho_\chi} - \frac{12}{\rho_\chi^2}\right)\right\}, \\ b_m = -\frac{3}{40}\{F(\rho_\chi) + \zeta_\chi G(\rho_\chi)\}, \\ F(r) = 19r + \frac{19}{2} - \frac{50}{r} + \frac{6}{r^2}, \qquad G(r) = 19 + \frac{1}{2r} - \frac{6}{r^2}, \\ \tau = \frac{153}{20} - (\rho_\chi - 2)A(\rho_\chi) - \zeta_\chi B_\tau(\rho_\chi), \\ A(r) = \frac{3}{80}\left(3 + \frac{184}{r} + \frac{140}{r^2} - \frac{24}{r^3}\right), \qquad B_\tau(r) = \frac{3(r+2)(3r-4)(3-r)}{40r^3}. \end{gathered}$$

For $2 \le r < 3$, $0 < A(r) < 39/8 < 5$. Also $(r+2)/r^3 \le 1/2$ and $(3r-4)(3-r) \le 25/12 < 9/4$, so $0 < B_\tau(r) < 1/10$. Consequently

$$\frac{15}{2} < \frac{153}{20} - \frac{1}{76} - \frac{1}{80} < \tau < \frac{153}{20} < 8.$$

All the remaining constants are evaluated at $\chi = 20/19$. They satisfy

$$a_2 > \frac{1}{3}, \quad p > 0, \quad 0 < h < \frac{1}{8}, \quad \frac{15}{2} < \tau < 8,$$

and

$$0 < a_l < \frac{1}{8}, \quad b_l < -\frac{1}{3}, \quad 13 < B < \frac{53}{4}, \quad a_m > \frac{1}{3}, \quad b_m > -2, \quad \tau_m > 0.$$

Here $a_2 > 9/25$ follows from $\chi+2 > 3$, $D_\chi < 17/8$ and $1+(1-\lambda_0)\chi^2 > 17/16$; the bound $a_l < 1/8$ is equivalent to $13\chi^2 - 9\chi - 5 = -25/361 < 0$, while $h = 969/8000 < 1/8$. For the bound on b_l, the function $44 - 76/r + 12/r^2$ increases from 9 on $r \ge 2$, and $7/r - 12/r^2 > 0$ there. Consequently $b_l < -27/80 < -1/3$. Also $F(2) = 24$ and $F'(r) = 19 + 50/r^2 - 12/r^3 < 32$ for $r \ge 2$. As $G(r) < 20$ and $\zeta_\chi < 1/8$, we obtain

$$b_m > -\frac{3}{40}\left(24 + \frac{1}{10} + \frac{5}{2}\right) = -\frac{399}{200} > -2,$$

using $32/380 < 1/10$. The function $3(4c+1)(76c-9)/(80c)$ has value $201/16$ at $c = 1$ and derivative strictly between 11 and 12 for $c \ge 1$. Hence $13 < 201/16+11/19 < B < 201/16+12/19 < 53/4$. The inequality $a_m > 1/3$ is equivalent to $-13\chi^2 + 54\chi - 40 > 0$; this quadratic is increasing on $[1,2]$ and has value 1 at 1. The signs of p and τ_m are immediate from their definitions.

The first resolvent is strictly convex in x: as $D_\chi < 9/4$, $D_\chi x + 2 \ge 6$,

$$c_2'' > \frac{2}{3} - \frac{2 \cdot 8(9/4)^2}{6^3} - \frac{2(1/8)}{2^3} = \frac{25}{96} > 0.$$

The initial values satisfy $c_2(2) = -1135271/2067200 > -5/9$ and $c_2'(2) > -2/5$. Its lower tangent therefore gives $c_2 > -28/45 > -5/8$.

For l, the function $G(x) = 2a_l x + B/(x+2)^2$ decreases, since $G' < 1/4 - 26/(25/6)^3 < 0$. Thus $G < 1/2 + 53/64 = 85/64$. Also $D_\chi x + 2 < 20/3$ gives $\tau D_\chi/(D_\chi x + 2)^2 > 27/80$. Hence

$$l' = G + b_l - \frac{\tau D_\chi}{(D_\chi x + 2)^2} < \frac{85}{64} - \frac{1}{3} - \frac{27}{80} = \frac{631}{960} < \frac{2}{3}.$$

As $l(2) = 5637/10336 < 5/9$, this proves $l < 2/3$. The last resolvent is convex, since $m'' = 6a_m x + 2b_m + 2\tau_m D_\chi^2/(D_\chi x + 2)^3 > 0$. At $x = 2$, $m = -891/2720 > -1/3$ and $m' > -3/8$; thus $m > -19/48 > -2/5$.

It remains to apply these bounds to the four α-kernels in (G.32). On the present box,

$$\alpha^{-2} \ge \frac{36}{49}, \quad (\alpha+\beta)^{-2} \le \frac{1}{9}, \quad \frac{V_0 - \alpha^2}{N^2} \le \frac{7}{81}, \quad \frac{\alpha}{N^2} \le \frac{7}{486}.$$

For the third bound the quotient decreases in both $V_0 \geq 8$ and $\alpha^2 \geq 1$, since $V_0 > 3\alpha^2$; its largest value is $7/81$. The coefficient bounds on the lower face now give

$$\mathcal{J}_\alpha < -\frac{1}{49} + \frac{2}{45} + \frac{56}{1215} + \frac{7}{1944} < \frac{3}{40}.$$

On the upper face they give

$$\mathcal{J}_\alpha < -\frac{9}{98} + \frac{5}{72} + \frac{14}{243} + \frac{14}{1215} < \frac{3}{40}.$$

This proves the required derivative bound on both faces and completes the maximum-duration sign.

The resolvent form. We prove that $\alpha \mapsto \mathcal{T}(q_b)$ is strictly concave on the enlarged box

$$2 \leq x \leq \frac{13}{6}, \qquad 1 \leq \alpha \leq \frac{7}{6}, \qquad x \leq \beta \leq \frac{20x}{19},$$
$$u = \frac{19}{25}, \qquad q_b = \lambda_0 q_c, \quad \lambda_0 = \frac{15}{16}.$$

Throughout this argument x, β, u are fixed when differentiating in α. Put

$$S = x + 1, \qquad V_0 = x(x+2), \qquad N = \alpha^2 + V_0,$$
$$e = S(u-1) + \frac{1}{S}, \qquad c_0 = 1 + 2x - 2\beta x,$$

and write $r = r_A = \alpha/(\alpha+\beta)$ and $X = e + (u-1)/r + r$. Inserting $q_b = \lambda_0 \alpha^2/N$ in the scale coefficients and collecting powers of α gives

$$\Theta(\alpha) := \mathcal{T}(q_b) = X + \frac{\widehat{C}_4\alpha^4 + \widehat{C}_3\alpha^3 + \widehat{C}_2\alpha^2 + \widehat{C}_1\alpha + \widehat{C}_0}{N^2},$$
$$\widehat{C}_j = \lambda_0 C_j - 2\lambda_0(1-\lambda_0)D_{4-j}.$$

Here the coefficients are independent of α:

$$D_0 = u + e, \qquad D_1 = 2\beta(2u + e - 1),$$
$$D_2 = \beta^2(4u-1) + 2uc_0 + e(c_0 - 1) - x - 2,$$
$$D_3 = 2\beta(2uc_0 - e - 2x - 1),$$
$$D_4 = x^2 + x + 1 + c_0(uc_0 - e - 2x - 1),$$

and

$$C_4 = -1 - e, \qquad C_3 = 4\beta u,$$
$$C_2 = -(1 + 2u + 3e)V_0 + 2\beta^2(4u-1) + 2uc_0 - e - (2x+1),$$
$$C_1 = -4\beta V_0 w,$$
$$w = -\frac{(1-u-e)V_0 + 2uc_0 - e - (2x+1)}{V_0},$$
$$C_0 = \big(3 + e - 2c_0(u+e)\big)V_0 + 2D_4.$$

The relation between $\widehat{C}_j, C_j, D_j$ also follows by first writing the scale expression as $X+\lambda_0 Y+2\lambda_0^2 Z$: the numerator of Z is $D_0\alpha^4+D_1\alpha^3+D_2\alpha^2+D_3\alpha+D_4$, whereas that of $Y+2Z$ is $C_4\alpha^4+C_3\alpha^3+C_2\alpha^2+C_1\alpha+C_0$.

Bounds for the coefficients. The box gives

$$-\frac{1}{2} < e < -\frac{1}{3}, \qquad -5 < c_0 \le -3, \qquad u+e > \frac{1}{4},$$
$$8 \le V_0 < \frac{19}{2}, \qquad \frac{3}{7} < \frac{\alpha}{\beta} \le \frac{7}{12}.$$

In particular $D_0, D_1 > 0$ and $D_3 < 0$. We next show

$$0 < D_2 < 3, \qquad D_4 < V_0^2, \qquad |C_2| < \frac{V_0}{2}, \qquad w < \frac{5}{8}, \qquad C_0 > V_0^2.$$

For D_2, differentiation in β gives

$$\partial_\beta D_2 = 2\big((4u-1)\beta - x(2u+e)\big) > 0.$$

Thus its lower bound is reduced to $\beta = x$. With $d = 1-u = 6/25$, its value there is

$$D_2 = 2dx^3 - x^2 + (1-6d)x + 4 - 2d - \frac{4}{S}.$$

Its derivative is $6dx^2 - 2x + 1 - 6d + 4/S^2 > 0$ for $x \ge 2$: the polynomial part is increasing and its value at 2 is $33/25$. Since $D_2(2) = 86/75$, this proves positivity with the constant norm $u = 19/25$. For the upper bound, $u < 7/9$, $2u+e > 1$ and $c_0 < 0$ give

$$D_2 < \frac{19}{9}\beta^2 - 2x\beta + x - \frac{1}{2} < \frac{x^2}{4} + x - \frac{1}{2} < 3.$$

Indeed $(19/9)(\beta/x) - 2 \le 2/9$ and $\beta/x < 9/8$, so the quadratic part is less than $x^2/4$; also $x^2 < 5$ and $x < 9/4$.

To bound C_2, temporarily allow $3/4 \le u \le 19/25$, retaining $e = S(u-1) + 1/S$. Then

$$\partial_\beta C_2 = 4\big((4u-1)\beta - ux\big) > 0, \qquad \partial_u C_2 = 8\beta^2 - 4\beta x - 3x^3 - 11x^2 - 7x + 1 < 0.$$

For the last sign, the quadratic in β increases for $\beta \ge x$; at $\beta = 9x/8$ its value is $-3x^3 - (43/8)x^2 - 7x + 1 < 0$. At $\beta = x$ and $u = 19/25$ one has

$$C_2 + \frac{V_0}{2} = 3dx^3 + \left(7d - \frac{1}{2}\right)x^2 + (7d-6)x - d - 2 + \frac{2}{S}.$$

This is $4/15$ at $x = 2$ and increases thereafter: its derivative is $9dx^2 + (14d-1)x + 7d - 6 - 2/S^2 > 0$. This proves the lower bound for C_2. For the upper bound, decrease u to $3/4$ and increase β to $9x/8$. Writing $e_* = -S/4 + 1/S > -1/2$ gives

$$C_2 \le -\left(\frac{5}{2} + 3e_*\right)V_0 + \frac{27}{16}x^2 + x + \frac{1}{2} - e_* < -V_0 + \frac{27}{16}x^2 + x + 1 < \frac{V_0}{2};$$

the last inequality follows from $-(3/16)x^2 + 2x - 1 > 0$ on the box.

The function w increases in β, since $\partial_\beta w = 4u/(x+2) > 0$. At $\beta = 20x/19$, keeping $u = 19/25$, its expression becomes

$$w = \frac{3}{5} + \frac{(x-2)(-6x^2+29x-3)}{25V_0}.$$

Here the quadratic factor is at most 32, because $32 - (-6x^2+29x-3) = (3x-7)(2x-5) \geq 0$. Also $x - 2 \leq x/13$ and $V_0 \geq 4x$. Consequently

$$w \leq \frac{3}{5} + \frac{8}{325} < \frac{3}{5} + \frac{1}{40} = \frac{5}{8}.$$

Finally,

$$D_4 = (x^2+x+1) + uc_0^2 - ec_0 - (2x+1)c_0 > 7 + \frac{27}{4} - \frac{5}{2} + 15 = \frac{105}{4}.$$

Since $3 + e - 2c_0(u+e) > 4$, this gives $C_0 > 4V_0 + 105/2 > V_0^2$; indeed $V_0(V_0-4) < (19/2)(11/2) < 105/2$. For the required upper bound on D_4, discard the negative term $-ec_0$. The remaining three terms are less than $10, 20, 30$, respectively, using $S < 7/2$, $u < 4/5$ and $|c_0| < 5$. Thus $D_4 < 60 < V_0^2$.

It follows that

$$\widehat{C}_4 < -\lambda_0/2, \qquad \widehat{C}_3 < 4\lambda_0\beta u < 3\beta, \qquad \widehat{C}_1 > \lambda_0 C_1,$$
$$|\widehat{C}_2| < \frac{V_0}{2} + \frac{V_0}{16} = \frac{9V_0}{16} < \frac{4V_0}{7}, \quad \widehat{C}_0 > \lambda_0(2\lambda_0 - 1)V_0^2 > \frac{7}{9}V_0^2.$$

For the bound on $\widehat{C}_2$, use $D_2 < 3 < V_0/2$ and $1 - \lambda_0 = 1/16$. The last bound uses $C_0 > V_0^2$, $D_4 < V_0^2$, $\lambda_0 > 8/9$ and $2\lambda_0 - 1 = 7/8$.

The curvature kernels. Put $t = \alpha^2/V_0$, so $0 < t \leq 49/288 < 3/17$. Differentiating the five resolvents gives

$$N^4(\alpha^4/N^2)'' = 12V_0\alpha^2(V_0 - \alpha^2),$$
$$N^4(\alpha^3/N^2)'' = 2\alpha(3V_0^2 - 8V_0\alpha^2 + \alpha^4),$$
$$N^4(\alpha^2/N^2)'' = 2(V_0^2 - 8V_0\alpha^2 + 3\alpha^4),$$
$$N^4(\alpha/N^2)'' = 12\alpha(\alpha^2 - V_0),$$
$$N^4(1/N^2)'' = 4(5\alpha^2 - V_0).$$

Define the nonnegative functions

$$\phi_1 = \frac{t^2(3-8t+t^2)}{(1+t)^4}, \qquad \phi_2 = \frac{t^2(1-t)}{(1+t)^4},$$
$$\psi = \frac{t|1-8t+3t^2|}{(1+t)^4}, \qquad \chi = \frac{t(1-5t)}{(1+t)^4}.$$

The baseline satisfies $\alpha^3 X''/(2\beta) = -(1-u) - r^3$. The bounds just proved therefore give

$$\frac{\alpha^3}{2\beta}\Theta'' < -\frac{6}{25} - r^3 + F(t), \qquad F(t) = 3\phi_1 + \frac{27}{2}\phi_2 + \frac{1}{3}\psi - \frac{2}{3}\chi.$$

In detail, the $\widehat{C}_3$ contribution is less than $3\phi_1$. The $\widehat{C}_1$ and $\widehat{C}_4$ contributions together are at most

$$\lambda_0 \left(24w - \frac{9}{7}\right) \phi_2 < \frac{27}{2}\phi_2.$$

The inequality for $\widehat{C}_1$ has the stated direction because $(\alpha/N^2)'' < 0$, even if $\widehat{C}_1$ changes sign. The two remaining contributions are at most $(4/7)(7/12)\psi = \psi/3$ and $-2(7/9)(3/7)\chi = -2\chi/3$.

We claim that $F(t) < 3t/2$ throughout $0 < t \le 3/17$. If $t \le 1/7$, then $|1 - 8t + 3t^2| \le 1 - 5t$, so $\psi \le \chi$ and

$$\frac{3}{2} - \frac{3\phi_1 + (27/2)\phi_2}{t} = \frac{3}{2}\frac{1 - 11t + 31t^2 + 2t^3 + t^4}{(1+t)^4} > 0,$$

because $1 - 11t + 31t^2 = (1 - 11t/2)^2 + 3t^2/4$. If $1/7 \le t \le 3/17$, then $1 - 8t + 3t^2 < 0$ and

$$\frac{3}{2} - \frac{F(t)}{t} = \frac{5\big((1-4t)(1-5t) - t^2\big) + 6t^3 + 3t^4}{2(1+t)^4} > 0.$$

Here $(1-4t)(1-5t) - t^2 \ge (5/17)(2/17) - (3/17)^2 > 0$.

To compare r^3 and t, put $y = \alpha/\sqrt{V_0}$. The box implies $3/10 < y < 1/2$ and $\beta/\sqrt{V_0} < 4/5$. For the latter bound, use $\beta/x < 16/15$ and $x/(x+2) < 9/16$. The concave function $5y - (y+4/5)^3$ is positive at both endpoints of $[3/10, 1/2]$, since $(11/10)^3 < 3/2$ and $(13/10)^3 < 5/2$. It follows that

$$\frac{r^3}{t} = \frac{y}{(y + \beta/\sqrt{V_0})^3} > \frac{y}{(y + 4/5)^3} > \frac{1}{5}.$$

We conclude that

$$\frac{\alpha^3}{2\beta}\Theta'' < -\frac{6}{25} - \frac{t}{5} + \frac{3t}{2} = -\frac{6}{25} + \frac{13t}{10} < 0,$$

because $t < 3/17 < 12/65$. This proves strict concavity with the constant norm $u = 19/25$ on the entire enlarged box.

The endpoint tangents of $\mathcal{T}$. Set $\alpha = 1$, $\beta = kx$, $1 \le k \le 20/19$, $S = x + 1$, $B = 1 + \beta$, $G = S - \beta(x-1)$ and $q = \lambda_0/S^2$. The residual factors as

$$\mathcal{T} = -2ab + (\beta - x)C,$$

$$a = 1 - 2qG, \qquad b = \frac{6S}{25} - \frac{1}{S} - q(2x + 1 - 2uG),$$

$$C = -\frac{6}{25} - \frac{1}{SB} + 4q - 2q^2(S + B).$$

All x-derivatives below keep k fixed. On $k = 1$, $a' = 8\lambda_0 x/S^3 > 0$ and

$$b'' = \frac{2[\lambda_0(4u-1)(2x-1) - S]}{S^4} > 0.$$

The sign uses $\lambda_0(4u-1) = 153/80 > 1$ and $2x - 1 \geq S$. At $x = 2$, $a = 19/24 > 3/4$, $b = 29/1200 > 0$ and $b' = 61/900 > 1/15$. Thus $\mathcal{T}_x = -2(a'b + ab') < -1/10$.

On $k = 20/19$ the same formula for b'' has the additional positive term $6\lambda_0 u(k-1)(x-1)$ inside the bracket. At $x = 2$,

$$a = 371/456 > 13/16, \qquad b = 3/400, \qquad b' = 97/1800 > 4/75.$$

Also $a' > 8\lambda_0 x/S^3 > 1/2$ on the whole interval, since $15x - S^3$ decreases and is positive at $x = 13/6$. Hence $2(a'b + ab') > 3/400 + 13/150 > 3/32$. For the correction $E = (k-1)xC$,

$$\frac{E'}{k-1} = -\frac{6}{25} + \frac{kx^2-1}{S^2B^2} - \frac{4\lambda_0(x-1)}{S^3} + \frac{4\lambda_0^2 A}{S^5},$$
$$A = (2x-1)S + (k-1)x(x-1).$$

Since $\lambda_0 k = 75/76 < 1$ and $x \geq 2$, $\lambda_0 A < (x-1)S^2$. The final positive term is absorbed by the preceding negative one. As $(kx^2-1)/B^2 < 1$, this gives $E' < (-6/25 + 1/9)/19 < -1/160$. Thus $\mathcal{T}_x < -1/10$ on this face too.

To control the other endpoint of the tangent in α, differentiate first at fixed x, β. For general α, the exact factorization is

$$\begin{aligned}
\mathcal{T} &= -2ab + \Delta_0 + q\Delta_1 + 2q^2\Delta_2,\\
a &= 1 - qg, \qquad b = -e - q((2S-1)/\alpha^2 - ug),\\
\Delta_0 &= (u-1)/r + r - e,\\
\Delta_1 &= 4\beta/\alpha - 1 + (5-4S)/\alpha^2 - (1-\alpha^{-2})e,\\
\Delta_2 &= \varepsilon^2 + \varepsilon S/\alpha^2 + S^2/\alpha^4 + g\varepsilon e - 1/r^2.
\end{aligned}$$

Here $q = \lambda_0\alpha^2/(\alpha^2 + x(x+2))$. At $\alpha = 1$ write $\mathcal{T}_\alpha = P + \Delta$, where $P = -2(a_\alpha b + ab_\alpha)$. We show $P' < -1/12$ and $\Delta' < 2/3$.

Put $C_1 = -a_\alpha$ and $E_1 = b_\alpha$, evaluated at $\alpha = 1$. Their simple resolvents are

$$C_1 = 2\lambda_0\left[\frac{k}{S} + \frac{k+1}{S^2} - \frac{2(3k+1)}{S^3} + \frac{4k}{S^4}\right],$$
$$E_1 = 2\lambda_0\left[\frac{ku}{S} + \frac{u(k+1)}{S^2} - \frac{2(3ku+u-1)}{S^3} + \frac{4ku-1}{S^4}\right].$$

Since $B \geq 3$ and $0 < G \leq 1$, $C_1 = 2q(B - 2G/S^2) > 0$. They satisfy $0 < C_1 < 2/3$, $C_1' < 0$, $E_1 > 1/2$ and $E_1' > -1/6$. For completeness, $C_1' = -2\lambda_0 H/S^5$, where

$$H = kS^3 + 2(k+1)S^2 - 6(3k+1)S + 16k.$$

At $S = 3$, $H = 7k$, $H' = 21k + 6$ and $H'' > 0$, proving the derivative sign; $C_1(3) = 2\lambda_0(22k+3)/81 < 2/3$. Likewise $E_1'' = 4\lambda_0 H_1/S^6$ with

$$H_1 = kuS^3 + 3u(k+1)S^2 - 12(3ku+u-1)S + 10(4ku-1).$$

At $S = 3$, $H_1 = 26 - u(14k + 9) > 0$, $H_1' = 9ku + 6u + 12 > 0$, and $H_1'' > 0$. Hence $E_1' \geq E_1'(3) = -14\lambda_0(ku+2)/243 \geq -49/324 > -1/6$. Since $E_1(3) \geq 5/9$ and $S - 3 \leq 1/6$, also $E_1 \geq 19/36 > 1/2$.

The derivative b' increases in x, since $b'' > 0$, and decreases in k, since $\partial_k b' = -2\lambda_0 u(3S - 4)/S^3 < 0$. Consequently $b'(x, k) \leq b'(13/6, 1) < 1/12$. Both b and b' are positive: $b'' > 0$ on both faces, their minima in k are at $20/19$, and the initial values there are positive. For b, this also follows from $b_k = -2\lambda_0 u(S - 1)(S - 2)/S^2 < 0$. For all $1 \leq k \leq 20/19$, $a' = 2\lambda_0[S + k(3x - 1)]/S^3 \geq 8\lambda_0 x/S^3 > 1/2$. Also $a \geq a(2, 1) = 19/24 > 0$, since G decreases in k. Finally $a < 11/12$, because $G \geq 173/342 > 1/2$ and $S^2 < 11$ give $1 - a = 2\lambda_0 G/S^2 > 15/176 > 1/12$. Consequently

$$P' = 2(C_1'b + C_1b' - a'E_1 - aE_1') < 2\left(\frac{1}{18} - \frac{1}{4} + \frac{11}{72}\right) = -\frac{1}{12}.$$

It remains to bound Δ'. Put $\delta = k-1$ and $G_0 = S-x(x-1)$. Differentiating the three correction terms above gives the displacement identity

$$\Delta = \Delta_c + \frac{6}{25}\delta x + \left[\frac{kx}{(1 + kx)^2} - \frac{x}{S^2}\right] + \delta F + \delta Q + \delta^2 R,$$

$$\Delta_c = \frac{6x}{25} + \frac{x}{S^2} + q(4x - 2 - 4\lambda_0) + K,$$

$$K = -2qe(2a_0 - 1), \quad a_0 = 1 - 2qG_0,$$

$$F = xq(4 - 8/S^2), \qquad Q = 4xq^2L,$$

$$R = -4x^2q^2(1 - 2/S^2), \quad L = -2S + 4/S - 1 - 2e(x - 1).$$

The three bounds $F' < 1/50$, $Q' < 3/8$, $R' < 1/16$ follow directly:

$$F' = \frac{4\lambda_0(-x^3 - x^2 + 7x - 1)}{S^5} \leq \frac{5}{324} < \frac{1}{50},$$

$$Q' = \frac{4\lambda_0^2[(1 - 3x)L + xSL']}{S^5} < 4\lambda_0^2 \frac{5}{243}\frac{61}{12} < \frac{3}{8},$$

$$R' < \frac{8\lambda_0^2 x(x - 1)}{S^5} \leq \frac{25}{432} < \frac{1}{16}.$$

Here the first numerator decreases from one; for the second, $L' < 0$, $L < 0$ and $-L < 61/12$ by completing the square in $2x - 12x^2/25$. The ratios $(3x-1)/S^5$ and $x(x-1)/S^5$ decrease for $x \geq 2$. The derivative of the square bracket in the displacement identity is nonpositive: $h(y) = y(1 - y)/(1 + y)^3$ decreases on $2 \leq y < 5/2$. The positive displacement derivative is therefore less than

$$\frac{1}{19}\left(\frac{6}{25} + \frac{1}{50} + \frac{3}{8}\right) + \frac{1}{16 \cdot 19^2} < \frac{1}{28}.$$

This is absorbed by $(x/S^2)' = -(x - 1)/S^3 < -1/28$. Finally $[q(4x - 2 - 4\lambda_0)]' < 4/15$ and $K' < 1/8$. For the latter, write $K = 2\lambda_0 jA$, where $j = -e/S^2 < 1/20$, $j' < 1/90$, $0 < A = 2a_0 - 1 < 1$ and $A' \leq 10/9$.

Indeed $j < (6/25 - 1/11)/3 < 1/20$; $j' = (3 - 6S^2/25)/S^4$ decreases from $7/675 < 1/90$, and $A' = 16\lambda_0 x/S^3 \le 10/9$. It follows that

$$\Delta' < \frac{6}{25} + \frac{4}{15} + \frac{1}{8} < \frac{2}{3}.$$

We have proved $(\mathcal{T}_\alpha)_x < 7/12$, so on both faces

$$(\mathcal{T} + \mathcal{T}_\alpha/6)_x < -\frac{1}{10} + \frac{7}{72} < 0.$$

At $x = 2$ the factored formulas give, respectively, $\mathcal{T} < -1/30$, $\mathcal{T}_\alpha < 1/5$ on $\beta = x$, and $\mathcal{T} < -1/60$, $\mathcal{T}_\alpha < 1/10$ on $\beta = 20x/19$. Thus both $\mathcal{T}(1)$ and $\mathcal{T}(1) + \mathcal{T}_\alpha(1)/6$ are negative throughout the x interval. Concavity in α bounds $\mathcal{T}(\alpha)$ by the larger of these two values, proving the second shape sign.

These two shape signs, together with the duration tangent and monotonicity in q, prove $\mathcal{R}(19/25; p, q) < 0$ on $0 \le q \le q_b$, $2q \le p \le r_A$, $p > 0$.

The shear sign. Here the small-shape domain (G.24) and the scale bound $q > \lambda_0 q_c$, with $\lambda_0 = 15/16$, suffice. Write $R = R_x$, $T = T_s$, $y = T/R$ and $D_s = 2\beta x - x - \alpha\beta$. Since

$$\nu = \frac{R^2 - T^2}{3 + T^2}, \qquad \sqrt{3}h = R + (1+\nu)T - 2\mathcal{U}_1 D_s \frac{q}{\alpha^2(1-q)},$$

the scale bound gives, with $Z = -\sqrt{3}h$,

$$Z > \frac{2\mathcal{U}_1 D_s \lambda_0}{x(x+2) + (1-\lambda_0)\alpha^2} - R - (1+\nu)T.$$

The two losses in the first term are small for separate reasons. From $\beta = x(1+\delta)$ and $\alpha \le 1 + 3\delta$,

$$D_s \ge 2x(x-1) + 2x(x-2)\delta - 3x\delta^2 \ge 2x(x-1)(1-e), \qquad e = \frac{3}{2 \cdot 19^2}.$$

Also $\alpha^2/[x(x+2)] \le f_0 = (22/19)^2/8$. Hence

$$\frac{Z}{R} > 2\zeta \frac{3 + R^2}{3 - R} - 1 - (1+\nu)y, \qquad \zeta = \frac{\lambda_0(1-e)}{1 + (1-\lambda_0)f_0} > \frac{12}{13}. \tag{G.33}$$

For the last comparison, cross-multiplication reduces it to $3 > 195e + 12f_0$. The bounds $e < 1/240$ and $f_0 < 7/40$ give $195e + 12f_0 < 13/16 + 21/10 < 3$.

It remains to compare the two shape terms. Put $c = \alpha/(1+\delta)$. The shape bounds give $1 \le c \le 11/10$ and

$$y = \frac{(x-c)(x+1)}{(x+c)(x-1)} \in [27/31, 1].$$

Indeed, this ratio decreases in c and increases in x for $c \ge 1$, since its x-derivative is $2(c-1)(c+x^2)/[(x+c)^2(x-1)^2]$. Its minimum occurs at $x = 2$, $c = 11/10$. Define

$$f(s, y) = \frac{(3+s)y + \frac{7}{2}(1-y^2)}{3 + y^2 s}.$$

For $s \ge 1/9$ and $27/31 \le y \le 1$, both derivatives are nonpositive:

$$f_s = \frac{y(1-y^2)(3-\frac{7}{2}y)}{(3+y^2 s)^2} \le 0, \qquad f_y = \frac{(3+s)(3-sy^2-7y)}{(3+y^2 s)^2} < 0.$$

Consequently $f(R^2, y) \le f(1/9, 27/31) = 854/741 < 15/13$, whereas $(3 + R^2)/(3 - R) \ge 7/6$. Thus

$$2\zeta \frac{3+R^2}{3-R} > \frac{28}{13} > 1 + f(R^2, y) = 1 + (1+\nu)y + \frac{7\nu}{2R^2}.$$

Together with (G.33) this gives $ZR > 7\nu/2$, including at $\delta = 0$.

G.4.7. *Gate S6. The central shear.* Write $R = R_x$, $T = T_s$, $D_s = 2\beta x - x - \alpha\beta$ and $A_s = D_s/[x(x+2)]$. The lower bound $\alpha \ge 1 + 13\delta/5$ gives

$$A_s \le \frac{2(x-1)}{x+2} + \frac{2x - 18/5}{x+2}\,\delta \le \frac{2(x-1)}{x+2} + \frac{\delta}{5},$$

since $x \le 13/6 < 20/9$. The upper bound $\alpha \le 1 + 3\delta$ gives

$$R - T = \frac{2(x\alpha - \beta)}{(x+1)(\beta+\alpha)} \le \frac{4x}{(x+1)^2}\,\delta \le \delta.$$

Here $T > 0$ because $\alpha \le 22/19 < 2 \le x \le \beta$. Thus $\nu \ge 0$ and $Z_c = -\sqrt{3}h_c = 2\mathcal{U}_1 A_s - R - (1+\nu)T$ imply

$$Z_c \le \frac{2R^2(1+R)}{3-R} + \left(1 + \frac{2\mathcal{U}_1}{5}\right)\delta.$$

Now $R \le 7/19$ gives $2(1+R)/(3-R) \le 26/25$ and $2\mathcal{U}_1 < 8/5$, while $\delta \le 1/19 < R^2/2$ follows from $R \ge 1/3$. Consequently

$$Z_c < \left(\frac{26}{25} + \frac{33}{50}\right) R^2 = \frac{17}{10} R^2 < \sqrt{3} R^2,$$

and hence $-h_c < R^2$.

The tangent curve. Fix $\nu \ge 0$ and $K > \nu$, and put $\mathcal{D} = \sqrt{K^2 - \nu^2}$ and $\tilde{h}(\delta) = (K\cos\delta - \mathcal{D})/\sin\delta$ for $0 < \delta < \pi$. Then $K\sin\delta + \tilde{h}\cos\delta = \sqrt{\tilde{h}^2 + \nu^2}$, because $(K - \mathcal{D}\cos\delta)^2 - (K\cos\delta - \mathcal{D})^2 = \nu^2 \sin^2\delta$; moreover $\tilde{h}'(\delta) = (\mathcal{D}\cos\delta - K)/\sin^2\delta < 0$, and $\tilde{h}$ inverts the function $\delta(h)$ of Appendix G.3.7. Along the curve,

$$\frac{\mathrm{d}}{\mathrm{d}\delta}(-J_0\tilde{h}(\delta) - c\delta) = \frac{J_0\sqrt{\tilde{h}^2 + \nu^2} - c\sin\delta}{\sin\delta}.$$

We prove that the derivative is negative under the continuation bounds. Write $R = R_x$, $H = -\tilde{h} > 0$, $J = J_0$ and $d = 3/2 - J$. The norm ceiling gives

$$J \le \frac{9 + R^2}{2(3+R^2)}, \qquad 2d \ge \frac{2R^2}{3+R^2} > \frac{7}{11}R^2.$$

Here $R \le 7/19 < 3/8$, so $R^2 < 1/7$. Also $J \le 3/2$, $c < 3$ and the same ceiling gives $J \le 41/28 < \kappa_0$, hence $K = \nu + 2 + 2\kappa_0/J > 4 + \nu$. Put $a = \nu/H$, $b = \nu/K$ and $W = \sqrt{H^2 + \nu^2}$. The tangent-curve identity gives

$$\sin\delta = \frac{KW + \mathcal{D}H}{K^2 + H^2}.$$

Since $JK = 2\kappa_0 + 2J + J\nu$ and $c = \kappa_0 + 3/2$, the desired inequality $JW < c\sin\delta$ is equivalent to

$$2d > J\nu + \frac{JH^2}{K} + c\left(1 - \frac{\sqrt{1-b^2}}{\sqrt{1+a^2}}\right). \tag{G.34}$$

Each term on the right has its own bound. Write $\delta_s = \beta/x - 1$ for the displacement, to distinguish it from the angle parameter. The shape inequalities $0 < T \le R$ and $R - T \le \delta_s$ give $\nu \le 2R\delta_s/3$, independently of the shear. Since $J \le 3/2$, $\delta_s \le 1/19$ and $R \ge 1/3$,

$$J\nu \le R\delta_s < \frac{1}{6}R^2, \qquad \frac{JH^2}{K} < \frac{3}{8}R^4 < \frac{1}{16}R^2.$$

Here the second estimate uses $H < R^2$, $K > 4$ and $R^2 < 1/7$. Moreover, $H/K < R^2/4 < 1/8$, so $b^2 \le a^2/64$. The shear hypothesis gives $a \le R/2 < 3/16$, and $0 \le b < 1$. The elementary square-root bounds $\sqrt{1-b^2} \ge 1 - b^2$ and $(1+a^2)^{-1/2} \ge 1 - a^2/2$ yield

$$1 - \frac{\sqrt{1-b^2}}{\sqrt{1+a^2}} \le \frac{a^2}{2} + b^2 \le \left(\frac{1}{2} + \frac{1}{64}\right)a^2.$$

As $a \le R/2$ and $c < 3$, the last loss in (G.34) is less than $(3/4)(1/2 + 1/64)R^2 < 2R^2/5$. All three losses therefore sum to less than $(1/6 + 1/16 + 2/5)R^2 < 7R^2/11 < 2d$. This proves strict decrease, including when $\nu = 0$.

The additional angle. Put $H = -h_c > 0$, $J = J_0$, $W = \sqrt{H^2 + \nu^2}$ and $\mathcal{D} = \sqrt{K^2 - \nu^2}$. The switch bounds give $u \le \mathcal{U}_i$, so $b_i = 1 - u/\mathcal{U}_i \ge 0$. The norm ceiling and $R^2 \ge 1/9$ imply $J \le 41/28 < \kappa_0$; consequently $K > 4 + \nu$ and $\mathcal{D} > 4$. Write $t = (W + H)/(K + \mathcal{D})$, so that $\omega_c = 2\arctan t$. The continuation bounds give $\nu^2 < H^2/6$ and $H < R^2 < 1/7$; in particular $t \ge H/K$ and $t < 3H/K < 1$. An exact decomposition is

$$2t = \frac{2H}{K} + \frac{2\nu^2}{(K+\mathcal{D})(W+H)} + \frac{2H\nu^2}{K(K+\mathcal{D})^2}.$$

Its last term is less than $H^3/(3K^3) \le t^3/3$, whereas

$$2\int_0^t \frac{s^2}{1+s^2}\,\mathrm{d}s \ge \frac{t^3}{3} \qquad (0 < t < 1).$$

Subtracting this integral from $2t$ proves the bound

$$\omega_c < \frac{2H}{K} + \frac{2\nu^2}{(K+\mathcal{D})(W+H)} \le \frac{2H}{K} + \frac{\nu^2}{8H}.$$

The identity $JK - 2c = -\frac{3}{2}(b_1 + b_2)$ and $K > 4$ now give

$$JH - c\omega_c > -\frac{3}{8}(b_1 + b_2)H - \frac{c\nu^2}{8H}.$$

Let $Z = \sqrt{3}H$ and $\ell(z) = z - (c/\sqrt{3})\arctan(z/\sqrt{3})$. Combining this estimate with $H_0 = \sqrt{3}(R + T - (b_1 R + b_2 T)/2)$ yields

$$\mathcal{F}_c > \frac{c\pi}{3} + \sqrt{3}\left[\ell(R) + \ell(T) - \frac{b_1 R + b_2 T}{2} - \frac{(b_1 + b_2)Z}{8} - \frac{c\nu^2}{8Z}\right].$$

A norm floor from convexity. The change of variable $t = (s - x)/(s + x)$ in the bridge secant gives, with $q = \delta/(2 + \delta)$,

$$u - \frac{3}{4} = \frac{x}{2(\beta - 1)}\left[\int_0^R \frac{t^2}{(1 + t)^2}\,\mathrm{d}t + \int_0^q \frac{t^2}{(1 - t)^2}\,\mathrm{d}t\right].$$

Discard the second integral. On $[0, R]$ the probability density $3t^2/R^3$ has mean $3R/4$ and variance $3R^2/80$. The function $f(t) = (1 + t)^{-2}$ satisfies $f''(t) \geq 6/(1 + R)^4$. Applying Jensen to $f(t) - 3t^2/(1 + R)^4$ therefore gives

$$\int_0^R \frac{t^2}{(1 + t)^2}\,\mathrm{d}t > \frac{R^3}{3}\left[\frac{1}{(1 + 3R/4)^2} + \frac{9R^2}{80(1 + R)^4}\right].$$

The inequality is strict even when $\delta = 0$, since the adjusted function is strictly convex on the interior of $[0, R]$. Thus, with $M = 2R + (1 + R)\delta$,

$$u > u_*(R, \delta) := \frac{3}{4} + \frac{(1 + R)R^3}{6M}\left[\frac{1}{(1 + 3R/4)^2} + \frac{9R^2}{80(1 + R)^4}\right]. \tag{G.35}$$

Bounds for the central shear. Write $\alpha = 1 + e\delta$, with $13/5 \leq e \leq 3$, $\beta = x(1 + \delta)$ and $x = (1 + R)/(1 - R)$. Then

$$T = T(R, \delta, e) = \frac{2R + ((e + 1)R - (e - 1))\delta}{2 + (e + 1 - (e - 1)R)\delta}.$$

Put $D = 3 + R^2$, $w = (e - 1)(1 - R^2)$, $\gamma = e + 1 - (e - 1)R$, $L = 2 + \gamma\delta$, and

$$Z_0 = \frac{2R^2(1 + R)}{3 - R}, \qquad A = \frac{1 + 2R}{3(1 - R)},$$
$$Z_U = Z_0 + A\delta - \frac{3}{2}\delta^2, \qquad Z_L = Z_0\left(1 + \frac{11}{2}\delta\right).$$

We prove $0 < Z_L \leq Z_c \leq Z_U$ by bounding the curvature and initial slope of the exact shear. For $f(t) = t/(3 + t^2)$, direct substitution in $Z_c = 2\mathcal{U}_1 A_s - R - (1 + \nu)T$ gives

$$Z_c = Z_0 + B\delta - C\delta^2 - D(f(T) - f(R)),$$
$$B = \frac{D[(1 - e) + (e + 3)R]}{2(3 - R)}, \quad C = \frac{eD(1 - R)}{2(3 - R)}.$$

Thus, writing $Q = f''(T)T_\delta^2 + f'(T)T_{\delta\delta}$,

$$Z_{c,\delta\delta} = -2C - DQ, \qquad T_\delta = -\frac{2w}{L^2}, \qquad T_{\delta\delta} = \frac{4w\gamma}{L^3}.$$

The shape bounds give $48/35 < w \le 16/9$, $3 < \gamma \le 10/3$ and $9/31 \le T \le 7/19$. On this interval $35/121 < f' < 23/75$ and $7/40 < -f'' < 1/4$. Here f' decreases, $-f''$ increases, and $-f''(T) < 2T/3 < 1/4$; the other comparisons follow at the two endpoints. Since $w/(\gamma L) \le 8/27$, and $\gamma/(2+\gamma\delta)^3$ increases in γ,

$$Q = \frac{4w\gamma}{L^3}\left(f'(T) + \frac{w}{\gamma L}f''(T)\right) > 4\frac{48}{35}3\left(\frac{19}{41}\right)^3\left(\frac{35}{121} - \frac{2}{27}\right) > \frac{7}{20}.$$

For the other direction,

$$Q < 4w\left(\frac{23\gamma}{75L^3} - \frac{7w}{40L^4}\right) \le \frac{23w\gamma}{150} - \frac{7w^2}{160} \le \frac{104}{135}.$$

The middle expression decreases in $L \ge 2$; after setting $L = 2$, it increases in w and γ, giving the last bound. Also $D(1-R)/[2(3-R)]$ decreases in R, and its value at $7/19$ exceeds $3/8$. Hence $39/40 < C \le 7/6$. Consequently

$$3 < 2C + DQ < \frac{7}{3} + \frac{1132}{361}\frac{104}{135} < \frac{19}{4}, \qquad -\frac{19}{4} < Z_{c,\delta\delta} < -3.$$

The initial slope is

$$z_1 := Z_{c,\delta}(R, 0, e) = B + \frac{w(3-R^2)}{2D},$$
$$\frac{\partial z_1}{\partial e} = \frac{R(1-R)(3-6R-R^2)}{(3-R)D} > 0.$$

For an upper bound take $e = 3$, so $z_1 = D(3R-1)/(3-R) + G(R)$, where $G(R) = R^2 - 7 + 24/D$. With $P = 8D/(3-R)^2 - (1-R)^{-2}$,

$$(z_1 - A)' = \frac{2R(3R-1)}{3-R} + P + G' > 0.$$

Indeed $G' = 2R(1 - 24/D^2) > -11/10$, while $P' < 1 + 13/4 - 27/4 < 0$ and $P(7/19) > 11/10$. Thus $A - z_1$ decreases in R, and its value at $R = 7/19$, $e = 3$ is positive. This proves $z_1 < A$. For a lower bound put $N = (1-e) + (e+3)R$, so $0 \le N < 1/2$ and

$$B_R = \frac{RN}{3-R} + \frac{D(e+5)}{(3-R)^2} < \frac{17}{4}.$$

The remaining term $w(3-R^2)/(2D)$ decreases in R, whereas $Z_0' \ge 19/24$. Therefore $[z_1 - (11/2)Z_0]' < 17/4 - (11/2)(19/24) < 0$. Its smallest value is at $R = 7/19$, $e = 13/5$, and exceeds $1/8$. Integrating the curvature bounds from $\delta = 0$ now gives

$$Z_c \le Z_0 + A\delta - \frac{3}{2}\delta^2 = Z_U, \qquad Z_c \ge Z_L + \frac{\delta(1-19\delta)}{8} \ge Z_L.$$

The shape bound. Retain $\ell(z) = z - (c/\sqrt{3})\arctan(z/\sqrt{3})$, and define

$$\mathfrak{B}(u; R, T, \delta) = \ell(R) + \ell(T) - \frac{b_1 R + b_2 T}{2} - \frac{(b_1 + b_2) Z_U}{8} - \frac{c\,\nu^2}{8 Z_L},$$

using the upper and lower bounds just proved for the central shear $-\sqrt{3}h_c$. Since $b_i \geq 0$ and $0 < Z_L \leq Z \leq Z_U$, the preceding score estimate gives $\mathcal{F}_c > c\pi/3 + \sqrt{3}\,\mathfrak{B}(u; R, T, \delta)$. The function $\mathfrak{B}$ is affine in u with positive slope, so u may be lowered to u_*. Proposition G.51 bounds $\mathfrak{B}$ from below.

Concavity in the second shape ratio. For fixed R, δ, u, the terms of $\mathfrak{B}$ that depend on T are

$$\ell(T) - \frac{T}{2} + \frac{2uT}{3 + T^2} + \frac{uZ_U}{2(3 + T^2)} - \frac{c\nu^2}{8Z_L}.$$

The second derivative of the first three terms is

$$\frac{2T}{(3 + T^2)^3}\left[c(3 + T^2) + 2u(T^2 - 9)\right] < 0,$$

because $u \geq 3/4$, $c < 3$ and $T^2 < 1$. The next term has second derivative $3uZ_U(T^2 - 1)/(3 + T^2)^3 < 0$. Finally,

$$(\nu^2)'' = \frac{4(3 + R^2)}{(3 + T^2)^4}\left[9T^2 - 3R^2 + T^2(5R^2 - 3T^2)\right] > 0 :$$

the ratio bound $27/31 \leq T/R \leq 1$ from Appendix G.4.6 gives $R^2/3 < T^2 \leq R^2$. Since $Z_L > 0$, the final term also has negative second derivative. This proves strict concavity in T directly.

G.4.8. *The face signs.* The face values are explicit functions of the shape (x, α, β), the norm u, the scale q and the duration τ, built from the scale form (G.22): w_*, $E_{L,*}$ and R^*_τ are evaluated at the given u and q, Γ_* is the Gram expression at $(t_0, \mathcal{U}(t_0))$, and $T = r_A/q$ is the upper end of the duration interval. Write $c_0(x) = \frac{x(7x+1)}{4x^2+11x+1}$ and call

$$x + \frac{8x(\alpha - 1)}{9x + 8} \leq \beta \leq x + c_0(x)(\alpha - 1)$$

the *strip*; by Lemma G.35 it contains the actual shapes for $1 < x \leq 5$. In gates S0 to S4 the proof runs in the same order: the norm is raised to its ceiling (the face value is monotone in u), the duration is removed by (G.23) or by a tangent bound for the logarithm, then β, then the shape α or $1/\alpha$, then x, each by the sign of a first or second derivative, and the remaining polynomials in one variable are checked on their intervals. Gate S6 instead uses the exact-angle score and the integral norm floor of Appendix G.4.7; its minimization is proved directly below.

Proposition G.46 (Faces of gate S0)**.** *Let* $F(\tau, \ell) = (w_* - 1)\tau + \frac{1}{\tau} + \frac{u-1}{r_A} + r_A + 1 - q - E_{L,*} + \ell + \frac{1}{2}(w_* - \frac{3}{4})$.

(a) *On* $12/5 \le x \le 9$, $1 \le \alpha \le (5x+4)/12$, $x \le \beta \le x + c_0(x)(\alpha-1)$, *with* $q = q_c$ *and* $u = \hat{u}$, *the values* $F(2, \frac{5}{2T} - \frac{11}{9})$, $F(\frac{8T}{9}, -\frac{1}{9})$ *and* $F(T, 0)$ *are negative.*

(b) *On* $12/5 \le x \le 9$, $(22x+20)/55 \le \alpha \le x$, $x + \frac{4x(\alpha-1)}{5x+4} \le \beta \le x + c_0(x)(\alpha-1)$, *with* $q = 1/7$ *and* $u = \bar{u}$, *the same three values are negative.*

Proof sketch. (a) Each test is convex in β: as a function of β it is a quadratic with positive leading coefficient plus simple poles with positive residues. So it suffices to take $\beta = x$ and an enlarged upper face $\beta = x + c_j(\alpha-1)$, with constant slopes $c_j \in \{7/8, 1, 5/4, 3/2\}$ on the four ranges $[12/5, 5/2]$, $[5/2, 3]$, $[3, 5]$, $[5, 9]$ of x; these slopes dominate c_0 because c_0 increases, with $c_0' = (73x^2+14x+1)/(4x^2+11x+1)^2$. On $\beta = x$ the cleared numerators increase in α, and at $\alpha = 1$ they factor through three polynomials in $x-2$ that are positive for $x \ge 12/5$. On each upper face the cleared numerator $N_j(x, \alpha)$ has $\partial_x^2 N_j > 0$: its expansion in $X = x - z_j$, $A = \alpha - 1$ has degree eight, and bounding each coefficient of X^k below by discarding positive terms and evaluating negative ones at the largest A leaves a polynomial in X that is positive on the range. The values and first x-derivatives at the lower end of each range are polynomials of degree seven in α, positive by a second-order Taylor bound around the midpoint of the α-interval. Where the α-range ends on a slanted edge $x = 2\alpha - o_j$, a third derivative in α is positive, and the values on the edge follow.

(b) Here $\bar{u} \le \bar{V}(x) \le u_b := \frac{x+74}{100}$, by $96(4x^2+11x+1)^2 - (7x+1)^2(x+1)(35x+13) = -179x^4 + 5606x^3 + 11040x^2 + 1882x + 83 > 0$ on $[0, 9]$ and $u_b - \bar{V} = (x-1)(35x^2 - 152x + 213)/(100(x+1)(35x+13)) > 0$. The shape is split at $\alpha = m(x) = (2x+3)/5$. For $m \le \alpha \le x$ use u_b; with $\beta = x + c(\alpha-1)$ and $3/5 \le c \le 3/2$, a cubic comparison in β leaves twelve polynomials of degree eight whose signs follow from one sign change in the shape variable. For $(22x+20)/55 \le \alpha \le m$ use the sharper ceiling $\bar{u} \le \hat{u}$, which follows from $75(4x^2+11x+1)^2 - (7x+1)^2(x+1)(35x+13) = -515x^4 + 3758x^3 + 8331x^2 + 1420x + 62 > 0$ on $[0, 9]$ (its derivative divided by x^2 decreases, so its minimum is at an endpoint). Each test is then convex in β, and two shape derivatives on five fixed β-faces reduce every sign to a single rational evaluation. □

Proposition G.47 (Faces of gate S1)**.** *Let* $\Phi_\tau = \Gamma_*(q) - \frac{7}{96} R_\tau^*(q)$.

(a) *On* $1 < x \le 2$, $1 \le \alpha$, $6\alpha^2 \le x(x+2)$, β *in the strip and* $\beta \le \alpha x$, *with* $q = q_c$ *and* $3/4 \le u \le u_\ell$*:* $\Phi_\tau > 9193/50000$ *for* $2 \le \tau \le T$.

(b) *On* $1 < x \le 2$, $1 \le \alpha \le x$, $x(x+2) \le 6\alpha^2$, β *in the strip and* $\beta \le \alpha x$, *with* $q = 1/7$, $u_f \le u \le \bar{V}(x)$, *and* $u \le u_\ell$ *when* $x \ge 3/2$ *and* $\alpha \le 6/5$*:* $\Phi_\tau > 9193/50000$ *for* $2 \le \tau \le T$.

Proof sketch. By (G.23) and the bound $\log y \le 2z + \frac{2}{3}z^3$ ($z = \frac{y-1}{y+1}$, $0 < y \le 1$) at $y = 2/T$, it suffices to show $F_0 > 0$ and $F_1 > 0$, where $F_0 = \Gamma_* - \frac{7}{96}\bar{R}_2 -$

$\frac{9193}{50000}$, $F_1 = F_0 - \frac{7}{96}(T-2)(w_* - \frac{3}{4})$ and $\bar{R}_2$ is R_2^* with that bound for the logarithm.

(a) The cap $6\alpha^2 \le x(x+2)$ gives $x \ge \sqrt{7} - 1 > 8/5$ and $\alpha \le 2/\sqrt{3} < 7/6$, and the strip lies in $x + \frac{1}{2}(\alpha - 1) \le \beta \le x + \frac{10}{13}(\alpha - 1)$, by $13x(7x+1) - 10(4x^2 + 11x + 1) = (x-2)(51x+5) \le 0$. On $u \in [3/4, 19/25]$ one has $-\frac{1}{5} < \partial_u F_0 \le \partial_u F_1 < 0$, and along the ceiling $u = u_\ell(x, \beta)$ the total β-derivative of each F_j is less than $-101/318720$; so u may be raised to u_ℓ and β to $\beta_+ = x + \frac{10}{13}(\alpha - 1)$. For F_0 the total α-derivative is less than $-4553/2589600$ below the cap, so α may be raised to $\sqrt{x(x+2)/6}$, where the central and outer formulas coincide and the argument of (b) gives $F_0 \ge 167/144000000$. For F_1, raise u to $19/25$; a third derivative shows that the cost part of F_1, which enters with a negative sign, has no interior maximum in $\alpha \in [1, 7/6]$, so F_1 is least at $\alpha = 1$ or $\alpha = 7/6$, where it decreases in x and exceeds $1/50000$ at $x = 2$.

(b) Three branches cover the domain. For $1 < x \le 3/2$ both tests exceed $499/884000$: monotonicity in u and in β, then concavity in $1/\alpha$, reduce them to $\alpha = 1$ and $\alpha = x$, where derivative signs in one variable conclude. For $3/2 \le x \le 2$ and $6/5 \le \alpha \le x$, concavity in $1/\alpha$ reduces to $\alpha = 6/5$ and $\alpha = x$, with margins $1/2000$, $3/500$ and $1/15000$. For $3/2 \le x \le 2$ and $1 \le \alpha \le 6/5$, $F_0 \ge 167/144000000$ after one split at $x = 5/3$; for F_1, concavity in β and in $u \in [3/4, 19/25]$ reduce to the four constant pairs $(\frac{\beta - x}{\alpha - 1}, u) \in \{\frac{1}{2}, \frac{10}{13}\} \times \{\frac{3}{4}, \frac{19}{25}\}$, the second derivative in $1/\alpha$ is less than $-11/800$, and the two shape ends give $F_1 \ge 1597/144000000$. □

Proposition G.48 (Faces of gate S2)**.** *Let* $G_\tau = R_\tau^*(q) + (2 - \frac{7\tau}{4T})(w_* - \frac{3}{4})$.

(a) On $13/6 \le x \le 12/5$, $1 \le \alpha$, $6\alpha^2 \le x(x+2)$, β *in the strip, with* $q = q_c$ *and* $u = u_C$*:* $G_\tau < -1/6000$ *for* $2 \le \tau \le T$.

(b) On $13/6 \le x \le 12/5$, $\alpha \le x$, $x(x+2) \le 6\alpha^2$, β *in the strip, with* $q = 1/7$*, and* $u = u_s$ *if* $\alpha \le 3/2$, $u = \bar{V}(x)$ *if* $\alpha \ge 3/2$*:* $G_\tau < 0$ *for* $2 \le \tau \le T$.

Proof sketch. (a) The domain lies in $1 \le \alpha \le 4/3$, $x + \frac{5}{8}(\alpha - 1) \le \beta \le \beta_+ = x + \frac{24x+51}{128}(\alpha - 1)$. The function $\log \tau + 1/\tau$ is concave on $[2, \infty)$, its second derivative being $(2 - \tau)/\tau^3$; its tangent at $t(\alpha) = \frac{20}{3} - 4\alpha$ ($\alpha \le 7/6$), $t = 2$ ($\alpha \ge 7/6$) makes the bound for G_τ affine in τ, so $G_\tau \le \max\{F_0, F_1\}$ for two explicit tests. Both increase in β, so $\beta = \beta_+$; both have negative total x-derivative (below $-1/250$ and $-1/2000$), so $x = 13/6$. There the norm is raised to an affine upper bound in α, the logarithm is bounded by a quadratic, and four polynomials in one variable with nine terms each give $F_0, F_1 < -1/6000$.

(b) Here $G'_\tau \le (1 - \frac{7}{4T})(w_* - \frac{3}{4})$, so $G_\tau \le \max\{H_0, H_1\}$, the values at $\tau = 2$ and at the end of the affine bound. The domain gives $\alpha \ge \frac{36x+25}{84} > \frac{6}{5}$, by $84^2 x(x+2) - 6(36x+25)^2 = -720x^2 + 3312x - 3750 < 0$, and $\beta = x + c(\alpha - 1)$ with $\frac{5}{8} \le c \le \frac{24x+51}{128} < \frac{7}{8}$. For $\alpha \le 3/2$, each H_j is convex in $1/\alpha$ (second derivative greater than $2/5$). At $\alpha = \frac{36x+25}{84}$ it increases in β and is negative

at the largest slope, with margins $1/5000$ and $1/1000$ on $[13/6, 9/4]$ and $[9/4, 12/5]$; at $\alpha = 3/2$, $u_s \le \bar{V}$, H_j increases in u, and its value at $\bar{V}$ is negative. For $\alpha \ge 3/2$ the second derivative in $1/\alpha$ exceeds $1/2$, and the values at $\alpha = 3/2$ and $\alpha = x$ are negative for every slope. □

Proposition G.49 (Faces of gate S3)**.** *Put* $y = 2q/r_A$, $k = 5\alpha - 1$, $J = \frac{u-1}{r_A} + r_A + 1 - q - E_{L,*} + \log y$, $F_0 = J + (2 + \frac{4(4-3y)}{k})(w_* - \frac{3}{4})$ *and* $F_1 = J + (\frac{r_A}{q} + \frac{4}{k})(w_* - \frac{3}{4})$.

(a) On $2 \le x \le 13/6$, $\frac{15x+511}{475} \le \alpha \le \frac{89}{72}$, $\beta = x + \frac{25x}{15x+36}(\alpha - 1)$, *with* $q = q_c$ *and* $u = u_{CB}$*:* $F_0, F_1 < 0$.

(b) On $2 \le x \le 13/6$, $\alpha \le x$, $x(x+2) \le 6\alpha^2$, $\beta = x + C(\alpha - 1)$ *with* $\frac{8x}{9x+8} \le C \le \frac{115x}{304}$, *with* $q = 1/7$ *and* $u = u_O$*:* $F_0, F_1 < 0$.

Proof sketch. (a) Along the curve both tests decrease in α. Grouping three terms of the α-derivative, and bounding the norm and the transported quantities by their ranges, reduces its sign to the single quadratic inequality

$$\frac{65452}{825}q^2 - \frac{68203}{3750}q + \frac{29299}{30000} < 0 \qquad (63/500 \le q \le 1/7),$$

which holds because the quadratic is convex and negative at both ends. So both tests are at most their values at $\alpha = \frac{15x+511}{475}$, $\beta = \frac{20x}{19}$. There $\log y \le -\frac{40153}{180000} + \frac{2(5y-4)}{5y+4}$ for $0 < y \le 6/7$, and the tests become $-N_i(x-2)$ divided by a positive polynomial, where N_0 and N_1 have degrees 17 and 15 and all their coefficients are positive except the linear one. For N_1 the quadratic part has negative discriminant. For N_0, write c_j for its coefficients and $h = x - 2 \in [0, 1/6]$: the quadratic part decreases on $[0, 1/6]$, since its vertex $-c_1/(2c_2)$ exceeds $1/6$, and it is positive at $h = 1/10$; on $[1/10, 1/6]$ the cubic term $c_3h^3 \ge c_3/1000$ exceeds the absolute value of the quadratic part at $h = 1/6$. So $N_0, N_1 > 0$.

(b) The multipliers satisfy $n_0, n_1 \in [25/8, 7/2]$ for $\alpha \le 5/4$ and $n_0, n_1 \in [5/2, 7/2]$ for $\alpha \ge 5/4$, by elementary quadratic bounds, and the domain gives $\alpha \ge \alpha_* = \frac{5(x+1)}{13}$, since $\frac{x(x+2)}{6} - \alpha_*^2 = \frac{19x^2+38x-150}{1014} > 0$. For a constant n the function $R_n = J + n(w_* - \frac{3}{4})$ is affine in n and has second derivative greater than $1/4$ in $1/\alpha$. Its values $R_{25/8}, R_{7/2}$ at α_*, $R_{5/2}$ at $\alpha = 5/4$ and R_n at $\alpha = x$ are negative for every slope C, by polynomial signs in one and two variables of degree at most eight. Convexity in $1/\alpha$ and affinity in n conclude. □

Proposition G.50 (Face of gate S4)**.** *On* $2 \le x \le 13/6$, $0 \le \delta \le 1/19$, $\alpha = 1 + e\delta$ *with* $13/5 \le e \le 3$, $\beta = x(1+\delta)$, *and* $u = \bar{u}(x, \beta)$, *the functions* Φ *and* Ψ *of Proposition G.40 exceed* $1829/10000$.

Proof sketch. Six derivative signs reduce Φ and Ψ to three points: $-\Phi_\delta > 27/5000$ on the whole box; $\Psi_e/(x\delta) > 0$, extended continuously to $\delta = 0$ (note $\Psi_e = \delta\Psi_\alpha$); $-\Psi_x > 11/14000$ at $e = 13/5$; $-\Phi_x > 1/1750$ at $\delta = 1/19$; $-\Phi_e > 0$ at $x = 13/6$, $\delta = 1/19$; and $-\Psi_{\delta\delta} > 81/1000$ at $x = 13/6$, $e = 13/5$.

Hence $\Phi \ge \Phi(\frac{13}{6}, \frac{1}{19}, 3)$ and $\Psi \ge \min\{\Psi(\frac{13}{6}, 0, \frac{13}{5}), \Psi(\frac{13}{6}, \frac{1}{19}, \frac{13}{5})\}$. Each of the three values is a rational number minus $\frac{1}{7}\log y$, with y equal to $\frac{103488}{134749}$, $\frac{12}{19}$ and $\frac{2524608}{3353029}$, and the bound $\log y < 2(z + z^3/3 + z^5/5)$, $z = \frac{y-1}{y+1} < 0$, shows that they exceed 1829/10000 by more than 2053/25000000, 4017/20000000 and 303/100000000. □

Proposition G.51 (Face of gate S6)**.** *For* $1/3 \le R \le 7/19$, $0 \le \delta \le 1/19$ *and* $13/5 \le e \le 3$,

$$\mathfrak{B}\big(u_*(R,\delta); R, T(R,\delta,e), \delta\big) > \frac{9}{4000}.$$

Proof. Concavity in T first reduces e to its two endpoints. We prove that the bound decreases in δ and, at $\delta = 1/19$, is concave in R. These two properties reduce the comparison to four corners.

The decomposition. Put $v = 4u_* - 3$, $k(a) = \ell(a) - a^3/[2(3+a^2)]$ and $g(a) = a/[2(3+a^2)]$. Then

$$\mathfrak{B} = \mathcal{P} - \frac{Z_U(b_1+b_2)}{8} - \frac{c\nu^2}{8Z_L}, \qquad \mathcal{P} = k(R) + k(T) + v\big(g(R) + g(T)\big).$$

On $9/31 \le a \le 7/19$ the derivatives satisfy

$$k''(a) = \frac{a\big[2c(3+a^2) - 3(9-a^2)\big]}{(3+a^2)^3} < f(a) := \frac{9a(a^2-1)}{(3+a^2)^3}.$$

Here $f'(a) = -9(3a^4 - 14a^2 + 3)/(3+a^2)^4 < 0$ for $a^2 < 1/7$. Since $9/31 > 2/7$, we get $k''(a) < f(2/7) < -2/25$ throughout the required interval. Also $k'' < 0$ on $(0,1)$ and $k'(1/4) < 0$, so $k' < 0$ on the required interval. Directly from the increasing function g and the floor (G.35),

$$\begin{gathered} 0 < v < \frac{1}{25}, \qquad 0 < g'(a) < \frac{2}{13}, \qquad g''(a) < 0, \\ 0 < G := g(R) + g(T) < \frac{2}{17}. \end{gathered} \tag{G.36}$$

For the first bound, v decreases in δ, while $v(R,0)$ increases in R and is below $1/25$ even at $R = 3/8$. For the next two, use $a \ge 9/31 > 2/7$ and $T \le R \le 7/19$.

The pair term decreases with displacement. Write $L = 2 + \gamma\delta$, $\gamma = e + 1 - (e-1)R$, and

$$\chi = \frac{L^2}{2(e-1)(1-R)M}, \qquad M = 2R + (1+R)\delta.$$

Then $v_\delta = v\chi T_\delta$, $T_\delta < 0$, and

$$\mathcal{P}_\delta = T_\delta F, \qquad F = k'(T) + vg'(T) + v\chi G.$$

The function χ increases in δ and decreases in R and e, so $0 < \chi \le \chi(1/3, 1/19, 13/5) < 3$. These signs follow by logarithmic differentiation: $2\gamma/L - (1+R)/M > 0$; the R derivative has the two negative contributions

$-2(e-1)\delta/L$ and $(4R-2+2R\delta)/[(1-R)M]$; and the e derivative has numerator $(e-1)(1-R)\delta-2-2\delta<0$. Differentiating F gives

$$F_\delta = T_\delta\big[k''+2v\chi g'+v\chi^2 G+vg''\big]+v\chi_\delta G>0.$$

Indeed, by (G.36) the bracket is less than $-2/25+(1/25)(12/13+18/17)<0$.

It remains to bound F at $\delta=0$, where $T=R$ and

$$F_0 = k'(R)+v_0\left[g'(R)+\frac{1}{(e-1)(1-R)(3+R^2)}\right].$$

Here $v_0>5R^2/18$: discard the positive variance term and use that $(1+R)/[3(1+3R/4)^2]$ decreases and remains above $5/18$ at $R=3/8$. The bracket exceeds $3/8$, since its first term exceeds $1/7$ and its second is at least $27/112>6/25$. Finally $k'(R)>-5R^2/48$. One way to check this last bound is to put $s=R^2$:

$$H(s) := -\frac{k'(\sqrt{s})}{s} = \frac{-27s^2+80s-3}{54s(3+s)^2},$$
$$H'(s) = \frac{27s^3-241s^2+9s+9}{54s^2(3+s)^3}>0.$$

The derivative is positive for $s\le 1/7$ because $9-241/49>0$, and $H(1/7)<5/48$. Consequently $F_0>-(5/48)R^2+(5/18)R^2(3/8)=0$. Thus $\mathcal{P}_\delta<0$.

The shear loss increases with displacement. Write $Z=Z_U=Z_0+A\delta-3\delta^2/2$, where $Z_0=2R^2(1+R)/(3-R)$ and $A=(1+2R)/[3(1-R)]$. Let $b_{10}=b_1(R,0)$. Since $v_\delta<0$, $b_{1,\delta}>0$, and

$$b_1+b_2 = 2b_1-(1-b_1)\nu, \qquad b_{10}\ge\frac{2R^2}{3(3+R^2)}.$$

The latter follows from $v_0\le R^2/3$, or from the secant ceiling of Appendix G.4.1. Put $a_\nu=(e-1)R(1-R^2)/(3+R^2)$ and $N=(e+1)R-(e-1)$. The identity

$$\nu=\frac{a_\nu\delta(1+a\delta)}{1+b\delta+j\delta^2}, \qquad a=\frac{e+1}{2}-\frac{e-1}{4}(R+R^{-1}),$$
$$b=\gamma-a_\nu, \quad j=\frac{3\gamma^2+N^2}{4(3+R^2)}$$

has $1/3\le a\le 7/12$, $11/4\le b\le 10/3$ and $j>27/16$. In particular $b\ge a+2$, $j\ge 2a$ and $a(b-a)\le 77/48<j$. Comparison of the denominator with $(1+a\delta)(1+2\delta)$, and then differentiation, give

$$\nu\le\frac{a_\nu\delta}{1+2\delta}, \qquad \nu_\delta\le\frac{a_\nu}{(1+2\delta)^2}.$$

Since $Z_\delta>0$,

$$(Z\nu)_\delta\le\frac{a_\nu(Z_0+2A\delta)}{(1+2\delta)^2}.$$

Here the omitted terms in the numerator are $(2A-9/2)\delta^2-6\delta^3\le 0$. As $a_\nu/(2b_{10})\le d(R):=3(1-R^2)/(2R)$, it suffices to show

$$E(R,\delta):=(A-3\delta)(1+2\delta)^2-d(R)(Z_0+2A\delta)>0.$$

This expression decreases in δ, since $E_\delta=A(4-2d)-3+(8A-24)\delta-36\delta^2<0$. At $\delta=1/19$, it increases in R. Indeed $A'\ge 9/4$, $(dA)'=(2-R^{-2})/2<0$, and

$$(dZ_0)'=\left(\frac{3R}{3-R}(1-R)(1+R)^2\right)'<2.$$

For the last bound use that $(1-R)(1+R)^2$ decreases from $32/27<4/3$ and $3-R>5/2$. Thus $E_R>9/4-2>0$. At $R=1/3$ its positivity follows from

$$\left(\frac{5}{6}-\frac{3}{19}\right)\left(\frac{21}{19}\right)^2>\frac{4}{5}>4\left(\frac{1}{9}+\frac{5}{57}\right).$$

It follows that $[Z(b_1+b_2)]_\delta\ge 2Z_\delta b_{10}-(Z\nu)_\delta>0$.

The anisotropy loss increases with displacement. For $\delta>0$,

$$\frac{\nu_\delta}{\nu}=\frac{4T(3+R^2)}{L\delta(3+T^2)(R+T)}>\frac{1}{4\delta}\ge\frac{19}{4},\qquad \frac{Z_{L,\delta}}{Z_L}\le\frac{11}{2}.$$

Use $T>1/4$, $L<3$, $3+T^2<4$ and $R+T<1$ for the strict bound. Hence ν^2/Z_L increases. Its derivative is zero at $\delta=0$. Together with the preceding two losses, this proves that $\mathfrak{B}$ strictly decreases in δ on the whole box.

Concavity at the largest displacement. Now fix $\delta=1/19$. Put $p=(e-1)/(39+e)\le 1/21$, so $T=(R-p)/(1-pR)$. Then $1<T_R<21/20$ and $0<T_{RR}<1/9$; for the first lower bound use $2R>p(1+R^2)$. The upper bounds follow from $pR<1/56$. The norm floor has the useful factorization

$$v=qh,\quad q=\frac{16(1+R)}{3(4+3R)^2}+\frac{3R^2}{80(1+R)^3},\qquad h=\frac{38R^3}{39R+1}.$$

On $1/3\le R\le 3/8$, the two summands q_1,q_2 satisfy $q_1'<-1/10$, $0<q_2'<1/100$, $q_1''<1/9$ and $q_2''<0$. Also $q_1\le 64/225$, $q_2<1/400$, so $q<23/80$. These bounds follow directly from their positive linear denominators. Moreover $0<h<R^2<1/7$ and $0<h''=(76/39)[1-(39R+1)^{-3}]<2$. Both hq_1 and hq_2 increase, and $q'<0$, so

$$0<v_R<\frac{23}{80}h'(7/19)<\frac{1}{5},\qquad v_{RR}<\frac{23}{40}+\frac{1}{63}<\frac{3}{5}.$$

Using these estimates, keeping both terms in $G_R=g'(R)+g'(T)T_R$, gives

$$\mathcal{P}_{RR}<-\frac{4}{25}+\frac{3}{5}\frac{2}{17}+2\frac{1}{5}\frac{2}{13}\left(1+\frac{21}{20}\right)+\frac{1}{25}\frac{2}{13}\frac{1}{9}<\frac{1}{25}.$$

The discarded terms have factors $k'(T)$ or g'' and are negative.

For $a=R$ or T, differentiation of $b=1-(3+v)/(3+a^2)$ gives

$$b_R=\frac{2aa_R(3+v)}{(3+a^2)^2}-\frac{v_R}{3+a^2}>\frac{1}{10},$$

and

$$b_{RR} = \frac{6(3+v)(1-a^2)a_R^2}{(3+a^2)^3} + \frac{2a(3+v)a_{RR} + 4aa_Rv_R}{(3+a^2)^2} - \frac{v_{RR}}{3+a^2} > \frac{1}{4}.$$

Indeed $a \geq 9/31$, $a_R \geq 1$, $a_{RR} \geq 0$ and $a^2 < 1/7$. The two lower comparisons are $6(9/31)/(22/7)^2 - 1/15 > 1/10$ and $18(6/7)/(22/7)^3 - 1/5 > 1/4$. Since $Z > 1/9$, $Z_R > 3/4$ and $Z_{RR} > 0$, while $b_1 + b_2 > 0$, we obtain

$$[Z(b_1+b_2)]_{RR} > 2\frac{3}{4}\frac{1}{5} + \frac{1}{9}\frac{1}{2} = \frac{16}{45}.$$

Here Z_0 is increasing and strictly convex, $Z_0'(1/3) = 19/24$, and the $A\delta$ term has positive first and second derivatives.

The remaining anisotropy term has small curvature because ν itself is small. Write $w = \nu/R = pABC$, where

$$A = 1-R^2, \quad B = 2-p(R+R^{-1}), \quad D = 3(1-pR)^2+(R-p)^2, \quad C = D^{-1}.$$

The bounds $A \leq 8/9$, $7/4 < B < 2$, $0 < B' \leq 3/7$, $D > 8/3$, $0 < D' < 4/5$ and $D'' < 21/10$ give $C < 3/8$, $C' < 0$ and $w' < 0$. For the last sign, $A'B + AB' < -7/6 + 3/7 < 0$. Since $C'' \geq -D''C^2$, differentiation gives

$$w'' \geq -p\left[2BC + \frac{2pAC}{R^3} + ABD''C^2 + 4RB'C + 2AB'D'C^2\right] > -\frac{1}{5}.$$

The five bracketed terms are respectively below $3/2$, 1, $3/5$, $1/4$ and $1/10$. Also $w \leq 2/63 < 1/30$. At this displacement,

$$\frac{\nu^2}{Z_L} = w^2 f, \qquad f = \frac{38}{49}\frac{3-R}{2(1+R)}.$$

Here $0 < f \leq 38/49$, $f' < 0$ and $f'' > 0$. All contributions to $(w^2 f)''$ except $2ww''f$ are nonnegative, so $(\nu^2/Z_L)'' > -38/3675$. As $c < 3$, the total score curvature satisfies

$$\mathfrak{B}_{RR} < \frac{1}{25} - \frac{2}{45} + \frac{19}{4900} = -\frac{1}{1764} < 0.$$

The minimum in R is therefore at an endpoint.

The four endpoint comparisons. Weighted Jensen in $\ell(z) = (1-c/3)z + (c/3)\int_0^z t^2/(3+t^2)\,\mathrm{d}t$ gives

$$\ell(z) > j(z) := (1-c/3)z + \frac{cz^3}{27(1+z^2/5)} \qquad (z > 0).$$

At $\delta = 1/19$ the four pairs (R,T) are $(1/3, 23/77)$, $(1/3, 9/31)$, $(7/19, 163/487)$ and $(7/19, 16/49)$. Replacing each ℓ by j makes these four comparisons rational, and each resulting value exceeds $9/4000$. For example, at $(7/19, 163/487)$ the four terms satisfy

$$j(R) + j(T) > \frac{2753}{200000}, \qquad \frac{b_1R + b_2T}{2} < \frac{403}{40000},$$
$$\frac{(b_1+b_2)Z_U}{8} < \frac{33}{25000}, \qquad \frac{c\nu^2}{8Z_L} < \frac{3}{25000}.$$

Their difference is 9/4000, and the other three substitutions give strict lower bounds above 9/4000. □

Appendix H. Periods with at most five pieces

By Theorem 3.3, a competitive normal vertex period whose state stays in the compact domain has at most five pieces. This section computes the cost of every normal vertex period with at most five pieces and shows that it is at least A_{oct}, with equality only for the octagon's trajectory. These are the terminal comparison and equality statements used in Section 5.2 to prove Theorem 1.2 and part (b) of Theorem 1.1.

The tangent closure (Lemma B.4) gives four scalar equations in the factors. With at most five factors, these equations either have no solution or determine the factors up to one or two parameters. The twist $X(T) = R^{-1}X(0)R$ then determines the initial state, and the cost becomes an explicit function of the parameters. The comparison must be sharp: the four-piece family contains the octagon, so its minimum must be located exactly. The five-piece family approaches the octagon at the boundary of its domain, where one factor tends to 1 and the period degenerates to four pieces. The strict cost inequality therefore has no uniform positive margin near this boundary.

H.1. **Closed vertex cycles.** The terminal comparison uses only the data of a period at its switches. We write them in the triangular coordinates $M \mapsto CMC^{-1}$, $C = \left(\begin{smallmatrix}\sqrt{3} & 1\\ \sqrt{3} & -1\end{smallmatrix}\right)$, in which the tangent factors of Lemma B.1 and the rotation R become

$$\widetilde{T}_0(q) = \begin{pmatrix} q & 0 \\ 0 & q^{-1} \end{pmatrix}, \qquad \widetilde{T}_1(q) = \begin{pmatrix} q^{-1} & q - q^{-1} \\ 0 & q \end{pmatrix}, \\ \widetilde{T}_2(q) = \begin{pmatrix} q^{-1} & 0 \\ q^{-1} - q & q \end{pmatrix}, \qquad \widetilde{R} = \begin{pmatrix} 0 & 1 \\ -1 & 1 \end{pmatrix} \tag{H.1}$$

Definition H.1 (Closed vertex cycle)**.** A *closed vertex cycle* with $N \geq 1$ pieces consists of labels $i_k \in \mathbb{Z}/3$, factors $q_k > 1$ and switch margins $m_k = (m_k(0), m_k(1), m_k(2))$ with positive entries, indexed by $k \in \mathbb{Z}$, such that for every k and j:

(1) (twist) $i_{k+N} = i_k + 1$, $q_{k+N} = q_k$, $m_{k+N}(j) = m_k(j-1)$, and $i_{k+1} \neq i_k$;

(2) (quadric) $m_k(0)m_k(1) + m_k(1)m_k(2) + m_k(2)m_k(0) = 4$;

(3) (one-arc map) with $i = i_k$, $a = m_k(i)$, $b = m_k(i+1)$, $c = m_k(i-1)$,

$$m_{k+1}(i) = a, \qquad m_{k+1}(i+1) = (a+b)q_k - a, \qquad m_{k+1}(i-1) = \frac{a+c}{q_k} - a; \tag{H.2}$$

(4) (tangent closure) $\widetilde{T}_{i_k}(q_k)\,\widetilde{T}_{i_{k+1}}(q_{k+1}) \cdots \widetilde{T}_{i_{k+N-1}}(q_{k+N-1}) = \widetilde{R}$.

Its *raw cost* is $\sum_{k=0}^{N-1} \rho(a_k, b_k, c_k, q_k)$, where a_k, b_k, c_k are the active, growing and decreasing margins of item (3) and

$$\rho(a, b, c, q) = a\big((a+b)(q-1) + (a+c)(1-q^{-1}) - a\log q\big). \tag{H.3}$$

The letter ρ denotes the raw arc cost only in this section and in Appendix H.6.

Lemma H.2 (The cycle of a normal period)**.** *Let P be a normal vertex period with N pieces. Its labels, its factors and the margins $m_k(j) = p_j$ of its twisted state at the switching times τ_k form a closed vertex cycle with N pieces, and*

$$\mathcal{C}(P) = \frac{\sqrt{3}}{4}\,(\textit{raw cost}), \qquad \textit{that is,} \qquad \frac{\mathcal{C}(P)}{\sqrt{12}} = \frac{\textit{raw cost}}{8}.$$

Proof. Positivity and the quadric are Lemma A.6; (H.2) and the cost of a piece, $(\sqrt{3}/4)\,\rho(a,b,c,q)$, are Lemma B.1. The twist $Y(t+T) = R^{-1}Y(t)R$ of the state shifts the margins, $p_j(R^{-1}XR) = p_{j-1}(X)$, which gives $m_{k+N}(j) = m_k(j-1)$. The tangent closure from the switch τ_k is Lemma B.4 for the period rebased at τ_k, conjugated by C. □

A period has cost below the octagon's exactly when the raw cost of its cycle is below $8\delta_{\mathrm{oct}}$. Two operations change a cycle without changing its raw cost: *rebasing*, which shifts all indices by a fixed integer, and *relabeling*, which adds 1 to every label and shifts the margins accordingly; rebasing a cycle with N pieces by r replaces its word by the last $N-r$ letters followed by the first r letters increased by one. Relabeling at most twice makes the first label 0.

H.2. **The words of at most five pieces.** A cycle with first label 0 is described by its word $i_0 i_1 \cdots i_{N-1}$: neighbouring letters differ, and by the twist the letter after i_{N-1} is $i_0 + 1 = 1$, so $i_{N-1} \neq 1$. For $N \leq 5$ there are $1+1+3+5+11 = 21$ such words.

Proposition H.3 (Words of at most five pieces)**.** *Let c be a closed vertex cycle with $N \leq 5$ pieces.*

(1) $N \geq 4$.

(2) If $N = 4$, every switch is forward: $i_{k+1} = i_k + 1$ for all k.

(3) If $N = 5$, either every switch is reverse, or some rebasing and relabeling of c has the word 0 1 0 2 0 *and the same raw cost.*

Proof. Relabel so that $i_0 = 0$ and compare the closure product of each of the 21 words with $\widetilde{R}$, entry by entry. Each entry of a product of the matrices (H.1) is a rational function of the factors, and for most words a single entry already contradicts $q_k > 1$. For the word 0, the upper left entry of $\widetilde{T}_0(q_0)$ is $q_0 \neq 0$. For 02, 020, 0202 and 02020 the upper right entry of the product vanishes, while that of $\widetilde{R}$ is 1; for 010 and 01010 the lower left entry vanishes instead of being -1; for 0210 and 02010 the upper left entry is a positive monomial in the factors instead of 0. For 012 two entries are needed: the lower row of the product is

$$\Big(-\frac{q_1(q_2^2-1)}{q_0 q_2},\ \frac{q_1 q_2}{q_0}\Big),$$

whose second entry forces $q_0 = q_1 q_2$, after which the first is $-(q_2^2-1)/q_2^2 \neq -1$. The words 0102, 01012 and 01212 fall in the same way, through the two

entries of the lower row: for 01212, the lower right entry gives $q_0q_2q_3 = q_1q_4(q_2^2+q_3^2-1)$, and then the lower left entry is $-1+q_1q_3/(q_0q_2q_4) \neq -1$. For 0212, the upper left and lower left entries give $q_2^2q_3^2 = q_2^2+q_3^2$ and $q_0q_2 = q_1q_3$, and then the lower right entry forces $q_2 = 1$. Table 4 lists the outcome. The surviving words are 0120, whose switches, including the one across the seam to the letter 1, are all forward; the five words 01020, 01202, 01210, 02012, 02120, which are the rebasings of 01020, normalized to start with 0; and 02102, whose five switches are all reverse. □

N	words with $i_0 = 0$, $i_{N-1} \neq 1$	outcome
1, 2, 3	0; 02; 010, 012, 020	excluded by the closure
4	0102, 0202, 0210, 0212	excluded by the closure
	0120 $(++++)$	the octagon's word
5	01010, 01012, 01212, 02010, 02020	excluded by the closure
	01020 $(+--++)$, 01202, 01210, 02012, 02120	rebasings of 01020
	02102 $(-----)$	entirely reverse (class R)

TABLE 4. The reduced words of at most five letters and their fate. The signs are those of the N switches of a period, the last one across the seam.

H.3. **Four pieces.** By Proposition H.3 and a relabeling, a four-piece cycle has the octagon's word 2 0 1 2. Its closure equations can be solved in closed form.

Lemma H.4 (Four-piece closure)**.** *If $p, q, r, z > 1$ and*

$$\widetilde{T}_2(p)\widetilde{T}_0(q)\widetilde{T}_1(r)\widetilde{T}_2(z) = \widetilde{R},$$

then $r = p$, $z = q$ and $p^2q^2 = p^2+q^2$.

Proof. The upper row of the product is $(-q(r^2z^2-r^2-z^2)/(prz),\ qz(r^2-1)/(pr))$, so $r^2z^2 = r^2+z^2$ and $qz(r^2-1) = pr$; together these give $pz = qr$. The lower right entry then gives $q^2r^2 = q^2+r^2$. Subtracting the two quadratic relations gives $(q^2-z^2)(r^2-1) = 0$, so $z = q$, and $pz = qr$ gives $r = p$. □

Write $a = 1/p$ and $b = 1/q$. The relation $p^2q^2 = p^2+q^2$ becomes $a^2+b^2 = 1$: the four factors are $(1/a, 1/b, 1/a, 1/b)$ for a point (a, b) of the open quarter circle, and every such point satisfies the closure. Put

$$s = a+b \in (1, \sqrt{2}\,], \qquad \mathcal{N} = (s-1)(s+3),$$

and define the *four-piece density*

$$\delta(a,b) = \frac{4s(s-1)+2(1+a)(1-b)\log a + 2(1-a)(1+b)\log b}{(s-1)(s+3)}. \tag{H.4}$$

Lemma H.5 (Four-piece cycles)**.** *Let c be a closed vertex cycle with word* 2 0 1 2*. There are $a, b > 0$ with $a^2 + b^2 = 1$ such that the factors are $(q_0, q_1, q_2, q_3) = (1/a, 1/b, 1/a, 1/b)$, the initial margins are*

$$(m_0(0), m_0(1), m_0(2)) = \frac{1}{\sqrt{\mathcal{N}}}\big(2(s-1),\ 2(1-a+b),\ 2(1+a-b)\big), \quad \text{(H.5)}$$

and the raw cost is $8\,\delta(a, b)$.

Proof. The factors come from Lemma H.4. For the margins, write (A, B, C) for the vector in (H.5) before the factor $\mathcal{N}^{-1/2}$; it satisfies $C + A = 4a$, $C + B = 4$ and $A + B = 4b$. Starting from arbitrary initial margins (x_0, x_1, x_2) and applying (H.2) four times, the twist $m_4(j) = m_0(j-1)$ gives two linear equations, $(1+a-b)x_0 = (s-1)x_2$ and $(1+a-b)x_1 = (1-a+b)x_2$; the quadric fixes the scale, and $x_2 > 0$ fixes the sign. Along the period the margins run through the permutations

$$(A, B, C) \to (B, A, C) \to (B, C, A) \to (A, C, B) \to (C, A, B)$$

of the same three numbers (each divided by $\sqrt{\mathcal{N}}$), so every switch state is strict-star, and the last one is the twist of the first. The active margins are C, B, C, B. For the first piece, (H.3) with $a_0 = C$, $b_0 = A$, $c_0 = B$ (each over $\sqrt{\mathcal{N}}$) and $q_0 = 1/a$ gives, by $C + A = 4a$ and $C + B = 4$,

$$\rho = \frac{C}{\mathcal{N}}\Big(4a\big(\tfrac{1}{a} - 1\big) + 4(1-a) - C\log\tfrac{1}{a}\Big) = \frac{8C(1-a) + C^2\log a}{\mathcal{N}},$$

the third piece costs the same, and the second and fourth cost $(8B(1-b) + B^2\log b)/\mathcal{N}$. On the circle, $C^2 = 4(1+a-b)^2 = 8(1+a)(1-b)$, $B^2 = 8(1-a)(1+b)$, and $(1+a-b)(1-a)+(1-a+b)(1-b) = 1-s+2ab = s(s-1)$. The sum of the four pieces is $8\delta(a, b)$. □

Theorem H.6 (The four-piece density)**.** *For $a, b > 0$ with $a^2 + b^2 = 1$,*

$$\delta(a, b) \ \geq\ \delta_{\mathrm{oct}} = \frac{8 - 4\sqrt{2} - \log 2}{2\sqrt{2} - 1},$$

with equality only at $a = b = 1/\sqrt{2}$.

The proof is in Appendix H.6.2. The density is symmetric in a and b, and its value at $a = b = 1/\sqrt{2}$ is δ_{oct}: there $s = \sqrt{2}$, $4s(s-1) = 8 - 4\sqrt{2}$, the logarithmic terms add up to $4(1-a^2)\log a = -\log 2$, and $\mathcal{N} = 2\sqrt{2} - 1$. The content of the theorem is that the symmetric point is the minimum. Writing $b \in (0, 1/\sqrt{2}\,]$ and $a = \sqrt{1-b^2} \geq b$, we show that $\psi(b) = \delta(\sqrt{1-b^2}, b)$ is strictly decreasing. Its derivative has the sign of $(s+1)\big(w(b) - w(a)\big) - s(s-1)(a-b)$, where $w(x) = (2x+2+2ab)\log x/(1-x)$, and w is increasing on $(0, 1)$ by the elementary bound $\log x \geq (x - x^{-1})/2$. The left panel of Figure 12 shows δ along the quarter circle.

Corollary H.7. *A closed vertex cycle with four pieces has raw cost at least $8\delta_{\mathrm{oct}}$. If its raw cost is $8\delta_{\mathrm{oct}}$, then every factor is $\sqrt{2}$, every switch is forward, and its initial margins are those of the octagon: the active margin*

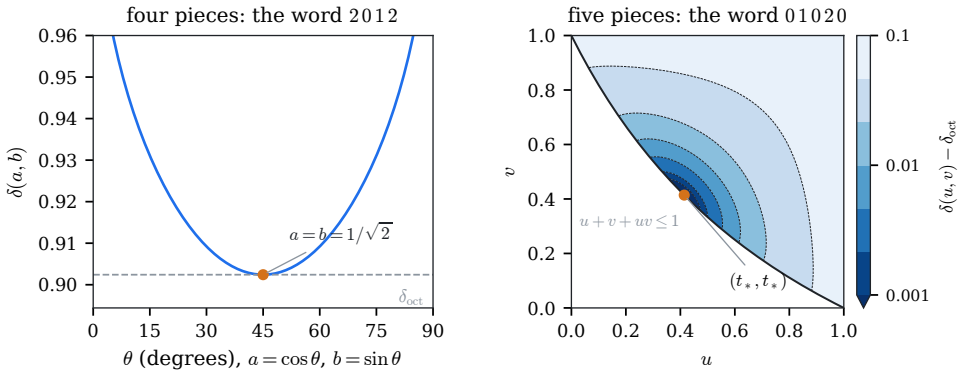


FIGURE 12. Left: the four-piece density $\delta(\cos\theta, \sin\theta)$ of (H.4); its minimum δ_{oct} is attained only at $\theta = 45°$, the octagon. Right: the excess $\delta(u,v) - \delta_{\mathrm{oct}}$ of the five-piece density (H.6) over its domain $u, v \in (0,1)$, $u + v + uv > 1$, on a logarithmic colour scale; it is positive everywhere and tends to 0 only at the boundary point (t_*, t_*), $t_* = \sqrt{2} - 1$, where the middle factor tends to 1. Both are computed from the cycle data; the script also checks the closure, the forced margins and the cost identity at random parameters.

$m_0(i_0)$ and the margin $m_0(i_0+2)$ equal $2/\sqrt{2\sqrt{2}-1}$, and $m_0(i_0+1) = 2(\sqrt{2}-1)/\sqrt{2\sqrt{2}-1}$.

Proof. Relabel to the word 2 0 1 2 and apply Lemma H.5 and Theorem H.6. At equality $a = b = 1/\sqrt{2}$, so every factor is $\sqrt{2}$, $s = \sqrt{2}$, $\mathcal{N} = 2\sqrt{2} - 1$, and (H.5) gives the stated margins. □

With active margin $a_k = 2/\sqrt{2\sqrt{2}-1}$ and factor $\sqrt{2}$, each piece lasts $(a_k/2)\log q_k = \log 2/(2\sqrt{2\sqrt{2}-1})$, the octagon's switching time t_{sw}.

H.4. **Five pieces.** By Proposition H.3, a five-piece cycle that is not entirely reverse can be rebased and relabeled to the word 0 1 0 2 0, whose switches have the signs $+--++$. Its closure has a two-parameter family of solutions. Put

$$P(t) = \frac{1+t^2}{2t}, \qquad Q(t) = \frac{1+t^2}{1-t^2} \qquad (0 < t < 1).$$

Lemma H.8 (Five-piece closure)**.** *If $p, q, r, s, z > 1$ and*

$$\widetilde{T}_0(p)\,\widetilde{T}_1(q)\,\widetilde{T}_0(r)\,\widetilde{T}_2(s)\,\widetilde{T}_0(z) = \widetilde{R},$$

then $p^2(q^2-1) = q^2$, $z^2(s^2-1) = s^2$ and $r^2 = (q^2-1)(s^2-1)$. Consequently there are $u, v \in (0,1)$ with $u + v + uv > 1$ and

$$(p, q, r, s, z) = \Big(P(u),\ Q(u),\ \frac{4uv}{(1-u^2)(1-v^2)},\ Q(v),\ P(v)\Big).$$

Proof. The three relations come from the entries of the product compared with $\widetilde{R}$. Every $q > 1$ is $Q(u)$ for $u = \sqrt{(q-1)/(q+1)} \in (0,1)$, and then $p^2 = q^2/(q^2-1)$ gives $p = P(u)$; likewise for s, z and v, and $r^2 = (Q(u)^2 - 1)(Q(v)^2 - 1)$ gives the middle factor. Finally $r > 1$ is equivalent to $u+v+uv > 1$ by the factorization $4uv - (1-u^2)(1-v^2) = (u+v+uv-1)(u+v+1-uv)$, whose second factor is positive. □

Put $d = (1-u)(1-v)$ and

$$F(t) = \Big(\frac{1+t}{1-t}\Big)^2 \log\frac{1+t^2}{2t} + \frac{1}{t^2}\log\frac{1+t^2}{1-t^2} + \log\frac{2t}{1-t^2},$$

and define the *five-piece density*

$$\delta(u,v) = \frac{2(d+2) - \frac{d}{2}\big(F(u)+F(v)\big)}{4-d}. \tag{H.6}$$

Lemma H.9 (Five-piece cycles)**.** *Let c be a closed vertex cycle with word $0\,1\,0\,2\,0$. There are $u, v \in (0,1)$ with $u+v+uv > 1$ such that the factors are those of Lemma H.8, the initial margins are*

$$(m_0(0), m_0(1), m_0(2)) = \frac{2}{\sqrt{d(4-d)}}\big((1+u)(1-v),\ d,\ (1-u)(1+v)\big),$$

and the raw cost is $8\,\delta(u,v)$.

Proof. As for four pieces, the twist gives two linear equations for the initial margins, which the quadric normalizes. With $A = (1+u)(1-v)$ and $B = (1-u)(1+v)$, the active margins of the five pieces are A, d/u, d, d/v, B, times $2/\sqrt{d(4-d)}$. The algebraic part of the raw cost (all but the terms $-a_k^2 \log q_k$) adds up to $16\,d(d+2)/(d(4-d))$, and the logarithmic part is

$$-\frac{4}{d(4-d)}\Big(A^2\log P(u) + \frac{d^2}{u^2}\log Q(u) + d^2\log r + \frac{d^2}{v^2}\log Q(v) + B^2\log P(v)\Big).$$

Since $A/d = (1+u)/(1-u)$, $B/d = (1+v)/(1-v)$ and $\log r = \log\frac{2u}{1-u^2} + \log\frac{2v}{1-v^2}$, the bracket is $d^2(F(u)+F(v))$, and the raw cost is $8\delta(u,v)$. □

Theorem H.10 (The five-piece density)**.** *For $u, v \in (0,1)$ with $u+v+uv > 1$, $\delta(u,v) > \delta_{\mathrm{oct}}$.*

The proof is in Appendix H.6.3; here is its structure. The function F enters only through $F(t) + 2\log(1-t)$, and this function of one variable is at most its value at $t_* = \sqrt{2}-1$, namely $(3+2\sqrt{2})\log 2 + 2\log(2-\sqrt{2})$. Summing over $t = u, v$ bounds $F(u)+F(v)$ above by a function of d alone, hence $\delta(u,v)$ below by a function of d. The constraint $u+v+uv > 1$ gives $d < d_* = (2-\sqrt{2})^2$, and at $d = d_*$ the lower bound equals δ_{oct} exactly. A tangent-line estimate for $d\log d$ at d_* shows that the bound exceeds δ_{oct} for $d < d_*$. The infimum of $\delta(u,v)$ is approached only as $u, v \to t_*$, where $d \to d_*$ and the middle factor r tends to 1: the five-piece period degenerates into the octagon's four pieces (right panel of Figure 12).

H.5. **The terminal theorem.**

Proof of Theorem 5.1. Let c be the cycle of the period (Lemma H.2). It is not entirely reverse, by Theorem 4.1, which holds for every number of pieces. By Proposition H.3, either it has four pieces, and then its raw cost is at least $8\delta_{\mathrm{oct}}$ by Corollary H.7, or it has five pieces and is a rebasing and relabeling of a cycle with word $0\,1\,0\,2\,0$, whose raw cost exceeds $8\delta_{\mathrm{oct}}$ by Lemma H.9 and Theorem H.10. By Lemma H.2, $\mathcal{C} \geq \frac{\sqrt{3}}{4} \cdot 8\delta_{\mathrm{oct}} = \sqrt{12}\,\delta_{\mathrm{oct}} = A_{\mathrm{oct}}$. □

No competitiveness and no compact domain are needed here. The equality case identifies the trajectory itself.

Proposition H.11 (Terminal equality)**.** *Let P be a normal vertex period with at most five pieces and cost A_{oct}. Then $T = t_{\mathrm{f}}$, and there is $r \in \mathbb{N}$ such that, for $t \in [0, T]$,*

$$R^{-r} g(t) R^{r} = g_{\mathrm{oct}}(t), \qquad R^{-r} X(t) R^{r} = X_{\mathrm{oct}}(t),$$

where $(g_{\mathrm{oct}}, X_{\mathrm{oct}})$ is the trajectory of the smoothed octagon (Proposition A.7).

Proof of Proposition H.11. A five-piece cycle that is not entirely reverse has raw cost strictly above $8\delta_{\mathrm{oct}}$, and the entirely reverse case is excluded by Theorem 4.1, so $N = 4$. By Corollary H.7 the period has four forward pieces, all factors $\sqrt{2}$, and the octagon's initial margins. A traceless matrix is determined by its margins, so after relabeling by a power of R the initial state is $\Phi(0, y_0)$ with $y_0^2 = (2\sqrt{2} - 1)/3$, and the control agrees almost everywhere with the octagon's control $\mathbf{e}_2\mathbf{e}_0\mathbf{e}_1\mathbf{e}_2$ on four pieces of length t_{sw}. Changing a control on a null set does not change a Carathéodory solution, and the octagon's trajectory is the unique solution with $g(0) = I$ and $X(0) = \Phi(0, y_0)$, by uniqueness of solutions of the state equations with a fixed vertex control on each piece. □

H.6. **The four- and five-piece estimates.** This subsection proves Theorems H.6 and H.10. The four-piece comparison uses monotonicity of a logarithmic weight; the five-piece comparison uses a concave integral kernel.

H.6.1. *A logarithm bound.*

Lemma H.12. *For $0 < x \leq 1$,* $\dfrac{x - x^{-1}}{2} \leq \log x$.

Proof. With $y = -\log x \geq 0$ the inequality reads $y \leq \sinh y$. The difference $\sinh y - y$ vanishes at 0 and has derivative $\cosh y - 1 \geq 0$. □

H.6.2. *The four-piece density.* Recall $\delta(a, b)$ from (H.4), on the open quarter circle $a, b > 0$, $a^2 + b^2 = 1$, with $s = a + b$.

Lemma H.13 (A monotone weight)**.** *For $c \geq 2$ the function $w_c(x) = (2x + c)\log x/(1 - x)$ is nondecreasing on $(0, 1)$. Consequently, for $0 < b \leq a < 1$,*

$$(2b + c)(1 - a)\log b \;\leq\; (2a + c)(1 - b)\log a.$$

Proof. The derivative is $w_c'(x) = \big((2+c)\log x + (2x+c)(1-x)/x\big)/(1-x)^2$. By Lemma H.12 and $2+c>0$, the numerator is at least

$$(2+c)\frac{x-x^{-1}}{2} + \frac{(2x+c)(1-x)}{x} = \frac{(c-2)(1-x)^2}{2x} \;\geq\; 0.$$

The second statement is $w_c(b) \leq w_c(a)$ multiplied by $(1-a)(1-b) > 0$. □

Proof of Theorem H.6. Since $\delta(a,b) = \delta(b,a)$, we may assume $b \leq a$, that is $b \in (0, 1/\sqrt{2}\,]$ and $a = \sqrt{1-b^2}$. Put $\psi(b) = \delta(\sqrt{1-b^2}, b)$. For $0 < b < 1/\sqrt{2}$, differentiating (H.4) with $a' = -b/a$ and simplifying on the circle gives, with $c = 2 + 2ab$,

$$\psi'(b) = \frac{2\Big((s+1)(w_c(b) - w_c(a)) - s(s-1)(a-b)\Big)}{a\,(s+1)\,(s+3)^2}. \tag{H.7}$$

Here $c \geq 2$ and $b < a$, so $w_c(b) \leq w_c(a)$ by Lemma H.13, while $s > 1$ and $a > b$ make the last term negative. Hence $\psi' < 0$ on $(0, 1/\sqrt{2})$, and ψ, which is continuous at $1/\sqrt{2}$, is strictly decreasing on $(0, 1/\sqrt{2}\,]$. For $b < 1/\sqrt{2}$ this gives $\delta(a,b) = \psi(b) > \psi(1/\sqrt{2}) = \delta_{\mathrm{oct}}$, the value computed after Theorem H.6. So $\delta(a,b) \geq \delta_{\mathrm{oct}}$ with equality only at $a = b$. □

H.6.3. *The five-piece density.* Recall F, $d = (1-u)(1-v)$ and $\delta(u,v)$ from (H.6), and put $t_* = \sqrt{2} - 1$, $d_* = (2-\sqrt{2})^2$ and $\mathcal{K} = (3+2\sqrt{2})\log 2$. Since $(3+2\sqrt{2})(2-\sqrt{2})^2 = 2$, we have $\mathcal{K} d_* = 2\log 2$.

We locate the maximum of F after a change of variables that splits it into two values of one concave function at two points of the unit circle.

Lemma H.14 (A concave function)**.** *Let $h(t) = -2t\log t/(1-t) - \log(1+t)$ for $0 < t < 1$. Then $x \mapsto h(\sqrt{x})$ is strictly concave on $(0,1)$. Consequently $h(\alpha) + h(\beta) \leq 2h(1/\sqrt{2})$ whenever $\alpha, \beta > 0$ and $\alpha^2 + \beta^2 = 1$, with equality only at $\alpha = \beta$.*

Proof. The logarithmic remainder has the integral representation

$$t - 1 - \log t = (1-t)^2 \int_0^1 \frac{s}{1-s+st}\,\mathrm{d}s.$$

Differentiate h and use this identity to obtain

$$\frac{h'(t)}{t} = 2\int_0^1 \frac{s}{1+t}\,\frac{1-s+s/t}{1-s+st}\,\mathrm{d}s.$$

For each fixed $s \in (0,1]$, the numerator $1 - s + s/t$ decreases with t, and both positive denominator factors increase. Thus the integrand, and hence $h'(t)/t$, strictly decreases on $(0,1)$. The derivative of $h(\sqrt{x})$ is $h'(\sqrt{x})/(2\sqrt{x})$, so this function is strictly concave. Apply its midpoint inequality at α^2 and β^2:

$$\tfrac{1}{2}h(\alpha) + \tfrac{1}{2}h(\beta) \leq h\big(\sqrt{(\alpha^2+\beta^2)/2}\big) = h(1/\sqrt{2}),$$

with equality only when $\alpha^2 = \beta^2$. □

Lemma H.15 (The envelope)**.** *For* $0 < t < 1$,

$$F(t) + 2\log(1-t) \;\le\; \mathcal{K} + 2\log(2-\sqrt{2}) = \mathcal{K} + \log d_*.$$

The bound is attained at $t = t_*$: there $(1+t^2)/(2t) = (1+t^2)/(1-t^2) = \sqrt{2}$, $2t/(1-t^2) = 1$ and $(1+t)/(1-t) = 1/t = \sqrt{2}+1$, so $F(t_*) = (3+2\sqrt{2})\log 2 = \mathcal{K}$, and $1 - t_* = 2 - \sqrt{2}$.

Proof. With $\alpha = 2t/(1+t^2)$ and $\beta = (1-t^2)/(1+t^2)$, so that $\alpha^2 + \beta^2 = 1$, expanding the logarithms of the products $1-\alpha = (1-t)^2/(1+t^2)$, $1+\alpha = (1+t)^2/(1+t^2)$, $1-\beta = 2t^2/(1+t^2)$ and $1+\beta = 2/(1+t^2)$ gives the identity

$$F(t) + 2\log(1-t) = \log 2 + h(\alpha) + h(\beta).$$

By Lemma H.14 this is at most $\log 2 + 2h(1/\sqrt{2})$. Since $\log(1/\sqrt{2}) = -\frac{1}{2}\log 2$ and $1 + 1/\sqrt{2} = (2+\sqrt{2})/2 = 1/(2-\sqrt{2})$,

$$h(1/\sqrt{2}) = \frac{\sqrt{2}\cdot\frac{1}{2}\log 2}{1 - 1/\sqrt{2}} - \log\frac{1}{2-\sqrt{2}} = (1+\sqrt{2})\log 2 + \log(2-\sqrt{2}),$$

so $\log 2 + 2h(1/\sqrt{2}) = (3+2\sqrt{2})\log 2 + 2\log(2-\sqrt{2})$. □

Lemma H.16. *If* $u, v \in (0,1)$ *and* $u+v+uv > 1$*, then* $d = (1-u)(1-v) < d_*$.

Proof. Put $x = 1-u$, $y = 1-v$ and $\omega = \sqrt{xy} \in (0,1)$. The hypothesis is $(2-x)(2-y) > 2$. Since $x + y \ge 2\omega$ by the arithmetic–geometric mean inequality,

$$(2-\omega)^2 \ge (2-x)(2-y) > 2.$$

Thus $2 - \omega > \sqrt{2}$, giving $d = \omega^2 < (2-\sqrt{2})^2 = d_*$. □

Proof of Theorem H.10. Adding Lemma H.15 at $t = u$ and $t = v$, and using $\log d = \log(1-u) + \log(1-v)$,

$$F(u) + F(v) \le 2\mathcal{K} + 2\log d_* - 2\log d.$$

Multiplying by $d/2 > 0$,

$$(4-d)\,\delta(u,v) = 2d + 4 - \tfrac{d}{2}\big(F(u)+F(v)\big) \;\ge\; 2d + 4 - \mathcal{K}d - d\log d_* + d\log d.$$

The tangent line of $d \mapsto d\log d$ at d_* gives $d\log d \ge d\log d_* + d - d_*$ (this is $\log(d/d_*) \ge 1 - d_*/d$), so

$$(4-d)\,\delta(u,v) \;\ge\; 4 - d_* + (3-\mathcal{K})\,d.$$

At $d = d_*$ the right side equals $\delta_{\mathrm{oct}}(4-d_*)$: indeed $2d_* + 4 - \mathcal{K}d_* = 16 - 8\sqrt{2} - 2\log 2$ and $4 - d_* = 2(2\sqrt{2}-1)$. Subtracting $\delta_{\mathrm{oct}}(4-d) = \delta_{\mathrm{oct}}(4-d_*) + \delta_{\mathrm{oct}}(d_*-d)$,

$$(4-d)\big(\delta(u,v) - \delta_{\mathrm{oct}}\big) \;\ge\; (\mathcal{K} - 3 - \delta_{\mathrm{oct}})(d_* - d).$$

The factor $d_* - d$ is positive by Lemma H.16. For the other factor put $r = \sqrt{2}$. The strict midpoint inequality for the convex function $x \mapsto 1/x$ on $[1, r]$ gives

$$\log 2 = 2\int_1^r \frac{\mathrm{d}x}{x} > \frac{4(r-1)}{r+1}.$$

Using $r^2 = 2$ and the exact value of δ_{oct}, we obtain

$$(2r-1)(\mathcal{K}-3-\delta_{\mathrm{oct}}) = 2(1+r)^2\log 2 - 5 - 2r > 8 - 5 - 2r = (r-1)^2 > 0.$$

Since $4 - d > 0$, this proves $\delta(u, v) > \delta_{\mathrm{oct}}$. □

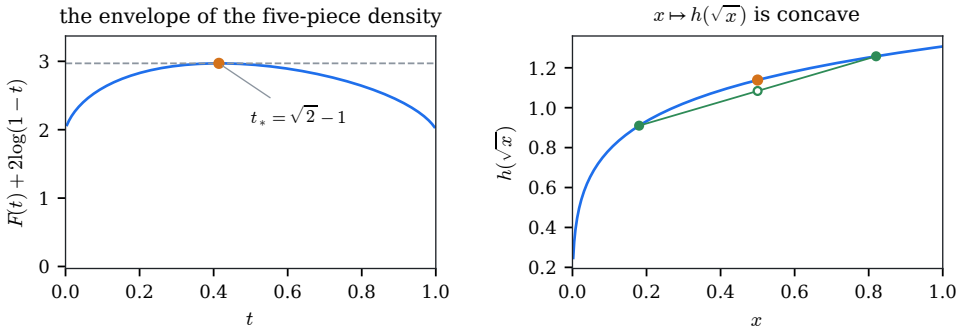


FIGURE 13. Left: $F(t) + 2\log(1-t)$ on $(0,1)$; its maximum $(3+2\sqrt{2})\log 2 + 2\log(2-\sqrt{2})$ is attained only at $t_* = \sqrt{2}-1$ (Lemma H.15). Right: the concave function $x \mapsto h(\sqrt{x})$ of Lemma H.14; a chord (green) lies below the graph, and the midpoint of the chord between α^2 and $\beta^2 = 1-\alpha^2$ lies below the value at $x = 1/2$ (orange). Computed; the script checks the identity $F(t) + 2\log(1-t) = \log 2 + h(\alpha) + h(\beta)$ and the concavity on a grid.

Appendix I. The equality case

Proposition H.11 identifies a period of cost A_{oct} with the octagon's trajectory after placing the disk in Reinhardt's normal position, rebasing at a switch and relabeling the control vertices by a power of R. To classify disks of density δ_{oct}, we apply this equality statement to the realization of the given disk and reverse these changes. This recovers the disk itself (Appendix I.1). We then prove the converse (Appendix I.2) and identify Hales's octagon K_{H} with Reinhardt's K_{oct} (Appendix I.3), completing the proofs of Theorem 1.1(a) and (c) in Section 5.4.

I.1. **Rigidity.** Let $s_0, \dots, s_5$ be the sixth roots of unity, so that $Rs_j = s_{j+1}$. The realization of a worst disk (Theorem A.10) keeps the link between its frame and its boundary.

Lemma I.1 (Realization with its contact link)**.** *Let K be a worst disk. There are an invertible linear map e of the plane, a balanced pair σ of eK in circle representation and a global minimizer (T, u, g, X) of the control problem such that eK is a worst disk, $\delta(K) = \mathcal{C}(T, X)/\sqrt{12}$, and for every j and every $t \in \mathbb{R}$*

$$\sigma_j(t) = g(t)\, s_j, \qquad \sigma_j(t+T) = \sigma_{j+1}(t).$$

The first identity is the definition of the frame of the realization, which is defined on all of $\mathbb{R}$; the second is the cyclic shift of the contacts by one sixth of the period. Together they force the frame to be twisted on all of $\mathbb{R}$, beyond the period $[0, T]$ on which the control problem is posed.

Lemma I.2 (The frame twist)**.** *If $\sigma_j(t) = g(t)s_j$ and $\sigma_j(t+T) = \sigma_{j+1}(t)$ for all j and t, then $g(t+T) = g(t)R$ for every $t \in \mathbb{R}$.*

Proof. $g(t+T)s_j = \sigma_j(t+T) = \sigma_{j+1}(t) = g(t)s_{j+1} = g(t)Rs_j$, and s_0, s_1 span the plane. □

The last ingredient recovers a disk from its boundary.

Lemma I.3. *Two compact convex sets with nonempty interior, both symmetric about the origin, that have the same frontier are equal.*

Proof. Let A, B be the two sets. The interior of B is connected, contains the origin, and does not meet the common frontier; it is covered by the disjoint open sets $\operatorname{int} A$ and $\mathbb{R}^2 \setminus \overline{A}$, so it lies in $\operatorname{int} A$, which contains the origin. By symmetry the interiors are equal, and a convex body is the closure of its interior. □

Proof of Theorem 5.3. Let K be a worst disk, and take e, σ and (T, u, g, X) as in Lemma I.1. By Lemma I.2, $g(t+T) = g(t)R$ on $\mathbb{R}$.

The cost is A_{oct}. Since K is a worst disk and $\delta(K_{\mathrm{H}}) = \delta_{\mathrm{oct}}$ (Proposition A.7), $\delta(K) \le \delta_{\mathrm{oct}}$, and part (b) of Theorem 1.1 gives $\delta(K) \ge \delta_{\mathrm{oct}}$. Hence $\mathcal{C}(T, X) = \sqrt{12}\,\delta(K) = A_{\mathrm{oct}}$.

The terminal equality. By Theorem A.30, the minimizer rebased at some $s \in [0, T)$ is a normal vertex period P with the same terminal time and cost, whose state stays in $\mathfrak{h}^{\star\star}$; its frame is $t \mapsto g(s)^{-1} g(s+t)$, because g is already twisted. By Theorem 3.3, P has at most five pieces, and by Proposition H.11, $T = t_{\mathrm{f}}$ and there is r with $R^{-r} g(s)^{-1} g(s+t) R^{r} = g_{\mathrm{oct}}(t)$ for $t \in [0, T]$. With $M = g(s) R^{r}$ this reads

$$g(s+t) = M\, g_{\mathrm{oct}}(t)\, R^{-r} \qquad (0 \le t < T).$$

Both sides satisfy $F(t+T) = F(t)R$: the left side by the twist of g, the right side because $g_{\mathrm{oct}}(t + t_{\mathrm{f}}) = g_{\mathrm{oct}}(t) R$ and R commutes with R^{-r}. Two curves with the same twist that agree on $[0, T)$ agree everywhere, so the identity holds for all $t \in \mathbb{R}$.

The arcs. Applying it to s_0 and using $R^{-r} s_0 = s_{-r}$,

$$\sigma_0(s+t) = M\, g_{\mathrm{oct}}(t)\, s_{-r} = M\, \sigma^{\mathrm{oct}}_{-r}(t) \qquad (t \in \mathbb{R}),$$

where $\sigma^{\mathrm{oct}}_j(t) = g_{\mathrm{oct}}(t) s_j$ are the contact curves of the octagon. In a balanced pair each contact curve traces the whole frontier once per period, so the frontier of eK is the range of σ_0, and the frontier of K_{H} is the range of $\sigma^{\mathrm{oct}}_{-r}$. Hence $\partial(eK) = M(\partial K_{\mathrm{H}}) = \partial(M K_{\mathrm{H}})$. Both eK and $M K_{\mathrm{H}}$ are convex disks symmetric about the origin, so $eK = M K_{\mathrm{H}}$ by Lemma I.3, and $K = e^{-1} M(K_{\mathrm{H}}) = e^{-1} g(s) R^{r}(K_{\mathrm{H}})$. □

The argument follows the realization of K throughout. Repeating the reduction of Appendices A to H for this trajectory identifies the disk's boundary; the lower bound determines its cost as A_{oct}.

I.2. **The converse.** The converse is the affine invariance of the greatest packing density.

Proof of Proposition 5.4. A translate, and more generally a congruent copy, of K has the same packings by congruent copies, so its greatest packing density is the same. By Fejes Tóth's theorem (Theorem 2.2), $\delta(K)$ is the greatest density $\mathrm{area}(K)/\det\Lambda$ of a lattice packing $K + \Lambda$. If $\det A > 0$, the lattice packings of K and of AK correspond through $\Lambda \mapsto A\Lambda$, and area and covolume are both multiplied by $\det A$, so the lattice densities agree. If $\det A < 0$, write $A = (AF)F$ with the reflection $F(x_0, x_1) = (x_0, -x_1)$: then FK is a congruent copy of K, and AF preserves orientation. The last statement follows from $\delta(K_{\mathrm{H}}) = \delta_{\mathrm{oct}}$ (Proposition A.7). □

I.3. **Reinhardt's construction.** Theorem 1.1 is stated for Reinhardt's octagon K_{oct} (Definition A.1), while the proof works with K_{H}, the region swept from the origin by the contact curve σ^{oct}_0 of the octagon's trajectory over one period $[0, 6t_{\mathrm{f}}]$. By Proposition A.7, K_{H} is also the convex hull of the six arcs $\sigma^{\mathrm{oct}}_j([0, t_{\mathrm{f}}])$. The two octagons differ by a vertical stretch of about 5.5% (Figure 2).

Proposition I.4 (Reinhardt's octagon is Hales's octagon)**.** *Let* $A = \mathrm{diag}(1, (2\sqrt{2}-1)/\sqrt{3})$. *Then* $A(K_{\mathrm{H}}) = K_{\mathrm{oct}}$.

Proof of Proposition I.4. Write $y_0^2 = (2\sqrt{2}-1)/3$ and $W = \exp(\sqrt{3}\,y_0 s)$ for $s \in [0, t_{\mathrm{sw}}]$, so that W runs over $[1, \sqrt{2}]$; recall that $\sigma_0^{\mathrm{oct}}(0) = s_0 = (1,0)$.

One eighth of the boundary. On the first switching interval the octagon's frame $g_0(s)$ is the flow of the control $\mathbf{e}_2$ from $\Phi(0, y_0)$; conjugated by A it acts by

$$Ag_0(s)A^{-1} = \begin{pmatrix} 1 & -\frac{W-1}{2\sqrt{2}-1} \\ 1-W^{-1} & 1-\frac{W-2+W^{-1}}{2\sqrt{2}-1} \end{pmatrix},$$

and $A\,g_0(t_{\mathrm{sw}})R^{-1}A^{-1}$ is the rotation by $-\pi/4$; both are direct computations from the closed form of the flow of $\mathbf{e}_2$. Since $g_{\mathrm{oct}}(t + t_{\mathrm{sw}}) = g_0(t_{\mathrm{sw}})R^{-1}\,g_{\mathrm{oct}}(t)\,R$, the curve $A\sigma_0^{\mathrm{oct}}$ is, on the first three switching intervals,

$$\begin{aligned} &(1,\ 1-W^{-1}) && \text{along side 0,}\\ &\left(\sqrt{2}+1-\tfrac{\sqrt{2}}{2}(W+W^{-1}),\ 1-\tfrac{\sqrt{2}}{2W}\right) && \text{around corner 0,}\\ &\left(\sqrt{2}-\tfrac{W}{2},\ \tfrac{W}{2}\right) && \text{along side 1.} \end{aligned}$$

The first piece runs from the midpoint $(1,0)$ of side 0 to the point $(1, 1-\sqrt{2}/2)$, and the third from $(\sqrt{2}-\frac{1}{2}, \frac{1}{2})$ to the midpoint $(\sqrt{2}/2, \sqrt{2}/2)$ of side 1. With d_k the distance to the kth side line, the middle piece has $d_7 = W/2$ and $d_2 = \sqrt{2}/(2W)$, so it lies on the hyperbola $d_7 d_2 = \sqrt{2}/4$; it starts at $(1, 1-\sqrt{2}/2)$ and ends at $(\sqrt{2}-\frac{1}{2}, \frac{1}{2})$, the points where the first and the third pieces end and start. The control on these three intervals is $\mathbf{e}_2$, $\mathbf{e}_0$, $\mathbf{e}_1$, and only the second bends the curve.

The rest. Three switching intervals later the frame relation gives $\sigma_0^{\mathrm{oct}}(t + 3t_{\mathrm{sw}}) = -h^3\sigma_0^{\mathrm{oct}}(t)$ with $h = g_0(t_{\mathrm{sw}})R^{-1}$, and $-Ah^3A^{-1}$ is the rotation by $\pi/4$. Over the period $6t_{\mathrm{f}} = 24t_{\mathrm{sw}}$ the curve $A\sigma_0^{\mathrm{oct}}$ therefore traces eight rotated copies of the three pieces, once around.

The two inclusions. Every point q of the three pieces lies in K_{oct}, and μq does not for $\mu > 1$: on the sides this is the inequality $d_k \geq 0$, and on the corner it is the hyperbola condition, which strictly fails past the arc. Since K_{oct} is invariant under the rotation by $\pi/4$, the same holds for every point $A\sigma_0^{\mathrm{oct}}(t)$. The set K_{oct} is star-shaped about 0, so $AK_{\mathrm{H}} \subseteq K_{\mathrm{oct}}$. Conversely, a point $p \neq 0$ of K_{oct} is a positive multiple λq of some $q = A\sigma_0^{\mathrm{oct}}(t)$, by the intermediate value theorem and $\sigma_0^{\mathrm{oct}}(t + 3t_{\mathrm{f}}) = -\sigma_0^{\mathrm{oct}}(t)$, and $\lambda \leq 1$ because $\mu q \notin K_{\mathrm{oct}}$ for $\mu > 1$. Hence $p \in AK_{\mathrm{H}}$. □

Each side of K_{oct} keeps a straight middle part of length $2-\sqrt{2}$, between the points of tangency $(1, \pm(1-\sqrt{2}/2))$ on side 0. The proposition transfers the facts known for K_{H} to K_{oct}; in particular, the affine images of the two octagons are the same sets.

Appendix J. The formal verification

The formalization of the main theorems uses Lean 4 [6] and Mathlib [31] and is available in the repository [35]. We describe its statements, verification and dependencies, then give the correspondence between the paper and the Lean declarations.

J.1. **The statements.** The file `Palomar/Challenge.lean` imports only Mathlib and states the three parts of Theorem 1.1 as the theorems `octagon_density`, `reinhardt` and `reinhardt_equality` of the namespace `ReinhardtChallenge`. Its packing definitions agree with Definition 2.1, which is Definition 2.1.1 of [13]. Points of the plane are maps $\{0,1\} \to \mathbb{R}$ and areas are Lebesgue measure. A convex disk is a compact convex set with nonempty interior, and a congruent copy of K is its image under a map $x \mapsto Ax + b$ with A orthogonal. A packing is a family of convex disks with pairwise disjoint interiors. Its density exists when the upper and lower limits, as $r \to \infty$, of the covered fraction of the disk of radius r about the origin are finite and equal. The greatest packing density $\delta(K)$ is the supremum over packings by congruent copies of K for which this density exists. A disk is centrally symmetric when there is a point c with $2c - x \in K$ for every $x \in K$. The smoothed octagon is given in Reinhardt's description (Definition A.1): with $d_k(x)$ the signed distance from x to the line of the k-th side of the regular octagon of inradius 1, it is the octagon $d_k \geq 0$ with the corner between sides k and $k+1$ cut off along the hyperbola $d_{k-1}d_{k+2} = \sqrt{2}/4$. The definitions are copied verbatim from the library, so the library's theorems have precisely the statements given in the challenge file.

J.2. **Verification and dependencies.** The three theorems are proved without additional hypotheses in the root module `SmoothedOctagon.lean`. Lean's kernel checks the proofs. Every theorem depends only on the standard axioms `propext`, `Classical.choice` and `Quot.sound`, as verified by an audit of every declaration in the library. The Comparator configuration `Palomar/comparator.json` checks that the root module proves the statements of `Palomar/Challenge.lean` with the same definitions. The proof is registered in the Palomar registry [26] as `PALOMAR-2026-11-09-000008`, at trust level high: Palomar's verifier checked it with the independent kernels nanoda and con-ron at commit `9ba6bf8` of [35]. Lean and Mathlib are pinned at version `v4.36.0-rc2`. The library has about 125,000 lines in 407 modules, and a full build takes about 3.5 CPU-hours.

The library depends on two others. The first [36] formalizes the parts of the book of Hales and Vajjha [13] that the proof uses, together with the classical results on which they depend, including Fejes Tóth's theorem (Theorem 2.2), the theorems of Minkowski and Mahler on critical lattices and circumscribed hexagons, the compact domain (Theorem A.12) and the lift of a global minimizer to a Pontryagin extremal (Theorem A.14). The second [34] proves the Pontryagin maximum principle [27] with free terminal

time, an integral cost and mixed boundary conditions, by the argument of Agrachev and Sachkov [1], and Filippov's theorem [11]; Brouwer's fixed-point theorem comes from [33].

J.3. **Dictionary.** Table 5 lists the Lean declarations corresponding to the numbered results and constructions of the paper. Names are relative to the namespace `SmoothedOctagon`; the prefixes `ReinhardtChallenge` and `SmoothedPolygons` mark the statements of the challenge file and the declarations of [36]. Multiple declarations in one row jointly establish the corresponding result. The rows are grouped by the appendix that proves the result; Theorems 1.1 and 1.2 and the definitions of Section 2 come first.

Table 5: The Lean declarations behind the numbered results.

Result	Lean declarations
Sections 1 and 2	
Theorem 1.2	`reinhardt_cost`, `controlLowerBound`
Theorem 1.1(a)	`ReinhardtChallenge.octagon_density`, `smoothedOctagon_spec`
Theorem 1.1(b)	`ReinhardtChallenge.reinhardt`, `reinhardtConjecture`
Theorem 1.1(c)	`ReinhardtChallenge.reinhardt_equality`, `reinhardt_equality_trajectory`
Definition 2.1	`ReinhardtChallenge.greatestPackingDensity`
Definition A.1	`ReinhardtChallenge.smoothedOctagon`
Definition A.5	`SmoothedPolygons.IsReinhardtMinimizer`
Theorem A.12	`SmoothedPolygons.theorem5_0_2`
Appendix A	
Definition A.13	`SmoothedPolygons.IsPontryaginExtremal`
Theorem A.14	`SmoothedPolygons.exists_isPontryaginExtremal_of_isReinhardtMinimizer`
Lemma A.15	`IsPontryaginExtremal.continuousOn_vertexHamiltonian`, `IsPontryaginExtremal.vertexHamiltonian_max_eq_zero`
Lemma A.16	`hamiltonian_vertices_equal_iff_reducedCostate_eq_zero`, `hamiltonian_maximizers_in_proper_edge`
Lemma A.17	`Reduction.clr_winding`, `Reduction.clr_angular`
Lemma A.18	`Reduction.clr_vertex_bounds`, `Reduction.clr_equilibrium_compact`
Proposition A.19	`Reduction.clr_budget`, `Reduction.clr_abs_lam_le`, `Reduction.clr_lam_sq_le`
Proposition A.20	`Reduction.covectorNormSq_le_singularGapSq`, `Reduction.switch_singularGapSq_pos`
Remark A.21	`Reduction.clr_frobSq_frame_le`, `Reduction.covectorNormSq_projective_bounds`
Proposition A.22	`Reduction.edgeBound_of_isReinhardtMinimizer`, `Reduction.Edge.edge_bangBang`
Lemma A.23	`Reduction.switch_exists_excluded_vertex`

Result	Lean declarations
Proposition A.24	`Reduction.exists_isPiecewiseVertex`
Definition A.25	`VertexRepresentation`
Lemma A.27	`Reduction.exists_rebase_vertexRepresentation`, `Reduction.isReinhardtMinimizer_rebase`
Definition A.26	`NormalPeriod`, `NormalPeriod.IsCompetitive`, `NormalPeriod.InCompactDomain`, `NormalPeriod.activeMargin`
Theorem A.28	`Reduction.lam_lt_zero_of_vertexRepresentation`, `Reduction.exists_normal_lift`
Lemma A.29	`Reduction.gn_piece`
Theorem A.30	`finiteNormalRepresentation`, `Reduction.cost_le_octagonCost`
Appendices B to D	
Lemma B.1	`vertexPiece_eq`, `vertexPiece_margin`, `vertexPiece_cost`, `NormalPeriod.piece_margin`, `NormalPeriod.cost_eq_sum`
Lemma B.2	`CostateCycle.factor_pp`, `CostateCycle.factor_mp`, `CostateCycle.factor_pm`, `CostateCycle.centre_factor`
Lemma B.3	`CostateCycle.rawCost_eq`, `CostateCycle.arc_eq`, `NormalPeriod.cost_eq_rawCost`
Lemma B.4	`NormalPeriod.tangentProduct_N`, `Term.triC_conj_tangentFactor`, `Term.triC_conj_R`
Definition B.5	`VertexRepresentation.IsForward`, `VertexRepresentation.IsReverse`, `VertexRepresentation.IsReverseRun`, `VertexRepresentation.IsForwardRun`
Lemma B.6	`NormalPeriod.ff_N_mod_three`
Definition 3.1	`VertexRepresentation.AllReverse`, `VertexRepresentation.AllForward`, `VertexRepresentation.ClassM3`, `VertexRepresentation.ClassM1`, `VertexRepresentation.ClassM2`
Proposition 3.2	`VertexRepresentation.partition`
Definition B.7	`costateL`, `costateA`, `costateB`, `costateAlpha`, `costateBeta`, `CostateCycle.sig`
Lemma B.8	`margin_mul_hamiltonian`, `costateB_perp`, `vertexPiece_costate`, `NormalPeriod.cut_transport`, `NormalPeriod.sig_le_of_mem`, `NormalPeriod.cut_sig_grow_le`, `NormalPeriod.cut_sig_decr_le`, `CostateCycle.sig_act_succ`, `NormalPeriod.costateCycle`
Definition B.9	`NormalPeriod.Ucal`, `CostateCycle.mu`, `NormalPeriod.mu_zero`, `CostateCycle.gam2`, `NormalPeriod.det_Ucal`, `triY`, `CostateCycle.Mloc`

Result	Lean declarations
Lemma B.10	`CostateCycle.yv_norm`, `CostateCycle.cross_pp`, `CostateCycle.cross_mm`, `CostateCycle.yv_pair`, `CostateCycle.outv_turn`, `CostateCycle.yv_add_N`, `CostateCycle.pre_N`
Lemma B.11	`CostateCycle.stationarity`, `CostateCycle.centre_stat`
Lemma B.12	`CostateCycle.swd_fwd`, `CostateCycle.swd_rev`, `CostateCycle.turn_secant`, `CostateCycle.turn_swd_ge`
Lemma B.13	`CostateCycle.gram_pp`, `CostateCycle.gram_mm`
Proposition B.16	`NormalPeriod.gam2_pos_of_mixed`, `NormalPeriod.wire_gam2_pos_of_allForward`
Proposition B.14	`NormalPeriod.cost_casimir`, `el_winding`
Corollary B.15	`NormalPeriod.gam2_lt_cap`
Theorem 3.3	`period_le_five`
Lemma C.2	`cstRevN_neg`
Lemma C.3	`cst_rev4_third_identity`, `cst_rev4_second_identity`
Lemma C.1	`CostateCycle.reverse_four_third`, `CostateCycle.reverse_four_second`
Theorem 4.1	`reverse_exclusion`
Lemma D.1	`CostateCycle.three_arc`, `CostateCycle.three_arc_not_min`, `CostateCycle.turn_up`, `CostateCycle.turn_down`, `cst_three_grow`, `cst_three_decr`
Proposition D.2	`reverseRun_two_and_centre_min`
Theorem 4.2	`minimumStructure_exclusion`
Corollary D.3	`NormalPeriod.rev_run`, `NormalPeriod.centre_min`, `NormalPeriod.swd_ge_of_mixed`
Lemma D.4	`el_three_gram`
Lemma D.5	`CostateCycle.cre_pos_pp`, `CostateCycle.cre_neg_mm`, `el_cross_bound`
Lemma C.4	`cst_log_pade`, `cst_zeta_gt`, `cst_eta_lt_zeta`
Lemma C.5	`octagonCost_ratio_lt`, `octagon_moment_ratio_lt_golden`, `octagonCost_lt_score`
Appendix E	
Theorem 4.3	`shortRuns_exclusion`
Lemma E.1	`VertexRepresentation.m1_pattern`
Lemma E.2	`NormalPeriod.m1_centreData`, `m1_CentreData`
Proposition E.3	`m1_angle_gap`
Corollary E.4	`m1_arccos_gt`, `m1_cos_two_pi_div_five`
Proposition E.5	`m1_winding`, `wind_winding_aux`, `wind_polar`, `wind_centre`, `wind_track_range`, `wind_seam`
Lemma E.6	`CostateCycle.m1_next_centre`, `CostateCycle.next_centre_local`, `CostateCycle.cross_mm`, `CostateCycle.pre_N`

Result	Lean declarations
Lemma E.7	`CostateCycle.euler_sum`, `CostateCycle.rawCost_eq`, `elGa_euler`
Proposition E.8	`m1_two_centre`, `m1_two_centre_bounds`, `m1_octagonCost_lt_two_centre`
Lemma E.9	`ag_half_sub_inv_le_log`, `ag_sigma_superadd`
Lemma E.10	`ag_phi_le`
Lemma E.11	steps of the proof of `m1_angle_gap`
Lemma E.12	`agSymmetric_gram_min`, `agSymRoot_existsUnique`, `agSymRoot_zero_spec`
Lemma E.13	`ag_core`
Lemma E.14	`m1_bc_bound`
Appendix F	
Theorem 4.4	`forward_exclusion`
Lemma F.1	`NormalPeriod.ff_N_mod_three`
Lemma F.2	`CostateCycle.ff_act_mul_lt`, `CostateCycle.ff_q_eq`, `CostateCycle.ff_rawCost`
Lemma F.3	`ff_seven_le`
Proposition F.4	`CostateCycle.ff_rows`, `CostateCycle.ff_floors`, `CostateCycle.ff_gram`, `NormalPeriod.ff_gam2_eq`, `ff_Win.toDim`
Proposition F.5	`NormalPeriod.ff_cost_casimir`, `NormalPeriod.wire_gam2_pos_of_allForward`, `NormalPeriod.wire_spectral`
(F.5)	`NormalPeriod.df_angle`
Corollary F.6	`NormalPeriod.ff_competitive_cap_exact`, `octagon_spectral_cap`
Lemma F.7	`ff_Dim.U_lt_three`, `fs_U_lt_three`, `fs_H5P_pos`
Lemma F.8	`ff_Dim.eleven_fifths_lt`, `ff_octagonCost_lt_sqrt`
Lemma F.13	`ff_budgetCost_strictConcaveOn`
Lemma F.9	`ff_Dim.budget`, `ff_Dim.h_le_c`, `ff_budget_g`, `ff_budget_linear`
Lemma F.10	`ff_Dim.neighbor`, `ff_resN_neg`
Lemma F.16	`fs_neighborValue_neg`, `fs_NS_Eval`
Lemma F.17	`fs_neighborSlope_neg`, `fs_NS_Dval`, `fs_resN_sym_of_floor`
Lemma F.18	`asym_continuous_row_neg`, `asym_original_row_neg`, `fs_resN_asym`
Lemma F.14	`fs_normValue_neg`, `fs_normTangent_neg`, `fs_resV_neg`, `fs_factor_curvature_nonneg`
Lemma F.15	`fwdSumPairedLog_le`, `fwdSumResidual32_original_neg`, `fs_resS_neg`
Lemma F.11	`ff_Dim.norms`, `ff_Dim.norms_nonneg`, `ff_resS_neg`, `ff_resV_neg`
Lemma F.12	`fcNormQ_polygon`, `fc_branch_curve_gt`, `fc_sum_curve_gt`, `ff_casimir_lower`

Result	Lean declarations
Appendix G	
Theorem 4.5	`nonshort_exclusion`
Lemma G.1	`NormalPeriod.wire_spectral`, `NormalPeriod.wire_cas_lt`, `NormalPeriod.dens_cas_gt`
Lemma G.2	`m2_centre`, `m2_scale`
Proposition G.4	`m2_small_forward_minimum`
Proposition G.3	`m2_min_centre`
Lemma G.5	`m2_reflect`, `m2_reflect_fwd_iff`, `m2_reflect_classM2`
Proposition G.6	`m2_window`
Proposition G.7	`m2_window`, `M2Window.ForwardNext`, `M2Window.ReverseNext`
Lemma G.8	`m2_comparison`
Lemma G.9	`nsrN_mono_k`, `nsrA_pos`
Lemma G.10	`nsr_window_basic`, `nsr_window_N_neg`, `nsr_high_t`, `nsr_scale_lt`, `nsr_scale_k`, `nsr_scale_of_t`, `nsr_centre_gap`, `nsr_centre_gap_k`, `nsr_window_dur`, `nsr_window_dur_x`, `nsr_reverse_order`, `nsr_reverse_order_k`
Proposition G.11	`nsr_reverse_core`
Theorem G.12	`m2_reverse_exclusion`
Proposition G.13	`nsf_low_ratio`, `nsf_small_gap_j`, `nsf_small_gap`
Theorem G.14	`m2_forward_exclusion`
Lemma G.16	`nsr_sec_swap`, `nsr_sec_start`, `nsr_m2F_eq_sec`, `nsr_sec_lt_chord`, `nsr_turn_floor`
Lemma G.17	`nsr_log_one_add_gt`, `nsr_secN_lt_vbar`, `nsr_log_lt_Lp`, `nsr_m2F_gt_Fl`, `mix_log_weighted_lt`, `nsr_log_one_add_jensen`, `nsr_secN_lt_jensen`, `nsf_secN_lt_VG`, `nsf_secN_lt_Vq`, `nsr_t_lt_T`
Lemma G.18	`nsrNv_nonpos`, `nsrNL_nonpos`, `nsr_N_pos_of_relax`
Lemma G.19	`nsr_centre_gap`, `nsr_centre_gap_k`, `nsr_gap_chord_pos`, `nsr_gap_normalize`
Lemma G.20	`mhtDelta_physical_pos`, `nsr_high_t_analytic`, `nsr_high_t'`
Lemma G.21	`nsr_unit_shape`, `nsr_cubic_pos`, `nsrU_signs`, `nsrUC0_pos`
Lemma G.22	`nsr_order_face`, `nsr_q_le_order`, `nsr_concave_min`, `nsr_order_one_pos`
Lemma G.23	`nsr_strip`
Lemma G.24	`nsr_low_unit`, `nsr_quartic_pos`
Lemma G.25	`nsf_lowRatio_core`, `nlr_unit_pos`
Lemma G.26	`nsf_LGN_core`, `nsf_LGB_core`, `nlt_unit_pos`, `nlt_shape_ge`, `hrl_lower_pos`, `hrs_shape_lt`
Lemma G.27	`nsf_chi_core`
Lemma G.28	`nsf_cas_slope`
Lemma G.29	`nsf_qelim`

Result	Lean declarations
Lemma G.30	`nsf_envelope`, `nsf_wU_floor`, `nsf_norm_derivative_neg`
Appendix G.3	
Definition G.31	`M2Short`
Proposition G.32	`m2_short_window`
(G.14)	`NormalPeriod.df_window`, `NormalPeriod.df_defect`
(G.16), (G.17)	`df_alg_m`, `df_alg_traces`, `SmallerShort.Point.AA`, `SmallerShort.Point.BB`
Theorem G.33	`m2_smallerShort_exclusion`
Lemma G.34	`M2Short.ss_C_le_B`, `SmallerShortData.u_le_V`, `ss_V_lt_limit`, `M2Short.ss_u_le`, `SmallerShortData.be_le_al_x`, `SmallerShortData.root_zero`, `SmallerShortData.q_le_qc`, `SmallerShortData.q_lt_seventh`
Appendix G.4.1	`ss_V_lt_VG`, `ss_V_le_Vbar`, `mix_log_weighted_lt`
Lemma G.35	`ss_strong_order`, `ss_upper_order_54`, `ss_upper_order_98`, `ss_small_upper_order`, `ss_quad_order`, `SmallerShortData.strong_box`
Proposition G.36	`M2Short.ratio_lt_twelve_fifths`, `SmallerShortData.ratio_lt_twelve_fifths`, `lgr_D_pos`, `lgr_pairing`
Proposition G.37	`M2Short.two_lt_ratio`, `SmallerShortData.two_lt_ratio`, `SmallerShortData.lr_cap_env`
Proposition G.38	`M2Short.ratio_lt_thirteen_sixths`, `SmallerShortData.ratio_lt_thirteen_sixths`
Proposition G.39	`M2Short.displacement_lt`, `SmallerShortData.displacement_lt`
Proposition G.40	`M2Short.upper_order_casimir_floor`, `SmallerShortData.upper_order_casimir_floor`, `SmallerShortData.cas_cap_env`
Proposition G.41	`M2Short.shear_neg`, `SmallerShortData.shear_neg`, `shr_residual_neg`, `shr_shear_sign`, `shr_norm_mono`, `shr_small_shape_factor`, `ss_A2_pos`
Proposition G.42	`M2Short.octagonCost_lt_completion`, `SmallerShortData.octagonCost_lt_completion`, `SmallerShortData.cost_angle`, `SmallerShortData.cost_continuation`, `cost_Fc_gt`, `octagonCost_lt_score`
Lemma G.43	`ss_first_contact`
Lemma G.44	`ss_gap_ratio_bounds`, `ss_gap_ratio_tilted`
Lemma G.45	`ss_reverseGram_mono`, `ss_rev_centre_scale`
Proposition G.46	`lgr_central_tau_two`, `lgr_central_duration`, `lgr_outer_tests`
Proposition G.47	`lr_central_face`, `lr_outer_face`

Result	Lean declarations
Proposition G.48	`ur_central_sign`, `ur_outer_sign_small`, `ur_outer_sign_large`
Proposition G.49	`dsp_central_alpha_proof`, `dsp_outer_face_proof`
Proposition G.50	`cas_floor_box`
(G.35)	`s6_norm_gt_variance_floor`, `cost_u_bounds`
Proposition G.51	`s6_final_score_antitone_displacement`, `s6_final_score_concave_R`, `s6_final_score_gt`
Appendices H and I	
Definition H.1	`Term.Cycle`, `Term.Cycle.rawCost`, `Term.rawArc`
Lemma H.2	`NormalPeriod.termCycle`, `NormalPeriod.cost_eq_rawCost`
Proposition H.3	`Term.Cycle.cycle_one_false`, `Term.Cycle.cycle_two_false`, `Term.Cycle.cycle_three_false`, `Term.Cycle.cycle_four_forward`, `Term.Cycle.cycle_five`
Lemma H.4	`Term.four_closure`
Lemma H.5	`Term.Cycle.four_param`
Theorem H.6	`Term.fourPiece_density_ge`, `Term.fourPiece_density_eq`, `ra_four_density`
Corollary H.7	`Term.Cycle.four_rawCost_ge`, `Term.four_equality_data`
Lemma H.8	`Term.five_closure`, `Term.Cycle.five_param`
Lemma H.9	`Term.Cycle.five_param`
Theorem H.10	`Term.fivePiece_density_gt`
Theorem 5.1	`terminal_cost`, `Term.rawCost_ge_of_le_five`
Proposition H.11	`terminal_equality`, `Term.four_of_rawCost_eq`, `Term.octagon_identification`
Theorem 5.2	`minimizer_cost_ge`
Lemma I.1	`eq_worstDisk_minimizer_realization`
Lemma I.2	`eq_frame_twist_of_contact`
Lemma I.3	`eq_disk_eq_of_frontier_eq`
Theorem 5.3	`worstDisk_eq_linear_image`
Proposition 5.4	`bridge_greatestPackingDensity_linear_image`, `bridge_greatestPackingDensity_translate`, `greatestPackingDensity_linear_image_octagon`
Proposition I.4	`cls_image_octagon`, `cls_exists_affine_iff`
Lemma H.12	`ra_half_sub_inv_le_log`
Lemma H.13	`ra_weight_monotoneOn`, `ra_weight_le`
Lemma H.14	`ra_fiveH_strictConcaveOn`, `ra_fiveH_circle_le`, `ra_fiveH_circle_eq_iff`
Lemma H.15	`ra_five_envelope`, `Term.fiveEnvelope`
Lemma H.16	a private lemma of `Term.fivePiece_density_gt`

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